Class 9
Physics
(Solutions)

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1. VECTORS
Solutions

LEVEL-I

1. Correct option: (B)
Scalars like temperature are the quantities that have only magnitude and no direction.
Therefore, the physical quantities which have only magnitude but no direction are called scalars.

2. Correct option: (C)
Vectors like force are the quantities that have both magnitude and direction.
Therefore, the physical quantity which are expressed in magnitude as well as direction are Called vectors.

3. Correct option: (D)
Vectors can be added, subtracted, multiplied by laws of vector combination.
Therefore, vectors cannot be added, subtracted and multiplied by ordinary laws of algebra.

4. Correct option: (A)
The magnitude of a vector is a positive number or zero. It doesn't have any sense of direction attached to it.
Therefore, the magnitude of a vector is a scalar.

5. Correct option: (B)
We know,
Force is a vector quantity.
Mass is a scalar quantity.
Velocity is a vector quantity.
Displacement is a vector quantity.
Therefore, the scalar quantity out of the given option is mass.

6. Correct option: (C)
We know, null vector is defined as having the magnitude of zero.
Therefore, a vector of zero magnitude is called null vector.

7. Correct option: (A)
We know, all non zero vectors are called proper vectors.
Therefore, $|\vec{A}| \neq 0$ is correct for proper vector.

8. Correct option: (D)
We know,
Force is a vector quantity.
Mass is a scalar quantity.
Distance is a scalar quantity.
Speed is a scalar quantity.
Work is a scalar quantity.
Energy is a scalar quantity.
Velocity is a vector quantity.
Displacement is a vector quantity.
Therefore, the pair containing a scalar and a vector quantity is speed and velocity.

9. Correct option: (C)
The negative of a vector is another vector having the same magnitude but opposite direction.
In option (A), both vectors point in same direction
In option (B), the vectors have different magnitude
In option (C), the vectors have same magnitude and opposite orientation
In option (D), the vectors have different magnitude
Therefore, the correct representative of negative vector for each other is shown in option c.

10. Correct option: (D)

Scalars are the quantities that have only magnitude and no direction.

We see 18 m due East, 20 m due South, 25 m due North all has a magnitude and direction both. But 40 m has only magnitude and no direction associated to it.

Therefore, the scalar quantity from the given option is 40 m.

11. Correct option: (C)

Given, $\vec{A} = 3\hat{i} - 12\hat{j} + 4\hat{k}$

Therefore, the magnitude of the vector is :

$|\vec{A}| = \sqrt{3^2 + (-12)^2 + 4^2}$

$|\vec{A}| = \sqrt{9 + 144 + 16}$

$|\vec{A}| = \sqrt{169}$

$|\vec{A}| = 13$

Thus, the magnitude of the vector $\vec{A} = 3\hat{i} - 12\hat{j} + 4\hat{k}$ is 13 units.

12. Correct option: (D)

Given, $0.5\hat{i} - 0.8\hat{j} + P\hat{k}$ is a unit vector.

We know, the magnitude of unit vector is unity.

$1 = \sqrt{0.5^2 + (-0.8)^2 + P^2}$

$1 = \sqrt{0.25 + 0.64 + P^2}$

$1 = \sqrt{0.89 + P^2}$

Solving for P , we get

$1 = 0.89 + P^2$

$P = \sqrt{1 - 0.89}$

$P = \sqrt{0.11}$

Therefore, if $0.5\hat{i} - 0.8\hat{j} + P\hat{k}$ is a unit vector, then $P = \sqrt{0.11}$.

13. Correct option: (A)

Given, $\vec{A} = 10\hat{i} + 10\hat{j}$

Therefore, the magnitude of the vector is :

$|\vec{A}| = \sqrt{10^2 + 10^2}$

$|\vec{A}| = \sqrt{2(100)}$

$|\vec{A}| = 10\sqrt{2}$

Therefore, the magnitude of the vector $\vec{A} = 10\hat{i} + 10\hat{j}$ is $10\sqrt{2}$ units.

14. Correct option: (B)

Given, $\vec{P} = 2\hat{i} - 5\hat{j} - 3\hat{k}, \vec{Q} = 6\hat{i} + 6\hat{j} + 4\hat{k}$

Therefore, the vector $\overrightarrow{PQ}$ is :

$\overrightarrow{PQ} = \vec{Q} - \vec{P}$

$\overrightarrow{PQ} = (6\hat{i} + 6\hat{j} + 4\hat{k}) - (2\hat{i} - 5\hat{j} - 3\hat{k})$

$\overrightarrow{PQ} = 4\hat{i} + 11\hat{j} + 7\hat{k}$

Therefore, the length of the vector $\overrightarrow{PQ} = 4\hat{i} + 11\hat{j} + 7\hat{k}$ is:

$|\overrightarrow{PQ}|=\sqrt{4^2+11^2+7^2}$

$|\overrightarrow{PQ}|=\sqrt{16+121+49}$

$|\overrightarrow{PQ}|=\sqrt{186}$

Therefore, if $\vec{P}=2\hat{i}+5\hat{j}-3\hat{k},\vec{Q}=6\hat{i}+6\hat{j}+4\hat{k}$, then $1=\sqrt{186}$

15. Correct option: (A)

Given, $|2\hat{i}+3\hat{j}+C\hat{k}|=7$

Now, using the magnitude of the vector, we get:

$|2\hat{i}+3\hat{j}+C\hat{k}|=7$

$7=\sqrt{2^2+3^2+C^2}$

$49=4+9+C^2$

$C=\sqrt{49-13}$

$C=6$

Therefore, if $|2\hat{i}+3\hat{j}+C\hat{k}|=7$, then $C=6$

16. Correct option: (C)

Given, $4\hat{i}-y\hat{j}+6\hat{k}$ is parallel to $8\hat{i}+3\hat{j}+12\hat{k}$

Now, we know, if two vectors $\vec{a},\vec{b}$ are parallel, we have:

$\vec{a}=c\vec{b}$, for some constant c.

Here, since $4\hat{i}-y\hat{j}+6\hat{k}$ is parallel to $8\hat{i}+3\hat{j}+12\hat{k}$, we should have:

$4\hat{i}-y\hat{j}+6\hat{k}=c(8\hat{i}+3\hat{j}+12\hat{k})$

Equating the right and left hand side, we get :

$4=8c$

$-y=3c$

$6=12c$ Therefore,

$c=1/2$

$y=-3/2$

Therefore, if $4\hat{i}-y\hat{j}+6\hat{k}$ is parallel to $8\hat{i}+3\hat{j}+12\hat{k}$, then $y=-3/2$.

17. Correct option: (D)

Given, $\vec{A}=2\hat{i}+3\hat{j}-4\hat{k},\vec{B}=6\hat{i}+3\hat{j}-\hat{k}$

Now, the sum of $\vec{A},\vec{B}$ is given by:

$\vec{A}+\vec{B}=2\hat{i}+3\hat{j}-4\hat{k}+6\hat{i}+3\hat{j}-\hat{k}$

$\vec{A}+\vec{B}=8\hat{i}+6\hat{j}-5\hat{k}$

Therefore, if $\vec{A}=2\hat{i}+3\hat{j}-4\hat{k},\vec{B}=6\hat{i}+3\hat{j}-\hat{k}$, then $\vec{A}+\vec{B}=8\hat{i}+6\hat{j}-5\hat{k}$.

18. Correct option: (C)

We know, the unit vector $\hat{n}$ along $\vec{a}$ is given by:

$$\hat{n}=\frac{\vec{a}}{|\vec{a}|}$$

Given, $\vec{a}=\hat{i}+\hat{j}$

Now, the unit vector along $\vec{a}=\hat{i}+\hat{j}$ is given by:

$$\hat{n} = \frac{1}{|\hat{i} + \hat{j}|}$$

$$\hat{n} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$$

Therefore, the unit vector along $\vec{a} = \hat{i} + \hat{j}$ is $\hat{n} = \frac{\hat{i} + \hat{j}}{\sqrt{2}}$.

19. Correct option: (B)

We know a unit vector along x axis is $\hat{i}$.

Now, given $\vec{A} + \vec{B}$ is a unit vector along the x axis where $\vec{A} = \hat{i} - \hat{j} + \hat{k}$

Therefore,

$\vec{A} + \vec{B} = \hat{\imath}$

$B = \hat{\imath} - (\hat{\imath} - \hat{\jmath} + \hat{k})$

$B = \hat{\jmath} - \hat{k}$

Therefore, if $\vec{A} + \vec{B}$ is a unit vector along the x axis where $\vec{A} = \hat{i} - \hat{j} + \hat{k}$, then $B = \hat{j} - \hat{k}$.

20. Correct option: (A)

Given, $\vec{A} = 3\hat{i} + 5\hat{j} - \hat{k}$

We know, a vector $\vec{B}$ of magnitude k in the direction of vector $\vec{A}$ is given by:

$\vec{B} = k\hat{n}$, where $\hat{n} = \frac{\vec{A}}{|\vec{A}|}$ is a unit vector along $\vec{A}$.

Here, the unit vector along $\vec{A} = 3\hat{i} + 5\hat{j} - \hat{k}$ is :

$$\hat{n} = \frac{3\hat{i} + 5\hat{j} - \hat{k}}{|3\hat{i} + 5\hat{j} - \hat{k}|}$$

$$\hat{n} = \frac{3\hat{i} + 5\hat{j} - \hat{k}}{|\sqrt{3^2 + 5^2 + 1}|}$$

$$\hat{n} = \frac{3\hat{i} + 5\hat{j} - \hat{k}}{|\sqrt{9 + 25 + 1}|}$$

$$\hat{n} = \frac{3\hat{i} + 5\hat{j} - \hat{k}}{|\sqrt{35}|}$$

Now, the vector along $\vec{A} = 3\hat{i} + 5\hat{j} - \hat{k}$ with magnitude 5 units is :

$$\vec{B} = 5\left(\frac{3\hat{\imath} + 5\hat{\jmath} - \hat{k}}{|\sqrt{35}|}\right)$$

$$\vec{B} = \left(\frac{15\hat{\imath} + 25\hat{\jmath} - 5\hat{k}}{|\sqrt{35}|}\right)$$

Therefore, the vector along $\vec{A} = 3\hat{i} + 5\hat{j} - \hat{k}$ with magnitude 5 units is $\left(\frac{15\hat{\imath}+25\hat{\jmath}-5\hat{k}}{|\sqrt{35}|}\right)$.

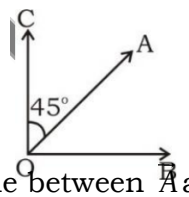
21. Correct option: (B)

Given, the angle between $\vec{A}$ and $\vec{C}$ is 45°

From the figure, the angle between $\vec{B}$ and $\vec{C}$ is 90°

Therefore, the angle between $\vec{A}$ and $\vec{B}$ is $90^\circ - 45^\circ = 45^\circ$.

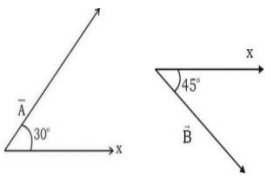
Therefore, if the angle between $\vec{A}$ and $\vec{C}$ is 45°, then the angle between $\vec{A}$ and $\vec{B}$ is 45°.



22. Correct option: (A)

From the figure, we see the angle between $\vec{A}$ and x axis is 30°

And, the angle between x axis and $\vec{B}$ is 45°



Hence, the angle between $\vec{A}$ and $\vec{B}$ can be found by putting the head of $\vec{B}$ to the head of $\vec{A}$.

Therefore, the angle between $\vec{A}$ and $\vec{B} = 45^\circ + 30^\circ = 75^\circ$

23. Correct option: (C)

Given, $2\hat{i} + 3\hat{j} + 4\hat{k}$

Therefore, the vector in the y-z plane is $3\hat{j} + 4\hat{k}$.

Hence the length of the vector in y-z plane is:

$\sqrt{3^2 + 4^2} = \sqrt{9 + 16}$

$\sqrt{3^2 + 4^2} = \sqrt{25}$

$\sqrt{3^2 + 4^2} = 5$

Therefore, if $2\hat{i} + 3\hat{j} + 4\hat{k}$, the length of the vector in y-z plane is 5 units.

24. Correct option: (A)

Given, the vector $\overrightarrow{AB}$ has initial point $A(1, 2, -2)$ and terminal point at $B(3, -2, 4)$.

Therefore, $\overrightarrow{AB}$ = Terminal point – Initial point

$\overrightarrow{AB} = 3\hat{i} - 2\hat{j} + 4\hat{k} - (\hat{i} + 2\hat{j} - 2\hat{k})$

$\overrightarrow{AB} = 2\hat{i} - 4\hat{j} + 6\hat{k}$

Further, the magnitude of $\overrightarrow{AB}$ is:

$|\overrightarrow{AB}| = \sqrt{2^2 + (-4)^2 + 6^2}$

$|\overrightarrow{AB}| = \sqrt{4 + 16 + 36}$

$|\overrightarrow{AB}| = \sqrt{56}$

Therefore, if the vector $\overrightarrow{AB}$ has initial point $A(1, 2, -2)$ and terminal point at $B(3, -2, 4)$, then $\overrightarrow{AB} = 2\hat{i} - 4\hat{j} + 6\hat{k}$ and the magnitude of $\overrightarrow{AB}$ is $\sqrt{56}$.

25. Correct option: (C)

Given, $2\hat{i} + 3\hat{j} + 8\hat{k}$

Now, we know, if two vectors $\vec{a}, \vec{b}$ are parallel, we have:

$\vec{a} = c\vec{b}$, for some constant c.

Here, for option c, we have $4\hat{i} + 6\hat{j} + 16\hat{k}$.

Also, $4\hat{i} + 6\hat{j} + 16\hat{k} = 2(2\hat{i} + 3\hat{j} + 8\hat{k})$

Therefore, $4\hat{i} + 6\hat{j} + 16\hat{k}$ is parallel to $2\hat{i} + 3\hat{j} + 8\hat{k}$.

LEVEL-II

1. Correct option: (A)

Density can be described by a single number.

Therefore, density is a scalar quantity.

2. Correct option: (C)

If two vectors are in the same plane, they are called coplanar vectors, otherwise they are called as non-coplanar vectors.

Therefore, vectors acting in different planes are called coplanar vectors.

3. Correct option: (B)

Unit vector by definition have magnitude one.

Therefore, vector whose magnitude is one is called is called unit vector.

4. Correct option: (A)

We know, the negative of a vector is another vector having the same magnitude but opposite direction.

Therefore, a vector having magnitude 5 and opposite to the vector $3\hat{i}-4\hat{j}$ is $-(3\hat{i}-4\hat{j})=-3\hat{i}+4\hat{j}$

5. Correct option: (D)

Given, $\frac{1}{2}\hat{\imath}-\frac{1}{2}\hat{\jmath}+a\hat{k}$is a unit vector.

We know, the magnitude of unit vector is unity. Hence,

$1=\sqrt{0.5^2+(-0.5)^2+a^2}$

$1=\sqrt{0.25+0.25+a^2}$

$1=\sqrt{0.50+a^2}$

Solving for a , we get

$1=0.50+a^2$

$a=\sqrt{1-0.50}$

$a=\sqrt{0.50}$

$a=\sqrt{\frac{1}{2}}$

Therefore, if $\frac{1}{2}\hat{\imath}-\frac{1}{2}\hat{\jmath}+a\hat{k}$is a unit vector, then $a=\sqrt{\frac{1}{2}}$.

6. Correct options: (A), (B), (C)

Solution:

We know,

length is a scalar

Time is a scalar

Temperature is a scalar

Displacement is a vector

Therefore, out of the given options, scalar quantities are length, time and temperature.

7. Correct option: (A), (B), (C)

We know,

Mass is a scalar with unit kg.

Time is a scalar with unit sec.

Density is a scalar with unit kg m^{-3}

Force is a vector quantity with unit Newton.

Therefore, the units that could be associated with a scalar quantity are kg, sec and kg m^{-3}.

8. Correct options: (B), (C), (D)

We know, two vectors are equal if they have the same magnitude and the same direction.

Therefore, two vectors of same physical quantity are unequal if they have different magnitudes but same direction, or if they have same magnitude but different directions, or they have different magnitudes and different directions.

9. Correct options: (A), (C), (D)

We know, a null vector has zero magnitude.

Now, acceleration of a body moving in a straight line with uniform velocity is zero, hence it is a null vector.

Since velocity vector of a body moving in a circular path with uniform speed has some non zero magnitude and changes direction continuously, it is not a null vector.

Since, position vector of the origin of the rectangular co-ordinate system is $0\hat{i}+0\hat{j}+0\hat{k}$, this is also a null vector.

Further, displacement vector of a stationary body, being equal to final and initial displacement is also a null vector.

Therefore, Acceleration of a body moving in a straight line with uniform velocity, position vector of the origin of the rectangular co-ordinate system and displacement vector of a stationary body are null vectors.

10. Correct options: (A),(B), (C), (D)
We know,
Force is a vector.
Torque is a vector.
Acceleration is a vector.
Velocity is a vector.

11. Correct option: (A)
Given, $0.3\hat{i}+0.4\hat{j}+a\hat{k}$ is a unit vector.
We know, the magnitude of unit vector is unity. Hence,
$1=\sqrt{0.3^2+0.4^2+a^2}$
$1=\sqrt{0.9+0.16+a^2}$
$1=\sqrt{0.25+a^2}$
Solving for a , we get
$1=0.25+a^2$
$a=\sqrt{1-0.25}$
$a=\sqrt{0.75}$
Therefore, if $0.3\hat{i}+0.4\hat{j}+a\hat{k}$ is a unit vector, then $a=\sqrt{0.75}$.

12. Correct option: (B)
Given, $2\hat{i}+3\hat{j}+\hat{k}$
We know, a unit vector $\hat{n}$ along $\vec{a}$ is given by: $\hat{n}=\frac{\vec{a}}{|\vec{a}|}$.
Hence, unit vector along $2\hat{i}+3\hat{j}+\hat{k}$ is given by:
$\hat{n}=\frac{2\hat{\imath}+3\hat{\jmath}+\hat{k}}{|2\hat{\imath}+3\hat{\jmath}+\hat{k}|}$
$\hat{n}=\frac{2\hat{\imath}+3\hat{\jmath}+\hat{k}}{|\sqrt{2^2+3^2+1^2}|}$
$\hat{n}=\frac{2\hat{\imath}+3\hat{\jmath}+\hat{k}}{|\sqrt{4+9+1}|}$
$\hat{n}=\frac{2\hat{\imath}+3\hat{\jmath}+\hat{k}}{|\sqrt{14}|}$
Therefore, the unit vector along $2\hat{i}+3\hat{j}+\hat{k}$ is $\frac{2\hat{i}+3\hat{j}+\hat{k}}{|\sqrt{14}|}$.

13. Correct option: (A)
Given, $\vec{A}=2\hat{i}-\hat{j}+3\hat{k}, \vec{B}=3\hat{i}-2\hat{j}-2\hat{k}$
We know, a unit vector $\hat{n}$ along $\vec{a}$ is given by: $\hat{n}=\frac{\vec{a}}{|\vec{a}|}$.
Hence, unit vector along $\vec{A}+\vec{B}$ is given by:
$\hat{n}=\frac{2\hat{\imath}-\hat{\jmath}+3\hat{k}+3\hat{\imath}-2\hat{\jmath}-2\hat{k}}{|2\hat{\imath}-\hat{\jmath}+3\hat{k}+3\hat{\imath}-2\hat{\jmath}-2\hat{k}|}$
$\hat{n}=\frac{5\hat{\imath}-3\hat{\jmath}+\hat{k}}{|5\hat{\imath}-3\hat{\jmath}+\hat{k}|}$
$\hat{n}=\frac{5\hat{\imath}-3\hat{\jmath}+\hat{k}}{|\sqrt{25+9+1}|}$

$\hat{n} = \frac{5\hat{i} - 3\hat{j} + \hat{k}}{|\sqrt{35}|}$

Therefore, if $\vec{A} = 2\hat{i} - \hat{j} + 3\hat{k}, \vec{B} = 3\hat{i} - 2\hat{j} - 2\hat{k}$ the unit vector along $\vec{A} + \vec{B}$ is $\frac{5\hat{i} - 3\hat{j} + \hat{k}}{|\sqrt{35}|}$.

14. Correct option: (C)

A particle is initially at point A(2,4,6)m and moves finally to the point B(3,2,-3)m

We know, if $\vec{r_1} = x_1\hat{i} + y_1\hat{j}, \vec{r_2} = x_2\hat{i} + y_2\hat{j}$ are two position vectors, then displacement vector is the difference of two position vectors:

$\Delta\vec{r} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j}$

Therefore, here the displacement vector is:

$\Delta\vec{r} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

$\Delta\vec{r} = (3-2)\hat{i} + (2-4)\hat{j} + (-3-6)\hat{k}$

$\Delta\vec{r} = \hat{i} - 2\hat{j} - 9\hat{k}$

Therefore, if a particle is initially at point A(2,4,6)m and moves finally to the point B(3,2,-3)m, then the displacement vector is $\hat{i} - 2\hat{j} - 9\hat{k}$.

15. Correct option: (C)

Given, position vector changes from $\hat{i} + 2\hat{j} + 3\hat{k}$ to $\vec{B} = 5\hat{i} + 4\hat{j} + 2\hat{k}$.

We know, if $\vec{r_1} = x_1\hat{i} + y_1\hat{j}, \vec{r_2} = x_2\hat{i} + y_2\hat{j}$ are two position vectors, then displacement vector is the difference of two position vectors:

$\Delta\vec{r} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j}$

Therefore, here the displacement vector is:

$\Delta\vec{r} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$

$\Delta\vec{r} = (5-1)\hat{i} + (4-2)\hat{j} + (2-3)\hat{k}$

$\Delta\vec{r} = 4\hat{i} + 2\hat{j} - \hat{k}$

Therefore, if position vector changes from $\hat{i} + 2\hat{j} + 3\hat{k}$ to $\vec{B} = 5\hat{i} + 4\hat{j} + 2\hat{k}$, then the displacement vector is $4\hat{i} + 2\hat{j} - \hat{k}$.

16. Correct option: (D)

Given, particle moves from $2\hat{i} + 3\hat{j} + 5\hat{k}$ to $\vec{B} = \hat{i} + \hat{j} + 3\hat{k}$.

We know, if $\vec{r_1} = x_1\hat{i} + y_1\hat{j}, \vec{r_2} = x_2\hat{i} + y_2\hat{j}$ are two position vectors, then magnitude of displacement vector is given by:

$|\Delta\vec{r}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)}$

Therefore, the magnitude of displacement is :

$|\Delta\vec{r}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$

$|\Delta\vec{r}| = \sqrt{(1-2)^2 + (1-3) + (3-5)}$

$|\Delta\vec{r}| = \sqrt{1 + 4 + 4}$

$|\Delta\vec{r}| = \sqrt{9}$

$|\Delta\vec{r}| = 3$

Therefore, if particle moves from $2\hat{i} + 3\hat{j} + 5\hat{k}$ to $\vec{B} = \hat{i} + \hat{j} + 3\hat{k}$., then the magnitude of displacement vector is 3m.

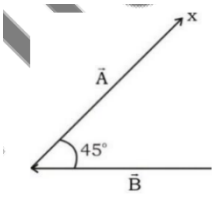
17. Correct option: (A)
We know, a vector don't changes if it slides parallel to itself.
And a vector whose magnitude is not zero is called a proper vector.
Therefore, both the statements are correct.

18. Correct option: (D)
We know, a vector whose magnitude 6 units along negative y axis is $-6\hat{j}$
Also, magnitude of the vector having initial point $P(x_1, y_1, z_1)$ and terminal point $Q(x_2, y_2, z_2)$ is $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$
Hence, statement 1 is false, but statement 2 is correct.

19. Correct match: (A) -> Q), (B) ->R); (C) -> S); (D)-> P)
We know,
Density is a scalar
Distance, work done possess magnitude
Force, displacement possess direction
Velocity is a scalar
Therefore, the correct match is (A) -> Q), (B) ->R); (C) -> S); (D)-> P)

20. Correct match: (A) -> T), (B) ->S); (C) -> Q); (D)-> R)
Given, $2\hat{i} + 3\hat{j} + 5\hat{k}$
(A) Therefore, the vector in the x-y plane is $2\hat{i} + 3\hat{j}$.
Hence the magnitude of the vector in x-y plane is:
$\sqrt{2^2 + 3^2}$
(B) Therefore, the vector in the x-z plane is $2\hat{i} + 5\hat{k}$.
Hence the magnitude of the vector in x-z plane is:
$\sqrt{2^2 + 5^2}$
(C) Therefore, the vector in the y-z plane is $3\hat{j} + 5\hat{k}$.
Hence the magnitude of the vector in y-z plane is:
$\sqrt{3^2 + 5^2}$
(D) Therefore, the vector in the xyz plane is $2\hat{i} + 3\hat{j} + 5\hat{k}$.
Hence the magnitude of the vector in y-z plane is:
$\sqrt{2^2 + 3^2 + 5^2}$
Therefore, the correct match is (A) -> T), (B) ->S); (C) -> Q); (D)-> R)

21. Correct number: 1
We see from the figure, the angle between $\vec{A}$ and $\vec{B}$ is 45°
Comparing this with $__ \times 45^\circ$, we see the required number is 1.



22. Correct number: 1
We know, the magnitude of a unit vector is unity.
Therefore, if a vector $\vec{A}$ is a unit vector, then $|\vec{A}| = 1$

Resolution Of Vectors

LEVEL-I

1. Correct option: (C)
We know, the direction cosines of a vector $\cos\alpha, \cos\beta, \cos\gamma$ satisfies the identity :
$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$
Therefore, if $\cos\alpha, \cos\beta, \cos\gamma$ are the direction cosines of a vector, then $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$.

2. Correct option: (B)

We know, the component of a vector $\vec{A}$ in the direction of another vector $\vec{B}$ is given by: $A\cos\theta$, where θ is the angle between $\vec{A}$ and $\vec{B}$.

Given, two vectors are perpendicular to each other. Thus, the angle between them is 90°. Therefore, the component of one vector in the direction of another vector is :

$A\cos 90^\circ = 0$

Therefore, if two vectors are perpendicular to each other, then the component of one vector in the direction of another vector is 0.

3. Correct option: (C)

We know, the direction cosines of a vector $\cos\alpha, \cos\beta, \cos\gamma$ satisfies the identity :

$\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$

Now, using $\sin^2\alpha + \cos^2\alpha = 1$, we get

$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 1 - \cos^2\alpha + 1 - \cos^2\beta + 1 - \cos^2\gamma$

$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 3 - \cos^2\alpha - \cos^2\beta - \cos^2\gamma$

Now using, $\cos^2\alpha + \cos^2\beta + \cos^2\gamma = 1$, we get

$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 3 - 1$

$\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$

Therefore, if a vector $\vec{P}$ makes angle α, β, γ with x, y and z axes respectively, then $\sin^2\alpha + \sin^2\beta + \sin^2\gamma = 2$.

4. Correct option: (A)

We know, the rectangular components of a vector $\vec{P}$ are given by:

$P\cos\theta$ and $P\sin\theta$, where θ is the angle between the vector $\vec{P}$ and the x axis.

Given, one of the rectangular components of a force P is $\frac{3P}{5}$.

Therefore,

$P\cos\theta = \frac{3P}{5}$

$\cos\theta = \frac{3}{5}$

Using $\sin^2\alpha + \cos^2\alpha = 1$, we get

$\sin^2\theta + \cos^2\theta = 1$

$\sin^2\theta = 1 - \cos^2\theta$

$\sin\theta = \sqrt{1 - \cos^2\theta}$

$\sin\theta = \sqrt{1 - \left(\frac{3}{5}\right)^2}$

$\sin\theta = \sqrt{\frac{16}{25}}$

$\sin\theta = \frac{4}{5}$

Therefore, the other rectangular component of the vector is :

$P\sin\theta = \frac{4P}{5}$

Therefore, if , one of the rectangular components of a force P is $\frac{3P}{5}$, then the other rectangular component of the vector is $\frac{4P}{5}$.

5. Correct option: (D)

We know, the horizontal and vertical components of a vector $\vec{P}$ are given by:
$P\cos\theta$ and $P\sin\theta$, where θ is the angle between the vector $\vec{P}$ and the x axis.
Given, Horizontal and vertical components are equal.
Therefore,

$$P\cos\theta = P\sin\theta$$

$$\tan\theta = 1$$

$$\theta = 45^\circ$$

Therefore, the angle with the vertical is $90^\circ - 45^\circ = 45^\circ$.
Therefore, if horizontal and vertical components are equal, then angle made by the vector to
the vertical is 45°.

6. Correct option: (B)

We know, the rectangular components of a vector $\vec{P}$ are given by:
$P\cos\theta$ and $P\sin\theta$, where θ is the angle between the vector $\vec{P}$ and the x axis.
Given, one of the two rectangular components of a force $\vec{F}$ is 25 N and it makes an angle of 60° with the force.
Therefore,

$$F\cos 60^\circ = 25$$

$$F\frac{1}{2} = 25$$

$$F = 50$$

Now, the other component of the force is :

$$F\sin 60^\circ = 50\frac{\sqrt{3}}{2}$$

$$F\sin 60^\circ = 25\sqrt{3}$$

Therefore, if one of the two rectangular components of a force is 25 N and it makes an angle of 60°
with force, then the magnitude of the other component is $25\sqrt{3}$ N.

7. Correct option: (C)

We know, the horizontal and vertical components of a vector $\vec{P}$ are given by:
$P\cos\theta$ and $P\sin\theta$, where θ is the angle between the vector $\vec{P}$ and the x axis.
Given, the force with magnitude 50 N acts at an angle of 30° with the x-axis, therefore the components of the force are given by:

$$F\,cos\,30^\circ = 50\frac{\sqrt{3}}{2}$$

$$F\,cos\,30^\circ = 25\sqrt{3}$$

$$F\,sin\,30^\circ = 50\frac{1}{2}$$

$$F\,sin\,30^\circ = 25$$

Therefore, the components of force of 50 N acting at an angle of 30° with x-axis is $25\sqrt{3}$ N, 25 N.

8. Correct option: (D)

We know, if a vector $\vec{P}$ makes angle α, β, γ with x, y and z axes respectively, then direction cosines of the vector are $\cos\alpha, \cos\beta, \cos\gamma$.

Now, $\hat{j}+\hat{k}$ makes an angle of 45° with x, y and z axes respectively.

Hence, the direction cosines of $\hat{j}+\hat{k}$ are:

$\cos\alpha = \cos(45^\circ)$

$\cos\alpha = \frac{1}{\sqrt{2}}$

$\cos\beta = \cos(45^\circ)$

$\cos\beta = \frac{1}{\sqrt{2}}$

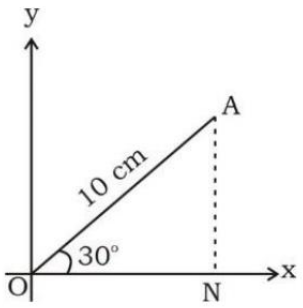
$\cos\gamma = \cos(45^\circ)$

$\cos\gamma = \frac{1}{\sqrt{2}}$

Therefore, direction cosines of $\hat{j}+\hat{k}$ is $\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}$.

9. Correct option: (A)

We know, the horizontal and vertical components of a vector $\vec{P}$ are given by:
$P\cos\theta$ and $P\sin\theta$, where θ is the angle between the vector $\vec{P}$ and the x axis.
From the given figure in question, we see the magnitude of the vector $\overrightarrow{OA}$ is 10 cm, and it makes an angle 30° with the axis.
Therefore,



$\overrightarrow{OA} = OA(\cos\theta\hat{i} + \sin\theta\hat{j})$

$\overrightarrow{OA} = 10(\cos 30^\circ \hat{i} + \sin 30^\circ \hat{j})$

$\overrightarrow{OA} = 10(\frac{\sqrt{3}}{2}\hat{i} + \frac{1}{2}\hat{j})$

$\overrightarrow{OA} = 5\sqrt{3}\hat{i} + 5\hat{j}$

Therefore, if OA = 10 cm, the vector $\overrightarrow{OA}$ written with the help of $\hat{i}$ and $\hat{j}$ is $\overrightarrow{OA} = 5\sqrt{3}\hat{i} + 5\hat{j}$.

10. Correct option: (B)

We know, the rectangular components of a vector $\vec{P}$ are given by:
$P\cos\theta$ and $P\sin\theta$, where θ is the angle between the vector $\vec{P}$ and the x axis.
Given, magnitude of the force is 20 N, and the first component makes an angle of 30° with the force.
Therefore the components of the force are:

$F\cos 30^\circ = 20\frac{\sqrt{3}}{2}$

$F\cos 30^\circ = 10\sqrt{3}$

$F\sin 30^\circ = 20\frac{1}{2}$

$F\sin 30^\circ = 10$

Therefore, if a force of 20 N is resolved into the perpendicular components such that the first component makes 30° with the force, the magnitudes of the components are $10\sqrt{3}, 10$ N.

11. Correct option: (C)

If two forces of equal magnitude but applied in opposite direction, then the body will remain in equilibrium, as the net force is zero.

Therefore, the minimum number of forces of equal magnitude in the same plane that can keep a particle in equilibrium is 2.

12. Correct option: (A)

Given, $\vec{p} = -\hat{i} + \hat{j} - \hat{k}$ and $\vec{q} = 5\hat{i} + 3\hat{j} - \hat{k}$ and $3\vec{p} + 2\vec{q} - \vec{r}$.

Now,

$\vec{r} = 3\vec{p} + 2\vec{q}$

$\vec{r} = 3(-\hat{i} + \hat{j} - \hat{k}) + 2(5\hat{i} + 3\hat{j} - \hat{k})$

$\vec{r} = 7\hat{i} + 9\hat{j} - 5\hat{k}$

Therefore, if $\vec{p} = -\hat{i} + \hat{j} - \hat{k}$ and $\vec{q} = 5\hat{i} + 3\hat{j} - \hat{k}$ and $3\vec{p} + 2\vec{q} - \vec{r}$, then $\vec{r} = 7\hat{i} + 9\hat{j} - 5\hat{k}$.

13. Correct option: (D)

We know,

Force is a vector

Velocity is a vector

Displacement is a vector

Current is a scalar

Since scalars don't follow law of vector addition, among the given physical quantities, current do not follow law's of vector addition.

14. Correct option: (A)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

15. Correct option: (B)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Given, two equal forces have their resultant equal to either.

Therefore,

$A = \sqrt{A^2 + A^2 + 2A^2\cos\theta}$

$A^2 = 2A^2(1 + \cos\theta)$

$\frac{1}{2} = 1 + \cos\theta$

$-\frac{1}{2} = \cos\theta$

$\theta = 120^\circ$

Therefore, if two equal forces have their resultant equal to either, then they are inclined at 120°.

16. Correct option: (D)

If the vectors are in different planes, for each vector we have three components. For zero resultant, we would need each component of the resultant vector to be zero.

Therefore, we would need at least 4 such vectors, such that the three components of the resultant of three vectors is cancelled in each dimension by the components of the fourth vector.

Therefore, the minimum number of vectors having different planes which can be added to give

zero resultant is 4.

17. Correct option: (C)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Given, sum of two unit vectors is also a vector of unit magnitude.

Therefore,

$1 = \sqrt{1+1+2\cos\theta}$

$1 = 2 + 2\cos\theta$

$\cos\theta = -\frac{1}{2}$

$\theta = 120^\circ$

Therefore the angle between the two unit vectors is 120°.

If one of the unit vector is reversed in direction, the angle between this reversed unit vector and the other unit vector is $180^\circ - 120^\circ = 60^\circ$

Therefore, magnitude of the difference of the two unit vectors is:

$\sqrt{A^2 + B^2 + 2AB\cos\theta} = \sqrt{1^2 + 1^2 + 2(1)(1)\cos 60^\circ}$

$\sqrt{A^2 + B^2 + 2AB\cos\theta} = \sqrt{1^2 + 1^2 + 2(1)(1)\frac{1}{2}}$

$\sqrt{A^2 + B^2 + 2AB\cos\theta} = \sqrt{1+1+1}$

$\sqrt{A^2 + B^2 + 2AB\cos\theta} = \sqrt{3}$

Therefore, if the sum of two unit vectors is also a vector of unit magnitude, then the magnitude of the difference of the two unit vectors is $\sqrt{3}$.

18. Correct option: (B)

Given, $\vec{P} + \vec{Q} = \vec{R}$ and $\vec{P} - \vec{Q} = \vec{S}$

Now, squaring each of these equations, we get:

$(\vec{P} + \vec{Q}).(\vec{P} + \vec{Q}) = \vec{R}.\vec{R}$

$\vec{P}.\vec{P} + \vec{P}.\vec{Q} + \vec{Q}.\vec{P} + \vec{Q}.\vec{Q} = \vec{R}.\vec{R}$

$P^2 + 2\vec{P}.\vec{Q} + Q^2 = R^2$

and

$(\vec{P} - \vec{Q}).(\vec{P} - \vec{Q}) = \vec{S}.\vec{S}$

$\vec{P}.\vec{P} - \vec{P}.\vec{Q} - \vec{Q}.\vec{P} + \vec{Q}.\vec{Q} = S^2$

$P^2 - 2\vec{P}.\vec{Q} + Q^2 = S^2$

Now, adding these two equations, we get:

$P^2 + 2\vec{P}.\vec{Q} + Q^2 + P^2 - 2\vec{P}.\vec{Q} + Q^2 = R^2 + S^2$

$2P^2 + 2Q^2 = R^2 + S^2$

Therefore, if $\vec{P} + \vec{Q} = \vec{R}$ and $\vec{P} - \vec{Q} = \vec{S}$, then $R^2 + S^2 = 2P^2 + 2Q^2$

19. Correct option: (B)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Given, two vectors $\vec{a}, \vec{b}$ are such that $|\vec{a} + \vec{b}| = |\vec{a}| + |\vec{b}|$

Therefore,

$|\vec{a}|+|\vec{b}|=\sqrt{a^2+b^2+2ab\cos\theta}$

$(a+b)^2=a^2+b^2+2ab\cos\theta$

$a^2+b^2+2ab=a^2+b^2+2ab\cos\theta$

$1=\cos\theta$

$\theta=0$

Therefore, if two vectors $\vec{a},\vec{b}$ are such that $|\vec{a}+\vec{b}|=|\vec{a}|+|\vec{b}|$ then the angle between $\vec{a}$ and $\vec{b}$ is 0.

20. Correct option: (D)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}|=\sqrt{A^2+B^2+2AB\cos\theta}$

Given, three vectors $\vec{A},\vec{B},\vec{C}$ are such that $\vec{A}+\vec{B}=\vec{C}$ and $|\vec{A}|=|\vec{B}|=|\vec{C}|$

Therefore,

$|\vec{C}|=\sqrt{A^2+B^2+2AB\cos\theta}$

$C=\sqrt{C^2+C^2+2C^2\cos\theta}$

$C=C\sqrt{2(1+\cos\theta)}$

Using $\cos^2\theta=\dfrac{1+\cos 2\theta}{2}$, we get

$C=C\sqrt{2(2)\cos^2(\theta/2)}$

$1=2\cos(\theta/2)$

$\dfrac{1}{2}=\cos(60^\circ)$

$60^\circ=\dfrac{\theta}{2}$

$\theta=120^\circ$

Therefore, if three vectors $\vec{A},\vec{B},\vec{C}$ are such that $\vec{A}+\vec{B}=\vec{C}$ and $|\vec{A}|=|\vec{B}|=|\vec{C}|$, then the angle between $\vec{a}$ and $\vec{b}$ is 120°.

21. Correct option: (C)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}|=\sqrt{A^2+B^2+2AB\cos\theta}$

Given, two forces with equal magnitude F with angle 60°

Therefore, the resultant is

$|\vec{R}|=\sqrt{F^2+F^2+2F^2\cos 60^\circ}$

$|\vec{R}|=\sqrt{2F^2+2F^2\dfrac{1}{2}}$

$|\vec{R}|=\sqrt{3F^2}$

$|\vec{R}|=\sqrt{3}F$

Therefore, if there are two forces with equal magnitude F with angle 60°, then their resultant is $\sqrt{3}F$.

22. Correct option: (A)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Given, the greatest and least resultant of two forces acting at a point is 10 N and 6 N respectively.

We know, the magnitude of the greatest and least resultant of two forces $\vec{F_1}$ and $\vec{F_2}$ are given by $F_1 + F_2$ and $F_1 - F_2$, when the forces act in the same direction and in the opposite direction, respectively.

Now, $F_1 + F_2 = 10$ and $F_1 - F_2 = 6$

Adding these two equations, we get :

$F_1 + F_2 + F_1 - F_2 = 10 + 6$

$2F_1 = 16$

$F_1 = 8$

Subtracting the two equations, we get:

$F_1 + F_2 - F_1 + F_2 = 10 - 6$

$2F_2 = 4$

$F_2 = 2$

When each force is increased by 3N, the new magnitude of the forces are $F_1 = 11$ and $F_2 = 5$.

Since these forces now makes an angle of 90° with each other, therefore the resultant is :

$|\vec{R}| = \sqrt{11^2 + 5^2 + 2(11)(5)\cos(90^\circ)}$

$|\vec{R}| = \sqrt{121 + 25}$

$|\vec{R}| = \sqrt{146}$

Therefore, if the greatest and least resultant of two forces acting at a point is 10 N and 6 N respectively and if each force is increased by 3 N, then the resultant of new forces, acting at a point by making an angle of 90° with each other is $\sqrt{146}$ N.

23. Correct option: (A)

We know, the sum of components of $\vec{A}$ and $\vec{B}$ is given by:

$R_x = A_x + B_x$

$R_y = A_y + B_y$

where R_x, A_x, B_x are the x components of $\vec{A}, \vec{B}$ and resultant $\vec{R}$ respectively.

Given, the magnitude of the x and y components of $\vec{P}$ are 7 and 6.

Also, the magnitudes of x and y components of $\vec{P} + \vec{Q}$ are 11 and 9 respectively.

Therefore,

$11 = 7 + Q_x$

$Q_x = 4$

$9 = 6 + Q_y$

$Q_y = 3$

Therefore, the magnitude of $\vec{Q}$ is :

$\sqrt{3^2 + 4^2} = 5$

Therefore, if the magnitude of the x and y components of $\vec{P}$ are 7 and 6 and the magnitudes of x and y components of $\vec{P} + \vec{Q}$ are 11 and 9 respectively, then the magnitude of $\vec{Q}$ is 5.

24. Correct option: Question is either wrong or incomplete.

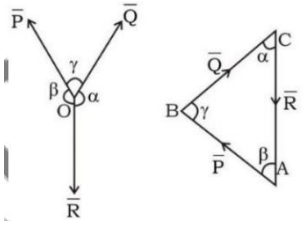
We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by
$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$
Given, two force vectors $2F$ and $\sqrt{2}F$ act, so that the resultant force is $10\sqrt{F}$
Therefore,
$10\sqrt{F} = \sqrt{(2F)^2 + (\sqrt{2}F)^2 + 2(2F)(\sqrt{2}F)\cos\theta}$
$10\sqrt{F} = \sqrt{4F^2 + 2F^2 + 4\sqrt{2}F^2\cos\theta}$
$10\sqrt{F} = F\sqrt{6 + 4\sqrt{2}\cos\theta}$
Since this equation has two unknowns F, θ, it cannot be solved.

25. Correct option: (A)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by
$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$
Given, two forces whose magnitudes are in the ratio 3 : 5 give a resultant of 28 N. Further, angle of their inclination is 60°.
Let the magnitude of two forces be $3x, 5x$ respectively.
Now,
$28 = \sqrt{(3x)^2 + (5x)^2 + 2(3x)(5x)\cos 60^\circ}$
$28 = \sqrt{9x^2 + 25x^2 + 30x^2\,\frac{1}{2}}$
$28 = \sqrt{49x^2}$
$28 = 7x$
$x = 4$
Therefore, the magnitude of the forces are:
$3x = 3(4) = 12$
$5x = 5(4) = 20$
Therefore, if two forces whose magnitudes are in the ratio 3 : 5 give a resultant of 28 N with angle of their inclination 60°, then the magnitude of the forces are 12N and 20N respectively.

26. Correct option: (C)

We know, Lami's theorem states that when three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two forces. Hence, if the forces acting on the body are A, B, C making angles α, β, γ with each other, then:
$$\frac{A}{\sin\alpha} = \frac{B}{\sin\beta} = \frac{C}{\sin\gamma}$$
Given, $\vec{P}, \vec{Q}, \vec{R}$ are three forces which simultaneously act at the point 'O' and keep it in equilibrium. Therefore, from Lami's theorem:
$$\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin\gamma}$$
Further, from the triangle on right, we see:
$$\frac{AB}{\sin\alpha} = \frac{BC}{\sin\beta} = \frac{AC}{\sin\gamma}$$
Dividing these two relations, we get:
$$\frac{P}{AB} = \frac{Q}{BC} = \frac{R}{AC} = K(\text{constant})$$



Therefore, if $\vec{P},\vec{Q},\vec{R}$ are three forces which simultaneously act at the point 'O' and keep it in equilibrium, then $\frac{P}{AB}=\frac{Q}{BC}=\frac{R}{AC}=K(\text{constant})$.

27. Correct option: (A)

We know, Lami's theorem states that when three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two forces. Hence, if the forces acting on the body are A,B,C making angles α,β,γ with each other, then:

$$\frac{A}{\sin\alpha}=\frac{B}{\sin\beta}=\frac{C}{\sin\gamma}$$

Given, $\vec{P},\vec{Q},\vec{R}$ are three forces which simultaneously act at the point 'O' and keep it in equilibrium. Therefore, from Lami's theorem:

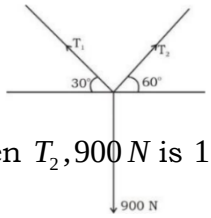
$$\frac{P}{\sin\alpha}=\frac{Q}{\sin\beta}=\frac{R}{\sin\gamma}$$

28. Correct option: (C)

We know, Lami's theorem states that when three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two forces. Hence, if the forces acting on the body are A,B,C making angles α,β,γ with each other, then:

$$\frac{A}{\sin\alpha}=\frac{B}{\sin\beta}=\frac{C}{\sin\gamma}$$

Given, 'P' is in equilibrium.



From the figure we see angle between T_1,T_2 is 90°, angle between $T_2,900\,N$ is 150°, angle Between $T_1,900\,N$ is 120°

Using Lami's theorem, we can write:

$$\frac{900}{\sin 90^\circ}=\frac{T_1}{\sin 150^\circ}=\frac{T_2}{\sin 120^\circ}$$

$$\frac{T_1}{T_2}=\frac{\sin 150^\circ}{\sin 120^\circ}$$

$$\frac{T_1}{T_2}=\frac{\sin 30^\circ}{\sin 60^\circ}$$

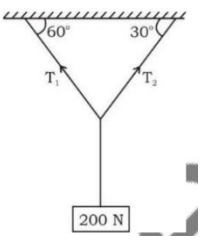
$$\frac{T_1}{T_2}=\frac{(1/2)}{(\sqrt{3}/2)}$$

$$\frac{T_1}{T_2}=\frac{1}{\sqrt{3}}$$

Therefore, if 'P' is in equilibrium, then $\frac{T_1}{T_2}$ is $\frac{1}{\sqrt{3}}$.

29. Correct option: (D)

We know, Lami's theorem states that when three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two forces. Hence, if the forces acting on the body are A,B,C making angles α,β,γ with each other, then:



$$\frac{A}{\sin\alpha}=\frac{B}{\sin\beta}=\frac{C}{\sin\gamma}$$

Given, 'P' is in equilibrium.

From the figure we see angle between T_1,T_2 is 90°, angle between $T_2,200\,N$ is 120°, angle

Between $T_1, 200\,N$ is 150°

Using Lami's theorem, we can write:

$$\frac{200}{\sin 90^\circ} = \frac{T_1}{\sin 120^\circ} = \frac{T_2}{\sin 150^\circ}$$

Hence,

$$T_1 = \sin 120^\circ \frac{200}{\sin 90^\circ}$$

$$T_1 = \sin 60^\circ \frac{200}{1}$$

$$T_1 = \sqrt{3}\frac{200}{2}$$

$$T_1 = 100\sqrt{3}$$

And,

$$T_2 = \sin 150^\circ \frac{200}{\sin 90^\circ}$$

$$T_2 = \sin 30^\circ \frac{200}{1}$$

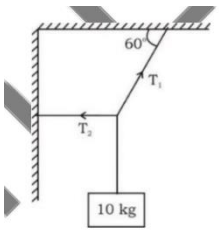
$$T_2 = 1\frac{200}{2}$$

$$T_2 = 100$$

Therefore, the values of T_1 and T_2 are $100\sqrt{3}\,N, 100\,N$ respectively.

30. Correct option: (A)

We know, Lami's theorem states that when three forces acting at a point are in equilibrium, then each force is proportional to the sine of the angle between the other two forces. Hence, if the forces acting on the body are A, B, C making angles α, β, γ with each other, then:



$$\frac{A}{\sin\alpha} = \frac{B}{\sin\beta} = \frac{C}{\sin\gamma}$$

Given, 'P' is in equilibrium.

From the figure we see angle between T_1, T_2 is 120°, angle between $T_2, 10\,kgwt$ is 90° , angle Between $T_1, 10\,kgwt$ is 150°

Using Lami's theorem, we can write:

$$\frac{10\,kgwt}{\sin 120^\circ} = \frac{T_1}{\sin 90^\circ} = \frac{T_2}{\sin 150^\circ}$$

Hence,

$$T_1 = sin\,90^\circ \frac{10kgwt}{sin\,120^\circ}$$

$$T_1 = \frac{10kgwt}{(\sqrt{3}/2)}$$

$$T_1 = \frac{20}{\sqrt{3}}kgwt$$

And,

$$T_2 = sin\,150^\circ \frac{10kgwt}{sin\,120^\circ}$$

$$T_2 = sin\,30^\circ \frac{200}{sin\,60^\circ}$$

$$T_2 = (1/2)\frac{10kgwt}{(\sqrt{3}/2)}$$

$T_2 = \frac{10}{\sqrt{3}} kgwt$

Therefore, the values of T_1 and T_2 are $\frac{20}{\sqrt{3}} kgwt$, $\frac{10}{\sqrt{3}} kgwt$ respectively.

LEVEL-II

1. Correct option: (B)

We know, the magnitude of the greatest and least resultant of two vectors $\overrightarrow{F_1}$ and $\overrightarrow{F_2}$ are given by $F_1 + F_2$ and $F_1 - F_2$, when the vectors act in the same direction and in the opposite direction, respectively.

Given, maximum and minimum magnitudes of the resultant of two vectors of magnitudes P and Q are found to be in the ratio 3 : 1.

Therefore,

$\frac{P+Q}{P-Q} = \frac{3}{1}$

$P + Q = 3P - 3Q$

$4Q = 2P$

$2Q = P$

Therefore, if maximum and minimum magnitudes of the resultant of two vectors of magnitudes P and Q are found to be in the ratio 3 : 1 , the correct relation is $2Q = P$.

2. Correct option: (C)

Given, eleven forces, each of magnitude 5N are acting on a particle simultaneously with each force making angle 30° with next one.

Now, let us suppose there are $\frac{360^\circ}{30^\circ} = 12$ such forces. Then, the resultant of all these forces will be zero, since the head of the first vector, and tail of the last vector will coincide in vector addition.

Therefore,

Vector sum of 12 forces of magnitude 5N each making angle 30° with next one = Vector sum of 11 forces of magnitude 5N each making angle 30° with next one + vector of magnitude 5N making an angle of 30° with 11th vector

Or,

0 = Vector sum of 11 forces of magnitude 5N each making angle 30° with next one + vector of magnitude 5N making an angle of 30° with 11th vector

Or, Vector sum of 11 forces of magnitude 5N each making angle 30° with next one = - (vector of magnitude 5N making an angle of 30° with 11th vector)

Therefore, if eleven forces, each of magnitude 5N are acting on a particle simultaneously with each vector making an angle 30° with next one, then the resultant of all the forces has magnitude 5N.

3. Correct option: (D)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Given, the greatest and least resultant of two forces acting at a point is 7 N and 3 N respectively.

We know, the magnitude of the greatest and least resultant of two forces $\overrightarrow{F_1}$ and $\overrightarrow{F_2}$ are given by $F_1 + F_2$ and $F_1 - F_2$, when the forces act in the same direction and in the opposite direction, respectively.

Now, $F_1 + F_2 = 7$ and $F_1 - F_2 = 3$

Adding these two equations, we get :

$F_1 + F_2 + F_1 - F_2 = 7 + 3$

$2F_1 = 10$

$F_1 = 5$

Subtracting the two equations, we get:

$F_1 + F_2 - F_1 + F_2 = 7 - 3$

$2F_2 = 4$

$F_2 = 2$

When each force is increased by 3N, the new magnitude of the forces are $F_1 = 8$ and $F_2 = 5$.

Since these forces now makes an angle of 60° with each other, therefore the resultant is :

$|\vec{R}| = \sqrt{8^2 + 5^2 + 2(8)(5)\cos(60^\circ)}$

$|\vec{R}| = \sqrt{64 + 25 + 40}$

$|\vec{R}| = \sqrt{129}$

Therefore, if the greatest and least resultant of two forces acting at a point is 7 N and 3 N respectively and if each force is increased by 3 N, then the resultant of new forces, acting at a point by making an angle of 60° with each other is $\sqrt{129}$ N.

4. Correct option: (B)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Given, two vectors $\vec{a}, \vec{b}$ are such that $|\vec{a} - \vec{b}| = |\vec{a}| + |\vec{b}|$

Therefore,

$|\vec{a}| + |\vec{b}| = \sqrt{a^2 + b^2 + 2ab\cos(180^\circ - \theta)}$

$(a+b)^2 = a^2 + b^2 - 2ab\cos\theta$

$a^2 + b^2 + 2ab = a^2 + b^2 - 2ab\cos\theta$

$-1 = \cos\theta$

$\theta = 180^\circ$

Therefore, if two vectors $\vec{a}, \vec{b}$ are such that $|\vec{a} - \vec{b}| = |\vec{a}| + |\vec{b}|$ then the angle between $\vec{a}$ and $\vec{b}$ is 180°.

5. Correct option: (C)

We want the resultant of two vectors to be 4N.

Now, if 2N and 2N act in same direction, the resultant will be 4N.

If 2N and 6N act in opposite direction, the resultant will be 4N.

If 4N and 8N act in opposite direction, the resultant will be 4N.

Therefore, for 3N and 8N the resultant cannot be 4N since the minimum and maximum resultant is limited to 5N and 13 N respectively.

6. Correct options: (B), (D)

Given, $\vec{A} + \vec{B}$ is perpendicular to $\vec{A} - \vec{B}$

Therefore,

$(\vec{A} + \vec{B}).(\vec{A} - \vec{B}) = 0$

$A^2 - B^2 = 0$

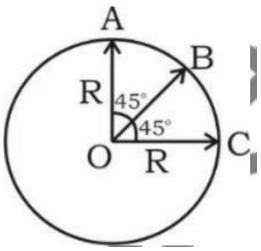
$A = B$

Also, the angle between $\vec{A}$ and $\vec{B}$ can have any value.

Therefore, if $\vec{A}+\vec{B}$ is perpendicular to $\vec{A}-\vec{B}$, then $A=B$ and the angle between $\vec{A}$ and $\vec{B}$ can have any value.

7. Correct options: (A)

We know, if two vectors are represented in magnitude and direction by the adjacent sides of a parallelogram drawn from a point, the diagonal passing through that point represents their resultant both in magnitude and direction.



Therefore, the resultant of $\overrightarrow{OA}$ and $\overrightarrow{OC}$ will be along $\overrightarrow{OB}$, with magnitude:

$C=\sqrt{OA^2+OC^2}$

$C=\sqrt{R^2+R^2}$

$C=R\sqrt{2}$

where R is the radius of the circle.

Therefore, the vector sum of this resultant along $\overrightarrow{OB}$ with $\overrightarrow{OB}$ is:

$R\sqrt{2}+R$

Therefore, the resultant of the vectors $\overrightarrow{OA}$, $\overrightarrow{OB}$ and $\overrightarrow{OC}$, if the radius of the circle is R is $R(1+\sqrt{2})$

8. Correct options: (A), (C)

We know, if two vectors $\vec{A},\vec{B}$ are perpendicular, then their dot product vanishes.

Given $|\vec{A}+\vec{B}|=|\vec{A}-\vec{B}|$

Squaring both sides:

$(\vec{A}+\vec{B})(\vec{A}+\vec{B})=(\vec{A}-\vec{B})(\vec{A}-\vec{B})$

$A^2+B^2+2\vec{A}.\vec{B}=A^2+B^2-2\vec{A}.\vec{B}$

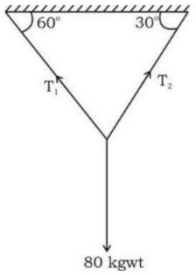
$\vec{A}.\vec{B}=0$

Therefore, if $|\vec{A}+\vec{B}|=|\vec{A}-\vec{B}|$, then angle between $\vec{A},\vec{B}$ is $90^\circ=\frac{\pi}{2}$.

9. Correct option: (A), (B)

We know, for equilibirum the net force is zero.

We see , three are three forces acting on the man, one is the gravitaonal force $F=80\,kgwt$



acting downwards, the other two are the tension forces T_1,T_2

Resolving the force into rectangular components, and balancing the force in vertical direction, we get:

$T_1\sin 60^\circ+T_2\sin 30^\circ=80\,kgwt$

$T_1\frac{\sqrt{3}}{2}+T_2\frac{1}{2}=80\,kgwt$

$\sqrt{3}T_1+T_2=160\,kgwt$

Similarly, resolving the force into rectangular components, and balancing the force in horizontal direction, we get:

$T_1\cos 60^\circ=T_2\cos 30^\circ$

$T_1\frac{1}{2}=T_2\frac{\sqrt{3}}{2}$

$T_1=\sqrt{3}T_2$

Substituting $T_1=\sqrt{3}T_2$ into $\sqrt{3}T_1+T_2=160\,kgwt$, we get:

$3T_1 + T_2 = 160\,kgwt$

$4T_2 = 160\,kgwt$

$T_2 = 40\,kgwt$

And thus $T_1 = 40\sqrt{3}\,kgwt$.

Therefore, if A man of 80 kgwt is supported by two cables as shown in the figure, then tension are $T_1 = 40\sqrt{3}\,kgwt$ and $T_2 = 40\,kgwt$.

10. Correct option: (30°)

We know, if two vectors $\vec{A}, \vec{B}$ are perpendicular, then their dot product vanishes.

Given, the resultant of two vectors $\vec{A}, \vec{B}$ s perpendicular to $\vec{A}$ and Magnitude of resultant $\vec{R}$ is equal to half magnitude of $\vec{B}$.

Therefore,

$\vec{A}.(\vec{A} + \vec{B}) = 0$

$A^2 + \vec{A}.\vec{B} = 0$

$A^2 + AB\cos\theta = 0$

$A = -B\cos\theta$

We also know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Therefore,

$\frac{B}{2} = \sqrt{B^2\cos^2\theta + B^2 + 2(-B\cos\theta)B\cos\theta}$

$\frac{B}{2} = \sqrt{B^2 - B^2\cos^2\theta}$

$\frac{B}{2} = B\sqrt{1-\cos^2\theta}$

$\frac{B}{2} = B\sin\theta$

$\frac{1}{2} = \sin\theta$

$\theta = 30^\circ$

Therefore, if the resultant of two vectors $\vec{A}, \vec{B}$ s perpendicular to $\vec{A}$ and Magnitude of resultant $\vec{R}$ is equal to half magnitude of $\vec{B}$, then the angle between $\vec{A}$ and $\vec{B}$ is 30°.

11. Correct option: (D)

We know, the horizontal and vertical components A_x, A_y of a vector $\vec{A}$ satisfies:

$A^2 = A_x^{\,2} + A_y^{\,2}$

Given, a force of F is inclined at an angle to the horizontal with 'X' as its horizontal component.

Therefore,

$F^2 = X^2 + Y^2$

$Y = \sqrt{F^2 - X^2}$

Therefore, if , a force of F is inclined at an angle to the horizontal with 'X' as its horizontal component, then the y component is $\sqrt{F^2 - X^2}$.

12. Correct option: (B)

We know, the rectangular components of a vector $\vec{P}$ are given by:

$P\cos\theta$ and $P\sin\theta$, where θ is the angle between the vector $\vec{P}$ and the x axis.

Given, a vector is inclined at θ to the horizontal and its vertical component is y.

Therefore,

$$P\sin\theta = y$$

$$P = \frac{y}{\sin\theta}$$

$$P = y\csc\theta$$

Therefore, if a vector is inclined at θ to the horizontal and its vertical component is y, then the magnitude of the vector is $y\csc\theta$.

13. Correct option: (C)

We know, the rectangular components of a vector $\vec{P}$ are given by:

$P\cos\theta$ and $P\sin\theta$, where θ is the angle between the vector $\vec{P}$ and the x axis.

Given, a vector is inclined at θ to the vertical and its horizontal component is H.

Therefore,

$$P\cos(90^\circ - \theta) = H$$

$$P\sin\theta = H$$

$$P = \frac{H}{\sin\theta}$$

$$P = H\csc\theta$$

Hence, the vertical component is :

$$P\sin(90^\circ - \theta) = \frac{H}{\sin\theta}\cos\theta$$

$$P\sin(90^\circ - \theta) = H\tan\theta$$

Therefore, if a vector is inclined at θ to the vertical and its horizontal component is H, then its vertical component is $H\tan\theta$.

14. Correct option: (C)

Given, two force 5 N and 3 N act on the object in the same direction.

Now, the magnitude of the resultant is simply the sum of the two forces = 5N+3N = 8N.

Therefore, if two force 5 N and 3 N act on the object in the same direction, then the magnitude
of the resultant is 8N.

15. Correct option: (A)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$$

Given, two forces each having magnitude F and angle between them is θ,

Therefore, the resultant is :

$$|\vec{R}| = \sqrt{F^2 + F^2 + 2F^2\cos\theta}$$

$$|\vec{R}| = F\sqrt{2(1+\cos\theta)}$$

Using $\cos^2\theta = \frac{1+\cos 2\theta}{2}$, we get:

$$|\vec{R}| = F\sqrt{2(2)\cos^2\left(\frac{\theta}{2}\right)}$$

$$|\vec{R}| = 2F\cos\left(\frac{\theta}{2}\right)$$

Therefore, the resultant of two forces each having magnitude F and angle between them is θ is $2F\cos\left(\frac{\theta}{2}\right)$.

16. Correct option: (B)

We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Given, two vectors having magnitude of 10 units each makes an angle of 60° between them.

Therefore the resultant is:

$|\vec{R}| = \sqrt{10^2 + 10^2 + 2(10^2)\cos 60^\circ}$

$|\vec{R}| = \sqrt{200 + 100}$

$|\vec{R}| = 10\sqrt{3}$

Therefore, if two vectors having magnitude of 10 units each makes an angle of 60° between them, then the magnitude of the resultant is $10\sqrt{3}$.

17. Correct option: (A)

We know, the resultant of two vectors A and B can be found using parallelogram law which states that the vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Further, if angle between $\vec{A}$ and $\vec{B}$ is zero, then

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB}$

$|\vec{R}| = (A + B)$

Therefore, the statement when $\vec{A} + \vec{B} = \vec{C}$ and $A + B = C$, then the angle between $\vec{A}$ and $\vec{B}$ must be zero is also correct.

18. Correct option: (C)

We know, the components of vectors are scalars.

Also, the y-component of a vector $4\hat{i} + 5\hat{j}$ is 5.

Therefore, the statement that the y-component of a vector $4\hat{i} + 5\hat{j}$ is 4 is false.

19. Correct match: (A) -> P), (B) -> R), (C) -> S), (D) -> Q)

(A) We know, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}| = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

Now, $\vec{A} - \vec{B} = \vec{C}$ and $A - B = C$

$A - B = \sqrt{A^2 + B^2 + 2AB\cos(180^\circ - \theta)}$

$(A - B)^2 = A^2 + B^2 - 2AB\cos\theta$

$A^2 + B^2 + 2AB = A^2 + B^2 - 2AB\cos\theta$

$-1 = \cos\theta$

$\theta = \pi$

(B) Here, $\vec{A} + \vec{B} = \vec{C}$ and $A - B = C$, hence

$A - B = \sqrt{A^2 + B^2 + 2AB\cos\theta}$

$(A - B)^2 = A^2 + B^2 + 2AB\cos\theta$

$A^2 + B^2 + 2AB = A^2 + B^2 + 2AB\cos\theta$

$1 = \cos\theta$

$\theta = 0$

(C) Here, $\vec{A}-\vec{B}=\vec{C}$ and $A^2+B^2=C^2$

$C=\sqrt{A^2+B^2+2AB\,cos(180°-\theta)}$

$C^2=A^2+B^2-2AB\,cos\,\theta$

$A^2+B^2=A^2+B^2-2AB\,cos\,\theta$

$0=cos\,\theta$

$\theta=\frac{\pi}{2}$

(D) Here, $\vec{A}+\vec{B}=\vec{C}$ and $A=B=C$

$C=\sqrt{A^2+B^2+2AB\,cos(\theta)}$

$C^2=C^2+C^2+2C^2\,cos\,\theta$

$1=2(1+cos\quad\theta)$

$\frac{-1}{2}=cos\,\theta$

$\theta=\frac{2\pi}{3}$

Therefore the correct match is : (A) -> P), (B) -> R), (C) -> S), (D) -> Q)

20. Correct match: (A) -> P), (B) -> Q), (C) -> T), (D) -> T)

We know, the rectangular components of a vector $\vec{P}$ are given by:

$P\cos\theta$ and $P\cos\theta$, where θ is the angle between the vector $\vec{P}$ and the x axis.

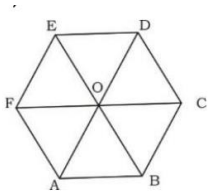
Therefore, Horizontal component of 'P' = $P\cos\theta$

And, Vertical component of 'P' = $P\cos\theta$

The angle between the x and y component is given by $tan\quad\theta=\frac{P_x}{P_y}$

Therefore, the correct match is (A) -> P), (B) -> Q), (C) -> T), (D) -> T)

21. Correct number: $3+\frac{\sqrt{3}}{4}$



We know, if two vectors are represented in magnitude and direction by the adjacent sides of a parallelogram drawn from a point, the diagonal passing through that point represents their resultant both in magnitude and direction.

Further, the parallelogram of law of vector addition of two vectors $\vec{A}$ and $\vec{B}$ is given by

$|\vec{R}|=\sqrt{A^2+B^2+2AB\cos\theta}$

Let the side of the hexagon be a.

Now, $\overrightarrow{AB}+\overrightarrow{AF}$ will be in the direction of $\overrightarrow{AD}$, with magnitude given by:

$|\vec{R}|=\sqrt{a^2+a^2+2a^2\cos120^\circ}$

$|\vec{R}|=\sqrt{a^2+a^2+2a^2\frac{-1}{2}}$

$|\vec{R}|=a$

Now, $\overrightarrow{AC}+\overrightarrow{AE}$ will be in the direction of $\overrightarrow{AD}$, with magnitude given by:

$|\vec{R}|=\sqrt{AC^2+AE^2+2(AC)(AE)\cos120^\circ}$

Now, for the magnitude AC, we see from the figure, AC is diagonal of the parallelogram ABCO. Since the diagonal of parallelograms are perpendicular bisector of each other, we have for the triangle OAB with $\angle OAB=60^\circ$:

$$\cos 30^\circ = \frac{AC/2}{AB}$$

$$\frac{\sqrt{3}}{2} = \frac{AC/2}{a}$$

$$AC = \frac{\sqrt{3}}{4}a$$

Therefore,

$$|\vec{R}| = \sqrt{AC^2 + AE^2 + 2(AC)(AE)\cos 120^\circ}$$

$$|\vec{R}| = \sqrt{\frac{3}{16}a^2 + \frac{3}{16}a^2 + 2\frac{3}{16}a^2\frac{-1}{2}}$$

$$|\vec{R}| = \frac{\sqrt{3}}{4}a$$

Now,

$$\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} = (\overrightarrow{AB} + \overrightarrow{AF}) + (\overrightarrow{AC} + \overrightarrow{AE}) + \overrightarrow{AD}$$

$$\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} = \overrightarrow{OD} + \frac{\sqrt{3}}{4}(\overrightarrow{OD}) + 2\overrightarrow{OD}$$

$$\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} = (3 + \frac{\sqrt{3}}{4})\overrightarrow{OD}$$

Therefore, comparing $\overrightarrow{AB} + \overrightarrow{AC} + \overrightarrow{AD} + \overrightarrow{AE} + \overrightarrow{AF} = ____\overrightarrow{OD}$, we see the correct number is $3 + \frac{\sqrt{3}}{4}$.

22. Correct number: 1

We know, if two vectors $\vec{A}, \vec{B}$ are perpendicular, then their dot product vanishes.

Given $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$

Squaring both sides:

$$(\vec{A} + \vec{B})(\vec{A} + \vec{B}) = (\vec{A} - \vec{B})(\vec{A} - \vec{B})$$

$$A^2 + B^2 + 2\vec{A}.\vec{B} = A^2 + B^2 - 2\vec{A}.\vec{B}$$

$$\vec{A}.\vec{B} = 0$$

Therefore, if $|\vec{A} + \vec{B}| = |\vec{A} - \vec{B}|$, then angle between $\vec{A}, \vec{B}$ is 90°. Comparing this with 9×10^x, we see the value of x is 1.

2. Motion in Plane
Solutions

LEVEL-I

1. Correct option: (B)

We know, for a projectile subjected to uniform gravitational field, the equation of the path of the projectile is given by:

$x = (v_0 \cos\theta)t$

$y = (v_0 \sin\theta)t - \frac{1}{2}gt^2$

where θ is the angle of projection, and v_0 is the initial velocity.

Eliminating t from the first two equations of the path of the projectile by solving for t in first equation and substituting in the second equation, we get:

$y = (v_0 \sin\theta)\frac{x}{v_0 \cos\theta} - \frac{1}{2}g\left(\frac{x}{v_0 \cos\theta}\right)^2$

$y = x\tan\theta - \frac{1}{2}g\left(\frac{x^2}{v_0^2 \cos^2\theta}\right)$

This is an equation of a parabola.

Therefore, the path of a projectile in a uniform gravitational field is always a parabola.

2. Correct option: (B)

We know, the projectile is subjected to uniform gravitational field, where the gravitational force acts vertically downward. Hence, the vertical velocity v_y and horizontal velocity v_x varies as:

$v_x = (v_0 \cos\theta)$

$v_y = (v_0 \sin\theta) - gt$

where θ is the angle of projection, and v_0 is the initial velocity.

We see, the horizontal velocity remains constant, and the vertical velocity changes continuously throughout the motion. In particular, the vertical velocity decreases continuously during the upward motion, owing to the gravitational force acting in opposite direction, till the topmost point, where the vertical velocity is zero.

Therefore, on reaching the maximum height, for a projectile vertical component of velocity is zero.

3. Correct option: (C)

We know, for a projectile subjected to uniform gravitational field, the gravitational force acts vertically downward. Hence, the vertical velocity v_y and horizontal velocity v_x varies as:

$v_x = (v_0 \cos\theta)$

$v_y = (v_0 \sin\theta) - gt$

where θ is the angle of projection, and v_0 is the initial velocity.

We see, the horizontal velocity remains constant, and the vertical velocity changes continuously throughout the motion. In particular, the vertical velocity decreases continuously during the upward motion, owing to the gravitational force acting in opposite direction, till the topmost point, where the vertical velocity is zero.

Therefore, at the highest point, we only have the horizontal velocity.

Further, the acceleration of the projectile is the gravitational acceleration g, which remains constant throughout the motion of the projectile, always directed to in vertically downward direction.
Therefore, at highest point of the projectile its velocity and acceleration are perpendicular to each other.

4. Correct option: (D)

We know, range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$ where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

And the time of flight T, the total time taken by the projectile in its journey, is given by,

$T = \frac{2v_0 \sin\theta}{g}$.

Further, height H, the maximum vertical distance covered by the projectile is given by:

$H = \frac{v_0^2 \sin^2\theta}{2g}$.

Given, height of the projectile is more.
Now, from the above equation of the height of the projectile, we see the height depends on the magnitude of the initial velocity of the projectile v_0 and the angle of projection θ.

The height increases whenever v_0 or θ increases (as θ increases, $\sin\theta$ also increases till $\theta = 90^\circ$, thereafter $\sin\theta$ decreases from $\theta = 90^\circ$ to $\theta = 180^\circ$).

Now, if v_0 or θ increases, time of flight also increases as it too depends on v_0 and θ, as seen by the second equation above.

But as θ increases, $\sin 2\theta$, and hence range, first increases till $\theta = 45^\circ$, thereafter $\sin\theta$ decreases as θ is further increased from $\theta = 45^\circ$ to $\theta = 90^\circ$). Therefore, it is not always true that increasing the angle of projection increases the range.
Therefore, time of flight and angle of projection are both more when height of the projectile is more.

5. Correct option: (C)

We know, for a projectile subjected to uniform gravitational field, the equation of the path of the projectile is given by:

$x = (v_0 \cos\theta)t$

$y = (v_0 \sin\theta)t - \frac{1}{2}gt^2$

where θ is the angle of projection, and v_0 is the initial velocity.

We also know, range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$.

Eliminating t from the first two equations of the path of the projectile by solving for t in first equation and substituting in the second equation, we get:

$$y = (v_0 \sin\theta)\frac{x}{v_0 \cos\theta} - \frac{1}{2}g\left(\frac{x}{v_0 \cos\theta}\right)^2$$

$$y = x\tan\theta - \frac{1}{2}g\left(\frac{x^2}{v_0^2 \cos^2\theta}\right)$$

Now, using $R = \frac{v_0^2 \sin 2\theta}{g} = \frac{2v_0^2 \sin\theta\cos\theta}{g}$, we get

$$y = x\tan\theta - \frac{1}{2}g\left(\frac{x^2}{v_0^2\left(Rg/(2v_0^2\sin\theta)\right)^2}\right)$$

$$y = x\tan\theta - \frac{1}{2}g\left(\frac{4x^2v_0^4\sin^2\theta}{v_0^2R^2g^2}\right)$$

$$y = x\tan\theta - \left(\frac{2x^2v_0^2\sin^2\theta}{R^2g}\right)$$

$$y = x\tan\theta - \left(\frac{2x^2v_0^2\sin^2\theta}{Rg}\right)\left(\frac{g}{2v_0^2\sin\theta\cos\theta}\right)$$

$$y = x\tan\theta - \left(\frac{x^2\sin\theta}{R\cos\theta}\right)$$

$$y = x\tan\theta - \left(\frac{x^2\tan\theta}{R}\right)$$

$$y = x\tan\theta\left(1 - \frac{x}{R}\right)$$

Therefore, equation of oblique projectile in terms of range is

$$y = x\tan\theta\left(1 - \frac{x}{R}\right)$$

6. Correct option: (C)

We know, height H , the maximum vertical distance covered by the projectile is given by:

$H = \frac{v_0^2\sin^2\theta}{2g}$, where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, complementary angle of θ is $90^\circ - \theta$.

Now, Ratio of maximum heights for complementary angle is:

$$\frac{H_\theta}{H_{90^\circ-\theta}} = \frac{\frac{v_0^2\sin^2\theta}{2g}}{\frac{v_0^2\sin^2(90^\circ-\theta)}{2g}}$$

$$\frac{H_\theta}{H_{90^\circ-\theta}} = \frac{\sin^2\theta}{\cos^2\theta}$$

$$\frac{H_\theta}{H_{90^\circ-\theta}} = \tan^2\theta$$

Therefore, for complementary angles of projection, the ratio of maximum heights is $\tan^2\theta$.

7. Correct option: (A)

We know, the time of flight T, the total time taken by the projectile in its journey, is given by $T = \frac{2v_0 \sin\theta}{g}$,where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, complementary angle of θ is $90^\circ - \theta$.

Now, ratio of time of flights for complementary angle is:

$$\frac{T_\theta}{T_{90^\circ-\theta}} = \frac{\frac{2v_0 \sin\theta}{g}}{\frac{2v_0 \sin(90^\circ - \theta)}{g}}$$

$$\frac{T_\theta}{T_{90^\circ-\theta}} = \frac{\sin\theta}{\cos\theta}$$

$$\frac{T_\theta}{T_{90^\circ-\theta}} = \tan\theta$$

Therefore, for complementary angles of projection, the ratio of time of flights is $\tan\theta$.

8. Correct option: (A)

We know, the time of flight T, the total time taken by the projectile in its journey, is given by, $T = \frac{2v_0 \sin\theta}{g}$,where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, the range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$.

Given, projectile have the same range R for two angles of projection. Let the angle of projections be θ_1 and θ_2.

Then,

$$\frac{v_0^2 \sin 2\theta_1}{g} = \frac{v_0^2 \sin 2\theta_2}{g}$$

$$\sin 2\theta_1 = \sin 2\theta_2$$

$$2\sin\theta_1 \cos\theta_1 = 2\sin\theta_2 \cos\theta_2$$

$$\sin\theta_1 \sin(90^\circ - \theta_1) = \sin\theta_2 \sin(90^\circ - \theta_2)$$

This equation is satisfied if $\theta_2 = \theta_1$ or when $\theta_2 = 90^\circ - \theta_1$.

Since the angle of projections are different, therefore:

$\theta_2 = 90^\circ - \theta_1$

Now, the time of flight for the two angles of projections are:

$$T_1 = \frac{2v_0 \sin\theta_1}{g}$$

$$T_2 = \frac{2v_0 \sin\theta_2}{g}$$

Multiplying these two equations, we get:

$$T_2T_1 = \frac{2v_0 \sin\theta_1}{g}\frac{2v_0 \sin\theta_2}{g}$$

$$T_2T_1 = \frac{4v_0^2 \sin\theta_1}{g}\frac{\sin(90^\circ - \theta_2)}{g}$$

$$T_2T_1 = \frac{4v_0^2 \sin\theta_1 \cos\theta_1}{g^2}$$

$$T_2T_1 = \frac{2v_0^2 \sin 2\theta_1}{g^2}$$

Using the formula for the range $R = \frac{v_0^2 \sin 2\theta}{g}$, we get:

$$T_2T_1 = \frac{2R}{g}$$

Therefore, if a projectile has the same range R for two angles of projection, with the times of flight in the two cases T_1 and T_2 respectively, then $T_2T_1 = \frac{2R}{g}$.

9. Correct option: (D)

Given, a particle is projected from level ground with velocity u.

We know, the projectile is subjected to uniform gravitational field, where the gravitational force acts vertically downward. Hence, the vertical velocity v_y and horizontal velocity v_x varies as:

$$v_x = (u\cos\theta)$$

$$v_y = (u\sin\theta) - gt$$

The horizontal velocity of the projectile remains unchanged throughout the motion, and the vertical velocity changes continuously.

At the end of the trajectory, the projectile has the same magnitude of the velocity, but since the projectile is now moving downwards, its vertical velocity is also directed downwards.

Therefore, the change in velocity of the projectile for a complete trajectory = Initial vertical velocity – Final vertical velocity

$$u\sin\theta - (-u\sin\theta) = 2u\sin\theta$$

Therefore, if a particle is projected from level ground with velocity u, the change in velocity of the

projectile for complete trajectory is $2u\sin\theta$.

10. Correct option: (C)

Given, a projectile is projected with a linear momentum p making an angle θ with the horizontal.

The horizontal velocity of the projectile remains unchanged throughout the motion, and the vertical velocity changes continuously.

At the end of the trajectory, the projectile has the same magnitude of the velocity, but since the projectile is now moving downwards, its vertical velocity is also directed downwards.

Therefore, the change in linear momentum of the projectile for a complete trajectory = Initial vertical momentum – Final vertical momentum

$p\sin\theta-(-p\sin\theta)=2p\sin\theta$

Therefore, if a projectile is projected with a linear momentum p making an angle θ with the horizontal, the change in momentum of the projectile on return to the ground is $2p\sin\theta$.

11. Correct option: (B)

We know, the potential energy U of an object at height h is given by:

$U=mgh$

where m is the mass of the object.

Further, the kinetic energy K of the object is given by:

$K=\frac{1}{2}mv^2$

where v is the velocity of the object.

Now, the height H, the maximum vertical distance covered by the projectile is given by

$H=\frac{v_0^2\sin^2\theta}{2g}$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, at the highest point, the vertical velocity is zero, and we only have the horizontal velocity $v_0\cos\theta$.

Therefore, ratio of kinetic energy to the potential energy at the highest point of the projectile is:

$\frac{K}{U}=\frac{mgh}{\frac{1}{2}mv^2}$

$\frac{K}{U}=\frac{mg\left(\frac{v_0^2\sin^2\theta}{2g}\right)}{\frac{1}{2}m\left(v_0\cos\theta\right)^2}$

$\frac{K}{U}=\frac{\sin^2\theta}{\cos^2\theta}$

$\frac{K}{U}=\tan^2\theta$

Therefore, at the highest point of the projectile, ratio of kinetic energy to the potential energy is $\tan^2\theta$.

12. Correct option: (A)

We know, the time of flight T, the total time taken by the projectile in its journey, is given by $T=\frac{2v_0\sin\theta}{g}$, where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

We also know, height H, the maximum vertical distance covered by the projectile is given by:

$H = \frac{v_0^2 \sin^2 \theta}{2g}$.

Given, a particle is fired at an angle of 45° and reaches the highest point in its path in $2\sqrt{2}$ s.

Therefore, using the expression for the time of flight, we get:

$2\sqrt{2} = \frac{2v_0 \sin 45^\circ}{g}$

$2\sqrt{2} = \frac{2v_0 (1/\sqrt{2})}{g}$

$v_0 = 2g$

Now, the maximum height of the projectile is:

$H = \frac{(2g)^2 \sin^2 45^\circ}{2g}$

$H = \frac{(2g)}{2}$

$H = g$

Therefore, if a particle is fired at an angle of 45° and reaches the highest point in its path in $2\sqrt{2}$ s, then the maximum height of the projectile is g.

13. Correct option: (B)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Given, the range of a projectile when launched at an angle 15° with the horizontal is 1.5 km.

Therefore,

$1.5(10^3) = \frac{v_0^2 \sin 30^\circ}{g}$

$1500 = \frac{v_0^2}{2g}$

$v_0^2 = 3000g$

Now, the range of the projectile when launched at an angle of 45° is:

$R = \frac{v_0^2 \sin 2\theta}{g}$

$R = \frac{(3000g) \sin 90^\circ}{g}$

$R = 3000\,\text{m}$

Therefore, if the range of a projectile when launched at an angle 15° with the horizontal is 1.5 km, then the range of a projectile when launched at an angle 45° is 3 km.

14. Correct option: (A)

We know, the time of flight T, the total time taken by the projectile in its journey, is given by $T = \frac{2v_0 \sin\theta}{g}$, where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Given, a body is projected from the ground with a velocity 10 m/s at an angle 30° to the horizontal.

Therefore, the time of the flight is:

$$T = \frac{2(10)\sin 30^\circ}{g}$$

$$T = \frac{2(10)(1/2)}{10}$$

$$T = 1\text{s}$$

Therefore, if a body is projected from the ground with a velocity 10 m/s at an angle 30° to the horizontal, then the time of the flight is 1s.

15. Correct option: (C)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$$R = \frac{v_0^2 \sin 2\theta}{g}$$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Given, two particles are projected with same velocity but at the angles of projection 35° and 55°

Therefore, the ratio of the horizontal ranges is:

$$\frac{R_1}{R_2} = \frac{\frac{v_0^2 \sin 2\theta_1}{g}}{\frac{v_0^2 \sin 2\theta_2}{g}}$$

$$\frac{R_1}{R_2} = \frac{\sin 2\theta_1}{\sin 2\theta_2}$$

$$\frac{R_1}{R_2} = \frac{\sin 70^\circ}{\sin 110^\circ}$$

We know, $\sin\theta = \sin(180^\circ - \theta)$

Hence,

$$\frac{R_1}{R_2} = \frac{sin\,70^\circ}{sin(180^\circ - 110^\circ)}$$

$$\frac{R_1}{R_2} = \frac{sin\,70^\circ}{sin\,70^\circ}$$

$$\frac{R_1}{R_2} = 1$$

Therefore, if two particles are projected with same velocity but at the angles of projection 35° and 55°, then the ratio of the horizontal ranges are 1:1.

16. Correct option: (C)

Given, the equation of trajectory of a projectile is:

$$y = 10x - \left(\frac{5}{9}\right)x^2$$

Now, the range of the projectile is the maximum horizontal distance travelled by the projectile, which is covered at the end of the path, where the projectile has finally landed in the ground.

Therefore, substituting $y = 0$ and solving for x in $y = 10x - \left(\frac{5}{9}\right)x^2$, we get:

$$0 = 10x - \left(\frac{5}{9}\right)x^2$$

$$0 = x(10 - \left(\frac{5}{9}\right)x)$$

Hence, either $x = 0$ or

$$10 - \frac{5}{9}x = 0$$

$$x = 90/5$$

$$x = 18$$

Since $x = 0$ is the starting point of the projectile, therefore the range of the projectile is 18 m.

Therefore, if the equation of trajectory of a projectile is $y = 10x - \left(\frac{5}{9}\right)x^2$, then the range of projectile is 18 m.

17. Correct option: (D)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$$R = \frac{v_0^2 \sin 2\theta}{g}$$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

We also know, height H, the maximum vertical distance covered by the projectile is given by:

$$H = \frac{v_0^2 \sin^2 \theta}{2g}.$$

Given, a body is projected with velocity v_0 such that its horizontal range and maximum vertical height is same.

Therefore,

$$\frac{v_0^2 \sin 2\theta}{g} = \frac{v_0^2 \sin^2 \theta}{2g}$$

$$2\sin\theta\cos\theta = \frac{\sin^2 \theta}{2}$$

$$4 = \tan\theta$$

We know, $\tan A = \frac{\text{Opposite side of } \angle A}{\text{Adjacent side of } \angle A}$.

Here, $4 = \tan\theta$

Therefore, the opposite side of θ has length 4 unit, and the adjacent side of θ has length 1 unit.

We know, from Pythagoras theorem,

$c^2 = a^2 + b^2$

where c is the length of the hypotenuse of the right-angled triangle, and a, b are the length of its other two sides.

Applying this theorem, we get:

$c^2 = 1^2 + 4^2$

$c = \sqrt{1^2 + 4^2}$

$c = \sqrt{17}$

We know, $\sin A = \frac{\text{Opposite side of } \angle A}{\text{Hypotenuse}}$. Putting the values, we get:

$\sin\theta = \frac{4}{\sqrt{17}}$

Now, the maximum vertical distance covered by the projectile is given by:

$H = \frac{v_0^2 (16/17)}{2g}$

.

$H = \frac{8v_0^2}{17g}$

Therefore, if a body is projected with velocity v_0 such that its horizontal range and maximum vertical height are same, then the maximum height is $\frac{8v_0^2}{17g}$.

18. Correct option: (A)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

We also know, height H, the maximum vertical distance covered by the projectile is given by:

$H = \frac{v_0^2 \sin^2\theta}{2g}$.

Given, a particle is projected with a velocity V such that its range on horizontal plane is twice of greatest height attained by it. Therefore,

$\frac{v_0^2 \sin 2\theta}{g} = 2\frac{v_0^2 \sin^2\theta}{2g}$

$2\sin\theta\cos\theta = \sin^2\theta$

$2 = \tan\theta$

We know, $\tan A = \frac{\text{Opposite side of } \angle A}{\text{Adjacent side of } \angle A}$.

Here, $2 = \tan\theta$

Therefore, the opposite side of θ has length 2 unit, and the adjacent side of θ has length 1 unit.

We know, from Pythagoras theorem,

$c^2 = a^2 + b^2$

where c is the length of the hypotenuse of the right-angled triangle, and a, b are the length of its other two sides.

Applying this theorem, we get:

$c^2 = 1^2 + 2^2$

$c = \sqrt{1^2 + 2^2}$

$c = \sqrt{5}$

We know, $\sin A = \frac{\text{Opposite side of } \angle A}{\text{Hypotenuse}}$. Putting the values, we get:

$\sin\theta = \frac{2}{\sqrt{5}}$

Also, $\cos A = \frac{\text{Adjacent side of } \angle A}{\text{Hypotenuse}}$

Therefore, $\cos\theta = \frac{1}{\sqrt{5}}$

Now, the range of the projectile is given by:

$R = \frac{V^2 \sin 2\theta}{g}$

$R = \frac{2V^2 \sin\theta \cos\theta}{g}$

$R = \frac{2V^2(2/\sqrt{5})(1/\sqrt{5})}{g}$

$R = \frac{4V^2}{5g}$.

Therefore, if a body is projected with velocity V such that its range on horizontal plane is twice of greatest height attained by it, then the range of the projectile is $\frac{4V^2}{5g}$.

19. Correct option: (D)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

We also know, height H, the maximum vertical distance covered by the projectile is given by:

$H = \frac{v_0^2 \sin^2 \theta}{2g}$.

Given, R is the maximum horizontal range of a projectile.

We know, maximum horizontal range is achieved when $\theta = 45^\circ$

Therefore,

$R = \frac{v_0^2 \sin 90^\circ}{g}$

$R = \frac{v_0^2}{g}$

Now, the maximum vertical distance covered by the projectile is given by:

$H = \frac{v_0^2 \sin^2 45^\circ}{2g}$.

$H = \frac{v_0^2}{4g}$

Using $R = \frac{v_0^2}{g}$, we get:

$H = \frac{R}{4}$

Therefore, if R is the maximum horizontal range of a projectile, then largest height attained by it is $\frac{R}{4}$.

20. Correct option: (B)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

The maximum range is thus given when $\theta = 45^\circ$.

Therefore,

$\tan\theta = \tan 45^\circ$

$\tan\theta = 1$

Therefore, if an object thrown at an angle θ with the horizontal and the range is maximum, then the value of $\tan\theta$ is 1.

21. Correct option: (D)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

We also know, height H , the maximum vertical distance covered by the projectile is given by:

$H = \frac{v_0^2 \sin^2 \theta}{2g}$.

Given, the horizontal range of the projectile is $4\sqrt{3}$ times of maximum height.

Therefore,

$R = 4\sqrt{3}H$

$\frac{v_0^2 \sin 2\theta}{g} = 4\sqrt{3}\frac{v_0^2 \sin^2 \theta}{2g}$

$2\sin\theta\cos\theta = 2\sqrt{3}\sin^2\theta$

$\frac{1}{\sqrt{3}} = \tan\theta$

$\theta = 30^\circ$

Therefore, if the horizontal range of the projectile is $4\sqrt{3}$ times of maximum height, the angle of

projection will be 30°.

22. Correct option: (D)

We know, height H , the maximum vertical distance covered by the projectile is given by:

$H = \frac{v_0^2 \sin^2 \theta}{2g}$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

We know, the time of flight T , the total time taken by the projectile in its journey, is given by $T = \frac{2v_0 \sin\theta}{g}$.

From first equation, we get:

$H = \frac{v_0^2 \sin^2 \theta}{2g}$

$\sin^2\theta = \frac{2Hg}{v_0^2}$

$\sin\theta = \frac{\sqrt{2Hg}}{v_0}$

Substituting $\sin\theta = \frac{\sqrt{2Hg}}{v_0}$ in $T = \frac{2v_0 \sin\theta}{g}$, we get:

$T = \frac{2v_0\left(\frac{\sqrt{2Hg}}{v_0}\right)}{g}$

$T = 2\sqrt{\frac{2H}{g}}$

Therefore, if a body projected at angle θ reaches a maximum height H, the total time of flight is

$2\sqrt{\frac{2H}{g}}$.

23. Correct option: (A)

We know, the time of flight T, the total time taken by the projectile in its journey, is given by $T = \frac{2v_0 \sin\theta}{g}$ where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Given, a ball is thrown by a boy is caught in 2 seconds later by another at some distance away on the same level, if angle of projection is 30°.

Therefore, the time of flight of the ball is 2s.

Hence,

$2 = \frac{2v_0 \sin 30^\circ}{g}$

$2 = \frac{2v_0(1/2)}{g}$

$v_0 = 2g$

$v_0 = 2(9.8)$

$v_0 = 19.6 m/s$

Therefore, if a ball thrown by a boy is caught in 2 seconds later by another boy at some distance away on the same level, and angle of projection is 30°, then the velocity of projection is $19.6\,m/s$.

24. Correct option: (C)

We know, for a projectile subjected to uniform gravitational field, the equation of the path of the projectile is given by:

$x = (v_0 \cos\theta)t$

$y = (v_0 \sin\theta)t - \frac{1}{2}gt^2$

where θ is the angle of projection, and v_0 is the initial velocity.

Eliminating t from the first two equations of the path of the projectile by solving for t in first equation and substituting in the second equation, we get:

$y = (v_0 \sin\theta)\frac{x}{v_0 \cos\theta} - \frac{1}{2}g\left(\frac{x}{v_0 \cos\theta}\right)^2$

$y = x\tan\theta - \frac{1}{2}g\left(\frac{x^2}{v_0^2 \cos^2\theta}\right)$

Now, the angle φ made by the direction of motion with the horizontal is given by:

$\tan\varphi = \frac{dy}{dx}$

$\tan\varphi = \tan\theta - \frac{gx}{v_0^2 \cos^2\theta}$

Given, a body is projected from a level ground with a velocity of $30\,\text{m/s}$ at 30° to the horizontal.

Therefore, the position of the projectile after 1.5 s the projection is:

$x = (30\cos 30^\circ)(1.5)$

$x = (30\frac{\sqrt{3}}{2})(1.5)$

$x = \frac{90\sqrt{3}}{4}$

Now, the angle made by its direction of motion 1.5 s the projection with horizontal is:

$$\tan\varphi = \tan 30^\circ - \frac{g\left(\frac{90\sqrt{3}}{4}\right)}{(30^2)\cos^2 30^\circ}$$

$$\tan\varphi = \frac{1}{\sqrt{3}} - \frac{15(2)}{30(\sqrt{3})}$$

$\tan\varphi = 0$

$\varphi = 0$

Therefore, if a body is projected from a level ground with a velocity of $30\,\text{m/s}$ at 30° to the horizontal, then the angle made by its direction of motion with horizontal after 1.5 s the projection is 0.

25. Correct option: (C)

We know, for the projectile subjected to uniform gravitational field, the vertical velocity v_y and horizontal velocity v_x varies as:

$v_x = (u\cos\theta)$

$v_y = (u\sin\theta) - gt$

The horizontal velocity of the projectile remains unchanged throughout the motion, and the vertical velocity changes continuously.

At the end of the trajectory, the projectile has the same magnitude of the velocity, but since the projectile is now moving downwards, its vertical velocity is also directed downwards.

Therefore, the change in velocity of the projectile when it reaches ground = Initial vertical velocity – Final vertical velocity

$u\sin\theta - (-u\sin\theta) = 2u\sin\theta$

Further, the time of flight T, the total time taken by the projectile in its journey, is given by $T = \frac{2u\sin\theta}{g}$

where u is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Given, a projectile at 30° to the horizontal reaches the ground in two seconds after its projection.

Therefore, the time of flight of the projectile is 2s. Hence,

$2 = \frac{2u\sin 30^\circ}{g}$

$u = 2g$

Therefore, the change in velocity of the projectile when it reaches ground is given by:

$2u\sin\theta = 2(2g)\sin 30°$

$2u\sin\theta = 2(2g)\frac{1}{2}$

$2u\sin\theta = 2g$

$2u\sin\theta = 2(9.8)$

$2u\sin\theta = 19.6\text{ms}^{-1}$

Therefore, if a projectile at 30° to the horizontal reaches the ground in two seconds after its projection, the change in velocity in that interval of time is $19.6\,\text{ms}^{-1}$.

26. Correct option: (B)

We know, for the projectile subjected to uniform gravitational field, the vertical velocity v_y and horizontal velocity v_x varies as:

$v_x = (u\cos\theta)$

$v_y = (u\sin\theta) - gt$

The horizontal velocity of the projectile remains unchanged throughout the motion, and the vertical velocity changes continuously.

At the end of the trajectory, the projectile has the same magnitude of the velocity, but since the projectile is now moving downwards, its vertical velocity is also directed downwards.

Therefore, the change in momentum of the projectile when it reaches ground = Final vertical momentum- Initial vertical momentum:

$-mu\sin\theta - mu\sin\theta - = -2mu\sin\theta$

Given, a body of mass 0.5 kg projected under gravity with a speed of 98 m/s at an angle of 30° with the horizontal.

Therefore, the change in momentum (in magnitude) of body is:

$2mu\sin\theta = 2(0.5)(98)\sin 30^\circ$

$2mu\sin\theta = 2(0.5)(98)(1/2)$

$2mu\sin\theta = 49.0\,\text{Ns}$

Therefore, if a body of mass 0.5 kg is projected under gravity with a speed of 98 m/s at an angle of 30° with the horizontal, then the change in momentum (in magnitude) of body is $49.0\,\text{Ns}$.

27. Correct option: (A)

We know, height H, the maximum vertical distance covered by the projectile is given by:

$$H = \frac{v_0^2\sin^2\theta}{2g}$$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Also, the potential energy U of an object at height h is given by:

$U = mgh$

where m is the mass of the object.

Given, a particle of mass 1kg fired with a speed of 10 m/s making an angle 30° with the horizontal.

Therefore, the potential energy at highest point is:

$$U = mg\frac{v_0^2 \sin^2\theta}{2g}$$

$$U = \frac{mv_0^2 \sin^2\theta}{2}$$

$$U = \frac{10^2 \sin^2 30^\circ}{2}$$

$$U = 12.5\,\text{J}$$

Therefore, if a particle of mass 1kg fired with a speed of $10\ \text{m/s}$ making an angle 30° with the horizontal, then the potential energy at highest point is $12.5\,\text{J}$.

28. Correct option: (B)

We know for a projectile in a uniform gravitational field, the vertical velocity v_y and horizontal velocity v_x varies as:

$$v_x = (v_0 \cos\theta)$$

$$v_y = (v_0 \sin\theta) - gt$$

where θ is the angle of projection, and v_0 is the initial velocity.

We see, the horizontal velocity remains constant, and the vertical velocity changes continuously throughout the motion. In particular, the vertical velocity decreases continuously during the upward motion, owing to the gravitational force acting in opposite direction, till the topmost point, where the vertical velocity is zero.

Given, a particle is projected making an angle 45° with horizontal having kinetic energy K.

Therefore,

$$\frac{1}{2}mv_0^2 = K$$

Now, the kinetic energy at the highest point is:

$$\frac{1}{2}m(v_0 \cos\theta)^2 = \frac{1}{2}mv_0^2 \cos^2 45^\circ$$

$$\frac{1}{2}m(v_0 \cos\theta)^2 = \frac{K}{2}$$

Therefore, if a particle is projected making an angle 45° with horizontal having kinetic energy K, then the kinetic energy at the highest point is $\frac{K}{2}$.

29. Correct option: (D)

We know for a projectile in a uniform gravitational field, the vertical velocity v_y and horizontal velocity v_x varies as:

$$v_x = (v_0 \cos\theta)$$

$$v_y = (v_0 \sin\theta) - gt$$

where θ is the angle of projection, and v_0 is the initial velocity.

We see, the horizontal velocity remains constant, and the vertical velocity changes continuously throughout the motion. In particular, the vertical velocity decreases continuously during the upward motion, owing to the gravitational force acting in opposite direction, till the topmost point, where the vertical velocity is zero.

Given, a particle is projected making an angle 60° with horizontal having kinetic energy K.

Therefore,

$$\frac{1}{2}mv_0^2 = K$$

Now, the kinetic energy at the highest point is:

$$\frac{1}{2}m(v_0\cos\theta)^2 = \frac{1}{2}mv_0^2\cos^2 60^\circ$$

$$\frac{1}{2}m(v_0\cos\theta)^2 = \frac{K}{4}$$

Therefore, if a particle is projected making an angle 45° with horizontal having kinetic energy K, then the kinetic energy at the highest point is $\frac{K}{4}$.

30. Correct option: (C)

We know for a projectile in a uniform gravitational field, the vertical velocity v_y and horizontal velocity v_x varies as:

$$v_x = (v_0\cos\theta)$$

$$v_y = (v_0\sin\theta) - gt$$

where θ is the angle of projection, and v_0 is the initial velocity.

We see, the horizontal velocity remains constant, and the vertical velocity changes continuously throughout the motion. In particular, the vertical velocity decreases continuously during the upward motion, owing to the gravitational force acting in opposite direction, till the topmost point, where the vertical velocity is zero.

Then, the vertical velocity increases for the downward motion, till it reaches the height it was originally released from, where the vertical velocity assumes its maximum value.

Therefore, the kinetic energy of the projectile is maximum at the point of projection and the point it reaches the ground.

Given, the range of a projectile is 100 m.

Therefore, its kinetic energy will be maximum after covering a distance equal to the range 100 m.

LEVEL-II

1. Correct option: (B)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$$R = \frac{v_0^2\sin 2\theta}{g}$$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Given, a projectile is projected with initial velocity $6\hat{\mathrm{i}} + 8\hat{\mathrm{j}}\,\mathrm{m/s}$.

We know, for a projectile with v_0 the magnitude of the initial velocity and θ the angle of projection, the horizontal velocity is $v_0\cos\theta$ and vertical velocity is $v_0\sin\theta$.

Comparing this with the initial velocity $6\hat{i}+8\hat{j}$, we thus get:

$v_0 \cos\theta = 6$

$v_0 \sin\theta = 8$

Therefore, the range of the projectile is:

$$R = \frac{2(v_0 \sin\theta)(v_0 \cos\theta)}{g}$$

$$R = \frac{2(6)(8)}{10}$$

$$R = 9.6\,m$$

Therefore, if a projectile is projected with initial velocity $6\hat{i}+8\hat{j}\,\text{m/s}$, then the range of the projectile is 9.6 m.

2. Correct option: (B)

We know, the height H, the maximum vertical distance covered by the projectile is given by:

$$H = \frac{v_0^2 \sin^2\theta}{2g}$$

where θ is the angle of projection, and v_0 is the initial velocity.

The vertical velocity decreases continuously during the upward motion, owing to the gravitational force acting in opposite direction, till the topmost point, where the vertical velocity is zero. Therefore, the velocity at the topmost point is just the horizontal velocity, $v_0 \cos\theta$.

Given, the speed of the projectile when it is at its greatest height is $\sqrt{\frac{2}{5}}$ it's speed at half the maximum height.

Now, using the relation relating velocity v to the distance covered s for constant acceleration motion $v^2 = u^2 + 2as$, we get:

$$v^2 = v_0^2 - 2g\frac{H}{2}$$

$$v^2 = v_0^2 - 2g\frac{v_0^2 \sin^2\theta}{4g}$$

$$v = \sqrt{v_0^2 - \frac{v_0^2 \sin^2\theta}{2}}$$

Since the speed of the projectile when it is at its greatest height is $\sqrt{\frac{2}{5}}$ it's speed at half the maximum height, so:

$$v_0 \cos\theta = \sqrt{\frac{2}{5}}\sqrt{v_0^2 - \frac{v_0^2 \sin^2\theta}{2}}$$

$$\cos^2\theta = \frac{2}{5}\left(1 - \frac{\sin^2\theta}{2}\right)$$

$$1 - \sin^2\theta = \frac{2}{5} - \frac{\sin^2\theta}{5}$$

$$5 - 5\sin^2\theta = 2 - \sin^2\theta$$

$$3 = 4\sin^2\theta$$

$$\sin\theta = \frac{\sqrt{3}}{2}$$

$$\theta = 60^\circ$$

Therefore, if the speed of the projectile when it is at its greatest height is $\sqrt{\frac{2}{5}}$ its speed at half the maximum height, then the angle of projection is 60°.

3. Correct option: (A)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$$R = \frac{v_0^2 \sin 2\theta}{g}$$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, the height H, the maximum vertical distance covered by the projectile is given by:

$$H = \frac{v_0^2 \sin^2\theta}{2g}$$

Given, an artillery piece consistently shoots its shell with same muzzle speed has a maximum range R. We want to find the elevation angle of gun to hit a target which is $R/2$ from the gun and on the same level.

The maximum range is given when $\theta = 45^\circ$. Therefore,

$$R = \frac{v_0^2}{g}$$

Now, the elevation angle of gun to hit a target which is $R/2$ from the gun and on the same level is given by:

$$\frac{v_0^2 \sin 2\theta}{g} = \frac{R}{2}$$

$$\frac{v_0^2 \sin 2\theta}{g} = \frac{v_0^2}{2g}$$

$$\sin 2\theta = \frac{1}{2}$$

$$\theta = 15^\circ$$

Therefore, if an artillery piece consistently shoots its shell with same muzzle speed and has a maximum range, then the elevation angle of gun to hit a target which is $R/2$ from the gun and on the same level is 15°.

4. Correct option: (A)

We know, the height H, the maximum vertical distance covered by the projectile is given by:

$$H = \frac{v_0^2 \sin^2 \theta}{2g}$$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, the time of flight T, the total time taken by the projectile in its journey, is given by $T = \frac{2v_0 \sin\theta}{g}$

Therefore, ratio of maximum height reached to the square of time of flight is:

$$\frac{H}{T^2} = \frac{\frac{v_0^2 \sin^2 \theta}{2g}}{\left(\frac{2v_0 \sin\theta}{g}\right)^2}$$

$$\frac{H}{T^2} = \frac{g}{8}$$

$$\frac{H}{T^2} = \frac{10}{8}$$

$$\frac{H}{T^2} = \frac{5}{4}$$

Therefore, ratio of maximum height reached to the square of time of flight is $\frac{5}{4}$.

5. Correct option: (C)

We know, for a projectile subjected to uniform gravitational field, the equation of the path of the projectile is given by:

$$x = (v_0 \cos\theta)t$$

$$y = (v_0 \sin\theta)t - \frac{1}{2}gt^2$$

where θ is the angle of projection, and v_0 is the initial velocity.

Given, a stone is projected from the ground with a velocity 14 m/s and one second later it clears a wall 2m high.

Therefore, using $y = (v_0 \sin\theta)t - \frac{1}{2}gt^2$, we get:

$$2 = (14\sin\theta) - \frac{1}{2}g$$

$$2 = (14\sin\theta) - 5$$

$$7/14 = \sin\theta$$

$$\theta = 30^\circ$$

Therefore, if a stone is projected from the ground with a velocity 14 m/s and one second later it clears a wall 2m high, then the angle of projection is 30°.

6. Correct option: (B)

We know, the height H, the maximum vertical distance covered by the projectile is given by:

$$H = \frac{v_0^2 \sin^2 \theta}{2g}$$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, the time of flight T, the total time taken by the projectile in its journey, is given by $T = \frac{2v_0 \sin \theta}{g}$

Given, the maximum height reached by the projectile is h.

Therefore, time of flight expressed in terms of the maximum height h is:

$$T = \frac{2v_0 \sin \theta}{g}$$

$$T = \frac{2\sqrt{2gh}}{g}$$

$$T = \sqrt{\frac{8h}{g}}$$

Therefore, if the maximum height reached by the projectile is h, then time of flight is $\sqrt{\frac{8h}{g}}$.

7. Correct option: (D)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$$R = \frac{v_0^2 \sin 2\theta}{g}$$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Given, a person throws a bottle into a dustbin at the same height as he is 2m away at an angle of 45°.

Therefore, the range of the bottle is 2 m, hence:

$$2 = \frac{v_0^2 \sin 90^\circ}{g}$$

$$v_0 = \sqrt{2g}$$

Therefore, if a person throws a bottle into a dustbin at the same height as he is 2m away at an angle of 45°, then velocity with which it was thrown is $\sqrt{2g}$.

8. Correct option: (C)

We know, the height H, the maximum vertical distance covered by the projectile is given by:

$$H = \frac{v_0^2 \sin^2 \theta}{2g}$$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

We know, the potential energy U of an object at height h is given by:

$U = mgh$

where m is the mass of the object.

Further, the kinetic energy K of the object is given by:

$K = \frac{1}{2}mv^2$

where v is the velocity of the object.

Given, the potential energy of the projectile at maximum height is $\frac{3}{4}$ times kinetic energy of projection. Therefore,

$mg\frac{v_0^2 \sin^2\theta}{2g} = \frac{3}{4}\left(\frac{1}{2}mv_0^2\right)$

$\frac{\sin^2\theta}{2} = \frac{3}{4}\left(\frac{1}{2}\right)$

$\sin\theta = \frac{\sqrt{3}}{2}$

$\theta = 60^\circ$

Therefore, if the potential energy of the projectile at maximum height is $\frac{3}{4}$ times kinetic energy of projection, then angle of projection is 60°.

9. Correct options: (A), (B), (C)

We know, the projectile is subjected to uniform gravitational field, where the gravitational force acts vertically downward. Hence, the vertical velocity v_y and horizontal velocity v_x varies as:

$v_x = (v_0 \cos\theta)$

$v_y = (v_0 \sin\theta) - gt$

where θ is the angle of projection, and v_0 is the initial velocity.

We see, the horizontal velocity remains constant, and the vertical velocity changes continuously throughout the motion.

Further, the acceleration remains constant, equal to g, throughout the motion.

Also, the total energy, the sum of kinetic energy and potential energy also remains constant throughout the motion.

Therefore, out of the given options, the physical quantities which remains constant during projectile motion are acceleration due to gravity, horizontal component of velocity and total energy of projectile.

10. Correct options: (A), (B), (C), (D)

We know, the projectile is subjected to uniform gravitational field, where the gravitational force acts vertically downward. Hence, the vertical velocity v_y and horizontal velocity v_x varies as:

$v_x = (v_0 \cos\theta)$

$v_y = (v_0 \sin\theta) - gt$

where θ is the angle of projection, and v_0 is the initial velocity.

We see, the horizontal velocity remains constant, and the vertical velocity changes continuously throughout the motion.

Since velocity changes continuously throughout the motion, therefore the linear momentum as well as the kinetic energy will also change. Further, since the height of the projectile is changing, the potential energy will also varies.

Therefore, the physical quantities which change during projectile motion are velocity, linear momentum, kinetic energy and potential energy.

11. Correct options: (A), (C), (D)

We know, the height H, the maximum vertical distance covered by the projectile is given by:

$$H = \frac{v_0^2 \sin^2 \theta}{2g}$$

where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, the time of flight T, the total time taken by the projectile in its journey, is given by $T = \frac{2v_0 \sin\theta}{g}$

And the range R, the maximum horizontal distance covered by the projectile, is given by:

$$R = \frac{v_0^2 \sin 2\theta}{g}$$

Given, two bodies are projected at angles 30° and 60° to the horizontal from the ground such that its maximum heights reached by them are equal.

Therefore,

$$\frac{v_1^2 \sin^2 \theta_1}{2g} = \frac{v_2^2 \sin^2 \theta_2}{2g}$$

$$\frac{v_1^2 \sin^2 30^\circ}{2g} = \frac{v_2^2 \sin^2 60^\circ}{2g}$$

$$\frac{v_1^2}{4} = \frac{3v_2^2}{4}$$

$$v_1 = \sqrt{3} v_2$$

The ratio of the time of flight is:

$$\frac{T_1}{T_2} = \frac{\frac{2v_1 \sin\theta_1}{g}}{\frac{2v_2 \sin\theta}{g}}$$

$$\frac{T_1}{T_2} = \frac{\sqrt{3}\sin 30^\circ}{\sin 60^\circ}$$

$$\frac{T_1}{T_2} = \frac{\sqrt{3}(1/2)}{(\sqrt{3}/2)}$$

$\frac{T_1}{T_2} = 1$

Therefore, the time of flight are equal.

Now, the ratio of the range is:

$\frac{R_1}{R_1} = \frac{\frac{v_1^2 \, sin\, 6\,0^\circ}{g}}{\frac{sin\, 1\,20^\circ}{g}}$

$\frac{R_1}{R_1} = \frac{\frac{3\sqrt{3}}{2}}{\frac{\sqrt{3}}{2}}$

$\frac{R_1}{R_1} = 3$

Therefore, if two bodies are projected at angles 30° and 60° to the horizontal from the ground such that the maximum heights reached by them are equal, then their time of flights are also equal, the ratio of initial speeds of projection is $\sqrt{3}$ and both take same time to reach maximum height.

12. Correct options: (A, C)

We know, the time of flight T, the total time taken by the projectile in its journey, is given by, $T = \frac{2v_0 \sin\theta}{g}$, where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, the range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$.

Given, projectile have the same range R for two angles of projection. Let the angle of projections be θ_1 and θ_2.

Then,

$\frac{v_0^2 \sin 2\theta_1}{g} = \frac{v_0^2 \sin 2\theta_2}{g}$

$\sin 2\theta_1 = \sin 2\theta_2$

$2\sin\theta_1 \cos\theta_1 = 2\sin\theta_2 \cos\theta_2$

$\sin\theta_1 \sin(90^\circ - \theta_1) = \sin\theta_2 \sin(90^\circ - \theta_2)$

This equation is satisfied if $\theta_2 = \theta_1$ or when $\theta_2 = 90^\circ - \theta_1$.

Since the angle of projections are different, therefore:

$\theta_2 = 90^\circ - \theta_1$

Now, the time of flight for the two angles of projections are:

$$T_1 = \frac{2v_0 \sin\theta_1}{g}$$

$$T_2 = \frac{2v_0 \sin\theta_2}{g}$$

Multiplying these two equations, we get:

$$T_2T_1 = \frac{2v_0 \sin\theta_1}{g}\frac{2v_0 \sin\theta_2}{g}$$

$$T_2T_1 = \frac{4v_0^2 \sin\theta_1}{g}\frac{\sin(90^\circ - \theta_2)}{g}$$

$$T_2T_1 = \frac{4v_0^2 \sin\theta_1 \cos\theta_1}{g^2}$$

$$T_2T_1 = \frac{2v_0^2 \sin 2\theta_1}{g^2}$$

Using the formula for the range $R = \frac{v_0^2 \sin 2\theta}{g}$, we get: $T_2T_1 = \frac{2R}{g}$

Now, dividing the two equations, we get:

$$\frac{T_1}{T_2} = \frac{\frac{2v_0 \sin\theta_1}{g}}{\frac{2v_0 \sin\theta_2}{g}}$$

$$\frac{T_1}{T_2} = \frac{\sin\theta_1}{\sin(90^\circ - \theta_1)}$$

$$\frac{T_1}{T_2} = \frac{\sin\theta_1}{\sin(90^\circ - \theta_1)}$$

$$\frac{T_1}{T_2} = \frac{\sin\theta_1}{\cos\theta_1}$$

$$\frac{T_1}{T_2} = \tan\theta_1$$

Therefore, if projectile have the same range R for two angles of projection , then T_1T_2 is directly proportional to the range R and $T_1T_2 = \frac{2R}{g}$, and $\frac{T_1}{T_2} = \tan\theta_1$.

13. Correct option: (C)

We know, when the projectile is subjected to uniform gravitational field, the initial vertical velocity v_y and horizontal velocity v_x are given by:

$$v_x = v_0 \cos\theta$$

$$v_y = v_0 \sin\theta$$

where θ is the angle of projection, and v_0 is the initial velocity.

Given, a body is projected with a velocity of 60 m/s at an angle 30° to horizontal.

Therefore, the initial velocity is:

$\vec{v} = v_0 \cos\theta \hat{i} + v_0 \sin\theta \hat{j}$

$\vec{v} = 60\cos 30^\circ \hat{i} + 60\sin 30^\circ \hat{j}$

$\vec{v} = 60\frac{\sqrt{3}}{2}\hat{i} + 60\frac{1}{2}\hat{j}$

$\vec{v} = 30\sqrt{3}\hat{i} + 30\hat{j}$

Therefore, if a body is projected with a velocity of 60 m/s at an angle 30° to horizontal, then initial velocity vector is $30\sqrt{3}\hat{i} + 30\hat{j}$.

14. Correct option: (B)

When the projectile is subjected to uniform gravitational field, the vertical velocity v_y and horizontal velocity v_x varies as:

$v_x = (v_0 \cos\theta)$

$v_y = (v_0 \sin\theta) - gt$

where θ is the angle of projection, and v_0 is the initial velocity.

Given, a body is projected with a velocity of 60 m/s at an angle 30° to horizontal.

Therefore, the velocity after 3s is:

$\vec{v} = v_0 \cos\theta \hat{i} + (v_0 \sin\theta - gt)\hat{j}$

$\vec{v} = 60\cos 30^\circ \hat{i} + (60\sin 30^\circ - 10(2))\hat{j}$

$\vec{v} = 60\frac{\sqrt{3}}{2}\hat{i} + (60\frac{1}{2} - 20)\hat{j}$

$\vec{v} = 30\sqrt{3}\hat{i} + 10\hat{j}$

Therefore, if a body is projected with a velocity of 60 m/s at an angle 30° to the horizontal, then velocity after 3s is $30\sqrt{3}\hat{i} + 10\hat{j}$.

15. Correct option: (A)

We know, for a projectile subjected to uniform gravitational field, the equation of the path of the projectile is given by:

$x = (v_0 \cos\theta)t$

$y = (v_0 \sin\theta)t - \frac{1}{2}gt^2$

where θ is the angle of projection, and v_0 is the initial velocity.

Given, a body is projected with a velocity of 60 m/s at an angle 30° to horizontal.

Therefore, the displacement after 2s is:

$\vec{r} = (v_0 \cos\theta)\hat{i} + ((v_0 \sin\theta)t - \frac{1}{2}gt^2)\hat{j}$

$\vec{r} = 60\cos 30^\circ (2)\hat{i} + (60\sin 30^\circ (2) - 10(2))\hat{j}$

$\vec{r} = 60\frac{\sqrt{3}}{2}2\hat{i} + (60\frac{1}{2}2 - 20)\hat{j}$

$\vec{r} = 60\sqrt{3}\hat{i} + 40\hat{j}$

Therefore, if a body is projected with a velocity of 60 m/s at an angle 30° to horizontal, then displacement after 2s is $60\sqrt{3}\hat{i} + 40\hat{j}$.

16. Correct option: (B)

We know, when the projectile is subjected to uniform gravitational field, the initial vertical velocity v_y and horizontal velocity v_x are given by:

$v_x = v_0 \cos\theta$

$v_y = v_0 \sin\theta$

where θ is the angle of projection, and v_0 is the initial velocity.

We know, the time of flight T, the total time taken by the projectile in its journey, is given by, $T = \frac{2v_0 \sin\theta}{g}$, where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, the range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$.

Given, an arrow is shot into the air. Its range is 200 m and its time of flight is 5 sec. Therefore,

$200 = \frac{v_0^2 \sin 2\theta}{g}$

$5 = \frac{2v_0 \sin\theta}{g}$

Using the first equation, we can write:

$200 = \frac{2v_0^2 \sin\theta\cos\theta}{g}$

$200g = 2(v_0 \sin\theta)(v_0 \cos\theta)$

Now, from the second equation, we have $\frac{5g}{2} = v_0 \sin\theta$. Substituting this into earlier equation, we get:

$200g = 2\frac{5g}{2}(v_0 \cos\theta)$

$v_0 \cos\theta = 40$

Therefore, if an arrow is shot into the air and has range 200 m and time of flight is 5 sec, then horizontal component of velocity of arrow is 40 m/s.

17. Correct option: (C)

We know, when the projectile is subjected to uniform gravitational field, the initial vertical velocity v_y and horizontal velocity v_x are given by:

$v_x = v_0 \cos\theta$

$v_y = v_0 \sin\theta$

where θ is the angle of projection, and v_0 is the initial velocity.

We know, the time of flight T, the total time taken by the projectile in its journey, is given by, $T = \frac{2v_0 \sin\theta}{g}$, where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, the range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$.

Given, an arrow is shot into the air. Its range is 200 m and its time of flight is 5 sec. Therefore,

$200 = \frac{v_0^2 \sin 2\theta}{g}$

$5 = \frac{2v_0 \sin \theta}{g}$

Using the first equation, we can write:

$200 = \frac{2v_0^2 \sin \theta \cos \theta}{g}$

$200g = 2(v_0 \sin \theta)(v_0 \cos \theta)$

Now, from the second equation, we have

$\frac{5g}{2} = v_0 \sin \theta$.

$25 = v_0 \sin \theta$

Therefore, if an arrow is shot into the air and has range 200 m and time of flight is 5 sec, then vertical component of velocity of arrow is 25 m/s.

18. Correct option: (C)

We know, when the projectile is subjected to uniform gravitational field, the initial vertical velocity v_y and horizontal velocity v_x are given by:

$v_x = v_0 \cos \theta$

$v_y = v_0 \sin \theta$

where θ is the angle of projection, and v_0 is the initial velocity.

We know, the time of flight T, the total time taken by the projectile in its journey, is given by, $T = \frac{2v_0 \sin \theta}{g}$, where v_0 is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Further, the range R, the maximum horizontal distance covered by the projectile, is given by:

$R = \frac{v_0^2 \sin 2\theta}{g}$.

Given, an arrow is shot into the air. Its range is 200 m and its time of flight is 5 sec.

In earlier two questions, we have calculated the horizontal and vertical component as:

$v_0 \cos \theta = 40$

$v_0 \sin \theta = 25$

Dividing these two equations, we get:

$\tan \theta = 25 / 40$

$\tan \theta = 5 / 8$

Therefore, if an arrow is shot into the air and has range 200 m and time of flight is 5 sec, then angle of projection with the horizontal is $\tan^{-1} 5/8$.

19. Correct match: (A) ->R), (B)->P), (C)->S), (D)-> Q)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$$R = \frac{u^2 \sin 2\theta}{g}$$

where u is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

(A) Given, the magnitude of the initial velocity of the projectile is 20 m/s, and angle of projection is 30°.

Therefore, the range is given by:

$$R = \frac{20^2 \sin 60}{g}$$

$$R = \frac{400(\sqrt{3}/2)}{10}$$

$$R = 20\sqrt{3}$$

(B) Given, the magnitude of the initial velocity of the projectile is 30 m/s, and angle of projection is 60°.

Therefore, the range is given by:

$$R = \frac{30^2 \sin 120^\circ}{g}$$

$$R = \frac{900(\sqrt{3}/2)}{10}$$

$$R = 45\sqrt{3}$$

(C) Given, the magnitude of the initial velocity of the projectile is 30 m/s, and angle of projection is 45°.

Therefore, the range is given by:

$$R = \frac{30^2 \sin 90^\circ}{g}$$

$$R = \frac{900}{10}$$

$$R = 90$$

(D) Given, the magnitude of the initial velocity of the projectile is 20 m/s, and angle of projection is 45°.

Therefore, the range is given by:

$$R = \frac{20^2 \sin 90^\circ}{g}$$

$$R = \frac{400}{10}$$

$$R = 40$$

Therefore, the correct match is (A) ->R), (B)->P), (C)->S), (D)-> Q)

20. Correct match: (A) ->R), (B)->S), (C)->P), (D)-> Q)

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$$R = \frac{v^2 \sin 2\theta}{g}$$

where v is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

We know, the height H, the maximum vertical distance covered by the projectile is given by:

$$H = \frac{v_0^2 \sin^2 \theta}{2g}$$

Given, a ball is projected from the ground with velocity v such that its range is maximum.

Now, the range is maximum when $\theta = 45^\circ$.

(A) When the projectile is subjected to uniform gravitational field, the vertical velocity v_y and horizontal velocity v_x varies as:

$$v_x = (v\cos\theta)$$

$$v_y = (v\sin\theta) - gt$$

where θ is the angle of projection, and v is the initial velocity.

Now, the horizontal velocity is constant throughout the motion.

Using $v_f{}^2 = v_i^2 + 2ay$ for constant acceleration motion, the vertical velocity at half of the maximum height is given by:

$$v_y{}^2 = (v\sin\theta)^2 - 2g\left(\frac{v_0^2 \sin^2 \theta}{4g}\right)$$

$$v_y{}^2 = (v\sin 45^\circ)^2 - 2g\left(\frac{v_0^2 \sin^2 45^\circ}{4g}\right)$$

$$v_y{}^2 = \frac{v^2}{2} - \frac{v^2}{4}$$

$$v_y{}^2 = \frac{v^2}{4}$$

Therefore, the magnitude of the velocity at half of the maximum height is:

$$v = \sqrt{v_x^2 + v_y^2}$$

$$v = \sqrt{(v\cos 45^\circ)^2 + \frac{v^2}{4}}$$

$$v = \frac{\sqrt{3}}{2}v$$

(B) Now, the vertical velocity at the maximum height is given by:

$$v_y{}^2 = (v\sin\theta)^2 - 2g\left(\frac{v_0^2 \sin^2 \theta}{2g}\right)$$

$$v_y{}^2 = (v\sin 45^\circ)^2 - 2g\left(\frac{v_0^2 \sin^2 45^\circ}{2g}\right)$$

$$v_y{}^2 = \frac{v^2}{2} - \frac{v^2}{2}$$

$$v_y{}^2 = 0$$

Therefore, the magnitude of the velocity at half of the maximum height is:

$$v = \sqrt{v_x^2 + v_y^2}$$

$$v = \sqrt{(v\cos 45^\circ)^2 + 0}$$

$$v = \frac{v}{\sqrt{2}}$$

(C) Now, the horizontal velocity of the projectile remains unchanged throughout the motion, and the vertical velocity changes continuously.

At the end of the trajectory, the projectile has the same magnitude of the velocity, but since the projectile is now moving downwards, its vertical velocity is also directed downwards.

Therefore, the change in velocity of the projectile for a complete trajectory = Initial vertical velocity – Final vertical velocity

$$v\sin\theta - (-v\sin\theta) = 2v\sin 45^\circ$$

$$v\sin\theta - (-v\sin\theta) = \sqrt{2}v$$

(D) Now, average velocity when it reaches the maximum height is given by the total distance travelled divided by the total time take.

The total distance travelled is $\sqrt{(R/2)^2 + H^2}$ and the total time taken is $T/2$, therefore, the average velocity is:

$$\frac{\sqrt{\left(\frac{v^2}{2g}\right)^2 + \left(\frac{v_0^2}{4g}\right)^2}}{\frac{2v_0}{2\sqrt{2}g}} = \frac{\frac{v_0^2}{2g}\sqrt{1+\frac{1}{4}}}{\frac{v_0}{\sqrt{2}g}}$$

$$\frac{\sqrt{\left(\frac{v^2}{2g}\right)^2 + \left(\frac{v_0^2}{4g}\right)^2}}{\frac{2v_0}{2\sqrt{2}g}} = \frac{v_0}{2}\sqrt{\frac{5}{2}}$$

Therefore, the correct match is (A) ->R), (B)->S), (C)->P), (D)-> Q)

21. Correct number: 4

We know, the range R, the maximum horizontal distance covered by the projectile, is given by:

$$R = \frac{v^2 \sin 2\theta}{g}$$

where v is the magnitude of the initial velocity of the projectile, and θ is the angle of projection.

Given, a projectile is thrown with an initial speed u and angle of projection 15°, to the horizontal has a range R. Therefore,

$$R = \frac{u^2 \sin 30^\circ}{g}$$

$$R = \frac{u^2}{2g}$$

If the same projectile is thrown at an angle of 45°, with speed 2u, then the range is:

$$R' = \frac{4u^2 \sin 90^\circ}{g}$$

$$R' = \frac{4u^2}{2g}$$

$$R' = 4R$$

Therefore, if a projectile is thrown with an initial speed u and angle of projection is 15° to the horizontal, the range is R and if the same projectile is thrown at an angle of 45°, with speed 2u, then the range will be increases by 4 times.

Horizontal Projection from The Top of a Tower

LEVEL-I

1. Correct answer: (B)

We know, for constant acceleration a, the distance travelled s is given by:

$$s = ut + \frac{1}{2}at^2$$

where u is the initial velocity and t is the time taken.

Given, a body is projected horizontally from the top of a tower. Hence, a constant gravitational force acts on the body vertically downward.

Since the body is released horizontally, the initial vertical velocity is zero. Taking the height of the tower to be h, we get, using $s = ut + \frac{1}{2}at^2$.

$$h = 0(t) + \frac{1}{2}gt^2$$

Or, solving for the time:

$$t = \sqrt{\frac{2h}{g}}$$

We see, it depends only on the height of the tower.

Therefore, if a body is projected horizontally from the top of a tower, then the time taken by it to reach the ground depends upon height of the tower.

2. Correct option: (C)

We know, for constant acceleration a, the distance travelled s is given by:

$$s = ut + \frac{1}{2}at^2$$

where u is the initial velocity and t is the time taken.

Given, two stones A and B thrown with a velocity 10 m/s ad 20 m/s from the top of a tower horizontally. Hence, a constant gravitational force acts on the stones vertically downward.

Since the stones are released horizontally, the initial vertical velocity is zero. Taking the height of the tower to be h, we get, using $s = ut + \frac{1}{2}at^2$.

$$h = 0(t) + \frac{1}{2}gt^2$$

Or, solving for the time:

$$t = \sqrt{\frac{2h}{g}}$$

We see, the time taken by the stone depends only on the height of the tower.
Since both stones are released from the same height, the time taken by stone A to reach the ground is the same as the time taken by stone B to reach the ground,

$t_1 = t_2$

Therefore, two stones A and B thrown with a velocity 10 m/s ad 20 m/s from the top of a tower horizontally reach the ground after time t_1 and t_2, then $t_1 = t_2$.

3. Correct option: (D)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$x = ut$

where u is the horizontal velocity of projection.

Given, a particle projected horizontally from the top of a table touches the ground at a distance 'd' from the edge of the table. Also, h is the height of the table.

Therefore, using $t = \sqrt{\frac{2h}{g}}$, the time taken by the particle to reach the ground is:

$$t = \sqrt{\frac{2h}{g}}$$

Since, the horizontal distance travelled by the projectile is d, we have:

$$d = u\left(\sqrt{\frac{2h}{g}}\right)$$

Therefore, the velocity of projection is:

$$u = d\left(\sqrt{\frac{g}{2h}}\right)$$

Therefore, if a particle projected horizontally from the top of a table touches the ground at a distance 'd' from the edge of the table and h is the height of the table, then the velocity of projection is $d\left(\sqrt{\frac{g}{2h}}\right)$.

4. Correct option: (C)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$x = ut$

where u is the horizontal velocity of projection.
Given, a body is projected horizontally with a velocity 10 m/s from the top of a tower of height 19.6 m.

Therefore, the time taken by the body to fall to the ground is:

$$t = \sqrt{\frac{2(19.6)}{9.8}}$$

$$t = \sqrt{4}$$

$$t = 2$$

Therefore, the distance at which it falls from the foot of the tower is:

$$y = 10(2)$$

$$y = 20\,m$$

Therefore, if a body is projected horizontally with a velocity 10 m/s from the top of a tower of height 19.6 m, the distance at which it falls from the foot of the tower is 20 m.

5. Correct option: (D)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$$x = ut$$

where u is the horizontal velocity of projection.

Given, a ball thrown horizontally from the top of a tower with a speed of 12.5 m/s strikes
the ground 75 m away from the tower.

Therefore, using $y = ut$, we have:

$$75 = 12.5(t)$$

$$t = 6$$

Now, using $t = \sqrt{\frac{2h}{g}}$, we get for the height of the tower:

$$6 = \sqrt{\frac{2h}{g}}$$

$$h = \frac{36g}{2}$$

$$h = 18(10)$$

$$h = 180\,m$$

Therefore, if a ball thrown horizontally from the top of a tower with a speed of 12.5 m/s strikes the ground 75 m away from the tower, then the height of the tower is 180 m.

6. Correct option: (A)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$$x = ut$$

where u is the horizontal velocity of projection.

Given, an aeroplane flying horizontally at an altitude of 490 m with a speed 180 kmph drops a bomb.

When the bomb is released, it is at a height of 490 m and will have the same horizontal speed as the aeroplane.
Therefore, the time taken by the bomb to reach the ground is:

$$t = \sqrt{\frac{2(490)}{9.8}}$$

$$t = \sqrt{100}$$

$$t = 10$$

Hence, the horizontal distance at which it hits the ground is:

$$y = 180\left(\frac{1000}{3600}\right)10$$

$$y = 500\,m$$

Therefore, if an aeroplane flying horizontally at an altitude of 490 m with a speed 180 kmph drops a bomb, the horizontal distance at which it hits the ground is 500 m.

7. Correct option: (D)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$$x = ut$$

where u is the horizontal velocity of projection.
Given, an aeroplane moving horizontally with a speed of 720 km/h, drops a food packet.
While flying at a height 396.9 m.
When the food packet is released, it is at a height of 396.9 m and will have the same horizontal speed as the aeroplane.
Therefore, the time taken by the food packet to reach the ground is:

$$t = \sqrt{\frac{2(396.9)}{9.8}}$$

$$t = \sqrt{81}$$

$$t = 9\,s$$

Therefore, if an aeroplane moving horizontally with a speed of 720 km/h, drops a food packet while flying at a height 396.9 m, the time taken by food packet to reach the ground is 9 s.

8. Correct option: (B)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$$x = ut$$

where u is the horizontal velocity of projection.
Given, at the height 80 m an aeroplane is moving with 150 m/s and a bomb is dropped from
it so as to hit a target.

Now, once the bomb is dropped, it will fall downward, simultaneously moving in the forward direction. Therefore, the distance from the target the bomb should be dropped is equal to the horizontal distance covered by the bomb while falling to ground.
Now, time taken by the bomb to reach the ground is:

$$t = \sqrt{\frac{2(80)}{10}}$$

$$t = 4$$

Therefore, the bomb should be dropped at a distance:

$$y = 150(4)$$

$$y = 600$$

Therefore, if at the height 80 m an aeroplane is moving with 150 m/s and a bomb is dropped from it so as to hit a target, then the distance from the target the bomb should be dropped is 600 m.

9. Correct option: (C)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$$x = ut$$

where u is the horizontal velocity of projection.
Given, a bomb is dropped from an aircraft travelling horizontally at 150 m/s at a height of
490 m.
When the bomb is released, it is at a height of 490 m and will have the same horizontal speed as the aeroplane.
Therefore, the time taken by the bomb to reach the ground is:

$$t = \sqrt{\frac{2(490)}{9.8}}$$

$$t = \sqrt{100}$$

$$t = 10$$

Hence, the horizontal distance at which it hits the ground is:

$$y = 150(10)$$

$$y = 1500\,m$$

Therefore, if an aeroplane flying horizontally at an altitude of 490 m with a speed 150 m/s drops a bomb, the horizontal distance travelled by the bomb before reaching the ground is 1500 m.

10. Correct option: (C)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$$x = ut$$

where u is the horizontal velocity of projection.

Given, a body is thrown horizontally from top of the tower of 5 m height and it touches the
ground of 10m away from the foot of the tower.
Therefore, the time taken by the body to reach the ground is:

$$t = \sqrt{\frac{2(5)}{10}}$$

$$t = 1$$

Now, as the body touches the ground 10 m away from the foot of tower, we have:

$$10 = u(1)$$

$$u = 10$$

Therefore, if a body is thrown horizontally from top of the tower of 5 m height and touches the ground 10m away from the foot of the tower, then initial velocity of the body is 10 m/s.

11. Correct option: (A)

We know, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$$v_x = u$$

$$v_y = -gt$$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.
Given, a body is projected horizontally from a height of 78.4 m with a velocity of 10 m/s.
Therefore, its velocity after 3 seconds is:

$$\vec{v} = 10\hat{i} - g(3)\hat{j}$$

$$\vec{v} = 10\hat{i} - 30\hat{j}$$

Therefore, if body is projected horizontally from a height of 78.4 m with a velocity 10 m/s, its velocity after 3 seconds is $10\hat{i} - 30\hat{j}$.

12. Correct option: None of the given option is correct

We know, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$$v_x = u$$

$$v_y = -gt$$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.
Given, a body is projected horizontally from the top of a tower with 20 m/s.
Therefore, its velocity after 4 seconds is:

$$\vec{v} = 20\hat{i} - g(4)\hat{j}$$

$$\vec{v} = 20\hat{i} - g(4)\hat{j}$$

$$\vec{v} = 20\hat{i} - 40\hat{j}$$

Therefore, the magnitude of the velocity is:

$$v = \sqrt{400 + 1600}$$

$$v = 44.72$$

Therefore, if a body is projected horizontally from the top of a tower with 20 m/s, its velocity after 4 seconds is 44.72 m/s.

13. Correct option: (B)

We know, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$v_x = u$

$v_y = -gt$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.

Further, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$t = \sqrt{\frac{2h}{g}}$

Given, a stone is thrown horizontally with a velocity g m/s from the top of tower of height g m.

Therefore, the time taken by the stone to reach the ground is:

$t = \sqrt{\frac{2g}{g}}$

$t = \sqrt{2}$

Therefore, the horizonal and vertical velocity of the stone after it reaches the ground are:

$v_x = g$

$v_y = -g\sqrt{2}$

Therefore, the speed of the stone with which it hits the ground is:

$v = \sqrt{v_x^2 + v_y^2}$

$v = \sqrt{g^2 + 2g^2}$

$v = g\sqrt{3}$

Therefore, if a stone is thrown horizontally with a velocity g m/s from the top of tower of height g m, then the speed of the stone with which it hits the ground is $g\sqrt{3}$ m/s.

14. Correct option: (B)

We know, the horizontal distance and vertical distance covered by the horizontally launched projectile in time t is given by:

$x = ut$

$y = \frac{1}{2}gt^2$

where u is the horizontal velocity of projection.

Eliminating t from these equations, we get:

$y = \frac{1}{2}g\left(\frac{x}{u}\right)^2$

$y = \frac{gx^2}{2u^2}$

Therefore, the angle θ, the direction of motion makes with the horizontal is given by:

$$\tan\theta = \frac{dy}{dx}$$

$$\tan\theta = \frac{gx}{u^2}$$

Given, a body is projected horizontally from certain height. After 2 seconds its direction of

motion makes 30° to the horizontal.

Now, the horizontal distance travelled by the projectile after 2s is:

$x = u(2)$

Therefore, the angle θ, the direction of motion makes with the horizontal is:

$$\tan 30^\circ = \frac{g(2u)}{u^2}$$

$$\frac{1}{\sqrt{3}} = \frac{g(2)}{u}$$

$$u = 2g\sqrt{3}$$

Therefore, if a body is projected horizontally from certain height and after 2 seconds its direction of motion makes 30° to the horizontal, then its initial velocity is $2g\sqrt{3}$ m/s.

15. Correct option: (C)

We know, the horizontal distance and vertical distance covered by the horizontally launched projectile in time t is given by:

$x = ut$

$$y = \frac{1}{2}gt^2$$

where u is the horizontal velocity of projection.

Eliminating t from these equations, we get:

$$y = \frac{1}{2}g\left(\frac{x}{u}\right)^2$$

$$y = \frac{gx^2}{2u^2}$$

Therefore, the angle θ, the direction of motion makes with the horizontal is given by:

$$\tan\theta = \frac{dy}{dx}$$

$$\tan\theta = \frac{gx}{u^2}$$

Given, a body is projected horizontally from certain height with a velocity of 19.6 m/s.

Therefore, the angle θ, the direction of motion makes with the horizontal after 2s is:

$$\tan\theta = \frac{g19.6(2)}{(19.6)^2}$$

$$\tan\theta = 1$$

$$\theta = 45^\circ$$

Therefore, if a body is projected horizontally from certain height with velocity 19.6 m/s, then after 2s, the angle made by its direction of motion to the horizontal is 45°.

16. Correct option: (C)

We know, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$v_x = u$

$v_y = -gt$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.

Given, a stone is projected horizontally from the top of a tower with a velocity 9.8 m/s of height 100 m.

Therefore, its velocity after 1 seconds is:

$\vec{v} = 9.8\hat{i} - g(1)^2\,\hat{j}$

$\vec{v} = 9.8\hat{i} - 9.8\,\hat{j}$

Therefore, the magnitude of the velocity is:

$v = \sqrt{9.8^2 + (9.8)^2}$

$v = 9.8\sqrt{2}$

Therefore, if a stone is projected horizontally from the top of a tower with a velocity 9.8 m/s of height 100 m, the magnitude of the velocity after 1s is $9.8\sqrt{2}$ m/s.

17. Correct option: (B)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$t = \sqrt{\frac{2h}{g}}$

Further, the horizontal distance covered by the projectile in time t is given by:

$x = ut$

where u is the horizontal velocity of projection.

Given, from the top of a building 78.4 m high, a ball is thrown horizontally hits the ground at a distance and the line joining the top of the building to the point where ball hits the ground makes an angle 45° with horizontal.

Now, using the right-angle triangle, the horizontal distance travelled by the ball till it hits the ground is:

$x = h\tan\theta$

$x = 78.4\tan 45^\circ$

$x = 78.4$

Also, the time it takes the ball to reach the ground is:

$t = \sqrt{\frac{2(78.4)}{9.8}}$

$t = 4$

Therefore, the initial velocity of the projectile is, using $x = ut$,

$78.4 = u(4)$

$u = 19.6$

Therefore, if from the top of a building 78.4 m high, a ball thrown horizontally hits the ground at a distance and the line joining the top of the building to the point where ball hits the ground makes an angle 45° with horizontal, then the initial velocity of the projectile is 19.6 m/s.

18. Correct option: (D)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$$x = ut$$

where u is the horizontal velocity of projection.

Given, a body projected horizontally with 10 m/s, from the top of a tower of height 20 m.

Therefore, the time taken by body to reach the ground is:

$$t = \sqrt{\frac{2(20)}{10}}$$

$$t = 2$$

Hence, the horizontal distance covered by the body till it reach the ground is:

$$x = 10(2)$$

$$x = 20$$

Now, the displacement of the body is given by subtracting the final position from the initial position:

$$\text{displacement} = (20\hat{i} + 0\hat{j}) - (0\hat{i} + 20\hat{j})$$

$$\text{displacement} = 20\hat{i} - 20\hat{j}$$

Now, the magnitude of the displacement is:

$$\sqrt{20^2 + 20^2} = 20\sqrt{2}$$

Therefore, if a body projected horizontally with 10 m/s, from the top of a tower of height 20 m, then on reaching the ground its displacement is $20\sqrt{2}$ m.

19. Correct option: (C)

We know, the horizontal distance and vertical distance covered by the horizontally launched projectile in time t is given by:

$$x = ut$$

$$y = \frac{1}{2}gt^2$$

where u is the horizontal velocity of projection.

Eliminating t from these equations, we get:

$$y = \frac{1}{2}g\left(\frac{x}{u}\right)^2$$

$$y = \frac{gx^2}{2u^2}$$

Therefore, the angle θ, the direction of motion makes with the horizontal is given by:

$$\tan\theta = \frac{dy}{dx}$$

$$\tan\theta = \frac{gx}{u^2}$$

Given, a body projected horizontally with 98 m/s from the top of a tower reaches the ground in 10 s.

Therefore, the horizontal distance covered by the body till it reaches the ground is:

$x = 98(10)$

$x = 980$

Therefore, the body strikes the ground at an angle θ such that:

$$\tan\theta = \frac{gx}{u^2}$$

$$\tan\theta = \frac{9.8(980)}{98^2}$$

$$\tan\theta = 1$$

$$\theta = 45^\circ$$

Therefore, if a body projected horizontally with 98 m/s from the top of a tower reaches the ground in 10 s, the body strikes the ground at an angle 45° .

20. Correct option: (D)

We know, for a horizontally launched projectile from a height h , the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$x = ut$

where u is the horizontal velocity of projection.

Given, two cliffs of height 120 m and 100.4 m are separated by a horizontal distance of 16 m.

Now, if car has to reach from first cliff to second cliff, then the vertical distance covered by the ball will be equal to the difference in the height of the building: $120 - 100.4 = 19.6$.

Therefore, the time taken by the car to reach the second cliff is:

$$t = \sqrt{\frac{2(19.6)}{9.8}}$$

$t = 2$

Now, the speed required to cover horizontal distance of 16 m is:

$16 = u(2)$

$u = 8$

Therefore, if two cliffs of height 120 m and 100.4 m are separated by a horizontal distance of 16 m, and if car has to reach from first cliff to second cliff, then the horizontal velocity of the car should be 8 m/s.

21. Correct option: (A)

We know, for a horizontally launched projectile from a height h , the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$x = ut$

where u is the horizontal velocity of projection.

Given, two paper screens A and B are separated by a distance 100 cm, a bullet pieces screen A and then screen B. The hole in B is 10 cm below in A.

Now, if bullet pierces screen A and then screen B, then the vertical distance covered by the bullet will be equal to the difference in the height of the two-paper screen, that is 10 cm or 0.1 m.

Therefore, the time taken by the bullet to pierce the second screen B is:

$$t = \sqrt{\frac{2(0.1)}{9.8}} = \sqrt{\frac{2}{98}} = \sqrt{\frac{1}{49}}$$

$$t = \frac{1}{7}$$

Now, the velocity at A the horizontal velocity, which is given by, using $x = ut$,

$$100 = u\frac{1}{7}$$

$$u = 700$$

Therefore, if two paper screens A and B are separated by a distance 100, and a bullet pieces screen A and then screen B and the hole in B is 10 cm below in A, then its velocity at A is 700 m/s.

22. Correct option: (B)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$x = ut$

where u is the horizontal velocity of projection.

Given, two tall buildings are separated by 20 m apart. A ball is thrown horizontally from a window 135 m above the ground in one building so that, it will enter a window 12.5 m above the ground.

Now, if ball has to reach the window in the second building, then the vertical distance covered by the ball will be equal to the difference in the height of the window in the two buildings: $135 - 12.5 = 122.5$.

Therefore, the time taken by the ball to reach the second window is:

$$t = \sqrt{\frac{2(122.5)}{9.8}}$$

$t = 5$

Now, the speed required to cover horizontal distance of 20 m is:

$20 = u(5)$

$u = 4$

Therefore, if two tall buildings are separated by 20 m apart and a ball is thrown horizontally from a window 135 m above the ground in one building so that, it will enter a window 12.5 m above the ground, then the horizontal velocity of the ball should be 4 m/s.

23. Correct option: (B)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$x = ut$

where u is the horizontal velocity of projection.

Given, from the top of a tower of 78.4 m, two stones are projected horizontally with 20 m/s and 30 m/s, in opposite directions.

Now, the distance taken by the stone to reach the ground is:

$$t = \sqrt{\frac{2(78.4)}{9.8}}$$

$$t = 4$$

Therefore, on reaching the ground their separation s is given by:

$$s = 20(4) + 30(4)$$

$$s = 80 + 120$$

$$s = 200$$

Therefore, if from the top of a tower of 78.4 m, two stones are projected horizontally with 20 m/s and 30 m/s, in opposite directions, then on reaching the ground their separation is 200 m.

24. Correct option: (C)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$x = ut$

where u is the horizontal velocity of projection.

Given, a staircase contains three steps each 10 cm high 20 cm wide.

Now, if a ball rolling off the upper most plane hit directly edge of the lowest plane, it has fallen a total distance of $3(10) = 30\,\text{cm}$, and covered a horizontal distance of $3(20) = 60$ cm.

Therefore, the time taken by the ball to hit the lowest plane is:

$$t = \sqrt{\frac{2h}{g}}$$

$$t = \sqrt{\frac{2(0.3)}{10}}$$

$$t = \frac{\sqrt{6}}{10}$$

Now, the minimum horizontal velocity of the ball is given by:

$x = ut$

$$0.6 = u\frac{\sqrt{6}}{10}$$

$$u = \sqrt{6}$$

Therefore, if a staircase contains three steps each 10 cm high 20 cm wide, then minimum horizontal velocity of a ball rolling off the upper most plane so as to hit directly edge of the lowest plane is $\sqrt{6}$ m/s.

25. Correct option: (C)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$$x = ut$$

where u is the horizontal velocity of projection.

Given, a ball rolls of the top of a stair way with a horizontal velocity 4.5 m/s. The steps are 0.2 meter high and 0.3 meter broad.

Now, let the ball strikes the n th step. Then, the ball has fallen a total distance of $0.2n$ m, and covered a horizontal distance of $0.3n$ m.

Therefore, the time taken by the ball to hit the n th stair is:

$$t = \sqrt{\frac{2(0.2n)}{g}}$$

$$t = \sqrt{\frac{4n}{100}}$$

$$t = \frac{\sqrt{n}}{5}$$

Now, since the horizontal velocity of the ball is 4.5 m/s, thus:

$$x = ut$$

$$0.3n = 4.5\frac{\sqrt{n}}{5}$$

$$n = 9$$

Therefore, if a ball rolls of the top of a stair way with a horizontal velocity 4.5 m/s and the steps are 0.2 meter high and 0.3 meter broad, then the ball strikes the 9th step.

26. Correct option: (B)

We know, the horizontal distance and vertical distance covered by the horizontally launched projectile in time t is given by:

$$x = ut$$

$$y = -\frac{1}{2}gt^2$$

where u is the horizontal velocity of projection.

Given, two particles are thrown horizontally in opposite directions with velocity u and $2u$ from top of a high tower.

Therefore, the displacement of the particles is:

$$\vec{d}_1 = ut\hat{i} - \frac{1}{2}gt^2\hat{j}$$

$$\vec{d}_2 = -2ut\hat{i} - \frac{1}{2}gt^2\hat{j}$$

Now, if the displacement vectors are perpendicular, then their dot product is zero, thus:

$\vec{d}_1.\vec{d}_2 = 0$

$(ut)(-2ut) + \left(\frac{1}{2}gt^2 \frac{1}{2}gt^2\right) = 0$

$-2u^2t^2 + \frac{g^2t^4}{4} = 0$

$t^2\left(-2u^2 + \frac{g^2t^2}{4}\right) = 0$

$t^2 = \frac{8u^2}{g^2}$

$t = \frac{2\sqrt{2}u}{g}$

Therefore, if two particles are thrown horizontally in opposite directions with velocity u and $2u$ from top of a high tower, then the time after which their displacement vectors are perpendicular is $\frac{2\sqrt{2}u}{g}$.

27. Correct option: (D)

We know, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$v_x = u$

$v_y = -gt$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.

Given, a stationary shell at certain height explodes into two fragments and move with velocities 1 m/s and 4 m/s horizontally in opposite directions.

Therefore, the velocity vector of the two fragments is:

$\vec{v}_1 = \hat{i} - gt\hat{j}$

$\vec{v}_2 = -4\hat{i} - gt\hat{j}$

Now, if the velocity vectors are perpendicular, then their dot product is zero, thus:

$\vec{v}_1.\vec{v}_2 = 0$

$-4 + g^2t^2 = 0$

$t^2 = \frac{4}{100}$

$t = \frac{2}{10}$

We know, the horizontal distance and vertical distance covered by the horizontally launched projectile in time t is given by:

$x = ut$

$y = -\frac{1}{2}gt^2$

where u is the horizontal velocity of projection.

Therefore, the displacement of the particles is:

$$\vec{d}_1 = \frac{1}{5}\hat{i} - \frac{1}{2}g\left(\frac{1}{5}\right)^2 \hat{j}$$

$$\vec{d}_1 = \frac{1}{5}\hat{i} - \frac{1}{5}\hat{j}$$

$$\vec{d}_2 = -\frac{4}{5}\hat{i} - \frac{1}{2}g\left(\frac{1}{5}\right)^2 \hat{j}$$

$$\vec{d}_2 = -\frac{4}{5}\hat{i} - \frac{1}{5}\hat{j}$$

Therefore, the distance between those two fragments at the moment when their velocity vectors become mutually perpendicular to each other is:

$$\sqrt{\left(\frac{1}{5}+\frac{4}{5}\right)^2 + \left(\frac{-1}{5}+\frac{1}{5}\right)^2} = 1$$

Therefore, if a stationary shell at certain height explodes into two fragments and move with velocities 1 m/s and 4 m/s horizontally in opposite directions, then the distance between those two fragments at the moment when their velocity vectors become mutually perpendicular to each other is 1 m.

28. Correct option: (D)

We know, the relative velocity of A as measured by an observer B is

$$\vec{v}_{AB} = \vec{v}_{AC} + \vec{v}_{CB}$$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, rain is falling vertically with a speed of 30 m/s and a woman rides a bicycle with a speed of 10 m/s in the north to south direction.

The direction in which she should hold her umbrella will be along the relative velocity of rain with respect to the woman. Hence, the relative velocity of rain with respect to the woman is:

$$\vec{v}_{RW} = \vec{v}_{RG} + \vec{v}_{GW}$$

where $\vec{v}_{RG}$ is the velocity of rain as measured by an observer in ground, $\vec{v}_{GW}$ is the velocity of ground as measured by woman. Therefore,

$$\vec{v}_{RW} = -30\hat{k} - 10\hat{i}$$

Therefore, the resultant velocity makes an angle θ with the horizontal such that:

$$\tan\theta = \frac{10}{30}$$

$$\theta = \tan^{-1}\frac{1}{3}$$

Therefore, if rain is falling vertically with a speed of 30 m/s and a woman rides a bicycle with a speed of 10 m/s in the north to south direction, then the direction in which she should hold her umbrella is $\tan^{-1}\frac{1}{3}$ with horizontal.

29. Correct option: (A)

We know, the relative velocity of A as measured by an observer B is

$$\vec{v}_{AB} = \vec{v}_{AC} + \vec{v}_{CB}$$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, a man can swim with a speed of 4 km/h in still water. The river is 1.0 km wide and it flows steadily at 3.0 km/h. Also, the boy makes his strokes normal to the river current.

Now, the relative velocity of the man with respect to the ground is:

$\vec{v}_{MG} = \vec{v}_{MR} + \vec{v}_{RG}$

where $\vec{v}_{MR}$ is the velocity of man as measured by river, $\vec{v}_{RG}$ is the velocity of river as measured by an observer on ground.

Substituting the given values, we get:

$\vec{v}_{MG} = 4\hat{j} + 3\hat{i}$

Hence, the component of his velocity along the direction normal to the bank is 4 km/h.

Since the width of the lake is 1km, the time taken by the man to cross the river is: $1/4$ hrs.

Therefore, he goes a distance down the river by:

$3(1/4)$ km = $750\,m$

Therefore, if a man can swim with a speed of 4 km/h in still water and the river is 1.0 km wide and it flows steadily at 3.0 km/h and the boy makes his strokes normal to the river current, then he goes a distance down the river by 750 m when he reaches the other bank.

30. Correct option: (D)

We know, the relative velocity of A as measured by an observer B is

$\vec{v}_{AB} = \vec{v}_{AC} + \vec{v}_{CB}$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, in a harbour, the wind is blowing at the speed of 72 km/h and the flag on the mast of a boat anchored in the harbour flutters along the N-E direction. Also, the boat starts moving at a speed of 51 km/h to the north.

The flag will move in a direction of the relative velocity of wind with respect to the boat.

Now, the velocity of wind with respect to the boat is:

$\vec{v}_{WB} = \vec{v}_{WH} + \vec{v}_{HB}$

where $\vec{v}_{WH}$ is the velocity of wind as measured by an observer on harbour, $\vec{v}_{HB}$ is the velocity of harbour as measured by an observer on boat.

Let $\hat{i}$ points in the East direction, and $\hat{j}$ in the north direction. Since the flag on the mast of a boat anchored in the harbour flutters along the N-E direction, this is the speed of the wind with respect to an observer on harbour, and we get:

$\vec{v}_{WB} = \frac{72}{\sqrt{2}}(\hat{i} + \hat{j}) - 51\hat{j}$

$\vec{v}_{FG} \cong \frac{72}{\sqrt{2}}\hat{i}$

Therefore, if in a harbour, the wind is blowing at the speed of 72 km/h and the flag on the mast of a boat anchored in the harbour flutters along the N-E direction and the boat starts moving at a speed of 51 km/h to the north, then the direction of the flag on the mast of the boat will be due East.

31. Correct option: (D)

Given, two boys are standing at the points A and B on the ground, where AB = a. The boy at B starts running in a direction perpendicular to AB with velocity v_1.The boy at A starts running simultaneously with velocity v and catches the other in time t.

Now, AB and the point where A catches B makes a right-angled triangle with base AB and the vertical distance travelled by B.

Therefore, using the Pythagoras theorem, we can write:

$c^2 = a^2 + b^2$

$(vt)^2 = a^2 + (v_1 t)^2$

$t^2 = \frac{a^2}{v^2 - v_1^2}$

$c^2 = a^2 + b^2$

$(vt)^2 = a^2 + (v_1 t)^2$

$t = \frac{a}{\sqrt{v^2 - v_1^2}}$

Therefore, if two boys are standing at the points A and B on the ground, where AB = a and the boy at B starts running in a direction perpendicular to AB with velocity v_1 and if the boy at A starts running simultaneously with velocity v, then he catches the other in time $\frac{a}{\sqrt{v^2 - v_1^2}}$.

32. Correct option: (B)

We know, the relative velocity of A as measured by an observer B is

$\vec{v}_{AB} = \vec{v}_{AC} + \vec{v}_{CB}$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, a river flows at the rate of 3 km/hr, and a person can row a boat at a speed of 5 km/hr in still water. Further, the difference between the time taken to cross the river by the shortest path and the quickest path is 4 minutes.

Now, the shortest path to cross the river is just the straight path across the river in vertical direction, for which we require the resultant velocity of the boat with respect to the ground to point in the vertical direction. Assuming the boat flows in a direction θ to the flow of the river, the resultant velocity of the boat with respect to the ground is:

$\vec{v}_{BG} = \vec{v}_{BR} + \vec{v}_{RG}$

$\vec{v}_{BG} = 5\cos\theta\hat{i} + 5\sin\theta\hat{j} + 3\hat{i}$

$\vec{v}_{BG} = (5\cos\theta + 3)\hat{i} + 5\sin\theta\hat{j}$

Since this velocity needs to be along the vertical direction, we thus have:

$\tan 90^\circ = \frac{5\sin\theta}{5\cos\theta + 3}$

$\infty = \frac{5\sin\theta}{5\cos\theta + 3}$

$5\cos\theta + 3 = 0$

$\cos\theta = -3/5$

Now, the magnitude of the velocity of the resultant is:

$v_{BG} = |(5\cos\theta + 3)\hat{i} + 5\sin\theta\,\hat{j}|$

$v_{BG} = \sqrt{(5\cos\theta + 3)^2 + (5\sin\theta)^2}$

$v_{BG} = \sqrt{5^2 + 3^2 + 2(5)(3)\cos\theta}$

$v_{BG} = \sqrt{25 + 9 + (30)(-3/5)}$

$v_{BG} = \sqrt{16}$

$v_{BG} = 4$

Let the width of the river be w . Hence the time taken in hours to cross the river via the shortest path is:

$\frac{w}{4}$

Now, the resultant velocity of the boat with respect to the ground is $\vec{v}_{BG} = (5\cos\theta + 3)\hat{i} + 5\sin\theta\,\hat{j}$. Therefore, the component of the velocity vertical to the flow of the river is $5\sin\theta$. Now, for the quickest path to cross the river, we require this velocity to be the maximum. This will happen when $\theta = 90^\circ$, so time taken to cross the river by quickest path is:

$\frac{w}{5}$

We are given the difference between the time taken to cross the river by the shortest path and the quickest path is 4 minutes, or 4/60 hours, thus

$\frac{w}{4} - \frac{w}{5} = \frac{4}{60}$

$15w - 12w = 4$

$3w = 4$

$w = 4/3\,km$

$w = 1\frac{1}{3}km$

Therefore, if a river flows at the rate of 3 km/hr, and a person can row a boat at a speed of 5 km/hr in still water such that the difference between the time taken to cross the river by the shortest path and the quickest path is 4 minutes, then the width of the river is $1\frac{1}{3}km$.

33. Correct option: (A)

We know, the relative velocity of A as measured by an observer B is

$\vec{v}_{AB} = \vec{v}_{AC} + \vec{v}_{CB}$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, to a man running upwards on the hill, the rain appears to fall vertically downwards

with 4 m/s. The velocity vector of the man with respect to the earth is $2\hat{i} + 3\hat{j}$ m/s.

So, the relative speed of the rain with respect to the man is:

$\vec{v}_{RM} = \vec{v}_{RG} + \vec{v}_{GM}$

where $\vec{v}_{RG}$ is the velocity of rain as measured by an observer on ground, $\vec{v}_{GM}$ is the velocity of ground as measured by the man.

Substituting the values, we get:

$-4\hat{j} = \vec{v}_{RG} - 2\hat{i} - 3\hat{j}$

$\vec{v}_{RG} = 2\hat{i} - \hat{j}$

Now, if the man starts running down the hill with the same speed, his velocity vector will point in the opposite direction: $-2\hat{i} - 3\hat{j}$ m/s. Hence, the relative speed of the rain with respect to the man is

$\vec{v}_{RM} = \vec{v}_{RG} + \vec{v}_{GM}$

$\vec{v}_{RM} = 2\hat{i} - \hat{j} + 2\hat{i} + 3\hat{j}$

$\vec{v}_{RM} = 4\hat{i} + 2\hat{j}$

Therefore, if to a man running upwards on the hill, the rain appears to fall vertically downwards with 4 m/s and the velocity vector of the man with respect to the earth is $2\hat{i} + 3\hat{j}$ m/s, and if the man starts running down the hill with the same speed, then the relative speed of the rain with respect to the man is $4\hat{i} + 2\hat{j}$.

34. Correct option: (D)

We know, the relative velocity of A as measured by an observer B is

$\vec{v}_{AB} = \vec{v}_{AC} + \vec{v}_{CB}$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, a boat crosses a river of width 200m in the shortest time and is found to experience a drift of 100 m in reaching the opposite bank. The time taken now is t.

Now, assuming the boat flows in a direction θ to the flow of the river, the resultant velocity of the boat with respect to the ground is:

$\vec{v}_{BG} = \vec{v}_{BR} + \vec{v}_{RG}$

$\vec{v}_{BG} = v_b \cos\theta \hat{i} + v_b \sin\theta \hat{j} + v_r \hat{i}$

$\vec{v}_{BG} = (v_b \cos\theta + v_r)\hat{i} + v_b \sin\theta \hat{j}$

where v_b is the magnitude of the velocity of boat with respect to the river, and v_r is the magnitude of the velocity of river with respect to the ground.

Now, the resultant velocity of the boat with respect to the ground is $(v_b \cos\theta + v_r)\hat{i} + v_b \sin\theta \hat{j}$. Therefore, the component of the velocity vertical to the flow of the river is $v_b \sin\theta$. Now, for the quickest path to cross the river, we require this velocity to be the maximum. This will happen when $\theta = 90^\circ$, so shortest time taken to cross the river is:

$\frac{200}{v_b} = t$

Since in this case, there is a drift of 100m on reaching the opposite bank, thus:

$\frac{100}{t} = v_r$

Now, the resultant velocity of the boat with respect to the ground is $(v_b \cos\theta + v_r)\hat{i} + v_b \sin\theta \hat{j}$. The shortest path to cross the river is just the straight path across the river in vertical direction, for which we require the resultant velocity of the boat with respect to the ground to point in the vertical direction.

Since this velocity needs to be along the vertical direction, we thus have:

$$\tan 90^\circ = \frac{v_b \sin\theta}{v_b \cos\theta + v_r}$$

$$\infty = \frac{v_b \sin\theta}{v_b \cos\theta + v_r}$$

$$v_b \cos\theta + v_r = 0$$

$$\cos\theta = -v_r / v_b$$

Now, the magnitude of the velocity of the resultant is:

$$v_{BG} = |(v_b \cos\theta + v_r)\hat{i} + v_b \sin\theta \hat{j}|$$

$$v_{BG} = \sqrt{(v_b \cos\theta + v_r)^2 + (v_b \sin\theta)^2}$$

$$v_{BG} = \sqrt{v_b{}^2 + v_r{}^2 + 2(v_b)(v_r)\cos\theta}$$

$$v_{BG} = \sqrt{v_b{}^2 + v_r{}^2 + 2(v_b)(v_r)(-v_r / v_b)}$$

$$v_{BG} = \sqrt{v_b{}^2 - v_r{}^2}$$

Substituting $\frac{200}{t} = v_b$ and $\frac{100}{t} = v_r$, we get:

$$v_{BG} = \sqrt{\left(\frac{200}{t}\right)^2 - \left(\frac{100}{t}\right)^2}$$

$$v_{BG} = \frac{10}{t}\sqrt{3}$$

Hence the time taken in hours to cross the river via the shortest path is:

$$\frac{200t}{10\sqrt{3}} = \frac{2t}{\sqrt{3}}$$

Therefore, if a boat crosses a river of width 200m in the shortest time and is found to experience a drift of 100 m in reaching the opposite bank and the time taken now is t, then time taken to cross the river by shortest path is $\frac{2t}{\sqrt{3}}$.

35. Correct option: None of the given answer is correct

We know, the relative velocity of A as measured by an observer B is

$$\vec{v}_{AB} = \vec{v}_{AC} + \vec{v}_{CB}$$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, a person travelling in a car along east with speed u observes the velocity of another car as $\sqrt{5}u$ and $\tan^{-1}(B)$ S of W.

Let $\hat{i}$ points along east and $\hat{j}$ points along north. Since the person observes velocity of another car as $\sqrt{5}u$ and $\tan^{-1}(B)$ S of W, we can write is as $-\sqrt{5}u(\cos B\hat{i} + \sin B\hat{j})$.

Now, the velocity of other car with respect to the ground is:

$$\vec{v}_{CG} = \vec{v}_{CP} + \vec{v}_{PG}$$

where $\vec{v}_{CP}$ is the velocity of car with respect to the person, $\vec{v}_{CP}$ is the velocity of person with respect to the ground.

Substituting the given values, we get:

$\vec{v}_{CG} = -\sqrt{5}u(cos\,B\,\hat{\imath} + sin\,B\,\hat{\jmath}) + u\hat{\imath}$

$\vec{v}_{CG} = u(-\sqrt{5}\,cos\,B + 1)\hat{\imath} - \sqrt{5}u\,sin\,B\,\hat{\jmath}$

Therefore, the magnitude of the velocity is:

$v_{CG} = \sqrt{\left(u(-\sqrt{5}\,cos\,B + 1)\right)^2 + \left(-\sqrt{5}u\,sin\,B\right)^2}$

$v_{CG} = u\sqrt{5\,cos^2\,B + 1 + 2\sqrt{5}\,cos\,B + 5\,cos^2\,B}$

$v_{CG} = u\sqrt{6 + 2\sqrt{5}\,cos\,B}$

We see we need the value of B to solve this question.

36. Correct option: (B)

We know, the relative velocity of A as measured by an observer B is

$\vec{v}_{AB} = \vec{v}_{AC} + \vec{v}_{CB}$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, a boat having a speed of 5 km/hr in still water, crosses a river of width 1 km along the shortest possible path in 15 minutes.

Now, the shortest path to cross the river is just the straight path across the river in vertical direction, for which we require the resultant velocity of the boat with respect to the ground to point in the vertical direction. Assuming the boat flows in a direction θ to the flow of the river, the resultant velocity of the boat with respect to the ground is:

$\vec{v}_{BG} = \vec{v}_{BR} + \vec{v}_{RG}$

$\vec{v}_{BG} = 5\cos\theta\hat{i} + 5\sin\theta\,\hat{j} + v_r\hat{i}$

$\vec{v}_{BG} = (5\cos\theta + v_r)\hat{i} + 5\sin\theta\,\hat{j}$

where v_r is the speed of river.

Since this velocity needs to be along the vertical direction, we thus have:

$$\tan 90^\circ = \frac{5\sin\theta}{5\cos\theta + v_r}$$

$$\infty = \frac{5\sin\theta}{5\cos\theta + v_r}$$

$5\cos\theta + v_r = 0$

$\cos\theta = -v_r / 5$

Now, the magnitude of the velocity of the resultant is:

$v_{BG} = |(5\cos\theta + v_r)\hat{i} + 5\sin\theta\,\hat{j}|$

$v_{BG} = \sqrt{(5\cos\theta + v_r)^2 + (5\sin\theta)^2}$

$v_{BG} = \sqrt{5^2 + v_r^{\,2} + 2(5)(v_r)\cos\theta}$

$v_{BG} = \sqrt{25 + v_r^{\,2} + (10v_r)(-v_r / 5)}$

$v_{BG} = \sqrt{25 - v_r^{\,2}}$

We are given the crosses the river of width 1 km along the shortest possible path in 15 minutes or ¼ hrs, thus:

$$\frac{1}{\sqrt{25-v_r^2}}=\frac{1}{4}$$

$$25-v_r^2=16$$

$$v_r=\sqrt{9}$$

$$v_r=3$$

Therefore, if a boat having a speed of 5 km/hr in still water, crosses a river of width 1 km along the shortest possible path in 15 minutes, then the speed of the river is 3 km/hr.

37. Correct option: (A)

We know, the relative velocity of A as measured by an observer B is

$$\vec{v}_{AB}=\vec{v}_{AC}+\vec{v}_{CB}$$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, a man swimming downstream overcame a float at a point M. After travelling distance D he turned back and passed the float at a distance D/2 from the point M.

Now, the velocity of the swimmer with respect to the water when he is swimming downstream is:

$$\vec{v}_{SD}=\vec{v}_{SW}+\vec{v}_{WG}$$

where $\vec{v}_{SW}$ is the velocity of the swimmer with respect to the still water and $\vec{v}_{WG}$ is the speed of the river. Since these velocity vectors lie in the same line, we can write for their magnitude as:

$$v_{SD}=v_{SW}+v_{WG}$$

When the swimmer turns back, he will be swimming against the flow of the river, that is, upstream. In this case the speed of the swimmer upstream is:

$$v_{SD}=v_{SW}-v_{WG}$$

Since he passed the float at a distance D/2 from the point M during his upstream journey, hence during this time, the float has moved a distance D/2 from the point M. The velocity of the float will be equal to the speed of the river. Therefore, equating the time taken by the swimmer to cover D/2 distance upward and by the float to cover D/2 distance downstream, we get:

$$\frac{D/2}{v_{WG}}=\frac{D/2}{v_{SW}-v_{WG}}$$

$$v_{WG}=v_{SW}-v_{WG}$$

$$\frac{v_{SW}}{v_{WG}}=2$$

Therefore, if a man swimming downstream overcame a float at a point M and after travelling distance D he turned back and passed the float at a distance D/2 from the point M, then the ratio of the speed of the swimmer with respect to still water to the speed of the river will be 2.

LEVEL-II

1. Correct option: (B)

We know, for a constant acceleration motion, the initial and final velocity, u, v respectively, are related to distance covered h through the equation:

$v^2 = u^2 + 2ah$

where a is the constant acceleration.

Given, a body is projected horizontally with a velocity u from certain height h and reaches the ground with velocity $4u$.

Since the horizontal velocity of the body when it reaches the ground is unchanged, hence its vertical velocity is:

$4u = \sqrt{u^2 + v^2}$

$16u^2 = u^2 + v^2$

$15u^2 = v^2$

Now, the vertically downward motion is the constant acceleration motion with uniform gravitational acceleration g. Also, since the body is projected horizontally, the initial vertical velocity is zero.

Therefore, the height is given by:

$15u^2 = 0 + 2gh$

$h = \frac{15u^2}{2g}$

Therefore, if a body is projected horizontally with a velocity u from certain height h and it reaches the ground with velocity $4u$, then the height is $\frac{15u^2}{2g}$.

2. Correct option: (A)

We know, the horizontal distance and vertical distance covered by the horizontally launched projectile in time t is given by:

$x = ut$

$y = \frac{1}{2}gt^2$

where u is the horizontal velocity of projection.

Eliminating t from these equations, we get:

$y = \frac{1}{2}g\left(\frac{x}{u}\right)^2$

$y = \frac{gx^2}{2u^2}$

Therefore, the angle θ, the direction of motion makes with the horizontal is given by:

$\tan\theta = \frac{dy}{dx}$

$\tan\theta = \frac{gx}{u^2}$

$\tan\theta = \frac{gt}{u}$

$u = \frac{gt}{\tan\theta}$

Given, a body is projected horizontally from the top of a tower. When the body strikes the ground its velocity is v and the direction of motion makes 30° with the horizontal.

Now, for using $v = u - gt$ for constant acceleration motion, we get:

$-v_y = 0 - gt$

$gt = v_y$

where negative sign shows the downward velocity.

Since when the body strikes the ground, its velocity is v, thus:

$v = \sqrt{u^2 + v_y^2}$

$v = \sqrt{\left(\frac{gt}{\tan\theta}\right)^2 + (gt)^2}$

$v = \sqrt{u^2 + v_y^2}$

$v = \sqrt{\left(\frac{gt}{\tan\theta}\right)^2 + (gt)^2}$

$v = t\sqrt{300 + 100}$

$v = 20t$

$t = \frac{v}{20}$

Substituting this for t in $u = \frac{gt}{\tan\theta}$, we get:

$u = \frac{10\sqrt{3}v}{20}$

$u = \frac{\sqrt{3}v}{2}$

Therefore, if a body projected horizontally from the top of a tower when strikes the ground has velocity v and the direction of motion makes 30° with the horizontal, then the velocity of projection is $\frac{\sqrt{3}v}{2}$.

3. Correct option: (B)

We know, for a constant acceleration motion, the initial and final velocity, u, v respectively, are related to distance covered h through the equation:

$v^2 = u^2 + 2ah$

where a is the constant acceleration.

Given, from the top of a tower of height h a body is projected horizontally with velocity u.

Therefore, taking the downward direction as positive, the vertical velocity of the object on reaching the ground is:

$v^2 = 0 + 2gh$

$v = \sqrt{2gh}$

Since the horizontal velocity remain the same throughout the motion, therefore, the change in velocity on reaching the ground is just the final vertical velocity $\sqrt{2gh}$.

Therefore, if from the top of a tower of height h a body is projected horizontally with velocity u, then magnitude of change in its velocity on reaching the ground is $\sqrt{2gh}$.

4. Correct option: (B)

We know, for a horizontally launched projectile, the equation of the path of the projectile is given by:

$x = ut$

$y = \frac{1}{2}gt^2$

where u is the horizontal launch velocity, and downward direction is taken to be positive.

Given, two thin wooden screens A and B are separated by 200 m. A bullet is travelling horizontally at a speed of $600\,\text{m/s}$, hits the screen A, penetrates through it and finally emerges out from B making holes in A and B.

Therefore, the time taken by the bullet to cover the horizontal distance of 200 m is given by:

$200 = (600)t$

$t = \frac{1}{3}$

Therefore, the difference in the heights of the holes in A and B is the vertical distance travelled by the bullet:

$y = \frac{1}{2}g\left(\frac{1}{3}\right)^2$

$y = \frac{1}{2}\frac{9.8}{9}$

$y = \frac{4.9}{9}$

$y = \frac{49}{90}$

Therefore, if two thin wooden screens A and B are separated by 200 m and a bullet is travelling horizontally at a speed of $600\,\text{m/s}$, hits the screen A, penetrates through it and finally emerges out from B making holes in A and B, then difference in the heights of the holes in A and B is $\frac{49}{90}$ m.

5. Correct option: (C)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$t = \sqrt{\frac{2h}{g}}$

Further, the horizontal distance covered by the projectile in time t is given by:

$x = ut$

where u is the horizontal velocity of projection.

Given, the ball strikes the n th step where each step has height h and width x.

Therefore, the time taken by the ball to hit the n th step is:

$t = \sqrt{\frac{2(nh)}{g}}$

Now, since each step has width x, thus:

$$nx = ut$$

$$u = \frac{nx}{t}$$

$$u = \frac{nx}{\sqrt{\frac{2(nh)}{g}}}$$

$$u = x\sqrt{\frac{gn}{2h}}$$

Therefore, if the ball strikes the n th step where each step has height h and width x, then initial velocity of the ball is $x\sqrt{\frac{gn}{2h}}$.

6. Correct option: (A)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, the horizontal distance covered by the projectile in time t is given by:

$$x = ut$$

where u is the horizontal velocity of projection.

Given, the ball strikes the 5th step where each step has height 20 cm and width 30 cm.

Therefore, the time taken by the ball to hit the 5th step is:

$$t = \sqrt{\frac{2(5(0.2))}{g}}$$

$$t = \sqrt{\frac{2}{g}}$$

Now, since each step has width 0.3 m, thus:

$$5(0.3) = u\sqrt{\frac{2}{g}}$$

$$u = \frac{1.5}{\sqrt{2/g}}$$

$$u = \frac{1.5}{\sqrt{1/5}}$$

$$u = 1.5\sqrt{5}$$

Therefore, the ball strikes the 5th step where each step has height 20 cm and width 30 cm, then initial velocity of the ball is $1.5\sqrt{5}$ m/s.

7. Correct option: (A)

We know, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$$v_x = u$$

$$v_y = -gt$$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.

Given, at certain height a body at rest explodes into two equal fragments which receives a
horizontal velocity 10 m/s each. Therefore, the velocity vector of the two fragments is:

$\vec{v}_1 = 10\hat{i} - gt\hat{j}$

$\vec{v}_2 = -10\hat{i} - gt\hat{j}$

Now, if the velocity vectors are perpendicular, then their dot product is zero, thus:

$\vec{v}_1.\vec{v}_2 = 0$

$-100 + g^2t^2 = 0$

$t^2 = \frac{100}{100}$

$t = 1$

Therefore, if at certain height a body at rest explodes into two equal fragments which receives a horizontal velocity 10 m/s each, then time interval after explosion for the velocity vectors of the two fragments become perpendicular to each other is 1s.

8. Correct option: (A)

We know, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$v_x = u$

$v_y = -gt$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.

Given, a bomb at rest explodes into two equal fragments receives a horizontal velocity u. Therefore, the velocity vector of the two fragments is:

$\vec{v}_1 = u\hat{i} - gt\hat{j}$

$\vec{v}_2 = -u\hat{i} - gt\hat{j}$

Now, if the velocity vectors are inclined at 60° to each other, then their dot product satisfies:

$\vec{v}_1.\vec{v}_2 = |\vec{v}_1||\vec{v}_2|\cos 60^\circ$

$-u^2 + g^2t^2 = (u^2 + g^2t^2)\frac{1}{2}$

$t^2 = \frac{3u^2}{g^2}$

$t = \frac{\sqrt{3}u}{g}$

Therefore, if a bomb at rest explodes into two equal fragments receives a horizontal velocity u, then time interval after which their velocity vectors will be inclined at 60° to each other is $\frac{\sqrt{3}u}{g}$ s.

9. Correct option: (A)

Given, a bomber flying horizontally with constant speed releases a bomb from an aeroplane.

When the bomb is dropped, it will have the same horizontal velocity as the aeroplane. For the ground-based observer, the bomb is falling vertically downward, and is also

simultaneously moving in horizontal direction. The path of bomb as seen by the observer on the ground is thus a parabola.
For the pilot, the bomb has the same horizontal velocity as the plane, therefore he only observes the bomb to be falling vertically downward. Therefore, the path of bomb as seen by a pilot is a straight line.

10. Correct options: (A), (B), (D)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$$v_x = u$$

$$v_y = -gt$$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.
Given, a body is projected horizontally with a velocity 20 m/s from the tower of height 78.4 m.
Therefore, the time of flight is:

$$t = \sqrt{\frac{2(78.4)}{9.8}}$$

$$t = \sqrt{16}$$

$$t = 4$$

Hence, range of the projectile is:

$$R = 20(4)$$

$$R = 80$$

Now, the velocity after 2s is:

$$\vec{v} = u\hat{i} - gt\hat{j}$$

$$\vec{v} = 20\hat{i} - 10(2)\,\hat{j}$$

$$\vec{v} = 20\hat{i} - 20\,\hat{j}$$

Therefore, the magnitude of the velocity after two seconds is:

$$v = \sqrt{(20)^2 + (-20)^2}$$

$$v = 20\sqrt{2}$$

Therefore, if a body is projected horizontally with a velocity 20 m/s from the tower of height 78.4 m, then time of flight is 4s, range of the projectile is 40 m, and the velocity after two seconds is $20\sqrt{2}$ m/s.

11. Correct option: (B)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}\,.$$

Given, a projectile is fired horizontally with a velocity 98 m/s from the hill of 490 m.
Therefore, time taken to reach the ground by the projectile is:

$$t = \sqrt{\frac{2(490)}{9.8}}$$

$$t = \sqrt{100}$$

$$t = 10$$

Therefore, if a projectile is fired horizontally with a velocity 98 m/s from the hill of 490 m, then time taken to reach the ground by the projectile is 10 s.

12. Correct option: (C)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, for a horizontally launched projectile from a height h, horizontal distance covered is given by:

$$x = ut$$

where u is the horizontal velocity of projection.

Given, a projectile is fired horizontally with a velocity 98 m/s from the hill of 490 m.

Therefore, time taken to reach the ground by the projectile is:

$$t = \sqrt{\frac{2(490)}{9.8}}$$

$$t = \sqrt{100}$$

$$t = 10$$

Now, the distance of target from the hill is:

$$y = 98(10)$$

$$y = 980$$

Therefore, if a projectile is fired horizontally with a velocity 98 m/s from the hill of 490 m, then the distance of target from the hill is 980 m.

13. Correct option: (B)

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$$v_x = u$$

$$v_y = -gt$$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.

Given, a projectile is fired horizontally with a velocity 98 m/s from the hill of 490 m.

Therefore, time taken to reach the ground by the projectile is:

$$t = \sqrt{\frac{2(490)}{9.8}}$$

$$t = \sqrt{100}$$

$$t = 10$$

Therefore, the velocity of the body when it strikes the ground is:

$\vec{v} = u\hat{i} - gt\hat{j}$

$\vec{v} = 98\hat{i} - 9.8(10)\,\hat{j}$

$\vec{v} = 98\hat{i} - 98\,\hat{j}$

Therefore, the magnitude of the velocity when it strikes the ground is:

$v = \sqrt{(98^2 + (-98)^2}$

$v = 98\sqrt{2}$

Therefore, if a projectile is fired horizontally with a velocity 98 m/s from the hill of 490 m, then the velocity of the body when it strikes the ground is $98\sqrt{2}$ m/s.

14. Correct option: (C)

We know, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$v_x = u$

$v_y = -gt$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.

Given, from a tower of 40 m, two bodies are simultaneously projected horizontally in opposite directions with velocities 2 m/s and 8 m/s respectively.

Therefore, the velocity vector of the two fragments is:

$\vec{v}_1 = 2\hat{i} - gt\hat{j}$

$\vec{v}_2 = -8\hat{i} - gt\hat{j}$

Now, if the velocity vectors are perpendicular, then their dot product is zero, thus:

$\vec{v}_1.\vec{v}_2 = 0$

$-16 + g^2t^2 = 0$

$t^2 = \frac{16}{100}$

$t = \frac{4}{10}$

$t = 0.4$

Therefore, if from a tower of 40 m, two bodies are simultaneously projected horizontally in opposite directions with velocities 2 m/s and 8 m/s respectively, then the time taken for the velocity vectors of the two bodies to become perpendicular to each other is 0.4 s.

15. Correct option: (D)

We know, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$v_x = u$

$v_y = -gt$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.

Further, for a horizontally launched projectile from a height h, horizontal distance covered is given by:

$y = ut$

Given, from a tower of 40 m, two bodies are simultaneously projected horizontally in opposite directions with velocities 2 m/s and 8 m/s respectively.
Therefore, the velocity vector of the two fragments is:

$\vec{v}_1 = 2\hat{i} - gt\hat{j}$

$\vec{v}_2 = -8\hat{i} - gt\hat{j}$

Now, if the velocity vectors are perpendicular, then their dot product is zero, thus:

$\vec{v}_1.\vec{v}_2 = 0$

$-16 + g^2t^2 = 0$

$t^2 = \frac{16}{100}$

$t = \frac{4}{10}$

$t = 0.4$

Therefore, the horizontal distance between the two bodies when their velocities are perpendicular to each other is:

$y = 2(0.4) + 8(0.4)$

$y = 0.8 + 3.2$

$y = 4.0$

Therefore, if from a tower of 40 m, two bodies are simultaneously projected horizontally in opposite directions with velocities 2 m/s and 8 m/s respectively, then the horizontal distance between the two bodies when their velocities are perpendicular to each other is 4 m.

16. Correct option: (C)

We know, for a horizontally launched projectile from a height h, horizontal distance and vertical distance covered is given by:

$x = ut$

$y = -\frac{1}{2}gt^2$

where u is the horizontal velocity of projection, and downward is negative.
Given, from a tower of 40 m, two bodies are simultaneously projected horizontally in opposite directions with velocities 2 m/s and 8 m/s respectively.
Therefore, the displacement vector of the two fragments is:

$\vec{d}_1 = 2t\hat{i} - \frac{1}{2}gt^2\hat{j}$

$\vec{d}_1 = -8t\hat{i} - \frac{1}{2}gt^2\hat{j}$

Now, if the displacement vectors are perpendicular, then their dot product is zero, thus:

$\vec{d}_1.\vec{d}_2 = 0$

$-16t^2 + \frac{1}{4}g^2t^4 = 0$

$t^2 = \frac{16(4)}{100}$

$t = \frac{8}{10}$

$t = 0.8$

Therefore, if from a tower of 40 m, two bodies are simultaneously projected horizontally in opposite directions with velocities 2 m/s and 8 m/s respectively, then time taken for the displacement vectors of the two bodies to become perpendicular to each other is 0.8 s.

17. Correct option: (A)

We know, the relative velocity of A as measured by an observer B is

$\vec{v}_{AB} = \vec{v}_{AC} + \vec{v}_{CB}$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, swimmer wishes to cross a 500m wide river flowing at 5 kmph. His speed with respect to water is 3 kmph. He heads in a direction making an angle θ with the flow.

Now, the relative velocity of the man with respect to the ground is:

$\vec{v}_{MG} = \vec{v}_{MR} + \vec{v}_{RG}$

where $\vec{v}_{MR}$ is the velocity of man as measured by river, $\vec{v}_{RG}$ is the velocity of river as measured by an observer on ground.

Substituting the given values, we get:

$\vec{v}_{MG} = (3\cos\theta\hat{i} + 3\sin\theta\hat{j}) + 5\hat{i}$

$\vec{v}_{MG} = (3\cos\theta + 5)\hat{i} + 3\sin\theta\hat{j}$

Hence, the component of his velocity along the direction normal to the bank is $3\sin\theta$.

Since the width of the lake is 500 m or 0.5 km, the time taken by the man to cross the river is:

$0.5/(3\sin\theta)\,\text{hour} = \dfrac{10}{\sin\theta}\,\text{min}$

Therefore, if a swimmer whose speed with respect to water is 3 kmph wishes to cross a 500m wide river flowing at 5 kmph and heads in a direction making an angle θ with the flow, then the time taken by the swimmer to cross the river is $\dfrac{10}{\sin\theta}\,\text{min}$.

18. Correct option: (D)

We know, the relative velocity of A as measured by an observer B is

$\vec{v}_{AB} = \vec{v}_{AC} + \vec{v}_{CB}$

where $\vec{v}_{AC}$ is the velocity of A as measured by C, $\vec{v}_{CB}$ is the velocity of C as measured by B.

Given, swimmer wishes to cross a 500m wide river flowing at 5 kmph. His speed with respect to water is 3 kmph. He heads in a direction making an angle θ with the flow.

Now, the relative velocity of the man with respect to the ground is:

$\vec{v}_{MG} = \vec{v}_{MR} + \vec{v}_{RG}$

where $\vec{v}_{MR}$ is the velocity of man as measured by river, $\vec{v}_{RG}$ is the velocity of river as measured by an observer on ground.

Substituting the given values, we get:

$\vec{v}_{MG} = (3\cos\theta\hat{i} + 3\sin\theta\hat{j}) + 5\hat{i}$

$\vec{v}_{MG} = (3\cos\theta + 5)\hat{i} + 3\sin\theta\hat{j}$

Hence, the component of his velocity along the direction normal to the bank is $3\sin\theta$.

Since the width of the lake is 500 m or 0.5 km, the time taken by the man to cross the river is:

$0.5/(3\sin\theta)\,\text{hour} = \frac{10}{\sin\theta}\,\text{min}$

This time will be shortest when the $\sin\theta$ will assumes its maximum value, that is, 1.

The shortest time will be $10\,\text{min}$.

Therefore, if a swimmer wishes to cross a 500m wide river flowing at 5 kmph and his speed with respect to water is 3 kmph, and heads in a direction making an angle θ with the flow, then the shortest time taken by the man to cross the river is $10\,\text{min}$.

19. Correct match: (A) -> R), (B) -> S), (C) -> T), (D) -> P)

(A) We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$v_x = u$

$v_y = gt$

where u is the horizontal velocity of projection.

Further, for a horizontally launched projectile from a height h, horizontal distance covered is given by: $y = ut$,where u is the horizontal velocity of projection.

Given, a body is projected horizontally from the top of a tower with velocity u.

Then, the maximum horizontal distance is:

$$y = u\left(\sqrt{\frac{2h}{g}}\right)$$

(B) Now, the vertical and horizontal velocity on reaching the ground is given by:

$v_x = u$

$$v_y = -g\sqrt{\frac{2h}{g}}$$

Therefore, velocity on reaching the ground is:

$$v = \sqrt{\left(g\sqrt{\frac{2h}{g}}\right)^2 + u^2}$$

$$v = \sqrt{2gh + u^2}$$

(C) The time taken by the body to reach the ground is $t = \sqrt{\frac{2h}{g}}$.

(D) As the projectile is launched horizontally, the initial vertical component of velocity is zero

Therefore, the correct match is: (A) -> R), (B) -> S), (C) -> T), (D) -> P)

20. Correct match: (A) ->R), (B) -> T), (C) -> P), (D) -> S)

(A) Given, a bomb at rest explodes into two fragments at a height h moves horizontally with

velocities u_1 and u_2.

We know, for a horizontally launched projectile from a height h, the vertical and horizontal velocity are given by:

$v_x = u$

$v_y = -gt$

where u is the horizontal velocity of projection, and negative sign shows the downward velocity.

Further, for a horizontally launched projectile from a height h, horizontal and vertical distance covered is given by:

$x = ut$

$y = -\frac{1}{2}gt^2$

Therefore, the velocity vector of the two fragments is:

$\vec{v}_1 = u_1\hat{i} - gt\hat{j}$

$\vec{v}_2 = -u_2\hat{i} - gt\hat{j}$

Now, if the velocity vectors are perpendicular, then their dot product is zero, thus:

$\vec{v}_1.\vec{v}_2 = 0$

$-u_1u_2 + g^2t^2 = 0$

$t^2 = \frac{u_1u_2}{g^2}$

$t = \frac{\sqrt{u_1u_2}}{g}$

Hence, time after velocity vectors is perpendicular is $\frac{\sqrt{u_1u_2}}{g}$.

(B) Now, the displacement vector of the two fragments is:

$\vec{d}_1 = u_1t\hat{i} - \frac{1}{2}gt^2\hat{j}$

$\vec{d}_2 = -u_2t\hat{i} - \frac{1}{2}gt^2\hat{j}$

Now, if the displacement vectors are perpendicular, then their dot product is zero, thus:

$\vec{d}_1.\vec{d}_2 = 0$

$-u_1u_2 + \frac{1}{4}g^2t^2 = 0$

$t^2 = \frac{4u_1u_2}{g^2}$

$t = \frac{2\sqrt{u_1u_2}}{g}$

Hence, time after displacement vectors is perpendicular is $\frac{2\sqrt{u_1u_2}}{g}$.

(C) Now, position of the fragments when velocity vectors are perpendicular are:

$\vec{d}_1 = u_1\left(\frac{\sqrt{u_1u_2}}{g}\right)\hat{\imath} - \frac{1}{2}g\left(\frac{\sqrt{u_1u_2}}{g}\right)^2\hat{\jmath}$

$\vec{d}_2 = -u_2\left(\frac{\sqrt{u_1u_2}}{g}\right)\hat{\imath} - \frac{1}{2}g\left(\frac{\sqrt{u_1u_2}}{g}\right)^2\hat{\jmath}$

Hence, separation between them when velocity vectors are perpendicular is:

$$d = \frac{\sqrt{u_1 u_2}}{g}(u_1 + u_2)$$

(D) Now, position of the fragments when displacement vectors are perpendicular are:

$$\vec{d}_1 = u_1\left(2\frac{\sqrt{u_1 u_2}}{g}\right)\hat{\imath} - \frac{1}{2}g\left(2\frac{\sqrt{u_1 u_2}}{g}\right)^2\hat{\jmath}$$

$$\vec{d}_2 = -u_2\left(2\frac{\sqrt{u_1 u_2}}{g}\right)\hat{\imath} - \frac{1}{2}g\left(2\frac{\sqrt{u_1 u_2}}{g}\right)^2\hat{\jmath}$$

Hence, separation between them when velocity vectors are perpendicular is:

$$d = \frac{2\sqrt{u_1 u_2}}{g}(u_1 + u_2)$$

Therefore, correct match is: (A) ->R), (B) -> T), (C) -> P), (D) -> S)

21. Correct number: 2

We know, for a horizontally launched projectile from a height h, the time taken by it to reach the ground is given by:

$$t = \sqrt{\frac{2h}{g}}$$

Further, for a horizontally launched projectile from a height h, horizontal distance covered is given by:

$x = ut$

where u is the horizontal velocity of projection.

Given, from the top of tower 80 m, two stones are projected horizontally with velocities 20 m/s, 30 m/s, in opposite directions. The distance between the two stones on reaching the ground is $n \times 10^2\ m$.

Now, time taken by each stone to reach the ground is:

$$t = \sqrt{\frac{2(80)}{10}}$$

$t = 4$

Therefore, the distance between the two stones on reaching the ground is:

$d = u_1 t + u_2 t$

$d = 20(4) + (30)(4)$

$d = 200$

Comparing this with $n \times 10^2\ m$, we see $n = 2$.

Therefore, if from the top of tower of height 80 m, two stones are projected horizontally with velocities 20 m/s and 30 m/s in opposite directions, then the distance between the two stones on reaching the ground is $2 \times 10^2\ m$.

3. Force and Laws of Motion

Solutions

Force

LEVEL – I

1. Correct option: (C) Both A and B.

The remark that follows alludes to the idea of forces in physics. A force is a push or a pull that can alter how an object moves. Depending on the kind and strength of the applied force, it has the ability to either move a stationary body or halt a moving one.

(A) A balanced force is a pair of forces that are opposing in direction and of equal magnitude as they act on an object. In this scenario, the object experiences no net force and either remains immobile or keeps moving in a straight path at a constant speed. The object will remain at rest, for instance, if you push it to the right with a force of 10 newtons and your friend pulls it to the left with a force of 10 newtons as well.

(B) Unbalanced Force: Two forces acting on an item that are not in opposition to one another or that are not equal in magnitude are referred to as unbalanced forces. Because there is a force acting on the object in this situation, its motion is altered. For instance, if you push something with a force of 15 newtons to the right and a buddy pulls it with a force of 10 newtons to the left, the forces are out of balance, and the thing will move to the right faster due to the net force of 5 newtons.

Accordingly, depending on their unique properties and the consequent net force exerted on the item, both balanced and unbalanced forces may be to blame for moving a stationary body or stopping a moving body.

2. Correct option: (A) balanced force.

A balanced force has the ability to alter size and shape. So a toothpaste tube changes shape when it is squeezed. An illustration of a balanced force is the force that caused this.

3. Correct option: (A) Balanced force act on the balloon.

A rubber balloon is held between the hands, and when pressed, it changes shape. This occurs because

A) The balloon is subject to balanced force.

B) The balloon is affected by an uneven force

C) The balloon experiences frictional forces.

D) The balloon experiences the effects of gravity

The balloon is compressed from two sides by unbalanced hand pressures. The balloon shape changes as a result.

As we All know, Newton's first law of motion states that unless an external force is applied, a body tends to be in a condition of inertia. Here, an uneven force is taken into account. This is due to the fact that only an uneven force can cause a body to move. Or, we may say that an unbalanced force tends to cause bodily motion because it isn't balanced by other forces.

The opposing force must counteract the balanced force in order to achieve balance. In other words, there is no motion caused by this force. But occasionally, it can cause an alteration in the body's shape. In the scenario under consideration, a rubber balloon encounters a balanced force when pressed between hands, which causes the rubber balloon to alter shape.

4. Correct option: (D) All the above.

A) Change in the body's speed: An unbalanced force might cause the body to accelerate or decelerate, changing its speed. The speed will rise if the force acts in the direction of the motion; it will decrease if the force works in the opposite direction.

B) Change in bodily shape: An imbalanced force can cause the body to deform or change its shape. This effect is especially important for deformable items like fluids or soft materials. The body's shape may change as a result of the force through compression, stretching, or bending.

C) Change in the body's motion direction: An unbalanced force has the power to alter the body's motion direction. The body may change its trajectory and move in a different direction when the force is applied at an angle to the original direction of motion.

In conclusion, all the effects listed in alternatives A, B, and C are caused by changes in the motion, speed, and shape of a body due to an unbalanced force. Therefore, D) All of the above is the right response.

5. Correct option: (B) Forward.

The passengers of a halted car are abruptly pushed forward when this happens.

Because of the inertia opposing the backward motion, when the car abruptly stops in the backward direction from a state of motion, the passengers feel a force applied to them in the forward direction.

As a result, when a moving car suddenly stops, the passengers are thrusting forward.

6. Correct option: (A) Backward.

The passengers of a moving car are abruptly yanked backwards as the car stops.

If a body is at rest or moving in a straight line at a constant speed, Newton's first rule of motion or inertia states that "unless an external force is applied to it, it will remain at rest or keep moving in a straight line at constant speed."

The backward jerk that occurs when a stationary car starts is felt by the passengers. The occupants experience a force being applied to them as the car begins and abruptly moves forward from a condition of rest due to inertia. The forward motion is resisted by the inertia.

7. Correct option: (B) inertia of motion.

The tangent to the wheel is the straight line along which the mud particles continue to move when they break free from the wheel. This is a result of motion's inertia.

8. Correct option: (C)mass of the object.

The first law of motion states that "A body continues to be in its state of rest or uniform motion along a given direction, until and unless an external force is applied on it." The law of inertia is another name for this.

A body's inability to alter its position of rest or uniform motion is known as inertia.

A body's mass can be thought of as its inertia in motion.

Inertia is therefore dependent on mass. The inertia of a body increases with mass, making movement of the body more challenging.

9. Correct option: (A) Newton’s first law of motion.

Newton's First Law of Motion offers the foundation for the solution to this issue. At first, both the leaves and the tree are dormant. However, when a tree is shook severely, the tree moves while the leaves tend to remain still. As a result, if we shake a tree limb aggressively while it is in the position of rest, some of the leaves may come off the tree.

10 The correct option (B) move forward.

Newton's first law of motion states that a body in motion will stay in motion even in the absence of an external force. Since the water in the tanker is not a component of the tanker, it is not stopped when the tanker is put on brakes because there is no outside force acting on it. As a result, the water in the tanker will advance. As long as there is no external force

acting on an object, it will continue to move in the same direction and at the same pace. In this scenario, using the brakes won't stop the water from going forward. If the tanker doesn't have a closed roof, the tanker will stop, but the water will keep flowing and could overflow.

11. Correct option: (D) mass.

The inertia of an object is determined by its mass. Therefore, inertia is measured by mass.

12. Correct option: (D) Keep it moving with uniform velocity.

According to Newton's first law, a body that is at rest or travelling in a straight path at a constant speed will continue to do so unless it is subjected to an external force. The law of inertia is the name given to this premise.

Therefore, a force is not needed to maintain the particle's uniform speed.

13. Correct option: (C) Inertia.

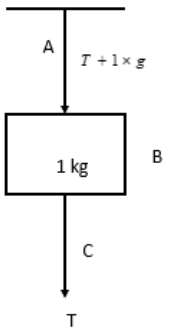
The body's inertia is described by Newton's first law of motion. According to this, everyone has a propensity to hold onto their state (whether it be rest or motion) because of their inertia.

According to the first law of motion, "a body at rest will stay at rest, and a body in motion will continue to move in a straight line until an external force acts on it."

14. Correct option: (A) Inertia of direction.

Due of directional inertia, passengers are thrown outward. Passengers will be thrown forward if the breaks are rapidly applied, and backward if the bus is suddenly accelerating.

15. Correct option: (B) The portion BC of the string will break.



Imagine that the string C experiences a force and develops a tension T.

The tension in piece AB for an equilibrium of a 1-kilogram block will be $T + 1 \times g$, hence the portion AB will always break first. When C receives an abrupt jolt, C creates an impulsive tension that is greater than the breaking tension initially, which breaks before this impulse can travel to A as a wave through a block.

16. Correct option: (A) (i) is true & (ii) is false.

Because the body of a bus passenger tends to travel forward due to inertia of motion, the passenger would jump off and run in the direction of the bus motion.

17. Correct option: (A) Inertia of rest.

When the bus accelerates rapidly, the passenger resists changing their condition of rest and continues to rest in the same manner, it will look like the person is regressing. The fact that No one can independently alter its state of motion.

18. Correct option: (B) Force.

The rate of change of momentum is directly inversely proportional to applied force, according to Newton's second law.

$F \propto \frac{m(v-u)}{t}$ from this, de deduce,

$F = ma$

So, Newton's 2nd law gives us measure of force.

19. Correct option: (C) Net force.

Mass is a scalar quantity; hence the direction of force and acceleration will be the same according to Newton's Second Law of Motion, which states that the net force on an object is equal to mass times its acceleration.

20. Correct option: (A) $2mv$**.**

The initial velocity of the body before collision $u = -V$

where the minus symbol denotes a movement towards the wall by the body.

Now initial momentum of the body is given by as, $p_i = mu = -mV$

The final velocity of the body after the collision, $v = V$

when the movement of the body away from the wall is shown positively.

Now, final momentum of the body is given as,

$p_f = mv = mV$

Change in momentum of body is given by as,

$\Delta P = P_f - P_i = mV - (-mV) = 2mV$

So, the change in momentum is $2mv$.

21. Correct option: (B) Proportional to the net force applied on it.

Newton's second law of motion states that "the change in momentum is in the direction of the applied force and the rate of change of momentum is directly proportional to the force applied." The force is expressed mathematically by Newton's second law of motion.

$$\text{Force} \propto \frac{dp}{dt}$$

22. Correct option: (B) Reduce the force exerted by the ball.

As the duration of the ball's slowing down increases, the impulse that the ball applies to the hand by moving the hand rearward decreases. As a result, the fielder uses less force to capture the ball with his hands.

23. Correct option: (B) $5\frac{\text{m}}{\text{s}^2}$**.**

$F = 5\text{N}$

$W = 9.8\text{N}$

Now we know that,

$W = mg$

$9.8 = m \times 9.8$

$m = 1\text{kg}$

Now force is given by as,

$F = ma$

$5 = 1 \times a$

$a = 5\frac{m}{s^2}$

So, the acceleration produced is $5\frac{m}{s^2}$.

24. Correct option: (B) $2\frac{\text{m}}{\text{s}^2}$**.**

$m = 2m$

$a = 4\frac{m}{s^2}$

Now force is given by as,

$F = ma$

$F = 4m$

If the car pulls another car of same mass,

Now force is given by as,

$F = 2ma'$

$a' = \frac{F}{2m}$

Now put the value of F in above equation,

$a' = \frac{4m}{2m}$

$a' = 2\frac{m}{s^2}$

So, the acceleration produced is $2\frac{\text{m}}{\text{s}^2}$.

25. Correct option: (C) South-west.

A force acting in the north-east is applied to an item. A second force should be applied in the south-west to counteract this one.

The second force should be applied in a 180° direction.

26. Correct option: (A) $10\sqrt{2}\text{kg}$**.**

Force vector, $\vec{F} = 6\hat{i} - 8j + 10k$

Now magnitude of the force is given by as,

$|F| = \sqrt{36 + 64 + 100} = 10\sqrt{2}N$

Now mass is given by as,

$m = \frac{F}{a} = \frac{10\sqrt{2}}{1} = 10\sqrt{2}\text{kg}$

So, the mass of the body is $10\sqrt{2}\text{kg}$.

27. Correct option: (C) 2N.

$m = 10\text{kg}$

$a = 0.2\frac{\text{m}}{\text{s}^2}$

Now, force is given by as,

$F = ma$

$F = 10 \times 0.2$

$F = 2\text{N}$

So, the magnitude of force acting on the body is 2N

28. Correct option: (A) 1800 N.

$m = 1800\text{kg}$

$u = 10\frac{\text{m}}{\text{s}^2}$

$s = 50\text{m}$

Now using the equation of motion,

$v^2 - u^2 = 2as$

$0 - 10^2 = 2a \times 50$

$a = \frac{-100}{100} = -1 \frac{\text{m}}{\text{s}^2}$

Now, force is given by as,

$F = ma$

$F = 1800 \times 1$

$F = 1800N$

So, the force acting on the ca is 1800 N.

29. Correct option: (B) 10 N.

Given:

$t = 4\text{sec}$

$s = 10\text{m}$

m=2kg

Now, using the second equation of motion,

$s = ut + \frac{1}{2}at^2$

$10 = 0 + \frac{1}{2}a \times 4$

$a = \frac{10}{2} = 5 \frac{m,}{s^2}$

Now force is given by as,

$F = ma$

$F = 2 \times 5$

$F = 10\text{N}$

So, the magnitude of the force is given by as 10N.

30. Correct option: (A) $12500\hat{i} + 6250\,j$ **.**

$m = 10g = 10 \times 10^{-3}\text{kg}$

F=$10\hat{i} + 5\,j$

Now using the force formula,

$ma = 10\hat{i} + 5\,j$

$10 \times 10^{-3} a = 10\hat{i} + 5\,j$

$\frac{10}{1000} a = 10\hat{i} + 5\,j$

$a = 1000\hat{i} + 500\,j$

$\frac{dv}{dt} = 1000\hat{i} + 500\,j$

$dv = \left(1000\hat{i} + 500\,j\right) dt$

$v = 1000t\hat{i} + 500\,j$

Now,

Velocity is $x-$ direction,

$$\frac{dx}{dt}=100t$$

$$x=\int_0^5 100tdt$$

$$x=\frac{100(5)^2}{2}\hat{i}$$

$$x=12500\hat{i}$$

Velocity is $y-$direction,

$$\frac{dy}{dt}=500t$$

$$x=\int_0^5 500tdt$$

$$x=\frac{500(5)^2}{2}\hat{i}$$

$$x=6350\,j$$

So, the position vector is $12500\hat{i}+6250\,j$.

31. Correct option: (B) 6250 N.

Given:

$m=10\text{g}=0.01\text{kg}$

$$\text{u}=250\frac{\text{m}}{\text{s}}$$

$s=5\text{cm}=0.05m$

$$v=0\frac{\text{m}}{\text{s}}$$

Using the equation of motion,

$v^2-u^2=2as$

$0^2-250^2=2\times0.05\times a$

$$a=\frac{62500}{0.1}=-625000\frac{\text{m}}{\text{s}^2}$$

The magnitude of force is given by as,

$F=ma$

$F=0.01\times625000$

$F=6250\text{N}$

So, the magnitude of the force is 6250 N.

32. Correct option: (A) 0°**.**

The weight (W) and the action (A) will both move in the same direction and side by side. As a result, there should be no angle between them.

On a table, a book is said to be lying. It is necessary to determine the angle between the weight of the book and how it acts on the table. The phrase "action of the book on the table" refers to the force that the book is exerting on the surface.

Action forces are often defined as forces that can operate independently, in response to a field, or as a result of biological beings like humans and animals. The weight of a body, electrostatic attraction and repulsion, and other forces are examples of action forces.
Reaction forces, on the other hand, can be thought of as forces that do not exist in the absence of an action force. They are there to counter whatever that a force of action has acted upon. The normal force, friction, tension, and other forces can be reaction forces.
The only force exerted on the table by the book now that it is lying (and not moving) is its weight. Consequently, this might be regarded as the book's action. The angle is zero since the book's weight and motion on the table both move in the same direction.

33. Correct option: (D) $2\frac{\text{m}}{\text{s}}$**.**

Acceleration of mass m_1, $a_1 = 6\frac{\text{m}}{\text{s}^2}$

Acceleration of mass m_2, $a_2 = 3\frac{\text{m}}{\text{s}^2}$

Now we know that force is given by as,

$F = ma$

$m = \frac{F}{a}$

$m_1 = \frac{F}{6}$ *or* $m_2 = \frac{F}{3}$

If the two masses are fixed together, and the same force is applied to the combination.
So, the net force is given by as,

$F = \left(m_1 + m_2\right)a$

$F = \left(\frac{F}{6} + \frac{F}{3}\right)a$

$F = \left(\frac{F + 2F}{6}\right)a$

$a = \frac{6}{3}$

$a = 2\frac{\text{m}}{\text{s}^2}$

So, the acceleration is $2\frac{\text{m}}{\text{s}}$.

34. Correct option: (C) $50\frac{\text{kg m}}{\text{s}}$**.**

Given:
Mass of the body, $m = 10\text{kg}$
Velocity, $v = 5\frac{m}{s}$
So, the linear momentum is given by as,
$p = mv$
$p = 10 \times 5$
$p = 50\frac{\text{kg}m}{s}$
So, the linear momentum is $50\frac{\text{kg}m}{s}$.

35 The correct option (B) $100\frac{\text{kg m}}{\text{s}}$ **.**

Initial velocity, $u = 0\frac{m}{s}$

Acceleration, $a = 4\frac{\text{m}}{\text{s}^2}$

Time, $t = 5\text{sec}$

Mass, $m = 5\text{kg}$

Now using the equation of motion,

$$v = u + at$$
$$v = 0 + 4 \times 5$$
$$v = 20\frac{m}{s}$$

Now , momentum is given by as,

$$p = mv$$
$$p = 5 \times 20$$
$$p = 100\frac{\text{kg}m}{s}$$

So, the momentum is $100\frac{\text{kg m}}{\text{s}}$.

36. Correct option: (A) Backward

According to Newton's third law, "Every action has an equal and opposite reaction," a forward jump causes a boat to experience backward force. As a result, the boat is moving backward.

37. Correct option: (D) All of the above.

According to Newton's Third Law of Motion, "There is always an equal and opposite reaction to every action."

Following are interpretations of Newton's Third Law of Motion.

1. There are always two forces at work. The forces exerted on a body A by a body B are equal and in opposition to each other.
2. The forces exerted on A by B and B by A act simultaneously.
3. Forces of action and reaction act on various bodies rather than the same body

38. Correct option: (B) (i) is false & (ii) is true.

The horse experiences some force from the ground as it applies some force to it. The horse moves as a result of this response force. As the horse advances, the force in turn pulls the cart. Therefore, the force of the earth is what causes the horse to move. Friction force is the name given to this force.

39. Correct option: (D) all the above .

According to Newton's third law of motion, every action has a corresponding and opposing reaction.

The third rule of motion of Newton is the physical concept that governs how a jet aircraft operates. There is an equal and opposite reaction to every action, according to Newton's third law of motion. A jet plane pushes air backward, and air pushes the jet plane backward in the other direction. When someone discharges a gun, a reaction force forces him to go backward.

The ground is pushed by the rocket during rocket lunching, and is then pushed back by the opposing force.

40. Correct option: (C) Third law of motion.

Swimming is based on Newton's third rule of motion since we move forward and experience equal forces from the water as we push it.

41. Correct option: (D) linear momentum.

The forces acting on the two objects are equal in size and react in opposition, according to Newton's third law. Such pressures result in one object gaining momentum while the other loses momentum, preserving the system's overall momentum in the absence of any external forces. As a result, the law of conservation of momentum follows from Newton's third law of motion. According to Newton's third law, an object B will exert a force on an object A that is identical in magnitude but directed in the opposite direction. Newton derived the law of conservation of momentum using this concept.

42. Correct option: (C) Gravitational force

Contact forces are the forces that apply to bodies when they come into physical contact. For instance, frictional force and the force applied when two bodies collide.

Non-contact forces are those that affect bodies without actually making physical contact with them. Examples include the gravitational and electrostatic forces.

So, the friction, Spring force and tension force are contact force and the gravitational force is non-contact force.

LEVEL -II

1. Correct option: (C) Upper part of the body continues to be in the state of motion, whereas the lower part of the body in contact with the seat comes to rest.

As a result of inertia. The lower body tends to come to rest when a fast bus abruptly stops, but the upper body typically continues to move uniformly. The passengers are consequently propelled forward. When a train suddenly stops, passengers in the moving train experience an instantaneous jolt forward because the upper half of their bodies is still in motion while the lower part of their bodies that are in contact with the seat are still at rest.

2. Correct option: (C) Newton's first law of motion.

An object's inertia tends to oppose any modification to either its condition of rest or state of motion. A carpet starts moving when it is struck with a stick. However, dust particles make an effort to fight off changes to their resting condition. Newton's first law of motion states that as the carpet moves, the dust particles remain at rest. As a result, due to inertia, the dust particle emerges through the carpet.

3. Correct option: (D) $2\times10^{7}\,\frac{\mathrm{N}}{\mathrm{m}^2}$**.**

Number of balls, $N=10000$

Mass of each ball, $m=1g=1\times10^{-3}\mathrm{Kg}$

Area, $A=1\mathrm{cm}^2=10^{-4}\mathrm{m}^2$

Velocity, $v=100\,\frac{\mathrm{m}}{\mathrm{s}^{-1}}$

Pressure is given by as,

$$P = \frac{F}{A} = \frac{ma}{A}$$

$$P = \frac{10000 \times 10^{-3}\left(v - (-v)\right)}{10^{-4}}$$

$$P = 2 \times 10^{7} \frac{\mathrm{N}}{\mathrm{m}^2}$$

So, the pressure on the surface is $2 \times 10^{7} \frac{\mathrm{N}}{\mathrm{m}^2}$.

4. Correct option: (A) 5000 N.

The mass of the vehicle, $m = 100\mathrm{k}g$

Initial velocity, $u = 5\frac{\mathrm{m}}{\mathrm{s}}$

Time required to stop the vehicle, $t = \frac{1}{10} = 0.1\mathrm{sec}$

Final velocity, $v = 0$,

From the newton second law of motion,

$$F = \frac{dp}{dt}$$

Here, $p = \mathrm{momentum}$

Now,

$$F = m\frac{v-u}{t}$$

$$F = 100\frac{[0-5]}{0.1}$$

$$F = -5000\mathrm{N}$$

Negative sign indicates the force is in the opposite direction.

So, the value of force is 5000 N.

5. Correct option: (B) $2mnu$ **.**

Maas of each ball= $m$kg

Velocity, $= u\frac{m}{s}$

In elastic collision we have ,

Change in momentum = $mu - (mu) = mu + mu = 2mu$

Now for N balls change in momentum is given by as,

$\Delta p = N \times 2mu$

Rate change of momentum $= \frac{\Delta p}{t} = \frac{N2mu}{1sec}$

Now according to Newton second law of motion,

$$F = \frac{\Delta p}{t}$$

$$F = 2Nmu$$

So, the force experienced by the surface is $2Nmu$.

6. Correct option: (D) $\sqrt{2}m_1 = m_2$**.**

We know that ,

$v \propto \sqrt{s}$

$\frac{v_1}{v_2} = \sqrt{\frac{2h}{h}} = \frac{\sqrt{2}}{1}$

For momentum,

$\frac{P_1}{p_2} = \frac{m_1 v_1}{m_1 v_2}$

$\frac{P_1}{p_2} = \frac{m_1}{m_2} \times \frac{\sqrt{2}}{1}$

$\sqrt{2}m_1 = m_2$

So, the required ratio is $\sqrt{2}m_1 = m_2$.

7. Correct option: (C) $\frac{a_1 a_2}{a_1 + a_2}$**.**

Now using the Newton law of motion,

Force on first body, $F_1 = m_1 a_1$

Force on the second body, $F_2 = m_2 a_2$

If the two bodies are combined, then force is,

$F = (m_1 + m_2)a$

$F = \left(\frac{F}{a_1} + \frac{F}{a_2}\right)a$

$F = \frac{a_1 a_2}{a_1 + a_2}$

So, the acceleration produced is $\frac{a_1 a_2}{a_1 + a_2}$.

8. Correct option: (C) $1\frac{\text{m}}{\text{s}}$**.**

Maas of the bullet, $m_b = 200g = 200 \times 10^{-3}\text{kg}$

Velocity of the bullet, $v = 5\frac{m}{\text{sec}}$

Maas of the gun, $m_g = 1\text{kg}$

So, the recoil velocity is given by as,

$v = \frac{m_b v_b}{m_g}$

$V = \frac{0.2 \times 5}{1}$

$V = 1\frac{m}{s}$

So, the velocity of the gun is $1\frac{\text{m}}{\text{s}}$.

9. Correct options: (B) and (C).

Both kg-m/s (kilogram-metre per second) and g-cm/s (gram-centimetre per second) are units of momentum. This demonstrates that momentum is a vector quantity with both magnitude and direction. It also displays the product of mass and velocity.

Momentum is the amount of motion a body can hold. Due to the object's mass and velocity, it is a measure of its motion. A body will have more momentum if its mass is greater and its speed is higher.

10. Correct options: (A) And (B).

The International System of Units (SI) uses the Newton (N) as the primary unit of force. The force needed to accelerate a mass of one kilogramme by one metre per second squared is known as a Newton ($N = \frac{\text{kg m}}{\text{s}^2}$).

A force measurement in the centimeter-gram-second (CGS) system is called a dyne (dyn). The force needed to propel a mass of one gramme by one centimetre per second squared is known as a dyne ($N = \frac{\text{kg m}}{\text{s}^2}$).

11. Correct option: (ABD)

12. Correct options: (A) and (D).

Forces in the vertical direction are zero because the mass is at rest and the resultant of the forces acting on it is equal to zero.

From the diagram, In the vertical direction:

$T_1 \sin 30^\circ + T_2 \sin 60^\circ - mg = 0$

$\frac{T_1}{2} + \frac{\sqrt{3}T_2}{2} = 5g$

$T_1 + \sqrt{3}T_2 = 100$

Similarly in horizontal direction:

$T_1 \cos 30^\circ - T_2 \cos 60^\circ$

$\frac{T_1}{T_2} = \frac{1}{\sqrt{3}}$

Solving above equation we get,

$T_1 + \sqrt{3} \times \sqrt{3}T_1 = 100$

$T_1 = 25\text{N}$

$T_2 = 25\sqrt{3}\text{N}$

So, the $T_1 = 25\text{N}$ and $T_2 = 25\sqrt{3}\text{N}$.

So, the correct option are (A) and (D).

13. Correct option: (B) 0.5 kg.

Using the newton law of force,

For mass m_1,

$F = m_1 a_1$

$5 = m_1 \times 10$

$m_1 = 0.5\text{kg}$

So, the mass of the first body is 0.5 kg.

14. Correct option: (C) 0.25 kg.

Using the newton law of force,

For mass m_2,

$F = m_2 a_2$

$5 = m_2 \times 20$

$m_1 = 0.25\,\text{kg}$

So, the mass of the second body is 0.25 kg.

15 the correct option (D) $6.67\frac{\text{m}}{\text{s}^2}$**.**

For mass m_1,

Using the newton law of force,

$F = m_1 a_1$

$5 = m_1 \times 10$

$m_1 = 0.5\text{kg}$

For mass m_2,

Using the newton law of force,

$F = m_2 a_2$

$5 = m_2 \times 20$

$m_1 = 0.25\,\text{kg}$

Now, the total mass= $m_1 + m_2 = 0.5 + 0.25 = 0.75\text{kg}$

Now acceleration when both the masses were tied,

$F = m_T a_T$

$a_T = \frac{5}{m_T} = \frac{5 \times 4}{3} = \frac{20}{2}\frac{m}{s^2} = 6.67\frac{m}{s^2}$

So, the acceleration is $6.67\frac{\text{m}}{\text{s}^2}$.

16 Correct option: (A)

17 Correct option: (B)

18 Correct option: (B)

19. Correct Matching is: (a)-(S), (b)-(R), (c)-(Q), (d)-(P).

The outcome of the forces acting on the mass while it is at rest is equal to zero, hence the forces acting in the vertical direction are.

A body's momentum is the amount of motion it is capable of. A body's momentum is calculated by multiplying its mass by its velocity. It's indicated by. When a body is at rest, its momentum is also zero because its velocity is zero.

A body's rate of momentum change occurs in the same direction as the applied force and is directly proportional to the applied force. The right way to phrase Newton's second law is as stated above.

According to the third rule, there is an equal and opposite reaction to every action (force) in nature. When object A pulls on item B, object B pulls back on object A with an equal and opposing force. In other words, interactions produce forces.

The correct option is (a)-(S), (b)-(R), (c)-(Q), (d)-(P).

20. The correct Matching is (a)-(R), (b)-(R),(c)-(P),(d)-(Q).

Given: The momentum of the bicycle, $P = 24\frac{\text{Kg m}}{\text{s}}$.

The momentum is given by as,

$P = mv$

$24 = mv$

Twice the mass and was moving at the same speed:

(A) The momentum of the bicycle doubles if its mass doubles and it continues to move at the same speed. $24\frac{\text{kg m}}{\text{s}}$ is the bicycle's initial momentum , thus let's say that it has a 24 kg mass. Its mass is 24 k g and its speed is 1. M / s . The bicycle has a momentum of $48\frac{\text{kg m}}{\text{s}}$ when the mass is doubled while maintaining the same speed.

(B) The momentum of the bicycle will be $48\frac{\text{kg m}}{\text{s}}$ the mass of the bicycle remains the same and the velocity doubles. The mass of the bicycle will remain 24 kg and the velocity is now $2\frac{\text{m}}{\text{s}}$.

$$p = mv = 24 \times 2 = 48\frac{\text{kg m}}{\text{s}}$$

(C) The bicycle will have a momentum $24\frac{\text{kg m}}{\text{s}}$ of if its mass is halved and velocity doubled.

$p = \left(\frac{24}{2}\right)(1 \times 2)$

$p = 24\frac{\text{kg m}}{\text{s}}$

(D) The bicycle will have a momentum of $12\frac{\text{kg m}}{\text{s}}$ if its mass is kept the same, but speed is halved.

$p = (24)\left(1 \times \frac{1}{2}\right)$

$p = 12\frac{\text{kg m}}{\text{s}}$

So, the correct matching is (a)-(R), (b)-(R),(c)-(P),(d)-(Q).

21. Correct Answer is:

$F = 100\text{dyn}$

$m = 5\text{g}$

$t = 10\text{sec}$

$u = 0$

u is the initial velocity.

Now using the newton's law of force,

$F = ma$

$a = \frac{100 \times 10^{-5}}{5 \times 10^{-3}}$

$a = 0.20 \frac{\text{m}}{\text{s}^2}$

Now, using the equation of motion,

$v = u + at$

$v = 0 + 0.20 \times 10$

$v = 2 \frac{\text{m}}{\text{s}}$

$v = 200 \frac{\text{cm}}{\text{s}}$

So, the velocity produced is $2 \times 100 \frac{\text{cm}}{\text{s}}$.

Free Body Diagram

LEVEL – I

1. Correct option: B) Its momentum is doubled.

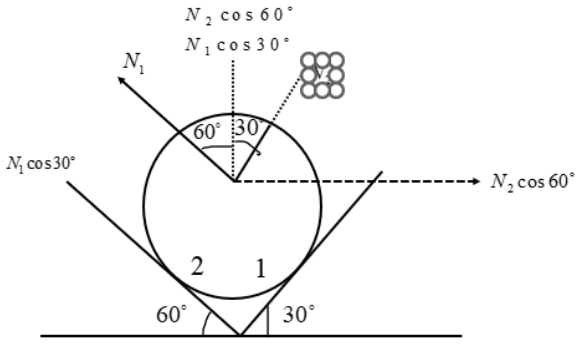
When the velocity is doubled, the kinetic energy is quadrupled. As a result, double the velocity twice the momentum. Momentum is proportional to velocity.

Momentum is given by,

$P = mv$

So, the momentum is directly proportional to velocity.

2. Correct option: (A) $N_1 = 50\text{N}, N_2 = 50\sqrt{3}\text{N}$**.**



From the diagram:

$\Rightarrow N_1 \cos 60^\circ = N_2 \sin 30^\circ$

$\sqrt{3} N_1 = N_2$

$\Rightarrow N_1 \cos 60^\circ + N_2 \cos 30^\circ = mg = 100$

$\frac{N_1}{2} + \sqrt{3} N_1 \times \frac{\sqrt{3}}{2} = 100$

$N_1 = 50\text{N}$

$N_2 = 50\sqrt{3}\text{N}$

So, N_1 is 50N and N_2 is $50\sqrt{3}\text{N}$.

3. Correct option: 1.4ms^{-1}**.**

Mass of the first body, $m_1 = 3\text{kg}$

Mass of the second body, $m_2 = 4\text{kg}$

Acceleration is given by,

$$a = \frac{(m_2 - m_1)}{(m_2 + m_1)} g$$

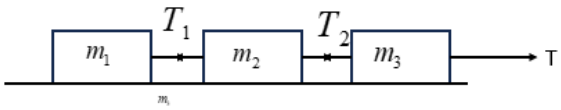
$$a = \frac{4-3}{4+3} \times 9.8$$

$$a = 1.4\text{ms}^{-1}$$

So, the acceleration of the system is 1.4ms^{-1}.

4 Correct option: (A)

5. Correct option: (C) $T_2 = \frac{(m_1 + m_2)}{m_1 + m_2 + m_3} T$



Let's consider a is acceleration,

$$T = (m_1 + m_2 + m_3) a \cdots\cdots (1)$$

Now , tension is given by,

$$T_1 = m_1 a$$

$$T_1 - T_1 = m_2 a \cdots\cdots (2)$$

$$T - T_2 = m_3 a$$

Form using equation (1) and (2), we get,

$$T_2 = a(m_1 + m_2)$$

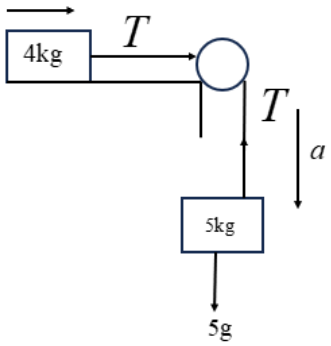
$$T_2 = \frac{T}{m_1 + m_2 + m_3}(m_1 + m_2)$$

So, the tension in the string between m_2 and m_3 is $T_2 = \frac{(m_1 + m_2)}{m_1 + m_2 + m_3} T$.

6. Correct option: (B) 5.44ms^{-2}**.**

$M_1 = 4\text{kg}$

$M_2 = 5\text{kg}$



Acceleration is given by,

$$a = \frac{M_1 g}{M_1 + M_2}$$

$$a = \frac{5 \times 9.8}{4 + 5}$$

$$a = 5.44 \text{ms}^{-2}$$

So, the acceleration of 5kg mass is 5.44ms^{-2}.

7. Correct option: (B) 200ms^{-2} **,** $16 \times 10^{-6} \text{Ns}$**.**

Given:

$F = 16\text{N}$

m=80g=0.08kg

Force is given by as,

$F = ma$

$16 = 0.08 \times a$

$a = 200 \text{ms}^{-2}$

Noe impulse is given by as,

Impulse = change in momentum

Impulse=m(v-u)

$\text{Impulse} = F \times t$

$\text{Impulse} = 16 \times 10^{-6}$

$\text{Impulse} = 16 \times 10^{-6} \text{Ns}$

So, the acceleration is 200ms^{-2} and impulse is $16 \times 10^{-6} \text{Ns}$.

8. Correct option: (A) $3.27 \frac{\text{m}}{\text{s}^2}$ **,** 6.54N**.**

Mass of the block, $\text{m}_1 = 2\text{kg}$

Mass of the body, $\text{m}_2 = 1\text{kg}$

So, the acceleration is given by,

$$a = \frac{m_2}{m_1 + m_2} \times g$$

$$a = \frac{1}{2 + 1} \times 9.8$$

$$a = 3.27 \frac{m}{s^2}$$

So, the tension is given by,

$T = m_1 a$

$T = 2 \times 3.25 = 6.54\text{N}$

So, the acceleration of block is $3.27 \frac{\text{m}}{\text{s}^2}$ and tension of the block is 6.54N.

9. Correct option: (D) 73.5 N.

Given:

$m_1 = 6\text{kg}$

$m_2 = 10\text{kg}$

Tension in the thread is given by,

$$T = \frac{2m_1 m_2}{m_1 + m_2} = \frac{2 \times 10 \times 6}{10 + 6} = 73.5N$$

So, the tension in the thread is 73.5N.

10. Correct option: (C) $\frac{g}{3}$.

$m_1 = 5\text{kg}$

$m_2 = 10\text{kg}$

Acceleration is given by,

$$a = \frac{m_2 - m_1}{m_1 + m_2} \times g$$

$$a = \frac{10-5}{10+5} \times g$$

$$a = \frac{g}{3}$$

So, the acceleration of the system is $\frac{g}{3}$.

11. Correct option: (C) $2\sqrt{3}$**.**

The tension in the given string can be divided into two portions. One component of the tension, directed in the opposite direction as the object's weight, counteracts gravitational force and maintains equilibrium. The magnitude of the other component of tension, which works in the opposite direction to another force represented as T_1, will be equal to T_1.

$T\sin\theta = 2\text{kg wt}$

$T\sin 30^\circ = 2\text{kg wt} \cdots\cdots (1)$

$T\cos\theta = T_1$

First equation gives.

m

So, T_1 will be,

$T = T\cos 30^\circ$

$T = 4\cos 30^\circ$

$T_1 = 2\sqrt{3}\text{kg wt}$

So, the tension T_1 in the horizontal string is $2\sqrt{3}\text{kg wt}$.

12. Correct option: (C) 4 kg and 4 kg

Because spring balances have no mass, the readings of both balances should be equal. If the springs are massless, tension is transmitted uniformly.

As a result, both will have the same reading, i.e. 4 kg and 4 kg.

13. Correct option: (B) 2.5 N.

$M = 5\text{kg}$

$m = 5\text{kg}$

F=6N

Now acceleration is given by as,

$$a = \frac{F}{M+m} = \frac{6}{7+5} = \frac{1}{2}\text{ms}^{-2}$$

Now the force on the lighter is given by,

$F' = ma$

$F' = 5 \times \frac{1}{2} = 2.5\text{N}$

So, the force on the lighter mass is 2.5N.

14. Correct option: (C) $\frac{mF}{M+m}$.

Mass of first block= m

Mass of the second block= M

Spring constant= k

Acceleration is given by,

$a = \frac{F}{M+m}$

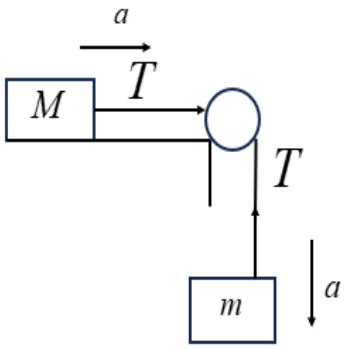
So, the force acting on mass is given by,

$F = ma$

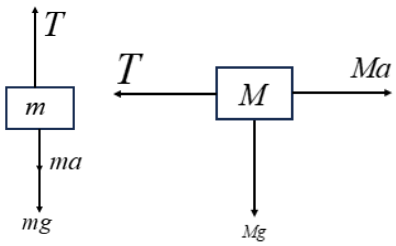
$F = \frac{mF}{M+m}$

So, the force on the block of mass $'m'$ is $\frac{mF}{M+m}$.

15. Correct option: (D) $\frac{Mm}{M+m}g$.



Free body diagram:



We get from the diagram,

$T = ma \cdots\cdots (1)$

$mg - T = ma \cdots\cdots (2)$

From the equation(1) and (2),

$mg - Ma = ma$

$a = \frac{mg}{M+m}$

Now, tension is given by,

$T = Ma$

$T = M\dfrac{mg}{M+m}$

$T = \dfrac{Mm}{M+m}g$

So, the tension in the string is $\frac{Mm}{M+m}g$.

16. Correct option: (B) Velocity.

For a given mass, $p \propto v$. If momentum is constant, then velocity must also be constant.

17. Correct option: (C) Linear momentum.

Because there is no external force acting on the rocket. The force with which the rocket releases the fuel, and the force with which the fuel applies to the rocket (action-reaction pair). Because there is no external force, we may use conservation of momentum because internal forces (fuel burning) do not modify the momentum of the entire system (rocket + fuel).

18. Correct option: (C) Conservation of linear momentum.

The Jet engine operates under the principle of Conservation of Linear Momentum. It generates a significant volume of gases from the combustion of fuel, which are permitted to escape backward through a jet. Because of its great velocity, the rearward rushing gas gains a lot of momentum.

19. Correct option: (A) It still moves with the same velocity.

The force is given by,

$F = ma$

$F = 0$ if the forces are balanced, implying $a = 0$. If there is no acceleration, the velocity remains constant. The solutions are constant velocity trajectories, which implies the aircraft will proceed in a straight line at a constant speed.

20. Correct option: (C) Acts against the rocket and push it up.

Rocket propulsion is based on Newton's third law, which states that every action has an equal and opposite reaction.

21. Correct option: (A) Linear momentum.

Newton's second and third laws result in linear momentum conservation.

22. Correct option: (B) Momentum.

If a powerful force F acts for a brief period of time dt, the impulse imparted I is,

$I = Fdt = \dfrac{dp}{dt}dt$

$I = dp =$ Changein momentum

23. Correct option: (A) Impulse.

In mechanics, tension, trust, and weight are all common forces, whereas impulse is not a force.

Impulse= Force$\times$Time Duration.

24. Correct option: (A) $2MV$.

Initial velocity of body, $u = -V$

where the minus sign signifies that the body is moving towards the wall.

Initial momentum of body, $p_i = Mu = -MV$

Final velocity of body, $v = V$

where a positive sign implies that the body is moving away from the wall.

Final momentum of the body, $p_f = Mv = MV$

Change in momentum, $\Delta p = p_f - p_i = MV - (-MV) = 2MV$

So, the change in momentum is$2MV$.

25. Correct option: (A) 0.5 sec.

Force, $F = 250N$

Momentum of the body, $\Delta p = 125\dfrac{\text{kg-m}}{\text{sec}}$

Change in momentum =Impulse

$\Delta p = F \times \Delta t$

$\Delta t = \dfrac{\Delta p}{F} = \dfrac{125}{250} = 0.5\text{sec}$

So, the time for which the force acts on the body is 0.5 sec.

26. Correct option: (C) 0.3 N-s.

Mass of the body, m = 150kg=150×10^{-3}kg

Acceleration, $a = 20\dfrac{\text{m}}{\text{s}^2}$

$\Delta t = 0.1\text{sec}$

Force is given by as,

$F = ma$

$F = 150 \times 10^{-3} \times 20$

$F = 3N$

Now, impulse is given by,

$\text{Impulse} = F.\Delta t$

$\text{Impulse} = 3 \times 0.1 = 0.3\text{N-s}$

So, the impulsive force is 0.3 N-s.

27. Correct option: (C) $1\dfrac{\text{m}}{\text{s}}$**.**

m_b = 200g=0.2kg

$v_b = 5\dfrac{m}{s}$

m_g = 1kg

The recoil velocity is given by,

$V = \dfrac{m_b v_b}{m_g}$

$V = \dfrac{0.2 \times 5}{1} = 1\dfrac{m}{s}$

So, the gun rebounds backwards with velocity $1\dfrac{\text{m}}{\text{s}}$.

28. Correct option: (C) 1.5×10^{-3}N-s**.**

Given:

Force, $F = 50\text{dynes}$

Time, $t = 3\text{sec}$

Impulse is given by as,

$\text{Impulse=Force} \times \text{Time}$

$\text{Impulse} = 50\times10^{5}\times3 = 1.5\times10^{-3}\text{N-s}$

So, the impulse force is 1.5×10^{-3}N-s.

29. Correct option: (D) $0.25\dfrac{\text{m}}{\text{s}}$**.**

Mass of the tank, $M = 100\text{kg}$

Mass of the body, $m = 0.25\text{kg}$

Velocity of tank, $V = 100\dfrac{\text{m}}{\text{s}}$

We have to find recoil velocity(v) of the tank.

$m \times v = M \times V$

$100 \times v = 0.25 \times 100$

$v = 0.25\dfrac{\text{m}}{\text{s}}$

So, the recoil velocity of the tank is $0.25\dfrac{\text{m}}{\text{s}}$.

30. Correct option: (C) 0.9 N-s.

$F = 600 - 2 \times 10^5 t$

Now force become is zero.

$0 = 600 - 2 \times 10^5 t$

$t = 3 \times 10^{-3}\,\text{sec}$

$t = 3\text{milliseconds}$

The bullet leaves the barrel in 3 milliseconds.

Therefore,

$F.idt = (6t \times 10^5 t)dt$

$J = \left[600t - (2 \times 10^5 \dfrac{t^2}{2})\right]$

$J = \left[600 \times (3 \times 10^{-3}) - (2 \times 10^5 \times \dfrac{(3 \times 10^{-3})^2}{2})\right]$

$J = 0.9\text{N-s}$

So, the average impulse imparted to the bullet is 0.9N-s.

31. Correct option: (C) 24 N.

Mass of the gun, $m = 10\text{kg}$

Mass of the bullet, $M = 20 \times 10^{-3}\text{kg}$

Velocity of the bullet, $v = 300\dfrac{\text{m}}{\text{s}}$.

Number of bullets fired, $n = 4$

Velocity of gun is given by,

Velocity of gun$= -20 \times 10^{-3}\text{kg} \times 4 \times \dfrac{300}{10}$

Velocity of gun$= = -\dfrac{12}{5}\dfrac{\text{m}}{\text{s}}$

Force is given by,

$F = ma = m\left(\dfrac{v-u}{t}\right)$

Mass is 10kg.

Initial velocity, $u = 0, t = 1\text{sec}$

Final velocity$= \dfrac{12}{5}$

Therefore,

Force is given by,

$F = 10 \times \frac{12}{5} = 24\text{N}$

So, the force required to hold the gun while firing is 24N.

LEVEL-II

1 Correct option: (D)

2. Correct option: (D) $T = 300\text{N}, T' = 200\text{N}$ **.**

Given:

Horizontal force, $F = 500\text{N}$

Mass of body A, $m_1 = 10\text{kg}$

Mass of the body B, $m_2 = 15\text{kg}$

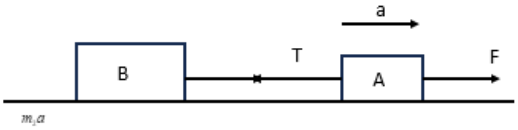
Total mass of the system, $m = m_1 + m_2 = 10 + 15 = 25\text{kg}$

According to newton's second law of motion,

$F = ma$

$a = \frac{F}{m} = \frac{500}{25} = 20\frac{\text{m}}{\text{s}^2}$

When force F is applied to body A, the following happens:

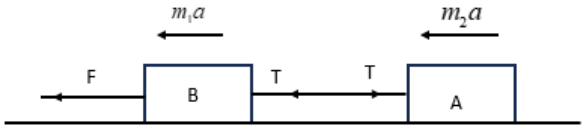


Using equation of motion,

$F - T = m_1 a$

$T = 500 - 10 \times 20 = 300\text{N} \cdots\cdots (1)$

When force F is applied to body B, the following happens:



Again using the equation of motion,

$F - T' = m_2 a$

$T' = F - m_2 a$

$T' = 500 - 15 \times 20 = 200\text{N}$

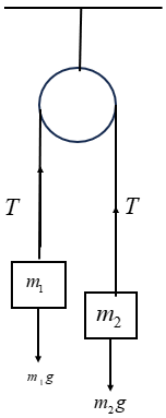
So, T is 300N and $T'$is 200N.

3. Correct option: (D) $0.2\frac{\text{m}}{\text{s}^2}$ **.**

$m_1 = 5\text{kg}$

$m_2 = 4.8\text{kg}$

Free body diagram:



From the free body diagram:

$m_1 g - T = m_1 a \cdots\cdots (1)$

$T - m_2 g = m_2 a \cdots\cdots (2)$

Using equation (1) and (2) we get,

$$a = \frac{m_1 - m_2}{m_1 + m_2} g$$

$$a = \frac{5 - 4.8}{5 + 4.8} \times 9.8$$

$$a = 0.2 \frac{\text{m}}{\text{s}^2}$$

So, the acceleration of the masses when left free to move is $0.2 \frac{\text{m}}{\text{s}^2}$.

4. Correct option: (C) $40 \frac{\text{km}}{\text{hour}}$.

Mass of the wagon, $m = 1000\text{kg}$

Initial velocity of wagon, $v = 50 \frac{\text{km}}{\text{h}}$

The system's initial linear momentum is given by as,

$$p_i = mv = 1000 \times 50 = 5 \times 10^4 \frac{\text{kg-km}}{\text{h}}$$

Let V be the final velocity of the system (waggon + mass dumped).

The system's final linear momentum is given by as,

$p_f = (m + M)V = (1000 + 250)V = 1250V$

According to conservation law of linear momentum,

$p_f = p_i$

$1250V = 5 \times 10^4$

$$V = \frac{5 \times 10^4}{1250} = 40 \frac{\text{km}}{\text{h}}$$

So, the velocity with which it moves is $40 \frac{\text{km}}{\text{h}}$.

5. Correct option: (C) $2.5\frac{\text{m}}{\text{s}}$**.**

Given:
Initial velocity, $u = 0$
Initial mass $= M$
After explosion,

$$m_1 = \frac{M}{4}$$

Velocity along x-direction:

$$v_1 = 3\frac{\text{m}}{\text{s}}i$$

Velocity along y-direction:

$$v_2 = 4\frac{\text{m}}{\text{s}}j$$

$$m_3 = M - \left(\frac{M}{4} + \frac{M}{4}\right) = \frac{M}{2}$$

$$v_3 = V$$

Because there is no external force,
Momentum before explosion=Momentum after explosion

$$M(0) = m_1v_1 + m_2v_2 + m_3v_3$$

$$0 = \frac{M}{4}(3)i + \frac{M}{4}(4)j + \frac{M}{2}(V)$$

$$0 = M\left[\frac{3}{4}i + 1j\right] + \frac{V}{2}$$

$$\frac{V}{2} = -\left[\frac{3}{4}i + 1j\right]$$

$$V = 2\times -\left[\frac{3}{4}i + 1j\right]$$

$$V = -\left[\frac{3}{2}i + 2j\right]$$

$$V = -\frac{3}{2}i - 2j$$

$$|V| = \sqrt{\left(\frac{3}{2}\right)^2 + 2^2} = 2.5\frac{\text{m}}{\text{s}}$$

So, the third piece will be thrown off with a velocity of $2.5\frac{\text{m}}{\text{s}}$.

6. Correct option: (B) 30N.
The ball's final velocity will be zero after catching.
The magnitude of momentum change is given by:

$$\left|\Delta\vec{p}\right| = \left|\overrightarrow{p_f} - \vec{p}_i\right|$$

$$\left|\Delta\vec{p}\right| = \left|m\vec{v} - m\vec{u}\right|$$

$$\left|\Delta\vec{p}\right| = \left|0.15\times 0 - 0.15\times 20\right| = 3\frac{\text{kg-m}}{\text{s}}$$

The average force applied by the ball is given by,

$F_{avg} = \frac{\Delta p}{\Delta t}$

$F_{avg} = \frac{3}{0.1}$

$\left|\vec{F}\right| = 30N$

As a result, the amount of the force produced by the ball on the players' hands is 30N.

7. Correct option: (A) $0.25\frac{\text{kg-m}}{\text{s}^2}$**.**

Mass of the object, $m = 0.05\text{kg}$

Velocity before the Collison, $V_1 = 5\frac{\text{m}}{\text{s}}$

Velocity after the collision is V_2.

We know that the net momentum of the item before and after the impact must be equal in order to meet the momentum conservation equation.

$p_1 = p_2 = \text{mass}\times\text{velocity} = 0.05\times 5 = 0.25\frac{\text{kg-m}}{\text{s}}$

We can now calculate the overall angle using the illustration.

$\theta = 30^\circ + 30^\circ = 60^\circ$

Using the cosine law, the magnitude of change in linear momentum is given by,

$\Delta p = \sqrt{p_1 + p_2 - 2p_1p_2\cos 60^\circ}$

$\Delta p = \sqrt{(0.25)^2 + (0.25)^2 - 2(0.25)^2\cos 60^\circ}$

$\Delta p = 0.25\frac{\text{kg-m}}{\text{s}^2}$

So, the change in momentum of the body is $0.25\frac{\text{kg-m}}{\text{s}^2}$.

8. Correct option: (A) mg.

Mass of the small block $= m$

Mass of the big block$= = M$

Force, $F = 3mg$

Angle, $\theta = 30^\circ$

Contact force is given by,

$F = \sqrt{(F_F)^2 + (F_N)^2} \cdots\cdots (1)$

Where, F_F = frictional force $\Rightarrow F_F = \mu mg \cdots\cdots (2)$

And F_N = Net force $\Rightarrow F_N = mg \cdots\cdots (3)$

Also, coefficient of friction $(\mu = 0)$ is zero because of smooth surface.

$F = \sqrt{(\mu mg)^2 + (mg)^2}$

$F = \sqrt{(mg)^2}$

$H = mg$

So, the contact force between those two blocks is mg.

9. Correct option: (D) 2ms^{-2}

Given:

$m_1 = 1\text{kg}$

$m_2 = 2\text{kg}$

$m_3 = 3\text{kg}$

The acceleration is given by,

$$a = \frac{F}{m_1 + m_2 + m_3}$$

$$a = \frac{12}{3+2+1} = 2\frac{\text{m}}{\text{s}^2}$$

So, he acceleration of the system is 2ms^{-2}.

10. Correct option: (C) $a_1 > a_2 > a_3$**.**

In case A:

Let be the string tension T_A and g be the acceleration due to gravity.

So, the using the equation of motion,

$T_A - mg = ma_1$

From the figure, the tension of on mass $2mg$ due to the downward acceleration due to gravity.

$2mg - mg = ma_1$

$a_1 = g$

In case: B

Let be the string tension T_B and g be the acceleration due to gravity.

So, the using the equation of motion,

$T_C - mg = (m+m)a_2$

From the figure, the tension of on mass $2mg$ due to the downward acceleration due to gravity.

$2mg = -mg = 2ma_2$

$$a_2 = \frac{g}{2}$$

In Case: C

Let be the string tension T_C and g be the acceleration due to gravity.

So, the using the equation of motion,

$T_B - mg = (m+2m)a_3$

From the figure, the tension of on mass $2mg$ due to the downward acceleration due to gravity.

$2mg - mg = 3ma_3$

$$a_3 = \frac{g}{3}$$

Therefore, $a_1 > a_2 > a_3$.

11 Correct option: (B) and (C)

12. Correct options: (A) and (B).

The momentum is given by,

$\vec{P} = m \times \vec{v}$

As a result, linear momentum is a measure of the amount of motion contained by the body. The force's impulse is the change in momentum caused by the force.As a result, Impulse reacts in the same direction as a change in momentum.

Acceleration is given by,

$\vec{a} = \frac{\vec{F}}{m} = \frac{1}{m}\frac{d\vec{p}}{dt}$

As a result, acceleration acts in the same direction as impulse.

Uniform circular motion has zero angular acceleration and so constant angular momentum. When an external force is present, linear momentum is not preserved.

13 Correct option: (B)

14 Correct option: (A)

15. Correct option: (B) $N = \frac{m_1 F_2 + m_2 F_1}{m_1 + m_2}$ **.**

The acceleration is given by,

$a = \frac{F_1 - F_2}{m_1 + m_2}$

$F_1 - N = m_1\left(\frac{F_1 - F_2}{m_1 + m_2}\right)$

$N = \frac{m_2 k_1 + m_1 k_2}{m_1 + m_{\partial 2}}$

16 Correct option: (D)

17 Correct option: (A)

18 Correct option: (D)

19 Correct Matching is: (a-R, b-S, c-P, d-Q)

20. The correct matching are (a)-(T), (b)-(P),(R),(c)-(S),(P),(d)-(Q).

(a) The conservation of momentum principle asserts that the amount of momentum inside a problem domain remains constant; momentum is not created nor destroyed, but only modified by the action of forces as defined by Newton's equations of motion.

(b) Allow a force F to act on a body of mass' m ' for a limited time' t '.

Impulse force is given by:

Impulse force = Ft

Therefore,

$\Rightarrow mat = m\left(\frac{v-u}{t}\right)t = m(v-u)$

As a result, the impulse of force equals the change in momentum.

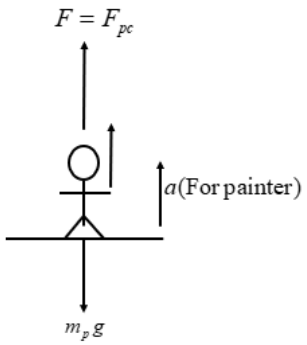
We can also write impulse force as,

$\text{Impluse} = \int F.dt$

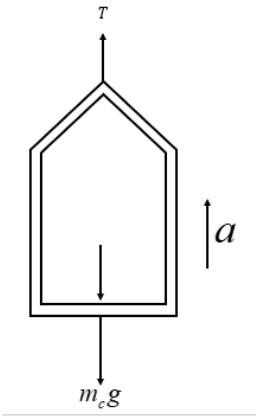
(c) The impulse (or change in momentum) exerted on the body is represented by the area under the force-time graph.

(d) The third law states that in nature, for every action (force), there is an equal and opposite response. When object A applies a force to object B, object B applies an equal and opposite force to object A. In other words, forces are the product of interactions.
So, The correct matching are (a)-(T), (b)-(P),(R),(c)-(S),(P),(d)-(Q).

21. Correct Answer is: 2



Free body diagram of painter



Free body diagram of crat

From the free body diagram:

$T + F - m_p g = m_p a \cdots\cdots (1)$

For crate:

$T - F - m_c g = m_c a \cdots\cdots (2)$

Subtract equation (2) from (2), we get,

$$a = \frac{F_1 - F_2}{m_1 + m_2}$$

$$F_1 - N = m_1 \left(\frac{F_1 - F_2}{m_1 + m_2}\right)$$

$$N = \frac{m_1 F_2 + m_2 F_1}{m_1 + m_2}$$

$$N = \frac{m_2 + m_1 k_2}{m_1 + m_{\partial 2}}$$

So, the acceleration of the painter is $2 ms^{-2}$.

Apparent Weight

LEVEL-I

1. Correct option: (A) Inertial frame of reference.

Newton's laws of motion apply in an inertial frame of reference. As a result, option (A) is right. Inertial frames of reference are also known as unaccelerated frames of reference, Newtonian frames of reference, or Galilean frames of reference.

2. Correct option: (A) Inertial.

An inertial frame is a vehicle that moves at a constant speed over a straight route. Newton's rules of motion are only valid in an inertial frame.

3. Correct option: (B) Non-inertial.

Because the Earth revolves both around the sun and on its own axis, it cannot be an inertial frame of reference. An accelerating object cannot be used as an inertial frame of reference, nor can a reference frame coupled to a non-inertial frame be used as an inertial frame of reference.

4. Correct option: (B) Non-inertial.

Unaccelerated frames are referred to as inertial frames, while accelerated frames are referred to as non-inertial frames. Because the earth moves in a circular motion around its axis, it must experience acceleration, making it a non-inertial frame of reference.

5. Correct option: (B) Pseudo.

According to Newton's second equation of motion, the force acting on an object is equal to its mass multiplied by its acceleration ($F = ma$). This law is written in the context of inertial frames, which are non-accelerating reference frames.

However, in real-world events, we frequently meet situations in which we are perceiving motion from an accelerating frame of reference. This type of frame is known as a non-inertial frame. Because Newton's second law was initially formulated for inertial frames, it does not directly apply to non-inertial frames.

We offer the concept of a pseudo force or fictitious force to make Newton's second law applicable in a non-inertial frame. Pseudo forces are added to account for the effects perceived in a non-inertial frame due to its acceleration, rather than being caused by physical interactions.

6. Correct option: (B) Cannot be an inertial frame because earth is revolving round the sun.

Because the Earth revolves both around the sun and on its own axis, it cannot be an inertial frame of reference. An accelerating object cannot be used as an inertial frame of reference, nor can a reference frame coupled to a non-inertial frame be used as an inertial frame of reference.

7. Correct option: (C) Remains same.

Weight, $F = mg$

Because of mass conservation, the net force in the downward direction remains constant.

8. Correct option: (A) First increases then decreases to zero.

The man can leap by putting force on the machine, and in return, he is pushed up by the machine's equal and opposite force, so the force exerted by man to jump adds up with the original force, causing his weight reading to rise and then fall.

9. Correct option: (A)

Gravity always accelerates in the downward direction.

When the lift is travelling downward with constant acceleration, that is, when the resultant acceleration is downward the,

$W = m(g - a)$

is the final weight, which is less than the initial weight mg. As a result, the individual feels weightless.

10. Correct option: (B) $3g$ **.**

The tension in the cord is given by,

$T = mg + ma$

$4mg = mg + ma$

$g = \frac{a}{3}$

$a = 3g$

The maximum acceleration with which the sphere can move up without breaking the cord is $3g$.

11. Correct option: (B) $R = mg + ma$ **.**

For non-inertial frame of reference:

$a = 0$

In addition, for the equation of motion, we must make it inertial.

This requires us to accelerate in the opposite direction.

Therefore, the equation of motion is,

$R = mg + ma$

When the lift goes upward, the body inside the lift is pushed downward, causing the corresponding R to act upward $= ma$. In addition, typical weight = mg. As a result, total $R = mg + ma$.

12. Correct option: (D) Accelerates upward with an acceleration equal to $4\frac{\text{m}}{\text{sec}}$**.**

A boy weighing 40 kilogrammes is standing in a lift. When the lift climbs upwards at a rate of $4\frac{\text{m}}{\text{sec}}$, the force felt by the boy's feet increases.

13. Correct option: (B) $\frac{g}{3}$**.**

When the lift is stationary, the weight of the man is, $W_0 = mg$

When the lift is travelling downward with acceleration a, the man's weight is,

$W = mg - ma$

$\frac{W_0}{W} = \frac{3}{2}$

$\frac{mg}{mg - ma} = \frac{3}{2}$

$2g = 3g - 3a$

$a = \frac{g}{3}$

So, the value of 'a' is $\frac{g}{3}$.

14. Correct option: (C) First increase and then decrease.

The acceleration due to gravity reduces as the individual ascends above the earth's surface. As a result, the reading on the spring balance will continue to fall.

The apparent weight of the body increases first due to upward acceleration, but eventually reduces due to gravity.

15. Correct option: (D) 0.

The tension equation derived from force balance.

$T = m(g - a)$

If the hoist rope snaps suddenly, the acceleration equals g, causing tension.

$T = m(g - g) = 0$

So, the tension exerted by the surface of lift is 0.

16. Correct option: B) 24 N.

When the lift is stationary, the spring force balances the weight as follows:

$kx = mg = 49\text{N}$

$\text{k} = \frac{49}{x} \cdots\cdots (1)$

And,

$mg = 49$

$m = \frac{49}{9.8} = 5\text{kg}$

When the lift goes downward at a rate of $5\frac{\text{m}}{\text{s}^2}$, we get:

$kx_2 = mg - 5 \times m$

$kx_2 = 49 - 5 \times 5 = 24N$

So there is a new reading in the spring balance. Which is 24N.

17. Correct option: (C) 10800N.

Mass of the lift, $m = 1000\text{kg}$

Acceleration, $a = 1\frac{m}{s^2}$.

$g = 9.8\frac{\text{m}}{\text{s}^2}$

The tension in a string is given as,

$T = mg + ma$

$T = 1000 \times 9.8 + 1000 \times 1$

$T = 10800\text{N}$

So, the tension built up in the string is 10800N.

18. Correct option: (C) 1200N.

Mas of man, $m = 80\text{kg}$

Acceleration of the lift, $a = 5\frac{\text{m}}{\text{s}^2}$

Acceleration due to gravity, $g = 10\frac{\text{m}}{\text{s}^2}$.

Normal reaction acting on the man is given by as,

$N = m(a + g)$

$N = 80(5 + 10) = 1200\text{N}$

So, the reading on the scale is 1200N.

19 The correct option (C) $2.5\frac{\text{m}}{\text{s}^2}$**.**

Mass of the monkey, $m = 20\text{kg}$

Let the monkey's acceleration be a.

Tension is given by,

$T = 25 \times 10 = 250\text{N}$

Using the equation of motion,

$ma = T - mg$

$T = m(a + g)$

$25 = 20(a + 10)$

$a = 2.5\frac{\text{m}}{\text{s}^2}$

So, the maximum acceleration with which the monkey can climb up along the rope is $2.5\frac{m}{s^2}$.

20. Correct option: (B) Mv**.**

Velocity of blood pumping= $v\frac{\text{m}}{\text{s}}$ =constant

Blood circulation rate= $\frac{dM}{dt} = M\frac{\text{kg}}{\text{sec}}$

Using the newton' second law of motion:

$\vec{F} = \frac{d\vec{p}}{dt}$

$F = \frac{dp}{dt} = \frac{d}{dt}(M.v) = v.\frac{dm}{dt} = v.M\text{N}$

F=MvN

So, The required force is Mv.

21. Correct option: (D) 11000N.

The string tension is given as follows.

$T = m(g + a)$

We obtain by replacing the values of mass and acceleration given in the question.

$T = 1000 \times (10 - 1)$

$T = 11000\text{N}$

So, the tension in the supporting cable is 11000N.

22. Correct option: A) 55 kg.

When the lift moves up at a steady speed, there is no change in acceleration, hence the weight remains constant.

23. Correct option: (D) $\frac{mg}{\cos\theta}$**.**

Because the block and wedge have no relative motion:

$ma\cos\theta = mg\sin\theta$

$\Rightarrow a = g\tan\theta$

The normal force exerted by the wedge on the block equals

$N = mg\cos\theta + ma\sin\theta$

$N = mg\cos\theta + m(g\tan\theta)\sin\theta$

$N = mg\cos\theta + mg\frac{\sin^2\theta}{\cos\theta}$

$N = mg\left(\frac{\cos^2\theta + \sin^2\theta}{\cos\theta}\right)$

$N = \frac{mg}{\cos\theta}$

So, the force exerted by the wedge on the block is $\frac{mg}{\cos\theta}$.

24. Correct option: (B) $4\frac{\text{m}}{\text{s}^2}$ **upwards.**

The gravitational force is 20000 N in the downward direction, while the tension force is 28000 N in the upward direction.

Using newton's second law of motion,

$\sum F = ma$

$T - mg = ma$

$a = \frac{T - mg}{m}$

$a = \frac{28000 - 2000 \times 9.8}{2000} = 4\frac{m}{s^2}$

So, the acceleration is $4\frac{\text{m}}{\text{s}^2}$ upwards.

25. Correct option: (C) 12000 N.

The lift is travelling upward with acceleration during the 11th second.

Acceleration is given by,

$a = \frac{0 - 36}{2} = -1.8\frac{m}{s^2}$

Tension in rope is given by,

$T = m(g - a)$

$T = 1500(9.8 - 1.8) = 12000\text{N}$

The tension in the rope pulling the lift at $t = 11\text{th}\,\text{sec}$ will be 12000N.

26. Correct option: (D) 36 metres.

We already know that the area beneath the velocity-time graph equals total displacement.

The displacement here is the vertical height at which he lifts the passengers.

So, the area of graph is given by,

$A = \left(\frac{1}{2} \times 2 \times 3.6\right) + (8 \times 3.6) + \left(\frac{1}{2} \times 2 \times 3.6\right)$

$A = 7.2 + 28.8$

$A = 36\text{m}$

So, the height to which the lift takes the passenger is 36 metres.

27. Correct option: (B) 25 N.

Body acceleration down the rough inclined plane $= g\sin\theta$

spring balance force applied is given by,

$F = mg\,sin\,\theta$

$F = 5 \times 10 \times sin\,90^\circ$

$F = 25N$

So, the spring balance measure is 25N.

28. Correct option: (C) 26 N.

Because the cage is closed, air provides the force, but the force is eventually delivered as normal force because air particles apply normal force to the cage's wall. As a result, the normal force necessary to create an acceleration of $2\frac{m}{s^2}$ is,

$F = ma = 1N$

As a result, the total weight registered by the spring is normally transferred to it.i.e, 25+1=26N.

Because the acceleration is in the upward direction, the normal reaction required is in the upward direction, and the equal and opposite normal will be put on the cage's base.

So, the spring balance will now record a weight of 26N.

LEVEL-II

1. Correct option: (D) 90000 N.

If the tension is T, the total upward force is, $T - \text{Weight} = T - 6000 \times g$.

Because weight is downward and we are computing net upward force, we subtracted the weight.
Newton's Second Law states that:
Force= Mass $\times$Acceleration
Acceleration, $a = 5\text{ms}^2$
Therefore,
$T - 6000g = 6000 \times 5$
$T = 30000 + 6000 \times 10$
$T = 90000\text{N}$
So, the tension in the cable is 90000N.

2. Correct option: (A) $t_1 \geq t_2$**.**

Time t_1 for the stationary lift= $\sqrt{\frac{2h}{g}}$

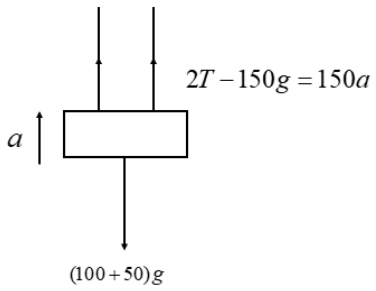
When lift is increasing at a steady rate, then

$t_2 = \sqrt{\frac{2h}{g+a}}$

Thus, $t_1 > t_2$.

3. Correct option: (B) 1125 N.

Let T denote string tension and a denote the acceleration of the (man + crate) system.
Total system mass (man + crate) $= (100+50)\text{g}=150\text{g}$
The following is a free body diagram of the entire system (man + crate):



From the free body diagram,
$2T = 150(a+g) = 150(5+10)$
$2T = 2250$
$T = 1125\text{N}$
So, the tension in the rope is 1125N.

4. Correct option: (C) $(M+m)g\tan\theta$**.**

a is acceleration in the horizontal direction of (wedge+block).
Acceleration is given by as,

$$a = \frac{F}{M+m}$$

Now, by analysing the block in relation to the wedge, which is a non-inertial frame of reference, we will apply fake force to the left of the block.
The forces along the incline should balance out because the block is stationary in relation to the wedge.
Write the equation of motion along the incline now.:

$\Rightarrow N_1 \cos 60^\circ = N_2 \sin 30^\circ$

$\sqrt{3}N_1 = N_2$

$\Rightarrow N_1 \cos 60^\circ + N_2 \cos 30^\circ = mg = 100$

$\frac{N_1}{2} + \sqrt{3}N_1 \times \frac{\sqrt{3}}{2} = 100$

$N_1 = 50\text{N}, N_2 = 50\sqrt{3}N$

So, the force is $(M+m)g\tan\theta$.

5. Correct option: (A) 240 N.

Acceleration of the monkey, $a = 4\frac{m}{s^2}$

We can construct the equation of motion using Newton's second law of motion as:

$mg - T = ma$

$T = m(g-a)$

$T = 40(10-4)$

$T = 240N$

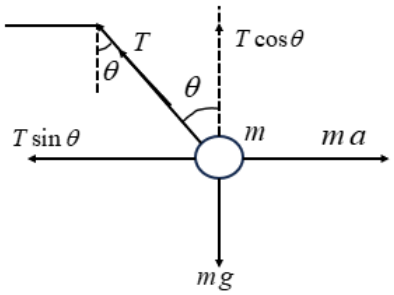
Since, $T < T_{\max}$ In this situation, the rope will not snap.

So, the tension in the rope is 240N.

6. Correct option: (C) Going down with constant acceleration.

When the apple is dropped, it is in free fall, which means that gravity is operating on it. According to the guy inside the elevator, the apple does not appear to be dropping. As a result, the man and the lift must likewise be descending with the action of gravity, i.e., a constant acceleration.

7. Correct option: (C) $\theta = tan^{-1}\left(\frac{a}{g}\right)$ in the direction of motion.



From the free diagram:

$T\sin\theta = ma \cdots\cdots (1)$

$T\cos\theta = mg \cdots\cdots (2)$

Using the equation (1) and (2), we get,

$$\frac{T\sin\theta}{T\cos\theta} = \frac{ma}{mg}$$

$$\tan\theta = \frac{a}{g}$$

$$\theta = \tan^{-1}\left(\frac{a}{g}\right)$$

So, the inclination of the string to the vertical is $\theta = \tan^{-1}\left(\frac{a}{g}\right)$ in the direction of motion.

8. Correct option: (A) (C)

9. Correct option: (ABD)

10. Correct options: (B) and (D).

Because the Earth revolves both around the sun and on its own axis, it cannot be an inertial frame of reference. An accelerating object cannot be used as an inertial frame of reference, nor can a reference frame coupled to a non-inertial frame be used as an inertial frame of reference.

11. Correct option: (A) 720 N.

$m = 60\text{kg}$

$a = 2\dfrac{\text{m}}{\text{s}^2}$

$g = 10\dfrac{\text{m}}{\text{s}^2}$

When the lift accelerates higher, the apparent weight of the individual in the lift is

$R = m(g + a)$

$R = 600(10 + 2) = 720\text{N}$

So, the weight of man when an elevator moves up is 720N.

12. Correct option: (B) 480 N.

$m = 60\text{kg}$

$a = 2\dfrac{\text{m}}{\text{s}^2}$

$g = 10\dfrac{\text{m}}{\text{s}^2}$

When the lift accelerates downward, the apparent weight of the man inside the lift is,

$R = m(g - a)$

$R = 60(10 - 2)$

$R = 480N$

So, the apparent weight of man when an elevator moves up with a retardation$2\frac{m}{s^2}$ is 480N.

13. Correct option: (A) Decreases by 20%.

$m = 60\text{kg}$

$a = 2\dfrac{\text{m}}{\text{s}^2}$

$g = 10\dfrac{\text{m}}{\text{s}^2}$

When the lift travels downwards:

$w_2 = m(g - a)$

$w_2 = 60(10 - 2)$

$w_2 = 480\text{N}$

Again,

$w_1 = mg$

$w_1 = 60 \times 10$

$w_1 = 600\text{N}$

Change the percentage of apparent weight now,

$$\%w = \frac{w_1 - w_2}{w_1} \times 100$$

$$\%w = \frac{600 - 480}{600} \times 100$$
$$\%w = 20\%$$

So, the charge percentage of apparent weight of a man is 20% Decrease.

14. Correct option: (B)

15. Correct option: (C)

16. Correct option: (B)

17. Correct option: (B) $g - a$ **.**

Lift acceleration in relation to the ground, $a_{\lg} = a$

Acceleration of ball with respect to ground, $a_{bg} = g$

Acceleration of the ball in relation to the person standing in the lift,

$a_{bl} = a_{bg} - a_{\lg}$

$a_{bl} = g - a$

So, the acceleration of the ball as observed by the man in the lift $g - a$.

18. Correct option: (A) g **.**

Lift acceleration in relation to the ground, $a_{\lg} = a$

Acceleration of ball with respect to ground, $a_{bg} = g$

So, the acceleration of the ball as observed by the man standing stationary on the Ground is g.

19. Correct Matching is: (a)-(Q), (b)-(R),(c)-(P),(d)-(S).

(a) Lift moves up with uniform acceleration a:

We can construct the equation of motion using Newton's second law of motion as:

$R - mg = ma$

$R = m(g + a)$

(b) b) Lift moves down with uniform acceleration a:

We can construct the equation of motion using Newton's second law of motion as:

$R + mg = ma$

$R = m(g - a)$

(c) Lift moves down with uniform velocity:

As the lift is down with uniform velocity, acceleration, $a = 0$

$R - mg = ma$

$R = m(g - 0)$

$R = mg$

(d) Freely falling lift :

When the lift is free to move under gravity, acceleration occurs. i.e. $a = g$

We can construct the equation of motion using Newton's second law of motion as:

$R - mg = ma$

$R = m(g - g)$

$R = 0$

The correct matching is (a)-(Q), (b)-(R),(c)-(P),(d)-(S).

20. Correct Matching is: (a)-Q), (b)-(P),(c)-(R),(d)-(P).

$m = 50\,\text{kg}$

$a = 1.2\,\frac{\text{m}}{\text{s}^2}$

(a) Goes up with acceleration $1.2\,\frac{\text{m}}{\text{s}^2}$:

Tension in string is given by:

$T = m(g + a)$

$T = \frac{50}{100}(9.8 + 1.2)$

$T = 0.55\text{N}$

(b) Goes down with deceleration $1.2\,\frac{\text{m}}{\text{s}^2}$:

Tension in string is given by:

$T = m(g - a)$

$T = \frac{50}{100}(9.8 - 1.2)$

$T = 0.43\text{N}$

(c) Goes up with uniform velocity:

$T = m(g)$

$T = \frac{50}{100}(9.8)$

$T = 0.49\text{N}$

(d) Goes down with an acceleration $1.2\,\frac{\text{m}}{\text{s}^2}$:

$T = m(g - a)$

$T = \frac{50}{100}(9.8 - 1.2)$

$T = 0.43\text{N}$

So, the correct matching is (a)-Q), (b)-(P),(c)-(R),(d)-(P).

21. Correct Answer is:5

$v_1 = 20\text{ms}^{-1}$

$m_1 = 10\text{kg}$

$\text{v}_2 = 0; m_2 = 6\text{kg}$

$\text{v}_3 = v; m_3 = 4\text{kg}$

Now using the conservation of momentum:

$m_1 v_1 = m_2 v_2 + m_3 v_3$

$10 \times 20 = 6 \times 0 + 4 \times v$

$4v = 200$

$v = 50\,\frac{\text{m}}{\text{s}}$

So, the velocity is $50\,\frac{\text{m}}{\text{s}}$ or $5 \times 10\,\frac{\text{m}}{\text{s}}$.

4. Work, Energy and Power

Solutions

LEVEL-I

1. Correct option: (D) Joule.

$F = 1\text{N}$

$S = 1\text{m}$

$\theta = 0^\circ$

Work done is given by as,

$W = F \times S \times \cos\theta$

$W = 1 \times 1 \times \cos 0^\circ$

$W = 1\text{joule}$

When a force of one Newton really moves a body a distance of one metre in the direction of the applied force, the work is said to be one joule.

2. Correct option: (D) All of the above.

When a force is applied to an item and the object is moved in the force's direction, work has been done. Let's analyse the following activities in light of this definition:

(A) A horse pulling a cart: In this case, work is being done. Work is accomplished as the cart moves in the direction of the force applied by the horse.

(B) A train being pulled by an engine: In this case, work is also being done. The train moves because of the force the machine applies to it. Work is being done as long as there is movement in the force's direction.

(C) A person lifting up a stone: Work is also being done here. In order to elevate the stone against gravity, the user exerts an upward push. Work is completed as the rock rises.

(D) All the above: This choice is the best one. Work is being done in all of the indicated activities (person lifting stone, engine hauling train, horse pulling wagon). In each instance, a force is applied, and there is movement in the force's direction, resulting in the achievement of labour as it is understood by science.

3. Correct option: (C) Force causes displacement of the body in its opposite direction.

When a force works in the opposite direction to the displacement, the work done is negative.

4. Correct option: (A) 294 J.

Given:

$m = 15\text{kg}$

$g = 9.8\frac{\text{m}}{\text{s}^2}$

$s = 2\text{m}$

Force is given by as,

Force = weight of box = $mg = 15 \times 9.8 = 147\text{N}$

Now work done is given by as,

$W = 147 \times 2 = 294\text{N}$

So, the work done by the coolie is 294 N.

5. Correct option: (B)1:4.

The velocity of the first body, $v_1 = v$

The velocity of the second body, $v_2 = 2v$

Kinetic energy is given by,

$$KE = \frac{mv^2}{2}$$

So, the kinetic energy is proportional to the square of the velocity.

$KE \propto v^2$

So, the ratio of the kinetic energy is given by,

$$\frac{KE_1}{KE_2} = \frac{v_1^2}{v_2^2} = \frac{v^2}{2v^2} = \frac{1}{4}$$

So, the ratio of the kinetic energies is 1:4.

6. Correct option: (C) F.S.

The work done is given by,

$W = F.S$.

7. Correct option: (D) Both (A) and (C).

The work that a force performs is unaffected by the body's mass (A) or initial velocity (C). The formula Work = Force Displacement $\times \cos\theta$, where is the angle between the force and displacement vectors, states that work is determined by the force applied and the displacement in the direction of the force. In this work equation, the mass and beginning velocity play no direct role.

8. Correct option: (C) Work.

The SI unit of energy and work is the joule (J). It shows how much work is done when one newton of force is applied over a one-meter displacement in the force's direction.

9. Correct option: (A) 2×10^3 J **.**

The momentum, $p = 20\text{kgms}^{-1}$

Mass of the body, $m = 0.1\text{kg}$

Kinetic energy is given by as,

$$K = \frac{1}{2}mv^2 \cdots\cdots (1)$$

Momentum is given by,

$m^{\circ}\ p = mv \cdots\cdots (2)$

From equation (1) and (2),

$$K = \frac{1}{2}m \times \frac{p^2}{m} = \frac{p^2}{2m}$$

$$K = \frac{(20)^2}{2\times 0.1} = 2000 = 2\times10^3\text{J}$$

So, the kinetic energy is 2×10^3 J .

10. Correct option: (B) Weight of the box.

The amount of work required to lift a box off the ground and keep it there depends on the box weight because using conservative force doesn't depend on the path followed.

Additionally, only conservative forces are used for lifting. Therefore, it relies on the starting and ending positions, as well as the force needed, which must be equivalent to the box weight.

11. Correct option: (C) It experiences an increase in energy through a mechanical influence.

To move a body, a force is applied. It travels a specific distance.

It gains more energy as a result of a mechanical influence.

Therefore, whenever a body experiences a rise in energy due to a mechanical impact, work is always being done on it.

12. Correct option: (B) Work done.

Work is measured as the area under the force-displacement curve.

It is the distance that the force's application caused the object to move (s). In this instance, displacement (s) is shown on the x-axis while force (f) is shown on the y-axis.

Work completed is a scalar quantity (w).

13. Correct option: (B) Velocity should be doubled.

Let the initial mass, $m_1 = m$

Initial the velocity of the body is,= v_1

The final maas of the body is, $m_2 = 4m$

Final Velocity of the body= v_2

Ao, the kinetic energy is same in both cases.

$$\frac{1}{2}m_1v_1^2 = \frac{1}{2}m_2v_2^2$$

$$\frac{1}{2}mv^2 = \frac{1}{2}\times 4m\times v_2^2$$

$$v_2^2 = \frac{1}{4}v^2$$

$$v_2 = \frac{1}{2}v$$

Therefore, if a body's mass is increased by four times, its velocity must be reduced by half in order to maintain a constant amount of kinetic energy.

14. Correct option: (B) 40 J.

Given:

$m = 5\text{kg}$

$F = 2\text{N}$

$t = 10\text{s}$

Initial velocity of the body is, $u = 0$

Acceleration body is given by as,

$$a = \frac{F}{m} = \frac{2}{5} = 0.4\text{ms}^{-2}$$

Now using the equation of motion,

$$s = ut + \frac{1}{2}at^2$$

$$S = 0 + \frac{1}{2}\times 0.4\times (10)^2$$

$$S = 0.2\times 100$$

$$S = 20m$$

Work done is given by,

$$W = F.s$$

$$W = 2\times 20$$

$$W = 40J$$

So, the work done is 40J.

15. Correct option: (D)

16. Correct option: (A) Positive.

If the force and displacement are applied in the same direction, then the work is positive. When force and displacement are perpendicular, no work is done. If something isn't moving,

then it must be. The work performed is negative if the item is travelling obliquely to the direction of the applied force.

17. Correct option: (B) Negative.

When force and displacement are parallel, there is no work done. When something isn't moving, it doesn't move. The work done is negative if the object is travelling in the opposite direction from the direction of the applied force.

18. Correct option: (C) $\left|\vec{F}\right|\left|\vec{S}\right|\cos\theta$.

Work is calculated as the product of force and displacement along the force's direction.

Work done = $\overrightarrow{Froce}.\vec{s}$

Work done= $Fs\cos\theta$

As a result, when force F acts on a body, the body is also moved a distance s in the horizontal direction, creating an angle between force F and the horizontal direction.

19. Correct option: (A) 12000J.

$F = 750\text{N}$

$s = 16\text{m}$

Work done is given by,

$W = F \times s$

$W = 750 \times 16 = 12000\text{J}$

So, the work done is 12000J.

20. Correct option: (D) 5000J.

$m = 200\text{Kgf}$

$g = 10\text{ms}^{-2}$

$\theta = 180^{\circ}$

$s = 2.5\text{m}$

We know that force is given by as,

$F = m \times g$

$F = 200 \times 10 = 2000\text{N}$

Now, work done is given by,

$W = F \times s$

$W = 2000 \times 2.5 = 5000\text{J}$

So. Correct option: 5000J.

21. Correct option:(C) 9.8J.

1kg.m=1kgf×1m=9.8N×1m=9.8J

22. Correct option: (C) No work is done on it.

As a result, when force F acts on a body, the body is also moved a distance s in the horizontal direction, creating an angle between force F and the horizontal direction.

23. Correct option: (C) Both A and B.

The units of work done are Joule and Nm.

Joules (J) are units of measure for work. Newtons (N) are the units of force. Metres (m) are used to quantify the distance travelled along the force's line of action.

$$W = F(N).s(m)$$

$$W = \text{Nm}$$

So, the unit of work done is Nm.

24. Correct option: (B) mgh.

Mass of the body is m.

Force by the gravity= Weight of mass= mg

Displacement of the body is h.
So, the work done is given by,
$\text{Wrok done}=\text{Force}\times\text{Displacement}$
$\text{Wrok done} = mg \times h$
$\text{Wrok done} = mgh$
So, the work done by the gravity is mgh.

25. Correct option: (D) 10^7**.**

$1\text{erg.}=10^7\,\text{joule}$
So, the number of ergs contained in one Joule is 10^7.

26. Correct option: (A) mgh**.**

Mass of the body is m.
Force by the gravity= Weight of mass= mg
Displacement of the body is h.
So, the work done is given by,
$\text{Wrok done}=\text{Force}\times\text{Displacement}$
$\text{Wrok done} = mg \times h$
$\text{Wrok done} = mgh$
So, the work done by force of the gravity is mgh.

27. Correct option: (C) Zero.

Work done is given by as,
$\text{Wrok done}=\text{Force}\times\text{Displacement}$
$\text{Wrok done} = F \times s$
As man is unable to move the wall, the wall's displacement is zero. [Applied force is not zero.
So, the work done is zero.

28. Correct option: (B) Greater than zero.

The normal reaction force is in the upward direction, but the person climbing the stairs is applying force in the downward direction. The displacement is also upward in this instance.
$v^2 = u^2 + 2as$
$(6)^2 = 0 + 2 \times 0.1$ As a result, the normal reaction and displacement are both in the same
$s = 180\text{m}$
direction, indicating that the force is exerting positive force.
whereas ascending the stairs, the person applies force in the downward direction, whereas the natural reaction force is in the upward direction. In this instance, the displacement is also upward. So long as the normal reaction and displacement are in the same direction, the force is doing positive work.

29. Correct option: (B) 5000J.

Given:
Mass, $m = 200\text{kgf}$
Height, $h = 2.5\text{m}$
Time, $t = 5\text{sec}$
Force is given by,
$F = mg$
$F = 200 \times 10 = 2000\text{N}$
Now work done is given by,

$W = F \times h$

$W = 2000 \times 2.5 = 5000\text{J}$

So, the work done is 5000J.

30. Correct option: (A) 12500 J.

Density of the water is $1000\frac{\text{kg}}{\text{m}^3}$.

Volume, $\rho = \frac{50}{1000}\text{m}^3$

The mass is given by,

$\text{Mass}(m) = \text{density} \times \text{volume}$

$m = 1000 \times \frac{50}{1000}$

$m = 50\text{kg}$

Work done is given by,

$\text{Wrok done(w)} = \text{Force} \times \text{Displacement}$

$w = F \times s$

$w = 500 \times 25 = 12500\text{J}$

So, the work done is 12500 J.

31. Correct option: (C) Both A & B.

(A) $1\text{Joule} = 1\text{N} \times 1\text{m}$

The energy that is transmitted to an object when a force of one Newton acts on it in the direction of the force's motion across a distance of one metre is equal to one joule.

(B) $1\text{N} = 10^7\text{erg}$

32. Correct option: (B) Negative.

A fielder provides a force that is opposite to the displacement of the ball when he catches it. As a result, the work is negative.

The angle between the force and the displacement is 180°.

$\text{Wrok done(w)} = \text{Force} \times \text{Displacement}$

$w = Fs\cos\theta$

$w = F.s\cos 180^\circ$

$w = -F.s$

So, the work done is negative.

33. Correct option: (C) Zero.

Since there is no displacement in the direction of the force of gravity or in the opposite direction, there is no work done against gravity.

The force of gravity acts on the coolie when he is carrying a burden on his head.

As a result, the force acts in a vertical direction while the displacement acts in a horizontal direction.

Consequently, the angle between the force's direction and the displacement's direction is 90°.

$\text{Wrokdone}(w) = \text{Force} \times \text{Displacement}$

$w = Fs\cos\theta$

$w = F.s\cos 90^\circ$

$w = 0$

So, the work done by coolie is 0.

34. Correct option: (A) 150J.

Power, $p = 50\text{W}$

Time taken, $t = 30\text{sec}$

Power is given by ,

$$\text{Power}(p) = \frac{\text{Workdone}(w)}{\text{Time}(t)}$$

$$50W = \frac{w}{30}$$

$$w = 150J$$

So, the work done by a machine is 150 J.

35. Correct option: (B) 3375J.

Mass of the body, $m = 50\text{kg}$

$h = 45 \times 15 = 675\text{cm} = 6.75\text{m}$

Time taken, $t = 9\text{sec}$

$\delta = 10\text{m}$

Potential energy (PE) is given by,

$$PE = mgh$$

$$PE = 50 \times 10 \times 6.75$$

$$PE = 3375J$$

36. Correct option: (C) Zero.

The work done is positive if the force and displacement are both in the same direction. The work done is 0 if force and displacement are perpendicular to one another. When neither the strength nor the direction of a force or a displacement are changing, a constant amount of work is completed.

The angle between the force's direction and the displacement's direction is 90°.

$\text{Wrokdone}(w) = \text{Force} \times \text{Displacement}$

$w = Fs\cos\theta$

$w = F.s\cos 90^\circ$

$w = 0$

So, the work done by the force is 0.

37. Correct option: (D) 10:1.

Mass of the truck, $m_1 = 5 \times 10^3\text{kg}$

Mass of the cart, $m_2 = 500\text{kg}$

Since speeds are constant, the relationship between kinetic energy and object mass is direct.

$$K \propto m$$

$$\frac{K_1}{K_2} = \frac{m_1}{m_2} = \frac{5 \times 10^3}{500} = \frac{10}{1}$$

$$K_1 = 10K_2$$

So, ratio of the kinetic energies is 10:1.

38. Correct option: (A) Power.

Power is the term for the rate at which work is done or energy is transferred.

So, the power is given by,

$$\text{Power}, p = \frac{\text{Work}(w)}{\text{time}(t)}$$

39. Correct option: (B) 5W.

Given:

Power, $p = 2000\text{Joule}$

Time taken, $t = 400\text{sec}$

So, the power is given by,

$\text{Power}=\frac{\text{Wrokdone}}{\text{time}}=\frac{2000\text{J}}{400\text{s}}=5\frac{J}{s}$

$5\frac{\text{J}}{\text{s}}$ in watt is given by,

$1\text{W}=1\frac{J}{s}$

$5\text{W}=5\frac{J}{s}$

So, the power is 5W.

40. Correct option: (C) 20s.

Given:

Power of bulb, $p = 150\text{w}$

Work done, $w = 3000\text{Joule}$

Power is given by,

$\text{Power}(p)=\frac{\text{Workdone}(w)}{\text{Time}(t)}$

$t=\frac{w}{p}$

$t=\frac{3000}{150}$

$t = 20s$

So, the time taken is 20s.

41. Correct option: (B) $P_1 > P_2$**.**

$W_1 = 500\text{J}$

$t_1 = 10\text{minute}=10\times 60\text{sec}$

$\text{W}_2 = 600\text{J}$

$t_2 = 20\text{minute}=20\times 60\text{sec}$

Power is given by,

$$P_1=\frac{W_1}{t_1}=\frac{500}{10\times 60}=0.83W$$

$$P_2=\frac{W_2}{t_2}=\frac{500}{20\times 60}=0.5W$$

Hence, $P_1 > P_2$.

42. Correct option: (C) Watt.

The rate of performing work is referred to as power.

$\text{Power}=\frac{\text{Work}}{\text{Time}}$

Joules and seconds are the SI units for work and time, respectively. So, the Joule per second, often known as Watt, is the SI unit of power.

43. Correct option: (C) Both A and B.

A machine's power consumption is influenced by two things: (i) the amount of work it completes, and (ii) the amount of time it takes to complete the task.

More power is used by a machine if it completes a given amount of work more quickly.

44. Correct option: (D) Work / time.

Power can be defined as the amount of work completed in a given amount of time. Watt (W), which is derived from joules per second (J/s), is the SI unit of power.

So, the power is given by ,

$$\text{Power}=\frac{\text{Work}}{\text{Time}}$$

45. Correct option: (A) 1watt.

The energy consumption rate of one joule per second is referred to as one watt.

$$1\text{W}=\frac{1\text{J}}{1\text{s}}$$

46. Correct option: (A) 30w.

Given:

$$W = 150J$$
$$t = 5sec$$

So, the power is given by as,

$$\text{Power}=\frac{\text{Work}}{\text{Time}}$$
$$\text{Power}=\frac{150}{5}$$
$$\text{Power}=30\text{w}$$

So, the power is 30w.

47. Correct option: (B) 375w.

Mass of the body, $m=50\text{kg}$

$h=45\times15=675\text{cm}=6.75\text{m}$

Time taken, $t=9\text{sec}$

$\delta=10\text{m}$

Potential energy(PE) is given by,

$$PE = mgh$$
$$PE = 50 \times 10 \times 6.75$$
$$PE = 3375J$$

$$\text{Power of the body } = \frac{\text{Potentialenergy}}{\text{Time}}$$
$$\text{Power}=\frac{3375}{9}$$
$$\text{Power}=375\text{W}$$

So, the power is 375w.

48. Correct option: (B) Power decreases.

The rate at which work is performed on an object is referred to as its power. Power is a temporal concept. It has to do with how quickly work is completed. One job or task can be completed slowly while the other is completed quickly. Since they are all the same job, the work is identical, but the power is different.

Power is given by,

$$\text{Power}=\frac{\text{Work}}{\text{Time}}$$

Because of the inverse relationship, power decreases as completion time increases.

49. Correct option: (C) Fv**.**

Power is given by,

$$\text{Power}=\frac{\text{Work}}{\text{Time}}$$

$$\text{Power} = \frac{F \times s}{t}$$
$$\text{Power} = Fv$$

So, the power is Fv.

50. Correct option: (D) 2×10^3w**.**

Mass of the water, $m = 100\text{kg}$

Height, $h = 20\text{m}$

Acceleration due to gravity, $g = 10\text{ms}^{-1}$

Time taken, $t = 10\text{sec}$

Work done against gravity is given by,

$W = mgh$

$W = 100\times20\times10 = 20000\text{J}$

We know that power is given by,

$$P = \frac{W}{t}$$
$$p = \frac{20000}{10} = 2000w = 2\times10^3 w$$

So, the power of a pump is 2×10^3w

51. Correct option: (D) 6250w.

Mass of the pump, $m = 50\,\text{kg}$

Height of water, $h = 100\text{m}$

Time taken, $t = 8\text{sec}$

Power is given by,

$$P = \frac{W}{t}$$
$$W = mgh$$
$$P = \frac{mgh}{t}$$
$$P = \frac{50\times10\times100}{8}$$
$$P = 6250w$$

So, the power of pump is 6250w.

52. Correct option: (B) Potential energy.

The potential energy is the energy that a body has while it is at rest and present at a specific height.

53. Correct option: (A) Kinetic energy.

Kinetic energy is the power that a body possesses as a result of its motion. Which is given by as,

$$KE = \frac{1}{2}mv^2$$

Here, m and v are mass and velocity of moving body.

54. Correct option: (A) 250J.

Given:

$$m = 5\text{kg}$$
$$h_1 = 5m$$
$$h_2 = 10m$$
$$g = 10\text{ms}^{-2}$$

Increase in potential is given by,

Increase in potential energy= final potential energy – Initial potential energy

Increase in potential energy $= mgh_1 - mgh_2$

Increase in potential energy $= mg(h_1 - h_2)$

Increase in potential energy $= 5 \times 10(10-5) = 250\text{J}$

So, the increase in potential energy is 250J.

55. Correct option: (C) $5\frac{\text{m}}{\text{s}}$**.**

Given:

Weight of the mass, $W = 600\text{N}$

$g = 10\text{ms}^{-2}$

Kinetic energy, $KE = 750\text{J}$

Work done by gravity is given by as,

$$W = mg$$
$$600 = m \times 10$$
$$m = 60\text{kg}$$

As, we know that kinetic energy given by,

$KE = \frac{1}{2}mv^2$

$750 = \frac{1}{2} \times 60 \times v^2$

$v = 5\frac{m}{s}$

So, the velocity is $5\frac{\text{m}}{\text{s}}$.

56. Correct option: (A) Commercial unit of electric energy.

Kilo watt hour (kWh) is the name of the commercial unit of electrical energy.

kWh can be changed into joules as follows:

$1\text{kWh}=3.6\times10^6\text{J}$

The kilowatt-hour (abbreviated kWh) is a unit of energy that represents the energy used to generate one kilowatt of power for one hour. Although it is not a formal standard unit in any system, the kilowatt-hour is frequently employed in electrical applications. The unit of electrical energy is the kWh, which is the same for household and commercial use.

57. Correct option: (A) $18\times10^6\text{J}$**.**

kWh can be changed into joules as follows:

$1\text{kWh}=3.6\times10^6\text{J}$

So, the 5kWh in joule is given as,

$5\text{kWh}=5\times3.6\times10^6\text{J}=18\times10^6\text{J}$

So, the 5kWh in joule is $18\times10^6\text{J}$.

58. Correct option: (C) 2.5 eV.

We can convert joule into ev this formula,

$1\text{ev} = 1.6\times10^{-19}\text{J}$

Now $4\times10^{-19}\text{J}$ into ev is given as,

$4\times10^{-19}\text{J}=\frac{4\times10^{-19}}{1.6\times10^{-19}}\text{ev}=2.5\text{ev}$

So, the $4\times10^{-19}\text{J}$ into ev is 2.5ev.

59. Correct option: (C) Electric motor.

The motor is the most used mechanism for converting electrical energy into mechanical energy. The majority of motors produce mechanical rotation in order to perform work using electromagnets.

60. Correct option: (A) m°.

Work done by external force is given by,

Work done by external forces = Change in kinetic energy

The initial Kinetic Energy of a particle is zero since it starts at rest.

Therefore,

$KE = Fd$

The fact that force is constant is a given.

We can observe that Kinetic energy is not mass-dependent.

So, the kinetic energy acquired by a body is directly proportional to m°.

61. Correct option: (A) Both parts will have numerically equal momentum.

Momentum would be conserved because there is no external force.

So, the momentum is given by,

$n \ \ p_i = 0$

After collision ball will move in x-direction,

$p_f = -m_1 v_1 + m_2 v_2$

Using the law of conservation,

$p_i = p_f$

$0 = m_1 v_2 + m_2 v_2$

$m_1 v_1 = -m_2 v_{\ 2} = k$

Where k is constant.

$v \propto \frac{1}{m}$

Both parts will have the same numerical momentum, but the lighter portion will have more velocity.

62. Correct option: (B) Potential energy.

The potential energy is the energy that a body possesses due to its position. As a result, a body positioned at a height above the ground has potential energy in relation to the ground.

63. Correct option: (A) Work energy theorem.

The work-energy theorem is a fundamental idea in physics that connects an object's work to its change in kinetic energy. It states that the work done by an object's net force is equal to the change in its kinetic energy. It can be stated mathematically as:

Work =Change in kinetic energy

$W = \Delta KE$

Where, W Represents the work done by the net force on the object.

ΔKE =The change in kinetic energy of the object.

Kinetic energy is the energy that an object has as a result of its motion.

When a force acts on an object and causes it to move, the work done by that force contributes to the change in kinetic energy of the object. If the work done is positive, the kinetic energy (speed of the item) increases; if the work done is negative, the kinetic energy (speed of the object) drops.

As a result, the phrase "The work done by a force in displacing a body measures the change in its kinetic energy" matches with the work-energy theorem notion, making option (A) the right response.

64. Correct option: (C) Sitting on the ground.

Potential energy of a body $=mgh$, hence potential energy is greatest when standing, assuming that gravity's acceleration and mass remain constant.

65. Correct option: (D) All of these.

Energy can only be converted from one form to another, according to the law of energy conservation; it cannot be created or destroyed. In contrast, some energy is wasted as heat energy while transferring energy to different forms.

66. Correct option: (A) $KE=\frac{p^2}{2m}$**.**

Kinetic energy is given by,

$KE=\frac{1}{2}mv^2\cdots\cdots(1)$

Momentum is given by,

$p=mv\cdots\cdots(2)$

So from the equation (1) and (2), we get,

$KE=\frac{m^2v^2}{2m}=\frac{p^2}{2m}$

So, the relation between kinetic energy and momentum is

$KE=\frac{p^2}{2m}$.

67. Correct option: (D) Both A and B.

In the International System of Units (SI), a joule is a unit of work or energy equal to the work done by a force of one newton acting through one metre.

68. Correct option: (B) 588J.

Given:

Mass of the object, $m=10\text{kg}$

Height, $h=6\text{m}$

$g=9.8\text{ms}^{-2}$

Potential energy is given by as,

$PE=mgh$

$PE=10\times9.8\times6=588\text{J}$

So, the energy possessed by the object is 588J.

69. Correct option: (A) At the maximum height.

Potential energy is given by,

$PE=mgh$

$PE\propto h$

When h is maximum, potential energy is maximum.

As an object's height increases, so does its potential energy. As a result, if the stone is hurled vertically and returns to the earth, its potential energy is greatest at the greatest height.

70. Correct option: (C) $\sqrt{x}$**.**

The relation between kinetic energy and momentum is given by,

$KE=\frac{p^2}{2m}$

$KE\propto p^2$

If Kinetic energy is increased x times, then momentum will increase by $\sqrt{x}$ times.

71. Correct option: (A)

72. Correct option: (B) $\frac{m_2}{m_1}$**.**

Masses of the bodies is m_1 and m_2 acted on by force F .

Acceleration due to mass m_1 , $a_1 = \frac{F}{m_1}$

Acceleration due to mass, $a_2 = \frac{F}{m_2}$

Velocity acquired during this time for mass m_1 ,

$v_1 = \frac{F}{m_1}t$

Velocity acquired during this time for mass m_2 ,

$v_1 = \frac{F}{m_2}t$

Ratio of the kinetic energy is given by,

$$\frac{E_1}{E_2} = \frac{\frac{1}{2}m_1 v_1^2}{\frac{1}{2}m_2 v_2^2}$$

$$\frac{E_1}{E_2} = \frac{m_1 \times \left(\frac{F}{m_1}t\right)^2}{m_2 \times \left(\frac{F}{m_2}t\right)^2}$$

$$\frac{E_1}{E_2} = \frac{m_2}{m_1}$$

So, the $\frac{E_1}{E_2}$ is $\frac{m_2}{m_1}$.

73. Correct option: (A) 1J.

Given:

Maas of the particle is, $m = 2mg = 2\times10^{-3}\text{g}$.

Distance, $d = 5\text{m}$

Time, H

Velocity is given by as,

$v = \frac{d}{t}$

$v = \frac{5}{5\times10^{-3}}$

$v = 10^3\frac{m}{s}$

Kinetic energy is given by as,

$KE = \frac{1}{2}mv^2$

$KE = \frac{1}{2}\times2\times10^{-3}\times(10^3)^2$

$KE = 1J$

So, the kinetic energy of the particle is 1J.

74. Correct option: (A) 25%.

Height from the which ball is dropped, $H = 2\text{m}$.

Height to which ball rebound, $h = 2\text{m}$

Potential energy at height H, $PE_1 = mgH$

Potential energy at height, $PE_2 = mgh$

Percentage loss in the potential energy is given by,

$$\%\Delta PE = \left[\frac{mg(H-h)}{mgH}\times 100\right]\%$$

$$\%\Delta PE = \left[\frac{mg(2-1.5)}{mg\times 2}\times 100\right]\%$$

$$\%\Delta PE = \left[\frac{0.5}{2}\times 100\right]\%$$

$$\%\Delta PE = 25\%$$

So, the percentage of energy lost is 25%.

75. Correct option: (A) 1:1.

Mass of first ball $KE = \frac{1}{2}mv^2$

$$KE = \frac{1}{2}(2m)v^2$$

$$KE = 2KE$$

Mass of second ball , $m_2 = 6g$

Ratio of kinetic energies , $\frac{K_1}{K_2} = \frac{3}{1}$

Kinetic energy is given by,

$$KE = \frac{1}{2}mv^2 \cdots\cdots (1)$$

Momentum is given by,

$$p = mv \cdots\cdots (2)$$

So from the equation (1) and (2), we get,

$$KE = \frac{m^2v^2}{2m} = \frac{p^2}{2m}$$

Now ratio of linear momentum is given by as,

$$\frac{K_1}{K_2} = \frac{p_1}{2m_1}\times\frac{2m_2}{p_2^2}$$

$$\frac{3}{1} = \frac{p_1}{p_2}\times\frac{6}{2}$$

$$\frac{p_1}{p_2} = \frac{1}{1}$$

$$\therefore p_1 : p_2 = 1:1$$

So, the ratio of their linear momenta 1:1.

76. Correct option: (B) 300J.

When the spring is stretched by two centimetres:

$$x = 2\text{cm}$$

$$E = 100J$$

Energy stored in spring,

$$E = \frac{1}{2}kx^2$$

$$100 = \frac{1}{2}k(2)^2$$
$$k = 50\frac{N}{\text{cm}}$$

Now, when the spring is stretched another 2 cm:

$x = 2+2 = 4\text{cm}$

Energy stored,

$$E = \frac{1}{2}k(x)^2$$
$$E = \frac{1}{2}(50)(4)^2 = 400J$$

So, the increase in energy, $\Delta E = 400-100 = 300\text{J}$

So, increase in energy is 300J.

77. Correct option: (A) Doubled.

The kinetic energy is given by,

$$KE = \frac{1}{2}mv^2$$

If mass is doubled:

$m = 2m$

Now, the kinetic energy is given by,

$$KE = \frac{1}{2}mv^2$$
$$KE = \frac{1}{2}(2m)v^2$$
$$KE = 2KE$$

Hence, the kinetic energy is doubled.

LEVEL – II

1. Correct option: (B)

Work done is given by,

$\text{Work done} = \text{Force} \times \text{displacement}$

Displacement, $d = 40\times 25 = 1000\text{cm}$

Force, $F = 20\times 10 = 200\text{N}$

Work done is given by,

$$W = 200\text{N}\times 10\text{m}$$
$$\text{W} = 2\times 10^3 J$$

So, the work done is $2\times 10^3\text{J}$.

2. Correct option: (A) The work done on the body against gravitational force is zero.

The angle between gravitational force and displacement is 90°.

So, the work done is zero.

$$W = FD\cos 90^\circ = 0$$

3. Correct option: (B) $\frac{1}{2}Fx$.

Work done given by,

$$W = \vec{F}.\vec{s} = F.x\cos\theta$$
$$W = Fx\cos 60^\circ$$
$$W = \frac{1}{2}Fx$$

So, the work done by the force is $\frac{1}{2}Fx$.

4. Correct option: (A) 2100J.

Because there are 30 steps of 20 cm each, the total distance would be 600 cm, or 6 m after conversion.

Given:

Force, $F = 350\text{N}$

Work is given by,

$W = F \times s$

$W = 350 \times 6 \times \cos 180^\circ = -2100\text{J}$

So, the work done against the gravity is 2100J.

5. Correct option: (B) 60°**.**

Work done, $W = 25\text{J}$

Force, $W = 25\text{J}$

Displacement, $s = 10\text{m}$

Work done is given by,

$W = Fs\cos\theta$

$\cos\theta = \frac{W}{Fs}$

$\cos\theta = \frac{25}{5 \times 10}$

$\cos\theta = \frac{1}{2}$

$\theta = 60^\circ$

So, the angle which the force makes with the direction of motion of the body is 60°.

6. Correct option: (A) Zero.

Because there is no displacement along or against the direction of gravity's pull, work done against gravity is zero.

7. Correct option:(C)100N.

Work done, $W = 6.4\text{kJ} = 6.4 \times 10^3\text{J}$

Distance moved, $s = 64\text{m}$

Work done is given by,

$\text{Work done} = \text{Force} \times \text{Displacement}$

$\text{W} = \text{F} \times S$

$6.4 \times 10^3 = F \times 64$

$F = \frac{6.4 \times 10^3}{64}$

$F = 100\text{N}$

So, the force applies is 100N.

8. Correct option: (B) $4 \times 10^4\text{J}$**.**

Mass of the water, $m = 50\text{kg}$

Force is given by as,

$F = mg$

$F = 50 \times 10 = 500\text{N}$

Work done is given by,

$W = 500 \times 80 = 4 \times 10^4\text{J}$

So, the work done by pump is $4 \times 10^4\text{J}$.

9. Correct option: (B) $1.1\times10^3\text{W}$**.**

Maas, $m=20\text{metricton}=20\times1000\text{kg}$

Distance, $s=20\text{m}$

Time, $t=1\text{h}=3600\,\text{sec}$

Power is given by,

$$\text{Power}=\frac{W}{t}=\frac{mg\times s}{t}=\frac{20\times1000\times10\times20}{3600}=1.1\times10^3\text{W}$$

So, the power is $1.1\times10^3\text{W}$.

10. Correct option: (D) $\frac{FS}{t}$**.**

Force $=F$

Displacement$=S$

Time$=t$

Power is given by as,

$$\text{Power}=\frac{\text{Work done}}{\text{time}}$$

$$\text{Power}=\frac{FS}{t}$$

So, the power is $\frac{FS}{t}$.

11. Correct option: (C) 48kg

Power is given by,

$$\text{Power}=\frac{\text{Workdone}}{\text{Time}}$$

$$P=\frac{W}{t}$$

$$P=\frac{(m_p+m_L)}{t}$$

$$1568=\frac{(50+m_p)\times9.8\times40}{25}$$

$$m_p=\frac{1568\times25}{40\times9.8}=50\text{kg}$$

So, the mass is 50kg.

The closest answer to 50kg is 48 kg.

12. Correct option: (A) 400.

Mass of the brick, $m=2.5\text{kg}$

Distance, $s=3.6\text{m}$

Average power, $m=2.5\text{kg}$

Force is given by,

$$F=mg$$

$$F=2.5\times9.8=24.5\text{N}$$

Work done is given by,

$$\text{Work done}=\text{Force}\times\text{distnace}$$

$$W=F\times s$$

$$W=24.5\times3.6=88.2\text{J}$$

Then, given his average rate of 9.8w (joules per second), compute the total work the man can do in one hour:

Total work in an hour= Average time × Time
Total work in an hour= $9.8 \times 3600 = 35280J$
The man works at an average rate of 9.8 watts, which translates to 9.8 joules of energy expended every second. As a result, dividing the total labour he can do in an hour by the work required to carry a single brick yields the number of bricks he can carry in one hour (3600 seconds):
$KE = \frac{1}{2}mv^2 = \frac{1}{2}m(at)^2$
$KE = \frac{1}{2} \times 10 \times 0.52 \times 10 = 125J$
So, the number of bricks is 400.

13. Correct option: (C) 2.

Power is given by,
$\text{Power}(p) = \frac{W}{t}$
Here, $W =$ wok done is constant.
Therefore,
$P \propto \frac{1}{t}$
The first man's output power to the second man's output power ratio is,
$\frac{P_1}{P_2} = \frac{t_2}{t_1} = \frac{20}{10} = \frac{2}{1}$
The first man's output power to the second man's output power ratio is $\frac{2}{1}$.

14. Correct option: (A) 3000 J.

Given:
Power supplied, $p = 100\text{w}$
Time, $t = 60\text{sec}$
Power used in stirring water =50% of the power supplied= $\frac{50}{100} \times 100\text{w} = 50\text{w}$
Work done is given by,
$\text{Work done} = \text{Power} \times \text{time}$
$\text{Work done} = 50 \times 60\text{s} = 3000J$
So, the work done on water in one minute is 3000J.

15. Correct option: (C)

Kinetic energy is given by,
$KE = \frac{p^2}{2m}$
Kinetic energy is the same for both the body,
$\frac{p_1^2}{2m_1} = \frac{p_2^2}{2m_2}$
$\frac{p_1^2}{p_2^2} = \frac{2m_1}{2m_2}$
$\frac{p_1^2}{p_2^2} = \frac{m_1}{m_2}$
$\frac{p_1}{p_2} = \frac{\sqrt{m_1}}{\sqrt{m_2}}$
So, the ratio $p_1 : p_2$ is equal $\frac{\sqrt{m_1}}{\sqrt{m_2}}$.

16. Correct option: (B) 12.5 m.

Initial kinetic energy, $KE_i = 490\text{J}$

Initial potential energy, $PE_i = 0$

Final kinetic energy, $KE_f = 0.5 \times 490$

Final Potential energy, $PE_f = mgh$

Using conservation law of energy ,

$$KE_i + PE_i = KE_f + PE_f$$
$$490 + 0 = (0.5 \times 490) + mgh$$
$$h = \frac{(490 - 245)}{2 \times 9.8}$$
$$h = 12.5m$$

So, the height is 12.5m

17. Correct option: (A) 44%.

Kinetic energy is given by,

$$K = \frac{1}{2}mv^2$$

And momentum is given by as,

$$P = mv$$

Now, the kinetic energy in terms of momentum is given by as,

$$K = \frac{p^2}{2m}$$

When p is increased by 20%.

The kinetic energy will be increased by,

$$\Delta K = \frac{(p \times 1.2)^2}{2m} = 1.44K$$
$$\frac{\Delta K}{K} \times 100 = 44\%$$

So, then K.E. increases by 44%.

18. Correct option: (C)

$k_1 = 3000\frac{\text{N}}{\text{m}}$

$k_2 = 1500\frac{\text{N}}{\text{m}}$

Kinetic energy is given by,

$KE = \frac{1}{2}kx^2$

Then the ratio of potential energies is given by,

$$\frac{KE_1}{KE_2} = \frac{\frac{1}{2}k_1x^2}{\frac{1}{2}k_1x^2}$$
$$\frac{KE_1}{KE_2} = \frac{k_1}{k_2}$$
$$\frac{KE_1}{KE_2} = \frac{3000}{1500}$$
$$\frac{KE_1}{KE_2} = \frac{2}{1}$$
$$KE_1 : KE_2 = 2:1$$

So, the ratio of potential energies is 2:1.

19. Correct option: (D) 8 J.

$F = 10\text{N}$

Force is given by ,

$$F = kx$$

$$k = \frac{F}{x} = \frac{10N}{10^{-3}m} = 10^4 \frac{N}{m}$$

Work done on the spring stretching = $\frac{1}{2}k(x_f^2 - x_i^2)$

Initially spring is at natural length i.e., $x_i = 0$

As a result of the work done in stretching through 40 mm,

$$W = \frac{1}{2}kx_f^2$$

$$W = \frac{1}{2} \times 10^4 \times (40 \times 10^{-3}m)^2$$

$$W = 8J$$

So, the work done in stretching the spring through 40 mm is 8J.

20. Correct option: (D) 125 J.

Mass of the body, $m = 10\text{kg}$

Because it is simultaneously worked upon by two forces 4N and 3N at right angles,

net force = $\sqrt{4^2 + 3^2} = 5\text{N}$

Acceleration is given by,

$a = \frac{F}{m} = \frac{5}{10} = 0.5\frac{\text{m}}{\text{s}^2}$

Kinetic energy is given by as,

$$KE = \frac{1}{2}mv^2 = \frac{1}{2}m(at)^2$$

$$KE = \frac{1}{2} \times 10 \times 0.52 \times 10 = 125J$$

So, the kinetic energy of the body at the end of 10 sec is 125J.

21. Correct option: (A) 0.1 joule.

Force constant, $k = 10\frac{\text{N}}{\text{m}}$

Initial stretch, $x = 0.20\text{m}$

Potential energy of object at initial stretch,

$$p_1 = \frac{1}{2}kx^2$$

$$p_1 = \frac{1}{2} \times 10 \times (0.2)^2$$

$$p_1 == 0.2J$$

Changed stretch, $x = 0.25\text{m}$

Potential stretch of object at changed stretch,

$p_2 = \frac{1}{2}kx^2$

$p_2 = \frac{1}{2} \times 10 \times (0.25)^2$

$p_1 == 0.3J$

Now, the difference in potential energy $= 10 \times 25 \text{ then K}=10$

$p_2 - p_1 = 0.3\text{J}-0.2\text{J}=0.1\text{J}$

So, the increase in potential energy is 0.1J.

22. Correct option: (C) n^2 times.

The relation between kinetic energy and momentum is given by,

$$KE = \frac{p^2}{2m}$$

$$KE \propto p^2$$

If momentum of body is increased 'n' times, then kinetic energy will increase by n^2 times.

23. Correct option: (D) Increase $\sqrt{2}$ times.

The relation between kinetic energy and momentum is given by,

$$K = \frac{p^2}{2m}$$

$$p = \sqrt{2mK}$$

$$p \propto \sqrt{K}$$

So, if a particle's kinetic energy is twice, its momentum becomes $\sqrt{2}$ times.

24. Correct option: (C) Becomes $\frac{1}{9}$th.

When mass was doubled, kinetic energy increased:

$m = 2m$

Therefore, the kinetic energy is given by,

$$K = \frac{1}{2}mv^2$$

$$K = \frac{1}{2} \times (2m) \times v^2$$

$$K = 2\left(\frac{1}{2}mv^2\right)$$

$$K = 2K$$

Hence kinetic energy becomes two times or doubled.

Kinetic energy when velocity is reduced by $\frac{1}{3}$rd :

$$K = \frac{1}{2}m\left(\frac{v}{3}\right)^2 = \frac{1}{9}\left(\frac{1}{2}mv^2\right) = \frac{1}{9}K$$

The kinetic energy reduces to $\frac{1}{9}$th of its initial value.

25. Correct option: (B) 9.8×10^5 J.

Mass of water, $m = 1000$kg

Height, $h = 100$m

Acceleration due to gravity,

$g = 9.8\text{ms}^{-2}$

Energy possessed is given by,

Potential energy $= mgh$

Potential energy $= 1000 \times 9.8 \times 100 = 9.8 \times 10^5$ J

So, the initial potential energy of water is 9.8×10^5 J.

26. Correct option: (B) 0.49J.

Mass of the ball, m=10g=0.01kg

Height, $h = 5$m

Rebounds height, $h' = 4$m

$g = 9.8\text{ms}^{-2}$

Potential energy is given by,

$$PE = mgh$$
$$PE = 0.01 \times 9.8 \times 5$$
$$PE = 0.49J$$

Initial Potential energy = Kinetic energy just before reaching the ground
So, the kinetic energy of the ball just before striking the ground is 0.49j.

27. Correct option: (A) 500 J.

Given: mass of the body=5kg
Potential energy at height 10m is given by,

$$PE = mgh = 5 \times 10 \times 10 = 500J$$

Potential energy at height 4m is given by,

$PE = mgh = 5 \times 10 \times 4 = 200\text{J}$

Loss in $PE = 500 - 200 = 300\text{J}$

For free fall, the total energy possessed by the organism at any point remains constant.
It is equal to sum of PE and KE.
At height 10m, at the top kinetic energy is zero. KE=0
Total energy= $PE + KE = 500 + 0 = 500\text{J}$
So, loss in potential energy is 500J.

28. Correct option: (C) 115 J.

We know that area under force-displacement curve gives work.
Work from, $x = 0$ to $x = 25$,
The work done is given by,

$W = \text{area } (x = 0 \text{ to } x = 5\text{m}) + \text{area } (x = 5\text{m to } x = 10\text{m}) + \text{area } (x = 10\text{m to } x = 15\text{m}) + \text{area } (x = 15\text{m to } x = 25\text{m})$

$$W = \frac{1}{2} \times 5 \times 4 + 4 \times 5 + \left(\frac{4+10}{2} \right) \times 5 + \frac{1}{2} \times 10 \times 10$$

$$W = 10 + 20 + 35 + 50 = 115\text{J}$$

So, the work done by the force is 115J.

29. Correct option: (A) $mg(h+x) = \frac{1}{2}kx^2$**.**

Using the energy conservation law, we know that,
Elastic potential energy contained in a compressed spring = change in gravitational potential energy

$$E_g = mgh(h + x)$$
$$E_g = \frac{1}{2}kx^2$$
$$mg(h + x) = \frac{1}{2}kx^2$$

30. Correct options: (B) and (C).

Because work is done in the opposite direction of the conservative force. As total mechanical energy is preserved, its potential energy grows and KE drops.

31. Correct options: (A), (B) and (D).

Using the work energy theorem,
$W_{net} = \Delta KE$
$(40 - 20)S = 40$
$S = 2m$
Work done by the gravity is given by,
$W = -20 \times 2 = -40\text{J} = 40 \times 2 = 80\text{J}$
Work done by tension= $40 \times 2 = 80\text{J}$.
So, Correct options: (A), (B) and (D).

32. Correct option: (C).

Because the particle is acted upon by a constant magnitude force that is constantly perpendicular to the particle's motion, the work done by the force in the direction of the velocity is zero. As a result, the magnitude of velocity will not change as a result of the force. Kinetic energy is given by,

$$KE = \frac{1}{2}mv^2$$

Because the mass and magnitude of velocity are always constant in this condition, the kinetic energy is also constant.

33. Correct options: (A and (C).

(A) The work done by the string is positive: The work done by the string on the block is positive if the string is pulled in the same direction as the displacement of the block.

(C) The work done by the string on the man is zero: Because the force exerted by the string on the man causes no appreciable displacement in the direction of the force, the work done by the string on the man is likely to be minimal or zero.

34. Correct option: (A)

35. Correct option: (B)

36. Correct option: (C)

37. Correct Matching is: (a)-(S), (b)-(R), (c)-(P), (d)-(Q).

(a) $1\text{ev} = 1.6\times10^{-19}\text{J}$

(b) $\text{K.W.H} = 3.6\times10^{6}\text{J}$

(c) Horespower=746W

(d) $1\text{erg} = 10^{-7}\text{J}$

So, the correct matching is (a)-(S), (b)-(R), (c)-(P), (d)-(Q).

38. Correct Matching is: (a)-(P), (b)-(S), (c)-(Q), (d)-(R).

a) A conservative force is a force used to move a particle from one point to another that is independent of the particle's route. It is only determined by the particle's beginning and final positions. Conservative forces include gravitational force and elastic spring forces.

b) Because the amount of work done by friction depends on the path, it is non-conservative. A potential energy can be associated with a conservative force but not with a non-conservative force.

c) The potential energy formula is affected by the force operating on the two objects. The formula for gravitational force is,

$PE = mgh$

d) The work-energy theorem states that the net work done on an object produces a change in the object's kinetic energy. Net work = change in kinetic energy = final kinetic energy - beginning kinetic energy is the formula.

39. Correct Matching is: (a-R, b-P, c-S, d-Q)

40. Correct Answer is: 4

Loss of energy is given by as,

Loss of energy = Kinetic energy -Potential energy

$$\text{Loss of energy} = \frac{1}{2}mv^2 - mgh$$

$$\text{Loss of energy} = \frac{1}{2}\times 1\times 20^2 - 1\times 10\times 18$$

$$\text{Loss of energy} = 20J$$

So, the energy lost is 20J. $4\times5.\text{n}=4$.

So, the value of n is 4.

41. Correct Answer is: 10

When a machine has a 25% efficiency, it produces 25 when the input is 100.
In the provided question, the input is either 1kW or 1000w.
Output=25% of input
So the output power is given by,

$$\text{Output power} = 25\% \; of \; 1\text{kw}$$

$$\text{Output power} = 250\text{w}$$

The output of the machine is 250w. 10×25 then K=10.

So, the value o f K is 10.

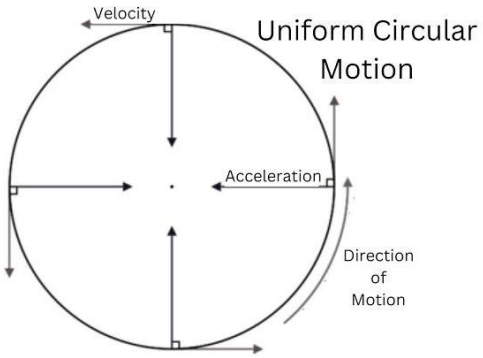
5. CIRCULAR MOTION

Solutions

LEVEL – I

Variables of Circular Motion:

1. Correct option: (C)



In a uniform circular motion, the linear speed remains constant, its linear velocity is not constant. It is because, the linear velocity of the object at any point of its circular path is directed along the tangent to the path and therefore its direction keeps on changing continuously. The change of linear velocity results in acceleration, which is always directed along the radius of the circle and towards its centre. Thus, uniform circular motion is a case of accelerated motion.

2. Correct option: (B)

In a circular motion, the magnitude of the radius vector or position vector is constant and equals to the radius of the circular path. Its direction is towards the circumference of the circle.

As the particle motes in a circular motion, along the circumference of a circle, the direction of the radius vector or position vector in circular motion changes continuously.

3. Correct option: (C)

Given: angular displacement (θ)=60˚

Radius of the circle(r)= 21 m

$\pi = \frac{22}{7}$

To calculate the distance covered from point A to B

Length of arc(AB) = $\frac{\theta}{360^\circ}$x2πr

Length of arc = $\frac{60^\circ}{360^\circ}$x2 x$\frac{22}{7}$x 21m

Length of arc = $\frac{1}{6}$x2 x$\frac{22}{7}$x 21m

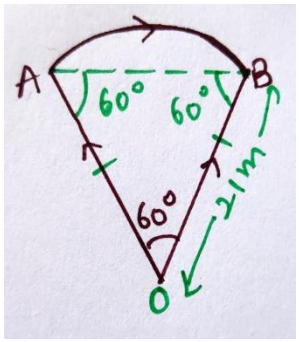
= 22m

Hence, distance covered = 22m

- **To calculate the displacement**

Displacement is the shortest distance between the initial and the final position of the object. It is equal to the difference between the final and initial positions of the object

Shortest distance= displacement =AB



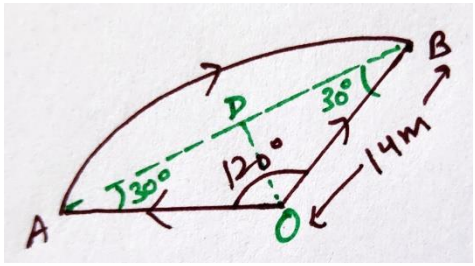
From the above figure
In triangle AOB
AO = BO = 21m (radius of the circle)
let ∠0AB = ∠0BA = x (base angles opposite to equal sides are equal)
By angle sum property of triangle
∠0AB + ∠0BA + 60° = 180°
x + x + 60° = 180°
2x = 180° - 60°
2x = 120°
x = 60°
∠0AB = ∠0BA =60°
As all the angles are equal so, it is an equilateral triangle which has all the sides equal
Shortest distance= displacement =AB= 21m

4. Correct option: (C)

Given: angular displacement (θ)=120°
Radius of the circle(r)= 14m
$\pi = \frac{22}{7}$

• To calculate the displacement

Displacement is the shortest distance between the initial and the final position of the object. It is equal to the difference between the final and initial positions of the object
Shortest distance= displacement =AB



From the above figure
In triangle AOB
AO = BO = 14m (radius of the circle)
let ∠0AB = ∠0BA = x (base angles opposite to equal sides are equal)
By angle sum property of triangle
∠0AB + ∠0BA + 120° = 180°
x + x + 120° = 180°
2x = 180° - 120°
2x = 60°
x = 30°
∠0AB = ∠0BA =30°
Drop a perpendicular OD from point O on length AB. Such that AD= DB

In triangle AOD

$\text{Cos}30^\circ = \frac{base}{hypotenuse}$

Base= AD

Hypotenuse= 14 m

$\text{Cos}30^\circ = \frac{AD}{14}$

$\frac{\sqrt{3}}{2} = \frac{AD}{14}$

$\frac{\sqrt{3}x14}{2} = AD$

$\sqrt{3}x7 = AD$

AB= 2 AD

$AB = \sqrt{3}x7x2 = 14\sqrt{3}$

Hence, displacement $AB = 14\sqrt{3}$

To calculate the distance covered from point A to B

Length of arc(AB) = $\frac{\theta}{360^\circ}x2\pi r$

Length of arc = $\frac{120^\circ}{360^\circ}x2\ x\frac{22}{7}x\ 14m$

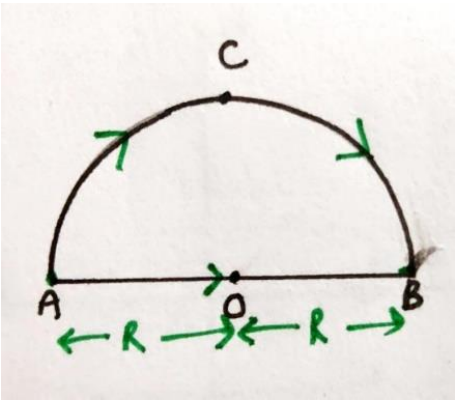
Length of arc = $\frac{1}{3}x2\ x\frac{22}{7}x\ 14m$

$=\frac{88}{3}m$

Hence, distance covered = $\frac{88}{3}m$

5. Correct option: (A)

Displacement is the difference between the final and initial positions of the object in a given time. Also, it is the shortest distance between two positions of the object.



Consider that while travelling the circular path, the boy completes half revolution along the path ACB.

Here, A is the initial position and B is the final position of the boy.

So, AB =shortest distance joining initial and final position of the boy= displacement of the boy

Given: radius of the circle = R

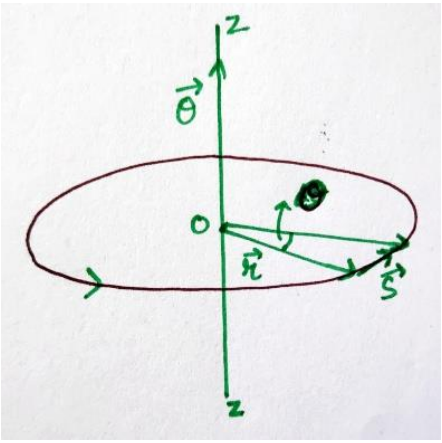
AB= diameter of the circle = 2R

Hence, displacement of the boy = 2R

6. Correct option: (A)

finite angular displacement is not a vector quantity, it is because, addition of two finite angular displacements does not obey commutative law. However, addition of two infinitesimally small angular displacements obeys commutative law and hence, an infinitesimally small angular displacement is a vector quantity.

7. Correct option: (D)



$\vec{r} \times \vec{s} = 0$ (both are linear)

$\vec{s} \times \vec{\theta} = 0$ (point of action is different hence, their cross product is zero)

$\vec{r} \times \vec{\theta} = -\vec{s}$

8. Correct option: (C)

angular displacement $(\theta) = \frac{\text{length of the arc(l)}}{\text{radius of the circle(r)}} - - - - - (1)$

Length of the arc in 1 rotation = circumference of the circle

$= 2\pi r$

Length of arc in $2\frac{1}{2}\ or \frac{5}{2}\ rotations = \frac{5}{2} \times 2\pi r = 5\pi r$

Putting value in equation 1

angular displacement $(\theta) = \frac{5\pi r}{r} = 5\pi$ radians

9. Correct option: (D)

Given: angular displacement $(\theta) = \frac{\pi}{2}$ radians

Radius of the circle= 20m

$\pi = \frac{22}{7}$

angular displacement $(\theta) = \frac{\text{length of the arc/linear distance(l)}}{\text{radius of the circle(r)}} - - - - - (1)$

Linear distance =angular displacement x radius

Linear distance = $\frac{\pi}{2} \times 20$ m

Linear distance = $\frac{22}{7 \times 2} \times 20$ m

Linear distance = 31.4m

10. Correct option: (A)

angular displacement $(\theta) = \frac{\text{length of the arc/linear distance(l)}}{\text{radius of the circl(r)}} - - - - - (1)$

Angular displacement is inversely proportional to radius.

Person 1 has less radius hence, more angular displacement.

Person 2 has more radius hence, less angular displacement.

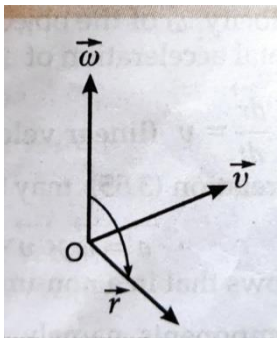
Hence, person 1 will cover larger angular displacement.

11. Correct option: (A)

In a uniform circular motion, the linear speed remains constant, its linear velocity is not constant. It is because, the linear velocity of the object at any point of its circular path is directed along the tangent to the path and therefore its direction keeps on changing continuously. The change of linear velocity results in acceleration, which is always directed along the radius of the circle and towards its centre. Thus, uniform circular motion is a case of accelerated motion.

12. Correct option: (A)

Relation between linear velocity and angular velocity.



$\vec{v} = \vec{r} \times \vec{\omega}$

Where $\vec{v}$ = linear velocity

$\vec{r}$ = radius

$\vec{\omega}$ = angular velocity

13. Correct option: (C)

Since particle describes equal angles in equal intervals of time, magnitude of velocity remains same. But direction changes since it is always tangential to circular path.

14. Correct option: (D)

Given: number of revolutions=120

Time taken= 1 minute = 60 sec

Angular speed $(\omega) = \frac{\text{angular displacement}(\theta)}{\text{time taken (t)}} - - - (1)$

Angular displacement in 1 revolution $=2\pi$

Angular displacement in 120 revolutions $=2\pi \times 120 = 240\pi$ radians

Putting the values in equation 1

Angular speed $(\omega) = \frac{240\pi \text{ radians}}{60 \text{ sec}}$

Angular speed $(\omega) = 4\pi$ rad/s

15. Correct option: (C)

Given: linear speed (v) = 10m/s

Radius of the circle(r) = 20cm = 0.20m (1cm = 0.01 m)

To find: angular speed (ω)

$v = r\,\omega - - - - - (1)$

Where v = linear speed

r = radius of the circle

ω = angular speed

Putting the given values in equation 1

$10 = 0.20x\,\omega$

$\frac{10}{0.20} = \omega$

$\frac{10x100}{20} = \omega$

$50 \text{ rad/s} = \omega$

Hence, angular speed = 50 rad/s

16. Correct option: (B)

Given: radius(r) of the circle made by second hand = 30cm

$\pi = 3.14$

Angular speed $(\omega) = \frac{\text{angular displacement}(\theta)}{\text{time taken (t)}}$

$\omega = \frac{\theta}{t} - - - -(1)$

where ω = angular speed
θ = angular displacement
t = time taken

We know that in 60 seconds, second hand completes 1 revolution
Time taken = 60 sec
Angular displacement in 1 revolution =2π
Putting values in equation 1

$\omega = \frac{2\pi}{60}$

$\omega = \frac{\pi}{30}$ rad/s

Also

$v = r\,\omega - - - - - (2)$

Where v = linear speed
r = radius of the circle
ω = angular speed

Putting the values in equation 2

$v = 30 \times \frac{\pi}{30}$ cm/s

$v = \pi$ cm/s

v = 3.14 cm/s

17. Correct option: (B)

Angular speed $(\omega) = \frac{\text{angular displacement}(\theta)}{\text{time taken (t)}}$

$\omega = \frac{\theta}{t} - - - - (1)$

where ω = angular speed
θ = angular displacement
t = time taken

We know that in 60 minutes, minute hand completes 1 revolution
Time taken = 60 minutes = 60 x 60 seconds = 3600 seconds
Angular displacement in 1 revolution =2π
Putting values in equation 1

$\omega = \frac{2\pi}{3600}$

$\omega = \frac{\pi}{1800}$ rad/s

18. Correct option: (A)

Given: linear velocity (v) = 1600cm/s = 16 m/s (1cm = 1/100 m)

Radius of the circle(r) = $\frac{diameter}{2} = \frac{400cm}{2}$ = 200cm = 2m

To find: angular velocity(ω)

$v = r\,\omega - - - - - (1)$

Where v = linear velocity
r = radius of the circle
ω = angular velocity

Putting the given values in equation 1

16= 2 x ω

$\frac{16}{2} = \omega$

8 rad/s = ω

Hence, angular velocity= 8 rad/s

19. Correct option: (D)

Velocity $= \frac{\text{displacement}}{\text{time interval}}$

Displacement is the difference between the final and initial positions of the object in a given time.

In a circular motion, when the particle completes 1 revolution it returns to its initial position. So, initial and final position are same.

Hence, displacement of the particle is 0.

So, average velocity = zero

20. Correct option: (A)

Given: number of revolutions(n) = 3500

Distance covered = 22 km= 22000 m (1km = 1000 m)

π = 3.14

Distance covered by the wheel in 1 revolution = πd (where d = diameter of the circle)

Distance covered in 1 revolution x number of revolutions(n) = total distance covered

$3.14 \text{x d x } 3500 = 22000\text{m}$

$d = \frac{22000}{3500 \text{ x } 3.14}\text{m}$

d = 2 m

Hence, diameter of the wheel = 2m

21. Correct option: (B)

Particle is moving with uniform speed = v

We know that

$v = r\,\omega$

Where v = linear speed

r = distance of the particle from the point of observation

ω = angular velocity

- **Angular velocity of particle P about point A**

$\omega_A = \frac{v}{r_A}$ -----(1)

Distance from the point A= AB=r_A = 2a

putting the values in equation 1 we get

$\omega_A = \frac{v}{2a}$

- **Angular velocity of particle P about point C**

$\omega_C = \frac{v}{r_C}$ -----(2)

Distance from the point C= BC=r_C = a

putting the values in equation 2 we get

$\omega_C = \frac{v}{a}$

$\frac{\omega_A}{\omega_C} = \frac{\frac{v}{2a}}{\frac{v}{a}} = \frac{1}{2}$= 1:2

22. Correct option: (C)

Given: linear velocity (v) = 5m/s

Radius of the circle(r) = 25 cm = 0.25 m (1cm= 0.01m)

To find: angular velocity(ω)

$v = r\,\omega$ – – – – – (1)

Where v = linear velocity

r = radius of the circle

ω = angular velocity

Putting the given values in equation 1

5= 0.25 x ω

$\frac{5}{0.25} = \omega$

$\frac{5 \text{ x}100}{25} = \omega$

$20 \text{ rad/s} = \omega$

Hence, angular velocity= 20 rad/s

23. Correct option: (B)

Given: uniform speed(v) = 22m/s

Radius of the circle(r) = 14 m

• To find the angular speed

We know that

v = r ω

Where v = linear speed

r = radius of the circle

ω = angular speed

$\omega = \frac{v}{r}$ -----(1)

putting the values in equation 1

$\omega = \frac{22}{14}$ rad/s

• To find the time period

$\omega = \frac{2\pi}{T}$ where T = time period

or,

$T = \frac{2\pi}{\omega} - - - -(2)$

Where ω = angular speed

T = time period (**Time period= time taken to complete 1 revolution.)**

Putting the given values in equation 2

$T = \frac{2 \text{ x } 22 \text{ x } 14}{7\text{x}22}$ $(\pi = \frac{22}{7})$

T = 4 sec

Hence, time period of the circle is 4 sec.

24. Correct option: (C)

Frequency= number of revolutions completed in 1 second.

Number of revolutions completed in 20 sec= 200

Number of revolutions completed in 1 sec = $\frac{200}{20} = 10$

Hence, frequency = 10 revolutions/sec or 10 Hz

25. Correct option: (C)

Frequency= number of revolutions completed in 1 second.

Time period= time taken to complete 1 revolution.

Frequency = $\frac{\mathbf{1}}{\mathbf{Time\ period}}$

A minute hand takes 1 hour or 3600 seconds to complete 1 revolution

So time period of minute hand = 3600 seconds

Frequency = $\frac{1}{3600}$ Hz

26. Correct option: (C)

Given**:** angular speed = 4π rad/sec

$\omega = 2\pi\nu - - - -(1)$

Where ω = angular speed

ν = frequency

Putting the given values in equation 1

$4\pi = 2\pi\nu$

$\frac{4\pi}{2\pi} = \nu$

2 Hz = ν

Hence, frequency = 2Hz

27. Correct option: (D)

Given angular displacement (θ)=30˚

Time taken= 2sec

180˚ = π radians

$1^{\circ} = \frac{\pi}{180} radians$

$30^{\circ} = \frac{\pi}{180} \text{x } 30\ radians$

$30^{\circ} = \frac{\pi}{6}\ radians$

$\pi = \frac{22}{7}$

Angular speed (ω) = $\frac{\text{angular displacement}(\theta)}{\text{time taken (t)}}$

$\omega = \frac{\theta}{t} ----(1)$

where ω = angular speed

θ = angular displacement

t = time taken

Putting values in equation 1

$\omega = \frac{\pi}{6 \text{ x } 2} rad/sec$

$\omega = \frac{\pi}{12}$ rad/s

$\omega = 2\pi\nu ----(2)$

Where **Or,**

Time period = $\frac{1}{\text{frequency}}$

Time period= $\frac{1}{\frac{1}{24}} =$ **24** sec

Hence, time period to complete one circle = 24 sec

28. Correct option: (A)

Given: linear velocity for car and scooter is same = v

Let the Radius of the wheel of car = r_C

Radius of the wheel of scooter = r_S

According to the question

$r_C = 2r_S$

We know that

$\omega = \frac{v}{r}$

Where v = linear speed

r = distance of the particle from the point of observation

ω = angular velocity

• **Angular velocity of car**

$\omega_C = \frac{v}{r_C}$ -----(1)

putting the values in equation 1 we get $\omega_C = \frac{v}{2r_S}$

• **Angular velocity of scooter**

$\omega_S = \frac{v}{r_S}$ -----(2)

$ratio = \frac{\omega_C}{\omega_S} = \frac{\frac{v}{2r_S}}{\frac{v}{r_S}} = \frac{1}{2}$= 1:2

29. Correct option: (A)

Given: initial angular velocity (ω_o) of flywheel =0 rad/s (because, flywheel is starting from rest)

Final angular velocity(ω) of flywheel= 45 rad/s

Time taken = 5s

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$\alpha = \frac{\omega - \omega_o}{t}$ *here*, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$\alpha = \frac{45-0}{5}\ rad/s^2$

α= 9 rad/s^2

angular acceleration = 9rad/s^2

30. Correct option: (A)

As the particle is moving in a circle of radius r with uniform speed(v).

We know that

$\omega = \frac{v}{r}$

Where v = linear speed

r = distance of the particle from the point of observation

ω = angular velocity

so angular speed will also be uniform/ constant

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$\alpha = \frac{\omega - \omega_o}{t}$ *here*, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

as there is no change in the angular speed/velocity so the angular acceleration will be zero.

31. Correct option: (A)

Given: initial velocity(u) of particle =5m/s

Final velocity(v) of particle= 6m/s

Radius of the circle = 20 cm = 0.20 m (1cm = 0.01m)

Time taken = 2s

• **Calculation of acceleration**

Acceleration= $\frac{\text{change in velocity}}{\text{time taken}}$

$a = \frac{v-u}{t}$ *here*, a= acceleration

u= initial velocity

v = final velocity

$a = \frac{6-5}{2}\ m/s^2$

$a = \frac{1}{2} m/s^2$

$a = 0.5 m/s^2$

acceleration $= 0.5 m/s^2$

• To calculate the angular acceleration

$\alpha = \frac{a}{r}$

where α= angular acceleration

a= linear acceleration

r = radius of the circular motion

putting the values in the above equation

$\alpha = \frac{0.5}{0.20} rad/s^2$

$\alpha = \frac{5x\ 100}{20\ x\ 10} rad/s^2$

$\alpha = 2.5\ rad/s^2$

hence, angular acceleration = 2.5 rad/s^2

32. Correct option: (C)

Given: initial angular velocity(ω_o) of engine =300 rpm = 10π rad/s

$1rpm = \frac{\pi}{30} rad/s$

$300rpm = \frac{\pi}{30} x\ 300\ rad/s$

$= 10\pi\ rad/s$

Final angular velocity(ω) of engine= 900 rpm= 30π rad/s

$1rpm = \frac{\pi}{30} rad/s$

$900rpm = \frac{\pi}{30} x\ 900\ rad/s$

$= 30\pi\ rad/s$

Time taken = 5s

• To calculate the number of revolutions

Number of revolutions(n) = $\frac{total\ angular\ displacement(\theta)}{angular\ displacement\ in\ 1\ revolution}$

Angular displacement in 1 revolution=2π

Number of revolutions(n) = $\frac{100\pi}{2\pi}$ = 50 revolutions

33. Correct option: (B)

Given: initial angular velocity(ω_o) of flywheel =300rpm= 10π rad/s

$1rpm = \frac{\pi}{30} rad/s$

$300rpm = \frac{\pi}{30} x\ 300\ rad/s$

$= 10\pi\ rad/s$

Final angular velocity(ω) of flywheel= 0 rad/s (flywheel comes to rest)

Angular acceleration= -4 rad/s^2 (negative sign for retardation)

To find: Time taken

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$\alpha = \frac{\omega - \omega_o}{t}$ *here,* α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$-4 = \frac{0 - 10\pi}{t}$

$t = \frac{0 - 10\pi}{-4}$

$t = 2.5\ \pi s$

hence, time taken = 2.5 πs

34. Correct option: (A)

Angular acceleration= $2 rad/sec^2$

Time taken = 10 sec

initial angular velocity(ω_o) = 0 rad/s (wheel starts from rest)

• **To calculate the angular displacement(θ)**

using the second equation of motion for circular motion

$\theta = \omega_o t + \frac{1}{2}\alpha t^2$ ---------(1) ω_o = initial angular velocity

α = angular acceleration of the object

t= time taken

θ= angular displacement of the object

putting the values in equation 1

$\theta = 0x10 + \frac{1}{2}x\ 2x(10)^2$

$\theta = 100$ rad

• **To calculate the number of revolutions**

Number of revolutions(n) = $\frac{total\ angular\ displacement(\theta)}{angular\ displacement\ in\ 1\ revolution}$

Angular displacement in 1 revolution=2π

$\pi = 3.14$

Number of revolutions(n) = $\frac{100}{2\pi} = \frac{100}{2\ x\ 3.14} = 15\ (approx)$

35. Correct option: (B)

angular displacement(θ)= 2.5 radians

Angular acceleration is constant

• **To calculate the angular acceleration**

Time taken = 1 sec

using the second equation of motion for circular motion

$\theta = \omega_o t + \frac{1}{2}\alpha t^2$ ---------(1) ω_o = initial angular velocity

α = angular acceleration of the object

t= time taken

θ= angular displacement of the object

putting the values in equation 1

$2.5 = 0x1 + \frac{1}{2}x\ \alpha x(1)^2$

$2.5\ x\ 2 = \alpha$

$5\ rad/s^2 = \alpha$

• To calculate the angle rotated in next second

Time taken = 2s

putting the values in equation 1

initial angular velocity(ω_o) = 0 rad/s (wheel starts from rest)

$\theta = 0x2 + \frac{1}{2}x\ 5x(2)^2$

$\theta = \frac{1}{2}x\ 5\ x\ 4$

$\theta = 10$ rad

angle rotated in the next second = angle rotated in 2 seconds – angle rotated in 1 second

$=10\ rad - 2.5\ rad$

$= 7.5$ rad

36. Correct option: (D)

Given: initial angular velocity(ω_o) of wheel =0rad/s (wheel starts from rest)

Final angular velocity(ω) of wheel= 100 rev/sec= 200π rad/s

1 rev/sec = 2π rad/s

100 rev/sec = 100x2π rad/s = 200π rad/s

- Time taken = 4s

To calculate the angular acceleration

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$$\alpha = \frac{\omega - \omega_o}{t}$$

here, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$$\alpha = \frac{200\pi - 0}{4}\,\text{rad}/s^2$$

$$\alpha = 50\pi\ \text{rad}/s^2$$

angular acceleration = $50\pi\ \text{rad}/s^2$

- **To calculate the angular displacement(θ)**

using the second equation of motion for circular motion

$\theta = \omega_o t + \frac{1}{2}\alpha t^2$ ---------(1)

putting the values in equation 1

$\theta = 0x4 + \frac{1}{2}x50\pi(4)^2$

$\theta = 25\pi \text{ x } 16$

$\theta = 400\pi$ rad

hence, angle rotated in 4 sec = 400π rad

37. Correct option: (B)

- **When the wheel starts from rest and uniformly accelerated for 10 seconds**

Given: initial angular velocity(ω_o) of wheel =0rad/s (wheel starts from rest)

Angular acceleration(α_1) = $4\text{rad}/\text{sec}^2$

time taken: 10 seconds

using the second equation of motion for circular motion

angular displacement $\theta_1 = \omega_o t_1 + \frac{1}{2}\alpha_1 t_1^2$ ---------(1)

putting the values in equation 1

$\theta_1 = 0x10 + \frac{1}{2}x4(10)^2$

$\theta_1 = 2 \text{ x } 100$

$\theta_1 = 200$ rad

Angular velocity after 10 seconds:

$\omega_1 = \omega_o + \alpha_1 t_1$

$\omega_1 = 4x\ 10$

$\omega_1 = 40\text{rad/s}$

- **For next 10 seconds the wheel rotated uniformly**

initial angular velocity(ω_o) of wheel = $\omega_1 = 40\text{rad/s}$

Angular acceleration= 0 rad/sec^2 (wheel is rotating uniformly)

time taken: 10 seconds

$\theta_2 = \omega_1 t_2 + \frac{1}{2}\alpha_2 t_2^2$

putting the values in equation 1

$\theta_2 = 40x10 + \frac{1}{2}x0(10)^2$

$\theta_2 = 400$ rad

• **The wheel decelerated for the next 10 seconds**

To calculate the angular acceleration

Final angular velocity $\omega_2 = 0$ rad/s (wheel is decelerating)

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$$\alpha_3 = \frac{\omega_2 - \omega_1}{t_3}$$

here, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$$\alpha_3 = \frac{0-40}{10}\,\text{rad}/s^2$$

$$\alpha_3 = -4\ \text{rad}/s^2$$

Angular displacement while decelerating :

$\theta_3 = \omega_1 t_3 + \frac{1}{2}\alpha_3 t_3^2$

putting the values in equation 1

$\theta_3 = 40\text{x}10 + \frac{1}{2}\text{x}(-4)\text{x}(10)^2$

$\theta_3 = 400 - 200$

$\theta_3 = 200$ rad

Total angular displacement = $\theta_1 + \theta_2 + \theta_3$ = 200+400+200 =800 rad

Hence, the total angular displacement = 800 rad

38. Correct option: (B)

Given: initial angular velocity(ω_o) of body =0 rev/s (body starts from rest)

Angular acceleration(α)= $2\text{rev}/s^2$

Angular displacement(θ)= 5 revolutions

To find: Final angular velocity(ω) of body

using the third equation of motion for circular motion

$\omega^2 - \omega_o^2 = 2\alpha\theta - - - - -(1)$ where ω_o = initial angular velocity

ω = final angular velocity

α = angular acceleration of the object

θ= angular displacement of the object

putting the values in equation 1

$\omega^2 - (0)^2 = 2\text{ x}2\text{ x }5$

$\omega^2 = 20$

$\omega = \sqrt{20}\,rev/s$ 20 = 2 x2 x5

$\omega = 2\sqrt{5}\,rev/s$

hence, angular velocity/speed = $2\sqrt{5}$ rev/s

39. Correct option: (C)

Given: initial angular velocity(ω_o) of stone=0rad/s (stone starts from rest)

Radius of the circle(r)= 0.5 m

Angular acceleration (α) = $3\text{rad}/s^2$

To find: Final angular velocity(ω) of stone

Time taken = 2s

• **To calculate the angular velocity**

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$$\alpha = \frac{\omega - \omega_o}{t}$$

here, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$$3 = \frac{\omega - 0}{2}$$

$$3 \text{ x } 2 = \omega$$

$$6 \text{ rad/sec} = \omega$$

• **To calculate the tangential speed**

We know that

v = r ω------(1)

Where v = linear speed

r = radius of the circle

ω = angular speed

putting the values in equation 1

v= 0.5 x 6 m/sec

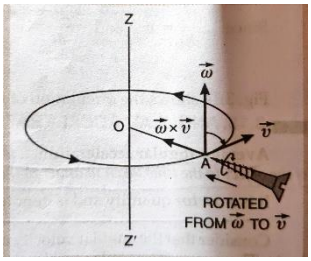
v= 3m/sec

hence, tangential speed = 3m/sec

40. Correct option: (B)

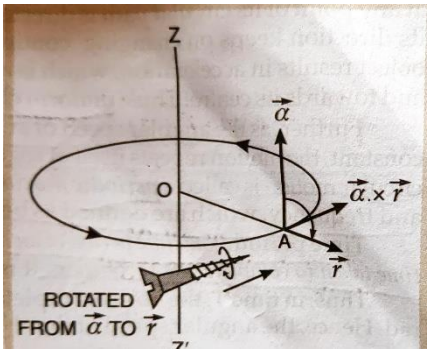
When an object is moving in a circular motion, its total acceleration consists of two components

• Radial component(centripetal acceleration) $a_c = \omega v = \frac{v^2}{r}$ (where v= linear velocity, r= radius of the circle)



• Transverse component of the acceleration(tangential/linear acceleration)

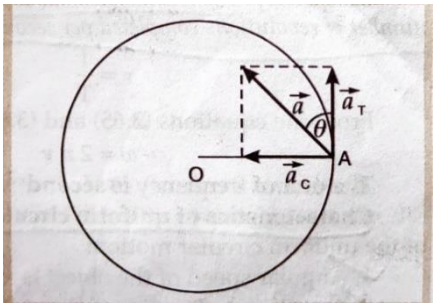
a_T or a $= r\alpha$ (where r= radius, α = angular acceleration)



• Magnitude of the resultant acceleration = $\sqrt{(a_c)^2 + (a_T)^2}$

$$= \sqrt{\left(\frac{v^2}{r}\right)^2 + (a)^2}$$

$$= \sqrt{\frac{v^4}{r^2} + (a)^2}$$



LEVEL-II

1. Correct option: (A)

We know that

$\omega = 2\pi\nu$ ------(1)

where ω = angular speed

ν = frequency

and Frequency (ν)= $\frac{\mathbf{1}}{\mathbf{Time\ period(T)}}$

putting in equation 1

$\omega = \frac{2\pi}{T}$----------(2)

also, centripetal acceleration $a_c = v\,\omega - - - -(3)$ (where v= linear speed, ω = angular speed)

form equation 2 and 3

$acceleration(a_c) = \frac{2\pi v}{T}$

2. Correct option: (D)

Angular speed (ω) = $\frac{\text{angular displacement}(\theta)}{\text{time taken (t)}}$

$\omega = \frac{\theta}{t} - - - -(1)$

where ω = angular speed

θ = angular displacement

t = time taken

as both the cars are completing the circle in same time so their angular speeds will be same if

$\omega_1 = angular\ speed\ of\ car\ 1$

$\omega_2 = angular\ speed\ of\ car\ 2$

Then $\omega_1 = \omega_2$

Ratio = $\frac{\omega_1}{\omega_2} = 1$

3. Correct option: (A)

Inertia of direction is the inability of the body to change its direction of motion by itself. That is, a body continues to move along the same straight line unless compelled by some external force to change it.

When A stone tied to one end of a string is whirled and the Stone is released suddenly, the stone flies off along the tangent to the circle. this is because the pull in the string was forcing the stone to move in a circle. as soon as the stone is released the pull vanishes. The stone in a bid to move along the straight line flies off tangentially.

4. Correct option: (A)

Given: distance between the earth and the moon(r) = 3.85×10^5km= 3.85×10^8m

Time taken by the moon to complete 1 revolution around the earth(Time period T) = 27.3 days

1 day = 86400 seconds

27.3 days = 27.3 x 86400 seconds

27.3 days = 2358720 seconds

We know that

Speed(v) = $\frac{distance}{time}$=$\frac{2\pi r}{t}$

Speed (v)= $\frac{2 \times 22 \times 3.85 \times 10^8}{7 \times 2358720}$= 1025.98 m/s

centripetal acceleration (a_c) of the moon $= \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

acceleration $= \frac{v^2}{r}$

$= \frac{(1025.98)^2}{3.58x10^8}$

$= 2.73 \times 10^{-3}$ m/s^2

hence acceleration of the moon with respect to the earth= 2.73×10^{-3} m/s^2

5. Correct option: (A)

$a_c = \frac{(5\pi)^2}{10}$ cm/s^2

$a_c = 2.5\pi^2$ cm/s^2

Hence, acceleration= $2.5\pi^2$ cm/s^2

6. Correct option: (A)

Given: radius of the earth(r)= 6400km

Time taken by earth to complete 1 rotation = 24 hours

linear velocity(v) = $\frac{distance}{time} = \frac{2\pi r}{t}$

Where v= linear velocity

r= radius of the circle

linear velocity(v)= $\frac{2x\ 22\ x\ 6400}{7\ x24}$

= 1676.19 km/h

7. Correct option: (D)

Given: linear speed(v)= 16m/s

Angle subtended(θ)=90˚= $\frac{\pi}{2} radians$

Length of arc (l) = 628 m

angular displacement (θ) = $\frac{\text{length of the arc/linear distance(l)}}{\text{radius of the circle(r)}} - - - - - (1)$

or

r = $\frac{\text{length of the arc/linear distance(l)}}{\text{angle subtended}(\theta)}$

r = $\frac{628\ x\ 2}{\pi} = \frac{628\ x\ 2}{3.14}$=400 m

centripetal acceleration (a_c) of the car $= \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

putting the values in the above equation

we get

$a_c = \frac{v^2}{r}$

$a_c = \frac{(16)^2}{400}$=0.64 m/s^2

Hence, centripetal acceleration of car = 0.64m/s^2

8. Correct option: (A)

Given: initial angular velocity of flywheel =ω_o

Time taken = 1minute

Final angular velocity(ω) of flywheel= $0.9\omega_o$

To find angular velocity at the end of second minute

- **For the first minute**

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$\alpha = \frac{\omega - \omega_o}{t}$ *here*, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$\alpha = \frac{0.9\omega_o - \omega_o}{1}$

$\alpha = -0.1\omega_o$ **-------(1)**

- **for the second minute**

Final angular velocity(ω) of flywheel= ω

$\alpha = \frac{\omega - \omega_o}{t}$ *here*, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$\alpha = \frac{\omega - 0.9\omega_o}{1}$

$\alpha = \omega - 0.9\omega_o$ **--------(2)**

as angular acceleration is constant

comparing equation 1 and 2

$\omega - 0.9\omega_o = -0.1\omega_o$

$\omega = -0.1\omega_o + 0.9\omega_o$

$\omega = 0.8\omega_o$

hence, angular velocity at the end of second minute = $0.8\omega_o$

9. Correct option: (C)

Given: initial angular velocity(ω_o) of wheel=0rad/s (wheel starts from rest)

Radius of the circle(r)= 0.5 m

Angular acceleration (α) = $3rad/s^2$

To find: Final angular velocity(ω) of stone

Time taken = 2s

- **To calculate the angular velocity**

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$\alpha = \frac{\omega - \omega_o}{t}$ *here*, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$3 = \frac{\omega - 0}{2}$

$3 \times 2 = \omega$

$6 \text{ rad/sec} = \omega$

- **To calculate the centripetal acceleration**

centripetal acceleration (a_c) of the car = $\frac{v^2}{r}$ -------(1)

where v= linear velocity

r= radius of the circle

we know that

$v = r\omega$------(2)

Where v = linear speed

r = radius of the circle

ω = angular speed

from equation 1 and 2

we have

centripetal acceleration (a_c) of the car = $r\omega^2$

putting the values in the above equation

we get, centripetal acceleration of the particle= $0.5x(6)^2 = 0.5 \times 36 = 18 \text{ m/s}^2$

10. Correct option: (C)

We know that

$\omega = \frac{v}{r}$

Where v = linear speed

r = distance of the particle from the point of observation

ω = angular velocity

As both the wheels are connected to the same belt so their linear velocity v will remain the same

- Angular velocity of wheel A

$\omega_A = \frac{v}{r_A}$ -------(1)

Radius of wheel A = r

putting the values in equation 1 we get

$\omega_A = \frac{v}{r}$

- **Angular velocity of wheel B**

$\omega_B = \frac{v}{r_B}$ -------(2)

Radius of wheel B = 3r

putting the values in equation 2 we get

$\omega_B = \frac{v}{3r}$

$ratio = \frac{\omega_A}{\omega_B} = \frac{\frac{v}{r}}{\frac{v}{3r}} = \frac{3}{1}$= 3:1

11. Correct option: (A, D)

In a uniform circular motion, the linear speed remains constant, its linear velocity is not constant. It is because, the linear velocity of the object at any point of its circular path is directed along the tangent to the path and therefore its direction keeps on changing continuously. The change of linear velocity results in acceleration, which is always directed along the radius of the circle and towards its centre. Thus, uniform circular motion is a case of uniformly accelerated motion.

Hence, in uniformly accelerated motion, the speed remain constant and the magnitude of acceleration remains constant.

12. Correct option: (B)

When the position vector sweeps out equal areas in equal intervals of time, then the angular speed of the particle remains constant.

- As the direction is changing continuously so velocity is not constant.
- Only the magnitude of centripetal acceleration remains constant but direction changes.
- Tangential acceleration will be zero because there is no change in the angular velocity and we know that

Transverse component of the acceleration(tangential/linear acceleration)

$a_T = r\alpha$ (where r= radius, α = angular acceleration)

As angular acceleration depends on change in angular velocity so it will be zero, which makes the tangential acceleration zero.

13. Correct options: (B), (C)

Given: initial angular velocity(ω_o) of particle=0 rad/s (particle is starting from rest)

Final angular velocity(ω) of particle=2 rad/s

Radius(r)= 0.4m

Time taken = 2s

• **To find the angular acceleration**

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$\alpha = \frac{\omega - \omega_o}{t}$ *here*, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$\alpha = \frac{2-0}{2}\ \text{rad}/s^2$

$\alpha = 1\ \text{rad}/s^2$

angular acceleration = $1\text{rad}/s^2$

• **To find the radial acceleration**

Centripetal/radial acceleration (a_c) = $r\omega^2$

where r = radius of the circle

ω = angular velocity

putting the values in the above equation

radial acceleration$(a_c) = 0.4\ \text{x}(2)^2 = 0.4\text{x}\ 4 = 1.6\ \text{m/s}^2$

• To calculate the resultant acceleration

Tangential/linear acceleration(a_T)= αr

Where $\alpha = angular\ acceleration$

r = radius of the circle

putting values in above equation

Tangential/linear acceleration$(a_T) = 1\ \text{x}\ 0.4 = 0.4\ \text{m/s}^2$

Magnitude of the resultant acceleration = $\sqrt{(a_c)^2 + (a_T)^2}$

Putting the values in the above equation

resultant acceleration = $\sqrt{(1.6)^2 + (0.4)^2} = \sqrt{2.56 + 0.16} = \sqrt{2.72}\ \ \text{m/s}^2$

14. Correct options(A), (B), (C), (D)

Given: radius of the circle= 20cm = 0.20 m

Linear speed (v) = 2t

• Linear speed at t = 3 sec

v= 2t

Putting the value of t in above equation

v= 2 x 3 = 6 m/s

• $\omega = \frac{v}{r}$

Where v = linear speed

r = distance of the particle from the point of observation

ω = angular speed

at t = 2 sec

v = 2 x 2 = 4 m/s

putting the values in the above equation

we get

$\omega = \frac{4}{0.20} = \frac{4\text{x}\ 100}{20} = 20\text{m/s}$

• linear acceleration a = $\frac{dv}{dt} = \frac{d(2t)}{dt} = 2\text{m/s}^2$

• Centripetal/radial acceleration (a_c) = $r\omega^2$

where r = radius of the circle

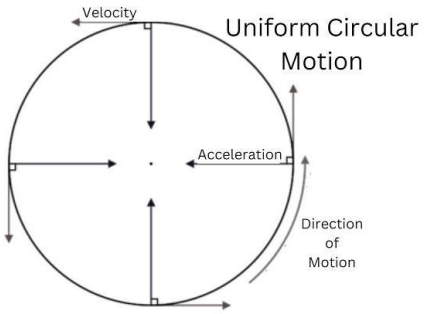
ω = angular velocity

putting the values in the above equation

Centripetal acceleration$(a_c) = 0.20\ \text{x}(20)^2 = 0.20\text{x}\ 400 = 80\ \text{m/s}^2$

15. Correct option (B), (C)

In a uniform circular motion, the linear speed remains constant, its linear velocity is not constant. It is because, the linear velocity of the object at any point of its circular path is



directed along the tangent to the path and therefore its direction keeps on changing continuously. The change of linear velocity results in acceleration, which is always directed along the radius of the circle and towards its centre. Thus, uniform circular motion is a case of accelerated motion.

An external force required to make a body move along circular path with uniform speed is called centripetal force.

A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

centripetal acceleration $(a_c) = \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

• as the momentum depends on velocity so it is not constant because the velocity changes in direction along the tangent.

• Kinetic energy remains constant because speed remains constant in uniform circular motion.

• In case of uniform circular motion, the angular velocity is constant.

So, angular acceleration $\alpha = \frac{d\omega}{dt} = 0$

As Tangential/linear acceleration$(a_T) = \alpha r = 0$

• centripetal acceleration $(a_c) = \frac{v^2}{r}$ is constant.

16. Correct options: (C), (D)

• Average velocity $= \frac{\text{displacement}}{\text{time interval}}$

During the period of 6 months the earth travels a distance equal to half a revolution.

Displacement = diameter of the orbit of earth around the sun.

Displacement is not zero, hence average velocity is not zero.

• Average speed $= \frac{\text{total distance}}{\text{total time}}$

During the period of 1 year earth travels 1 revolution around the sun in the elliptical orbit. Hence average speed is not zero.

• Acceleration$= \frac{\text{change in velocity}}{\text{time taken}}$

During the period of 1 year, net displacement is zero because the earth returns to its initial position, so velocity is zero. so, change in velocity is zero.

Hence, acceleration of the particle is also zero.

• Instantaneous acceleration of the earth = centripetal acceleration$(\frac{v^2}{r})$ pointing towards the sun.

17. Correct options: (C), (D)

When an object is moving in a circular motion, its total acceleration consists of two components

• Radial component(centripetal acceleration) $a_c = \frac{v^2}{r}$ (where v= linear velocity, r= radius of the circle)

• Transverse component of the acceleration(tangential/linear acceleration)

a_T or a $= r\alpha$ (where r= radius, α = angular acceleration)

• For a non-uniform circular motion radial component and the tangential component both are non zero.

$a_c \neq 0$, $a_T \neq 0$

• For a uniform circular motion

the angular velocity is constant.

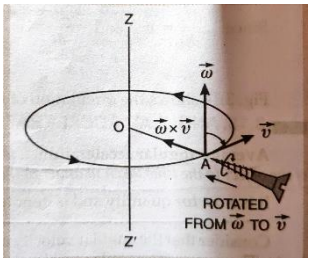
So, angular acceleration $\alpha = \frac{d\omega}{dt} = 0$

As Tangential/linear acceleration$(a_T) = \alpha r = 0$

centripetal acceleration $(a_c) = \frac{v^2}{r}$ is constant.

$a_c \neq 0$, $a_T = 0$

18. Correct option: (A), (B), (D)



• $\vec{a_c} = \vec{\omega} \times \vec{v}$----------(1)

Where $\vec{v}$ = linear velocity

$\vec{a_c}$ = centripetal acceleration

$\vec{\omega}$ = angular velocity

In the equation 1

All the three vectors are mutually perpendicular to each other

$\vec{a_c} \perp \vec{\omega}$

$\vec{\omega} \perp \vec{v}$

$\vec{a_c} \perp \vec{v}$

Hence, angle between centripetal acceleration and angular velocity is 90˚.

• In case of uniform circular motion, the angular velocity is constant.

So, angular acceleration $\alpha = \frac{d\omega}{dt} = 0$

As Tangential/linear acceleration$(a_T) = \alpha r = 0$

• In a uniform circular motion, the linear speed remains constant, its linear velocity is not constant. It is because, the linear velocity of the object at any point of its circular path is directed along the tangent to the path and therefore its direction keeps on changing continuously. The change of linear velocity results in acceleration, which is always directed along the radius of the circle and towards its centre. Thus, uniform circular motion is a case of accelerated motion.

An external force required to make a body move along circular path with uniform speed is called centripetal force.

A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

Centripetal/radial acceleration $(a_c) = \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

net acceleration is only radial acceleration.

19. Correct options(A), (C), (D)

$\overrightarrow{a_c} = \overrightarrow{\omega} \times \overrightarrow{v}$----------(1)

Where $\vec{v}$ = linear velocity

$\overrightarrow{a_c}$ = centripetal acceleration

$\overrightarrow{\omega}$ = angular velocity

In the equation 1

All the three vectors are mutually perpendicular to each other

$\overrightarrow{a_c} \perp \overrightarrow{\omega}$

$\overrightarrow{\omega} \perp \overrightarrow{v}$

$\overrightarrow{a_c} \perp \overrightarrow{v}$

As

Tangential/linear acceleration(a_T)= αr

as $\alpha = \frac{d\omega}{dt}$, direction of angular acceleration is same as angular velocity

so, direction of angular velocity and linear acceleration is same.

20. Correct option: (C)

Given: angular displacement $\theta = 4t^2$

radius of the circle= 1m

linear velocity (v) = angular velocity(ω) x radius (r)

v = ωr----------(1)

also $\omega = \frac{d\theta}{dt}$

using the given relation for angular displacement θ

$\omega = \frac{d(4t^2)}{dt} = 2x4t = 8t$--------(2)

at t = 2 sec

ω= 8 x 2 = 16 rad/s

putting the values in equation 1

we get

v= 16 x 1 m/s = 16 m/s

hence, linear velocity at t = 2sec= 16m/s

21. Correct option: (A)

• Centripetal/normal acceleration $(a_c) = r\omega^2$-------(3)

where r = radius of the circle

ω = angular velocity

$\omega = \frac{d\theta}{dt}$

using the given relation for angular displacement θ

$\omega = \frac{d(4t^2)}{dt} = 2x4t = 8t$

at t = 1 sec

ω = 8 x 1 = 8 rad/s

putting the values in equation 3

Centripetal/normal acceleration$(a_c) = 1\,x(8)^2 = 64\ m/s^2$

22. Correct option: (B)

Tangential/linear acceleration(a_T)= αr − − − − − − − (4)

Where $\alpha = angular\ acceleration$

r = radius of the circle

$\alpha = \frac{d\omega}{dt}$

using the relation for angular velocity from equation 2

$\alpha = \frac{d(8t)}{dt} = 8\ rad/s^2$

putting values in the equation 4

Tangential/linear acceleration(a_T)= 8 x 1 = 8 m/s^2

23. Correct option: (B)

resultant acceleration = $\sqrt{(a_c)^2 + (a_T)^2}$

Tangential acceleration(a_T)= 8 m/s^2

Centripetal/normal acceleration(a_c) = 64 m/s^2

Putting the values

net acceleration = $\sqrt{(64)^2 + (8)^2}$= $\sqrt{4096 + 64}$ = $\sqrt{4160}$ m/s^2 = $8\sqrt{65}$ m/s^2

24. Correct option: (C)

Given: angular displacement θ = 4t − 3t^2 + t^3

radius of the circle= 1m

linear velocity (v) = angular velocity (ω) x radius (r)

v = ωr----------(1)

also ω = $\frac{d\theta}{dt}$

using the given relation for angular displacement θ

ω = $\frac{d(4t-3t^2+t^3)}{dt}$= 4 − 6t + 3t^2-------(2)

at t = 2 sec

ω = 4 − 6x2 + 3(2)^2 = 4 − 12 + 12 = 4 rad/s

25. Correct option: (A)

$averag\ angular\ velocity$ (ω_{av}) = $\frac{\theta_f-\theta_i}{t_f-t_i}$----------(3)

Given: t_f= 4 sec

t_i= 0 sec

Time interval = 4 sec

θ_f = angular displacement at time t= 4 sec

Putting the values in given equation

θ_f = 4x 4 − 3x(4)^2 + (4)^3 = 16 − 48 + 64 = 32

θ_i= angular displacement at time t = 0 sec

θ_i = 4x 0 − 3x(0)^2 + (0)^3 = 0 − 0 + 0 = 0

Putting the values in the equation 3

$\omega_{av} = \frac{32-0}{4} = 8$ rad/s

26. Correct option: (C)

$\alpha = \frac{d\omega}{dt}$

where

α = angular acceleration

ω = angular velocity

using the relation for angular velocity from equation 2

$\alpha = \frac{d(4 - 6t + 3t^2)}{dt} = -6 + 6t$

at t = 4 sec

$\alpha = -6 + 6 \times 4 = -6 + 24 = 18\ rad/s^2$

27. Correct Matching is: (a)---(Q) (b)---(Q),(S) (c)---(Q), (S) (d)---(P), (S)

(a) Tangential/linear acceleration$(a_T) = \frac{dv}{dt}$ – – – – – – – (4)

$a_t = \frac{d(2t^2)}{dt}$

using the relation for angular velocity from equation 2

$a_t = 4t\ rad/s^2$

a_t *increases with time*

(b) Centripetal/radial acceleration $(a_c) = \frac{v^2}{r}$

$(a_c) = \frac{(2t^2)^2}{r} = \frac{4t^4}{r}$

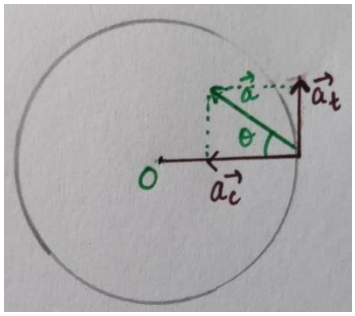
Centripetal/radial acceleration (a_c) increases with time and also depends on radius

(c) *angular speed* $\omega = \frac{v}{r} = \frac{2t^2}{r}$

Angular speed (ω) increases with time and also depends on radius

(d)

angle between the total acceleration



$\tan\theta = \frac{a_t}{a_c} = \frac{4t}{\frac{4t^4}{r}} = \frac{r}{t^3}$

$\theta = \tan^{-1}\frac{r}{t^3}$

Angle between the total acceleration decreases with the time and also depends on the radius.

28. Correct Matching is: (a)---(R) (b)---(S) (c)---(P) (d)---(Q)

(a)Angular displacement is measured in terms of angle θ

SI unit of angular displacement = radians

(b)linear displacement is measured in terms of units of length

SI unit of linear displacement= metre

(c) Angular velocity $(\omega) = \frac{\text{angular displacement}(\theta)}{\text{time taken (t)}}$

$\omega = \frac{\theta}{t}$

SI unit of θ = radian

SI unit of time = second

SI unit of angular velocity = radian/second or rad/s

(d)linear velocity= $\frac{\text{displacement}}{\text{time taken (t)}}$

SI unit of displacement = metre

SI unit of time = second

SI unit of linear velocity = metre/second or m/s

29. Correct Answer is: 6

Given: initial angular velocity(ω_o) of wheel=0rad/s (wheel starts from rest)

Angular acceleration (α) = 2rad/s^2

To find: Final angular velocity(ω) of stone

Time taken = 3s

- **To calculate the angular velocity**

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$\alpha = \frac{\omega - \omega_o}{t}$ *here*, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$2 = \frac{\omega - 0}{3}$

$2x\ 3 = \omega$

$6\ \text{rad/sec} = \omega$

angular velocity= 6 rad/sec

30. Correct Answer is: 1

Given: initial angular velocity(ω_o) of stone=0rad/s (stone starts from rest)

Radius of the circle(r)= 0.5 m

Angular acceleration (α) = 2rad/s^2

To find: Final angular velocity(ω) of stone

Time taken = 1s

- **To calculate the angular velocity**

Angular acceleration= $\frac{\text{change in angular velocity}}{\text{time taken}}$

$\alpha = \frac{\omega - \omega_o}{t}$ *here*, α= angular acceleration

ω_o = initial angular velocity

ω = final angular velocity

$2 = \frac{\omega - 0}{1}$

$2\ x1\ = \omega$

$2\ \text{rad/sec} = \omega$

- **To calculate the tangential speed**

We know that

$v = r\ \omega$------(1)

Where v = linear speed

r = radius of the circle

ω = angular speed

putting the values in equation 1

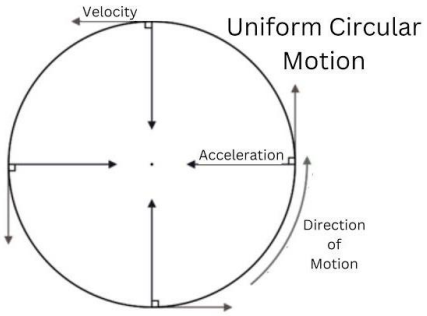
v= 0.5 x 2 m/sec

v= 1m/sec

hence, tangential speed = 1 m/sec

Centripetal force
LEVEL – I

1. Correct option: (A)



An external force required to make a body move along circular path with uniform speed is called centripetal force.

A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

centripetal acceleration $(a_c) = \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

2. Correct option: (D)

$\alpha = \frac{d\omega}{dt}$

where

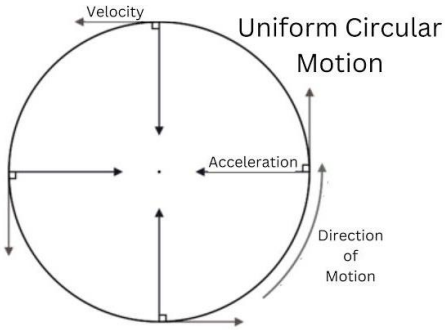
α = angular acceleration

ω = angular velocity

direction of angular acceleration is same as that of angular velocity.

it is along the perpendicular to the plane of the circle.

3. Correct option: (D)



An external force required to make a body move along circular path with uniform speed is called centripetal force.

A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

In a uniform circular motion, the linear speed remains constant, its linear velocity is not constant. It is because, the linear velocity of the object at any point of its circular path is directed along the tangent to the path and therefore its direction keeps on changing continuously. The change of linear velocity results in acceleration, which is always directed along the radius of the circle and towards its centre. Thus, uniform circular motion is a case of accelerated motion.

centripetal acceleration $(a_c) = \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

according to newton's second law of motion

F= m x a

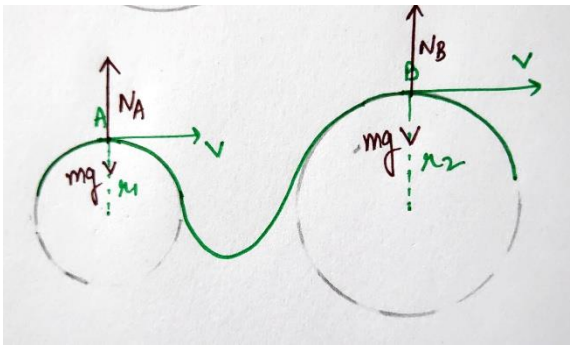
Centripetal force(F)= $\frac{mv^2}{r}$

4. Correct option: (D)

Centrifugal force is not a real force. It is called fictitious or pseudo force because its origin cannot be identified. It only arises in the observer's frame of reference.

As the particle is observed from the inertial frame of reference so centrifugal force is zero.

5. Correct option: (C)



The resultant of weight of the body and the normal reaction of the board provides the necessary centripetal force. Therefore,

mg-N= $\frac{mv^2}{r}$

- for the point A

$N_A = mg - \frac{mv^2}{r_1}$--------(1)

- for the point B

$N_B = mg - \frac{mv^2}{r_2}$--------(1)

As shown in the figure $r_1 < r_2$

Hence, $\frac{mv^2}{r_1} > \frac{mv^2}{r_2}$

Larger value will be subtracted from the N_A and smaller value will be subtracted from N_B

Hence, $N_A < N_B$

6. Correct option: (B)

Given: maximum force (tension T) = 400 N

Mass(m)= 1kg

Radius of the circular motion(r)= length of the string = 1m

To find: maximum linear velocity(v_{max})

The tension provides the necessary Centripetal force(F)= $\frac{mv_{max}^2}{r}$

$T_{max} = \frac{mv_{max}^2}{r}$

Putting the values in the above equation

$400 = \frac{1x\, v_{max}^2}{1}$

$400N = v_{max}^2$

$(20)^2 = v_{max}^2$

$20 = v_{max}$

Hence, maximum linear velocity = 20m/s

7. Correct option: (B)

Given: number of links= 125

Length of each link(l) = 1.25 m

Total mass of 125 links = 125

Mass of each link = $\frac{2}{125}$kg

angular velocity(ω)=3000rev/min=100π rad/s

1rpm = $\frac{\pi}{30}$rad/s

3000rpm = $\frac{\pi}{30}$x 3000 rad/s

= 100π rad/s

• To find the radius of the circular path

angular displacement (θ) = $\frac{\text{length of the arc(l)}}{\text{radius of the circle(r)}}$

length of arc= length of each link = 1.25 m

angular displacement in 1 round = 2π radians

radius (r) = $\frac{1.25}{2\pi}$

• To find the centripetal force

Centripetal force(F)= $\frac{mv^2}{r}$

v = rω

putting in above equation

Centripetal force(F)= $\frac{m(r\omega)^2}{r}$= $mr\omega^2$

Putting the values in the above equation

Centripetal force(F)= $\frac{2}{125}$x$\frac{1.25}{2\pi}$ x $(100\pi)^2$

Centripetal force(F)= $\frac{1}{100}$x$\frac{1}{\pi}$ x 100π x 100π

Centripetal force(F) on each link = 100π = 100 x 3.14 = 314 N

8. Correct option: (B)

Given: linear speed(v)= 16m/s

Angle subtended(θ)=180˚= π *radians*

Length of arc (l) =distance travelled = 1256 m

Mass of the car = 500kg

angular displacement (θ) = $\frac{\text{length of the arc/linear distance(l)}}{\text{radius of the circle(r)}}$ − − − − − (1)

or

r = $\frac{\text{length of the arc/linear distance(l)}}{\text{angle subtended}(\theta)}$

r = $\frac{1256}{\pi}$=$\frac{1256}{3.14}$= 400m

centripetal force (F_c) of the car = $\frac{mv^2}{r}$

where v= linear velocity

r= radius of the circle

m = mass of the car

putting the values in the above equation

we get

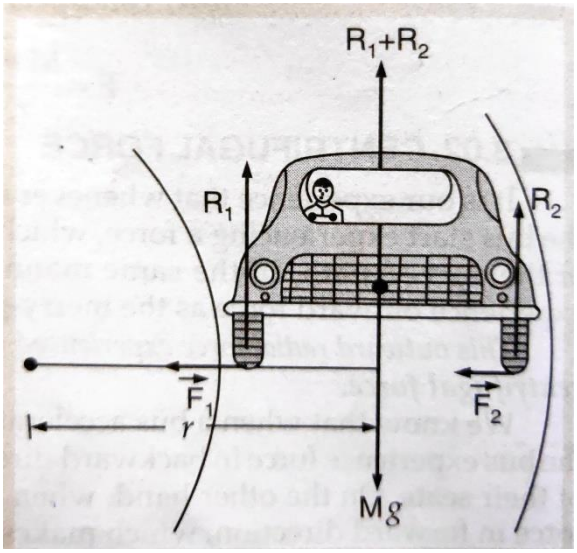
$F_c = \frac{mv^2}{r}$

$F_c = \frac{500x(16)^2}{400}$=320 N

Hence, centripetal force acting on the car = 320 N

9. Correct option: (A)

when a van is taking a circular turn on a level road. While taking the round, the tyres of the vehicles try to leave the road and go away from the centre of the curve.



The total force of friction provides the necessary centripetal force

$F_1 + F_2 = \frac{mv^2}{r}$

$F_1 = \mu R_1$

$F_2 = \mu R_2$

where F_1 *and* F_2 *are the forces of friction* , $r =$ *radius of the circular turn*

$\mu R_1 + \mu R_2 = \frac{mv^2}{r}$

$\mu(R_1 + R_2) = \frac{mv^2}{r}$----------(1)

Where

R_1 *and* R_2 *are the normal recations*

μ = coefficient of friction

the total normal reaction on the van balances the weight(mg) of the van

$R_1 + R_2 = mg$ (g= acceleration due to gravity = 9.8 m/s^2 or 10 m/s^2)

Putting the value in equation 1

$\mu\, mg = \frac{mv^2}{r}$--------(2)

equation 2 gives the maximum velocity/minimum radius of curvature with which van can take the circular turn of radius r(without skidding)

given: coefficient of friction $\mu = 0.5$

velocity (v)= 72kmph= 20m/s

$1km/h = \frac{5}{18}\, m/s$

$72km/h = 72x\, \frac{5}{18}\, m/s$

$= 20m/s$

To find: minimum radius of curvature

Putting the values in equation 2

$0.5x500x\, 10 = \frac{500(20)^2}{r}$

$5 = \frac{400}{r}$

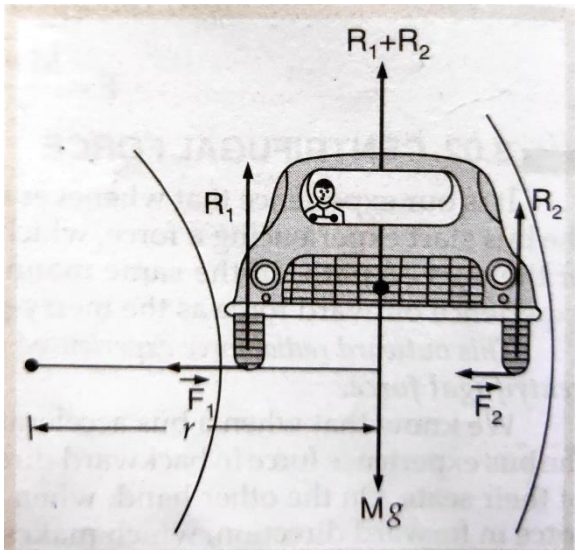
$r = \frac{400}{5} = 80m$

hence, minimum radius of curvature = 80 m

Vehicle on Level Circular Road and Banking Road :

10. Correct option: (C)

when a car is taking a circular turn on a level road. While taking the round, the tyres of the vehicles try to leave the road and go away from the centre of the curve.



The total force of friction provides the necessary centripetal force

$F_1 + F_2 = \frac{mv^2}{r}$

where F_1 *and* F_2 *are the forces of friction*

$r =$ *radius of the circular turn*

v=linear velocity

11. Correct option: (D)

Centripetal force(F)= $\frac{mv^2}{r}$

$v = r\omega$

where v= linear velocity

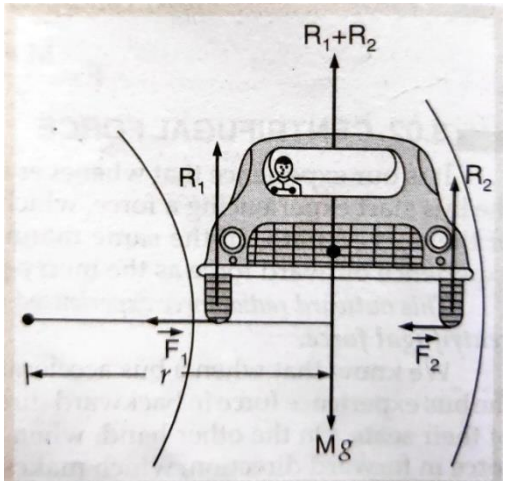
r= radius of the circular path

ω = angular velocity

putting in above equation

Centripetal force(F)= $\frac{m(r\omega)^2}{r} = mr\omega^2$

when a body is taking a circular turn on a level road. While taking the round, the body try to leave the path and go away from the center of the curve.

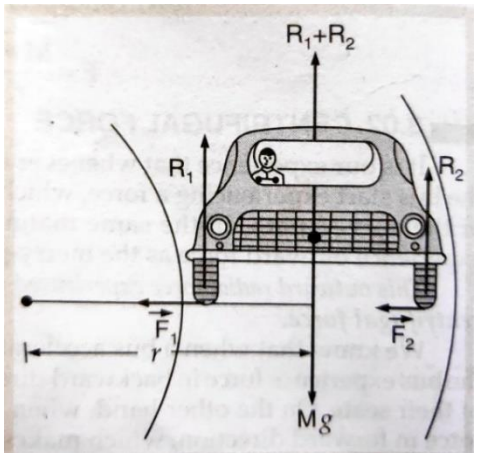


The total force of friction provides the necessary centripetal force
$F_1 + F_2 = mr\omega^2$
where F_1 *and* F_2 *are the forces of friction*
$F_1 = \mu R_1$
$F_2 = \mu R_2$
$\mu R_1 + \mu R_2 = mr\omega^2$
$\mu(R_1 + R_2) = mr\omega^2$------(1)
Where
R_1 *and* R_2 *are the normal recations*
μ = coefficient of friction
the total normal reaction on the body balances the weight(mg) of the body
$R_1 + R_2 = mg$ (g= acceleration due to gravity = 9.8 m/s^2 or 10 m/s^2)
Putting the value in equation 1
$\mu\, mg = mr\omega^2$
$\mu g = r\omega^2$--------(2)
equation 2 gives the maximum angular velocity/minimum radius of curvature with which body can take the circular turn of radius r(without skidding)
given: coefficient of friction $\mu = 0.5$
radius of circular path(r)= 5cm =0.05 m
putting the values in equation 2
$0.5 \times 10 = 0.05 \times \omega^2$
$\frac{0.5x10}{0.05} = \omega^2$
$\frac{5x10x100}{5x10} = \omega^2$
$100 = \omega^2$
$(10)^2 = \omega^2$
10 rad/s $= \omega$
Angular velocity = 10 rad/s

12. Correct option: (D)

Coin will slip when
$\mu g = r\omega^2$--------(1)
where μ = coefficient of friction
g= acceleration due to gravity = 10 m/s^2
ω= angular velocity
r= radius of the circular path
given: radius (r)= 9cm
putting the values in equation 1
$\mu g = 9 \times \omega^2$------(2)
to find: radius when the angular velocity is tripled
$\omega = 3\omega$
putting the values in equation 1
$\mu g = r \times (3\omega)^2$----------(3)
μg is constant
equating equation 2 and 3
$9\omega^2 = rx(3\omega)^2$
$9\omega^2 = rx\, 9\omega^2$
1 cm = r
So, the coin will just slip when radius from the centre = 1cm

13. Correct option: (A)



when a car is taking a circular turn on a level road. While taking the round, the tyres of the vehicles try to leave the road and go away from the centre of the curve.

The total force of friction provides the necessary centripetal force

$F_1 + F_2 = \frac{mv^2}{r}$

$F_1 = \mu R_1$

$F_2 = \mu R_2$

where F_1 *and* F_2 *are the forces of friction* , $r =$ *radius of the circular turn*

$\mu R_1 + \mu R_2 = \frac{mv^2}{r}$

$\mu(R_1 + R_2) = \frac{mv^2}{r}$----------(1)

Where

R_1 *and* R_2 *are the normal recations*

μ = coefficient of friction

the total normal reaction on the van balances the weight(mg) of the van

$R_1 + R_2 = mg$ (g= acceleration due to gravity = 9.8 m/s^2 or 10 m/s^2)

Putting the value in equation 1

$\mu\, mg = \frac{mv^2}{r}$

$\mu\, g = \frac{v^2}{r}$--------(2)

equation 2 gives the maximum velocity/minimum radius of curvature with which car can take the circular turn of radius r(without skidding)

given: coefficient of static friction $\mu_s = 2$

radius of the circular path = 45m

To find: maximum speed of the car

Putting the values in equation 2

$2x\,10 = \frac{(v)^2}{45}$

$20\,x\,45 = (v)^2$

$900 = (v)^2$

$(30)^2 = (v)^2$

v= 30 m/s

hence, maximum speed of the car = 30m/s= 108 km/h

$1m/s = \frac{18}{5}$ km/h

$30m/s = 30\,x\frac{18}{5}$ km/h

30m/s= 108 km/h

14. Correct option: (C)

Given: velocity(v) = 36km/h= 10m/s

$1\text{km/h} = \frac{5}{18}\text{ m/s}$

$36\text{km/h} = 36\text{x}\ \frac{5}{18}\text{ m/s}$

$= 10\text{m/s}$

Radius of the path(r)= 30m

To find: coefficient of friction

$\mu g = \frac{v^2}{r}$----------(1)

where v= linear velocity of the scooter

r= radius of the circular path

μ= coefficient of friction

g= acceleration due to gravity = 9.8 m/s^2 or 10 m/s^2)

equation 1 gives the maximum velocity/minimum radius of curvature with which scooter can take the circular turn of radius r(without skidding)

putting the values in equation 1

$\mu \text{ x } 10 = \frac{(10)^2}{30}$-

$\mu = \frac{100}{30\text{x } 10}$-

$\mu = 0.33$

hence, coefficient of friction = 0.33

15. Correct option: (C)

$\mu g = \frac{v^2}{r}$-

$v_{max}^2 = r \text{ x } \mu g$----------(1)

where v= linear velocity of the car

r= radius of the circular path

μ= coefficient of friction

g= acceleration due to gravity = 9.8 m/s^2 or 10 m/s^2)

equation 1 gives the maximum velocity/minimum radius of curvature with which car can take the circular turn of radius r(without skidding)

modifying equation 1

$v_{max} = \sqrt{\mu rg}$-------(2)

Given: $v_{max} = 20m/s$

$20 = \sqrt{\mu rg}$-------(3)

When the road becomes wet

$\mu = \frac{\mu}{2}$

$v'_{max} = \sqrt{\frac{\mu}{2} rg}$

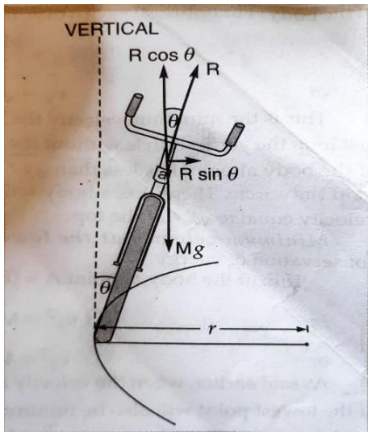
$v'_{max} = \frac{\sqrt{\mu rg}}{\sqrt{2}}$

Putting the value of $\sqrt{\mu rg}$ in equation 4 from equation 3

$v'_{max} = \frac{20}{\sqrt{2}} = 10\sqrt{2}\text{ m/s}$

Hence, the maximum speed without deviating the car = $10\sqrt{2}$ m/s

16. Correct option: (B)



Tangent of the angle made by the cyclist with the vertical is given by the equation below

$\tan\theta = \frac{v^2}{rg}$----------(1)

where θ= angle made by the cyclist with the vertical

v= linear velocity of the cyclist

r= radius of the circular path

g= acceleration due to gravity = 9.8 m/s^2 or 10 m/s^2

given: speed of the cyclist = $10\sqrt{2}$ m/s

radius of the circular path =20m

putting the values in equation 1

$\tan\theta = \frac{\left(10\sqrt{2}\right)^2}{20 \text{ x } 10} = \frac{100\text{x } 2}{200} = 1$

$\tan\theta = \tan 45^\circ$

so, $\theta = 45^\circ$

hence, cyclist makes an angle of 45° with the vertical.

17. Correct option: (C)

Given: angular velocity (ω)= 40rev/min

$1\text{rpm} = \frac{\pi}{30}\text{rad/s}$

$40\text{rpm} = \frac{\pi}{30}\text{x } 40 \text{ rad/s}$

$= \frac{4}{3}\pi \text{ rad/s}$

Mass(m)= 0.25kg

Radius of the circular motion(r)= 1.5m

To find: tension in the string

The tension provides the necessary Centripetal force(F)= $mr\omega^2$

$T = mr\omega^2$

Putting the values in the above equation

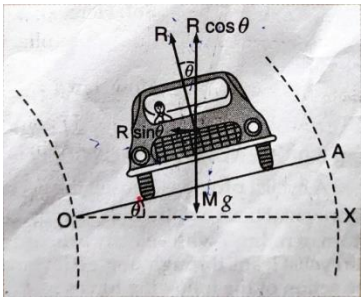
$T = 0.25\text{x}1.5\text{x}\left(\frac{4}{3}\pi\right)^2$ (given :$\pi^2 \simeq 10$)

$T = 0.375 \text{ x}\frac{16}{9} \text{ x } 10$

T = 6.66 N

Hence, tension in the string = 6.66 N

18. Correct option: (D)



Tangent of the angle of banking for a speed v and a curved track of radius r is given by the given below equation

$\tan\theta = \frac{v^2}{rg}$----------(1)

where θ= angle of banking

v= linear speed

r= radius of the circular path

g= acceleration due to gravity = 10 m/s^2

given: speed (v) =90km/h=25m/s

$1km/h = \frac{5}{18} m/s$

$90km/h = 90x \frac{5}{18} m/s$

$= 25m/s$

To find: radius of the circular path

putting the values in equation 1

$\tan45^\circ = \frac{(25)^2}{rx10} = \frac{625}{10x\,r}$

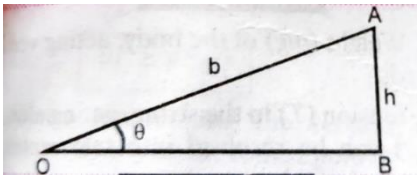
$1 = \frac{625}{10x\,r}$

$(\tan45^\circ = 1)$

$r = \frac{625}{10}$=62.5m

hence, radius of the circular path = 62.5 m

19. Correct option: (B)



Tangent of the angle of banking for a speed v and a curved track of radius r is given by the given below equation

$\tan\theta = \frac{v^2}{rg} = \frac{h}{b}$----------(1)

where h = height of the outer rail

b = distance between the rails

θ= angle of banking

v= linear speed

r= radius of the circular path

g= acceleration due to gravity = 9.8 m/s^2

given: speed (v) =48km/h=13.3m/s

$1km/h = \frac{5}{18} m/s$

$48km/h = 48x \frac{5}{18} m/s$

$= 13.3 m/s$

b = 1m

radius(r)= 400m

To find: height(h) of the outer rail

$\frac{v^2}{rg} = \frac{h}{b}$--------(2)

Or

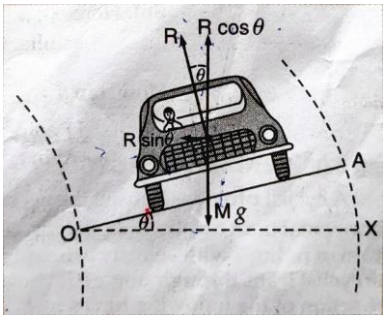
$h = \frac{b \times v^2}{rg}$----------(3)

putting the values in equation 3

$h = \frac{1 \times (13.3)^2}{400 \times 9.8} = 0.045$ m

hence, the outer rail must be raised by 0.045 m

20. Correct option: (C)



Tangent of the angle of banking for a speed v and a curved track of radius r is given by the given below equation

$\tan\theta = \frac{v^2}{rg}$----------(1)

where θ= angle of banking

v= linear velocity of the car

r= radius of the circular path

g= acceleration due to gravity = 9.8 m/s^2 or 10 m/s^2

given: speed of the car = $10\sqrt{10}$ m/s

radius of the circular path =$100\sqrt{3}$m

to find: angle of inclination

putting the values in equation 1

$\tan\theta = \frac{\left(10\sqrt{10}\right)^2}{100\sqrt{3} \times 10} = \frac{100 \times 10}{1000\sqrt{3}} = \frac{1}{\sqrt{3}}$

$\tan\theta = \tan 30^\circ$

so, θ= 30˚

hence, the angle of inclination = 30˚

21. Correct option: (B)

Tangent of the angle of banking for a speed v and a curved track of radius r is given by the given below equation

If the friction is not used

$\tan\theta = \frac{v^2}{rg}$----------(1)

where θ= angle of banking

v= linear speed

r= radius of the circular path

g= acceleration due to gravity = 10 m/s^2

given: radius (r)=50m

angle of banking(θ)= 30˚

To find: speed(v)
putting the values in equation 1

$$\tan 30^\circ = \frac{(v)^2}{50x10}$$

$$\frac{1}{\sqrt{3}} = \frac{(v)^2}{50x\,10}$$

$$(\tan 30^\circ = \frac{1}{\sqrt{3}}$$

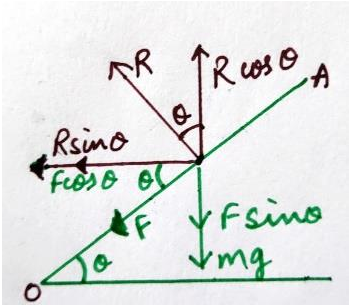
$$(v)^2 = \frac{500}{\sqrt{3}} = \frac{500}{1.73} = 289$$

$$(v)^2 = (17)^2$$

$$v=17\ m/s$$

hence, speed = 17 m/s

22. Correct option: (A)



The maximum permissible speed for the vehicle is much greater than the optimum value of the speed on a banked road. It is because, friction between road and the tyres of the vehicle also contributes to the required centripetal force.
As the vehicle moves along the circular banked road OA, force of friction between road and tyres of the vehicle i.e.
F= μR
acts in the direction AO.
The force of friction can be resolved into two components:
(i) μ R sin θ along vertical (in downward direction) and
(ii) μ R cos θ along horizontal (in inward direction)
Since there is no motion along vertical,

$$R\cos\theta = Mg + \mu R\sin\theta \text{---------(1)}$$

Let v_{max} be the maximum permissible speed of the vehicle. The centripetal force is now provided by the components R sin θ and μ R cosθ

$$R\sin\theta + \mu R\cos\theta = \frac{Mv_{max}^2}{r} \text{--------(2)}$$

From the equation 1 we have,

$$Mg = R\cos\theta(1 - \mu\tan\theta) - - - -(3)$$

Also, from the equation 2 we have

$$\frac{Mv_{max}^2}{r} = R\cos\theta\,(\mu + \tan\theta) \text{--------(4)}$$

Dividing the equation 4 by 3 , we have

$$\frac{v_{max}^2}{gr} = \frac{\mu+\tan\theta}{1-\mu\tan\theta}$$

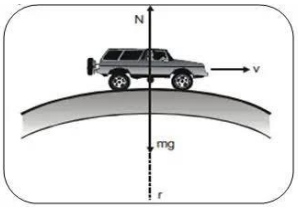
$$v_{max} = \left[\frac{gr(\mu+\tan\theta)}{1-\mu\tan\theta}\right]^{1/2}$$

$$v_{max} = \sqrt{rg\left[\frac{(\mu+\tan\theta)}{1-\mu\tan\theta}\right]}$$

Vertical Circular motion:

23. Correct option: (C)

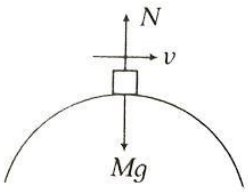
(A) Horizontal level bridge



weight of the body provides the necessary centripetal force. Therefore,

$N = mg = \frac{mv^2}{r}$--------(1)

(B) Convex bridge

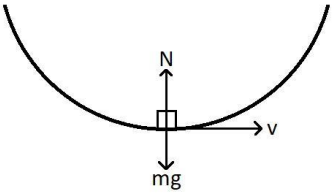


The resultant of weight of the body and the normal reaction of the bridge provides the necessary centripetal force. Therefore,

$mg - N = \frac{mv^2}{r}$

$N = mg - \frac{mv^2}{r}$--------(2)

(C) Concave bridge



The resultant of weight of the body and the normal reaction of the bridge provides the necessary centripetal force. Therefore,

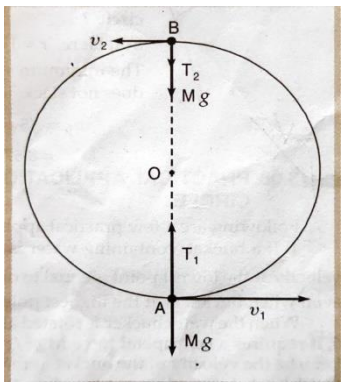
$N - mg = \frac{mv^2}{r}$

$N = mg + \frac{mv^2}{r}$--------(3)

From equation 1, 2 and 3

We can see that the normal reaction in case of concave bridge is the highest.

24. Correct option: (A)



At the lowest point A, a part of the tension T_1 balances the weight of the body and the remaining part provides the necessary centripetal force. Therefore,

$T_1 - mg = \frac{mv_1^2}{r}$

At the highest point, the tension in the string and the weight of the body together provide the necessary centripetal force. Hence,

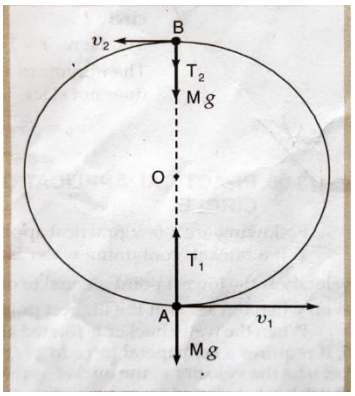
$T_2 + mg = \frac{mv_2^2}{r}$

Actual forces acting on the particle are mg and T

The resultant of mg and T will provide the necessary centripetal force.

Centrifugal force acting outwards on the particle is not a real force.

25. Correct option: (A)



Consider a body of mass m tied at the end of a string and whirled in a vertical circle of radius r. Let v_1 and v_2 be velocities of the body and $T_1$and T_2 be tensions the string at the lowest point A and the highest point B respectively. The velocities the body at points A and B will be directed along tangents to the circular path at these points, while tensions in the string will always act towards the fixed point O.

At the lowest point A, a part of the tension T_1 balances the weight of the body and the remaining part provides the necessary centripetal force. Therefore,

$T_1 - mg = \frac{mv_1^2}{r}$----------(1)

At the highest point, the tension in the string and the weight of the body together provide the necessary centripetal force. Hence,

$T_2 + mg = \frac{mv_2^2}{r}$----------(2)

Let us now find the minimum velocity, the body should possess at the lowest point, so that the string does not slack when it is at the highest point. the body is then said to just loop the vertical circle. It is obvious that the velocity at the lowest point will be minimum , when the velocity at the highest point is also minimum.

velocity at the highest point will be minimum, when the tension in the string is zero i.e

$T_2 = 0$

In that case, the whole of the centripetal force will be provided by the weight of the body. Therefore, in such a case, the equation 2 becomes

$$0 + mg = \frac{mv_2^2}{r}$$

Or

$$v_2 = \sqrt{gr}$$

This is the minimum velocity the body should possess at the top so that it can just loop the vertical circle without the slackening of the string. In case, the velocity of the body at point B is less than $\sqrt{gr}$, the string will slack and the body will not loop the circle. Therefore, a body will just loop the vertical circle, if it possesses velocity equal to $\sqrt{gr}$ at the top.

Minimum velocity at the lowest point:

According to the principle of conservation of energy,

K.E. of the body at point A = (P.E.+K.E.) of the body at point B

i.e $\frac{1}{2}mv_1^2 = mg(2r) + \frac{1}{2}mv_2^2$

$v_1^2 = 4gr + v_2^2$

As said earlier, when the velocity at the highest point is minimum, then velocity at the lowest point will also be minimum. Setting $v_2 = \sqrt{gr}$, we have

$v_1^2 = 4gr + gr$

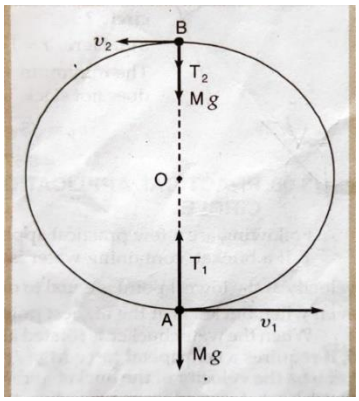
$v_1 = \sqrt{5gr}$----------(3)

The equation gives the magnitude of the velocity at the lowest point with which body can safely go round the vertical circle of radius r or can loop the circle of radius r.

As we can see from the equation 3

The minimum horizontal velocity to be given to the stone at the lowest point depends on the length of the string(r)

26. Correct option: (C)



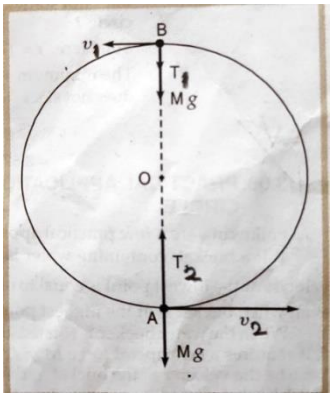
Consider a body of mass m tied at the end of a string and whirled in a vertical circle of radius r. Let v_1 and v_2 be velocities of the body and $T_1$and T_2 be tensions the string at the lowest point A and the highest point B respectively. The velocities the body at points A and B will be directed along tangents to the circular path at these points, while tensions in the string will always act towards the fixed point O.

At the lowest point A, a part of the tension T_1 balances the weight of the body and the remaining part provides the necessary centripetal force. Therefore,

$T_1 - mg = \frac{mv_1^2}{r}$----------(1)

Net force acting in the lowest position is T_1-mg.

27. Correct option: (D)



Consider a body of mass m tied at the end of a string and whirled in a vertical circle of radius r. Let v_2 and $v_1$be velocities of the body and T_2 and T1 be tensions the string at the lowest point A and the highest point B respectively. The velocities the body at points A and B will be directed along tangents to the circular path at these points, while tensions in the string will always act towards the fixed point O.

At the lowest point A, a part of the tension T2 balances the weight of the body and the remaining part provides the necessary centripetal force. Therefore,

$$T_2 - mg = \frac{mv_2^2}{r}$$

$$T_2 = \frac{mv_2^2}{r} + mg \text{ ----------(1)}$$

At the highest point, the tension in the string and the weight of the body together provide the necessary centripetal force. Hence,

$$T_1 + mg = \frac{mv_1^2}{r}$$

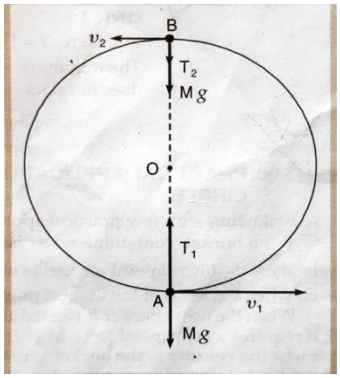
$$T_1 = \frac{mv_1^2}{r} - mg \text{----------(2)}$$

From equation 1 and 2 we can see

For the same speed the tension at the lowest point will be more than that at the highest point.

Hence, $T_1 < T_2$

28. Correct option: (B)



$$v_1 = \sqrt{5gr}$$

The equation gives the magnitude of the velocity at the lowest point with which body can safely go round the vertical circle of radius r or can loop the circle of radius r.

If $r' = \frac{r}{4}$

$$v_1' = \sqrt{\frac{5gr}{4}} = \frac{\sqrt{5gr}}{2} = \frac{v_1}{2}$$

Hence, velocity at the lowest point becomes half

29. Correct option: (D)

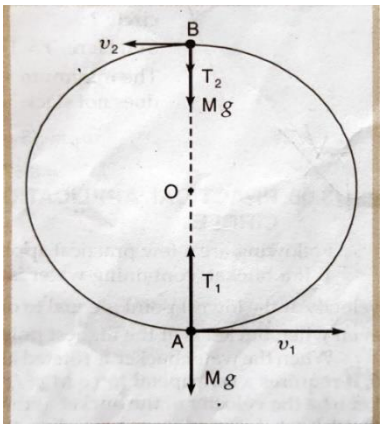
Given: maximum tension (T)= 140.6N

Mass(m)= 2kg

Radius of the circular motion(r)= length of the string = 2m

To find: maximum speed(v_{max}) at the lowest point

At the lowest point A, a part of the tension T_1 balances the weight of the body and the remaining part provides the necessary centripetal force. Therefore,



$T_1 - mg = \frac{mv_1^2}{r}$ ----------(1)

Speed will be maximum when the tension will be maximum

From equation 1

$T_{max} = \frac{mv_{max}^2}{r} + mg$

Putting the values in the above equation

$140.6 = \frac{2 \times v_{max}^2}{2} + 2 \times 9.8$ (g= acceleration due to gravity = 9.8 m/s^2)

$140.6 = v_{max}^2 + 19.6$

$140.6 - 19.6 = v_{max}^2$

$140.6 - 19.6 = v_{max}^2$

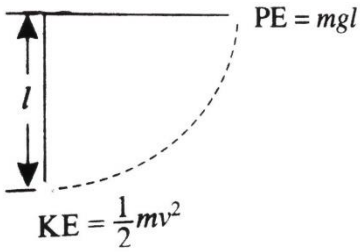
$121 = v_{max}^2$

$(11)^2 = v_{max}^2$

$11 = v_{max}$

Hence, maximum speed at the lowest point = 11 m/s

30. Correct option: (C)



According to the principle of conservation of energy,

Kinetic energy of the body at point A = potential energy of the body at height of suspension B

i.e $\frac{1}{2} mv^2 = mgl$

or $v = \sqrt{2gl}$

where v = velocity of the ball

Length of the thread = l

Acceleration due to gravity = g

Hence, minimum horizontal velocity to be given to the ball = $\sqrt{2gl}$

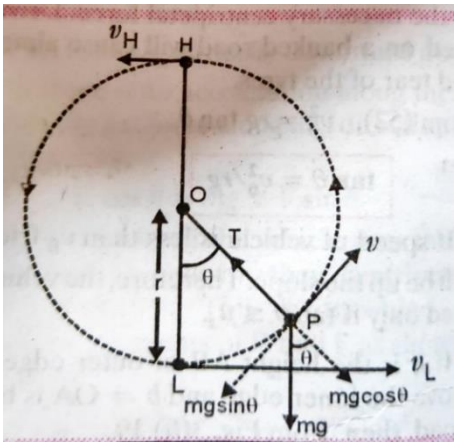
31. Correct option: (C)

Given: mass (m) = 5kg

Length of the string =radius (r) =1m

Vertical speed of the bob(v)= $4m/s$

Tension (T) in the string = 31 N



let us consider the motion of a small body tied to one end of a string and whirled in a vertical circle. At any time, let the body be at P at angular position θ. The forces acting on the body are

(i), Weight (mg) of the body, acting vertically downwards,

(ii) Tension (T) in the string, acting along PO. Now (mg) can be resolved into two rectangular components:

mg cos θ, opposite to T

mg sin θ; along tangent to the circle at P.

Net force on the body at P, acting along PO = $(T - mg\cos\theta)$.

This must provide the necessary centripetal force $(\frac{mv^2}{r})$ required by the body, where v is linear velocity of the body at P.

$(T - mg\cos\theta) = \frac{mv^2}{r}$

$T = \frac{mv^2}{r} + mg\cos\theta$----------(1)

Putting the given values in equation 1

$31 = \frac{5(4)^2}{1} + 5 \times 9.8x \cos\theta$ (g= $9.8 m/s^2$)

31= 5x16+ 49 cos θ

31= 80 +49 cosθ

31-80 = 49 cos θ

-49 = 49 cos θ

Cosθ = -1

Cosθ = cos 180˚

Θ = 180˚

Angle made with the vertical = 180˚

Hence, the body is at the highest point H

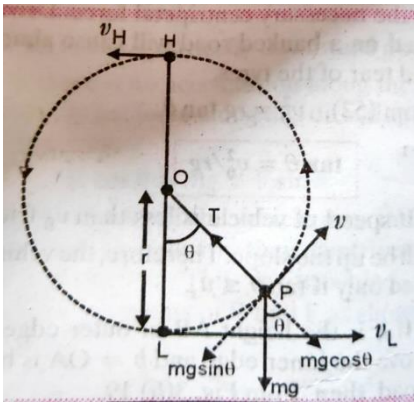
32. Correct option: (B)

Given: weight (mg) of the body = 10N

Length of the string = 1m

Angle θ made with the vertical downwards = 60 ˚

Vertical speed of the bob(v)= $\sqrt{9.8}m/s$



let us consider the motion of a small body tied to one end of a string and whirled in a vertical circle. At any time, let the body be at P at angular position θ. The forces acting on the body are

(i), Weight (mg) of the body, acting vertically downwards,

(ii) Tension (T) in the string, acting along PO. Now (mg) can be resolved into two rectangular components:

mg cos θ, opposite to T

mg sin θ; along tangent to the circle at P.

Net force on the body at P, acting along PO = $(T - mg\cos\theta)$.

This must provide the necessary centripetal force $(\frac{mv^2}{r})$ required by the body, where v is linear velocity of the body at P.

$(T - mg\cos\theta) = \frac{mv^2}{r}$

$T = \frac{mv^2}{r} + mg\cos\theta$----------(1)

Putting the given values in equation 1

$T = \frac{m(\sqrt{9.8})^2}{1} + 10 \times \cos 60^\circ$

$T = \frac{m\times 9.8}{1} + 10 \times \frac{1}{2}$

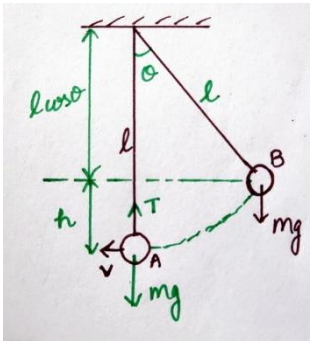
$T = mg + 5$ (given weight (mg) = 10N)

T = 10 +5

T= 15 N

Hence, tension in the string = 15N

33. Correct option: (C)



Given: breaking strength of the string = maximum tension = 2mg

To find: maximum angular amplitude from the vertical (θ)

From the given figure height h = l(1-cosθ)

According to the principle of conservation of energy,

Kinetic energy of the body at point A = potential energy of the body at point B

i.e $\frac{1}{2}\,mv^2 = mgh$

$\frac{1}{2}\,mv^2 = mgl(1-\cos\theta)$

$mv^2 = 2mgl(1-\cos\theta)$----------(1)

At lowest point A, the tension is maximum

So at point A

$T = \frac{mv^2}{l} + mg$

$T_{max} = 2mg = \frac{mv^2}{l} + mg$

Using value from equation 1

$2mg = \frac{2mgl(1-\cos\theta)}{l} + mg$

$2mg - mg = 2mg(1-\cos\theta)$

$Mg = 2mg(1-\cos\theta)$

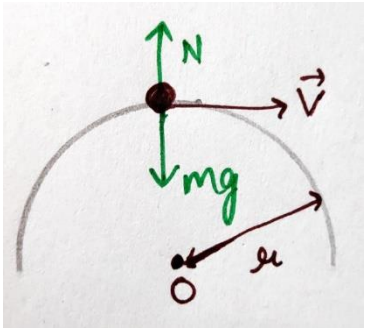
$\frac{1}{2} = 1-\cos\theta$

$\cos\theta = \frac{1}{2} = \cos 60^\circ$

$\theta = 60^\circ$

hence, maximum angular amplitude from the vertical = 60 °

34. Correct option: (A)



The resultant of weight of the body and the normal reaction of the hemisphere provides the necessary centripetal force. Therefore,

$mg - N = \frac{mv^2}{r}$-------(1)

The mass will not slide, when the normal reaction(N) =0

Equation 1 becomes

$mg = \frac{mv^2}{r}$

$g = \frac{v^2}{r}$

$v = \sqrt{gr}$----------(2)

given: radius (r)= 20 cm = 0.20 m

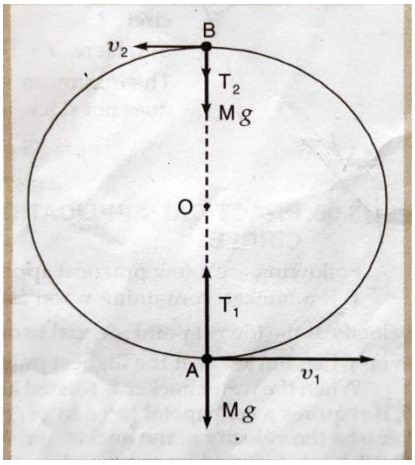
$v = \sqrt{9.8 \times 0.20}$

$v = \sqrt{1.96}$

$v = 1.4$ m/s

hence, the minimum velocity with which body will fly off = 1.4 m/s

35. Correct option: (A)



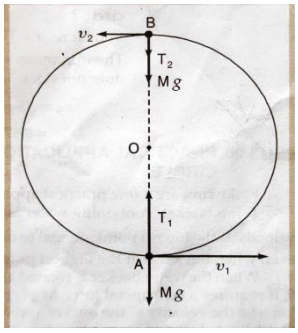
According to the principle of conservation of energy,
K.E. of the body at point A = (P.E.+K.E.) of the body at point B

i.e $\frac{1}{2}\,mv_1^2 = mg(2r) + \frac{1}{2}mv_2^2$

$\frac{1}{2}\,mv_1^2 - \frac{1}{2}mv_2^2 = mg(2r)$

$change\ in\ K.E = \frac{1}{2}\,mv_1^2 - \frac{1}{2}mv_2^2 = 2mgr$

36. Correct option: (B)



$v_2 = \sqrt{gr}$---------(1)

Equation 1 gives the minimum velocity the body should possess at the top so that it can just loop the vertical circle without the slackening of the string.

Minimum velocity at the lowest point:

According to the principle of conservation of energy,
K.E. of the body at point A = (P.E.+K.E.) of the body at point B

i.e $\frac{1}{2}\,mv_1^2 = mg(2r) + \frac{1}{2}mv_2^2$

$v_1^2 = 4gr + v_2^2$

As said earlier, when the velocity at the highest point is minimum, then velocity at the lowest point will also be minimum. Setting $v_2 = \sqrt{gr}$, we have

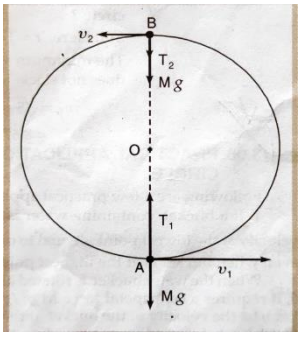
$v_1^2 = 4gr + gr$

$v_1 = \sqrt{5gr}$----------(2)

The equation 2 gives the magnitude of the velocity at the lowest point with which body can safely go round the vertical circle of radius r or can loop the circle of radius r.

Ratio $= \frac{v_1}{v_2} = \frac{\sqrt{5gr}}{\sqrt{gr}} = \sqrt{5}:1$

37. Correct option: (2mg)



Consider a body of mass m tied at the end of a string and whirled in a vertical circle of radius r. Let v_1 and v_2 be velocities of the body and T_1 and T_2 be tensions the string at the lowest point A and the highest point B respectively. The velocities the body at points A and B will be directed along tangents to the circular path at these points, while tensions in the string will always act towards the fixed point O.

At the lowest point A, a part of the tension T_1 balances the weight of the body and the remaining part provides the necessary centripetal force. Therefore,

$$T_1 - mg = \frac{mv_1^2}{r} \text{----------(1)}$$

At the highest point, the tension in the string and the weight of the body together provide the necessary centripetal force. Hence,

$$T_2 + mg = \frac{mv_2^2}{r} \text{----------(2)}$$

As the speed is constant so, the magnitude of centripetal force remains the same

So equation 1 and 2 become

$$T_1 - mg = \frac{mv^2}{r}-$$

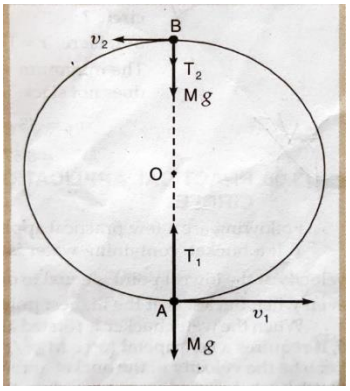
$$T_1 = \frac{mv^2}{r} + mg \text{----------(3)}$$

$$T_2 + mg = \frac{mv^2}{r}$$

$$T_2 = \frac{mv^2}{r} - mg \text{-----(4)}$$

Change in tensions = $T_1 - T_2 = \frac{mv^2}{r} + mg - \frac{mv^2}{r} + mg = 2mg$

38. Correct option: (C)



Consider a body of mass m tied at the end of a string and whirled in a vertical circle of radius r. Let v_1 and v_2 be velocities of the body and T_1 and T_2 be tensions the string at the lowest point A and the highest point B respectively. The velocities the body at points A and B will be directed along tangents to the circular path at these points, while tensions in the string will always act towards the fixed point O.

At the lowest point A, a part of the tension T_1 balances the weight of the body and the remaining part provides the necessary centripetal force. Therefore,

$$T_1 - mg = \frac{mv_1^2}{r}$$

At the highest point, the tension in the string and the weight of the body together provide the necessary centripetal force. Hence,

$$T_2 + mg = \frac{mv_2^2}{r}$$

Let us now find the minimum velocity, the body should possess at the lowest point, so that the string does not slack when it is at the highest point. the body is then said to just loop the vertical circle. It is obvious that the velocity at the lowest point will be minimum, when the velocity at the highest point is also minimum.

velocity at the highest point will be minimum, when the tension in the string is zero i.e

$$T_2 = 0$$

In that case, the whole of the centripetal force will be provided by the weight of the body. Therefore, in such a case, the equation becomes

$$0 + mg = \frac{mv_2^2}{r}$$

Or

$$v_2 = \sqrt{gr} \text{------(1)}$$

This is the minimum velocity the body should possess at the top so that it can just loop the vertical circle without the slackening of the string. In case, the velocity of the body at point B is less than $\sqrt{gr}$, the string will slack and the body will not loop the circle. Therefore, a body will just loop the vertical circle, if it possesses velocity equal to $\sqrt{gr}$ at the top.

Minimum velocity at the lowest point.

According to the principle of conservation of energy,

K.E. of the body at point A = (P.E.+K.E.) of the body at point B

i.e $$\frac{1}{2} mv_1^2 = mg(2r) + \frac{1}{2} mv_2^2$$

$$v_1^2 = 4gr + v_2^2$$

As said earlier, when the velocity at the highest point is minimum, then velocity at the lowest point will also be minimum. Setting $v_2 = \sqrt{gr}$, we have

$$v_1^2 = 4gr + gr$$

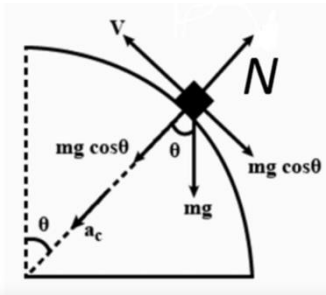
$$v_1 = \sqrt{5gr} \text{-------(2)}$$

The equation 2 gives the magnitude of the velocity at the lowest point with which body can safely go round the vertical circle of radius r or can loop the circle of radius r.

Hence, velocity of the body when the string is in horizontal position= $\sqrt{5gr}$

LEVEL-II

1. Correct option: (A)



the weight mg of the motorcycle can be resolved into two components:

(i) Mg cosθ acting towards the centre of the circular path

(ii) Mg sinθ acting along the tangent to the circular path

The component mg cosθ of the weight of the motorcycle and the normal reaction of the bridge provides the necessary centripetal force. Therefore,

$Mg \cos \theta - N = \frac{Mv^2}{r}$

$N = Mg\cos\theta - \frac{Mv^2}{r}$-------(1)

as the motorcycle ascends over the bridge, the angle of inclination decreases and becomes zero on the top of the bridge

as a result, cosθ will keep on increasing and value of the normal reaction N will also increase.

2. Correct option: (C)

Given: maximum tension (T)= 25 N

Mass(m)= 0.5kg

Radius of the circular motion(r)= length of the string = 0.5m

To find: maximum linear velocity(v_{max})

The tension provides the necessary Centripetal force(F)= $\frac{mv_{max}^2}{r}$

$T_{max}=\frac{mv_{max}^2}{r}$

Putting the values in the above equation

$25=\frac{0.5x\ v_{max}^2}{0.5}$

$25= v_{max}^2$

$(5)^2 = v_{max}^2$

$5 = v_{max}$

Hence, maximum linear speed = 5m/s

3. Correct option: (B)

Given: Radius of the circular motion(r) =1km =1000m

Speed(v)= 900km/h=250m/s

$1km/h = \frac{5}{18}\ m/s$

$900km/h = 900x\ \frac{5}{18}\ m/s$

$= 250m/s$

To find: centripetal acceleration

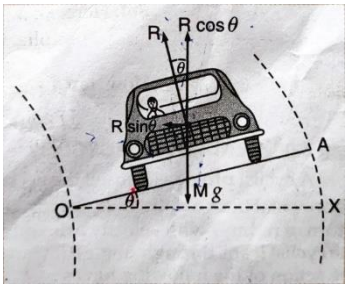
centripetal acceleration $(a_c) = \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

centripetal acceleration $(a_c) = \frac{(250)^2}{1000} = \frac{250x250}{1000}$=62.5 m/s^2

4. Correct option: (B)

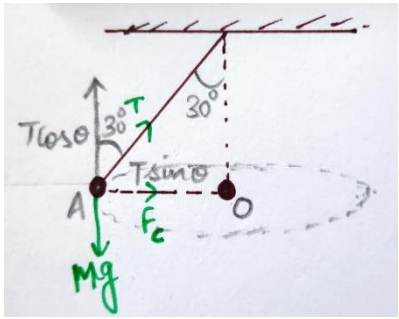


Consider a vehicle of weight Mg moving round a curved path of radius r with speed v on a road banked through angle θ. If OA is banked road and OX is a horizontal line, then angle AOX = θ is called angle of banking.

The vehicle is under the action of following forces:

(i) The weight Mg acting vertically downward i.e. perpendicular to the horizontal line OX.
(ii) The reaction R of the ground to the vehicle acting along normal to the banked road OA in upward direction.
The vertical component R cos θ of normal reaction R will balance the weight of the vehicle and the horizontal component R sin θ will provide the necessary centripetal force to the vehicle. Thus,
R cos θ = Mg--------(1)
R sin θ = $\frac{Mv^2}{r}$--------(2)
Dividing equation 2 by 1
$\frac{R\sin\theta}{R\cos\theta} = \frac{\frac{Mv^2}{r}}{Mg}$
$\tan\theta = \frac{v^2}{rg}$
$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$
Hence, the minimum angle of the road to be banked to take a safe turn =$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$

5. Correct option: (D)



it is acted upon by the following forces:
(i) The weight Mg of the suspended mass acting vertically downward.
(ii) Tension T in the string acting along the length of the string towards the point of suspension. The tension T in the string can be resolved into two components Le. T sinθ along AO and T cos θ along vertically upwards. The component T cos θ balances the weight M g of the suspended body and the component T sinθ provides the necessary centripetal force to move it along horizontal circular path. Thus,
T cos θ = Mg--------(1)
T sin θ = $\frac{Mv^2}{r}$--------(2)
Given: angle made with the vertical θ= 30˚
Radius(r)= $40\sqrt{3}m$
Dividing equation 2 by 1
$\frac{T\sin\theta}{T\cos\theta} = \frac{Mv^2}{rxMg}$
$\tan\theta = \frac{v^2}{rg}$
speed (v) = $\sqrt{gr\tan\theta}$
speed (v) = $\sqrt{10x40\sqrt{3}\tan30°}$ (g= 10 m/s^2)
$= \sqrt{400\sqrt{3}\,x\frac{1}{\sqrt{3}}}$
$=\sqrt{400}$

Speed (v) = 20 m/s
Hence, speed of the train = 20m/s

6. Correct option: (B)

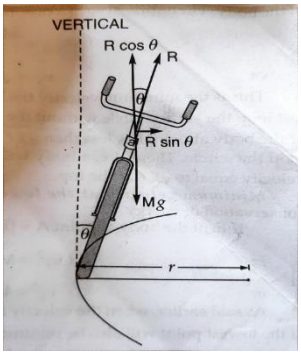
Given : mass of the man(m) = 70 kg
Radius of the circular path = 3m
Coefficient of friction μ = 0.30
Acceleration due to gravity(g)= 10 m/s^2
The normal reaction provides the necessary centripetal force;
Normal reaction (R) = $mr\omega^2$----------(1)
Where ω = angular speed
For equilibrium
$f_s = mg$
$\mu R = mg$
from equation 1
$\mu mr\omega^2 = mg$
$\mu r\omega^2 = g$

$$\omega = \sqrt{\frac{g}{\mu r}}$$

$$\omega = \sqrt{\frac{10}{0.30 \text{x } 3}} = \sqrt{\frac{1000}{90}} = 3.33\ rad/s$$

hence, minimum angular speed = 3.33 rad/s

7. Correct option: (B)



Tangent of the angle made by the cyclist with the vertical is given by the equation below

$$\tan\theta = \frac{v^2}{rg}$$----------(1)

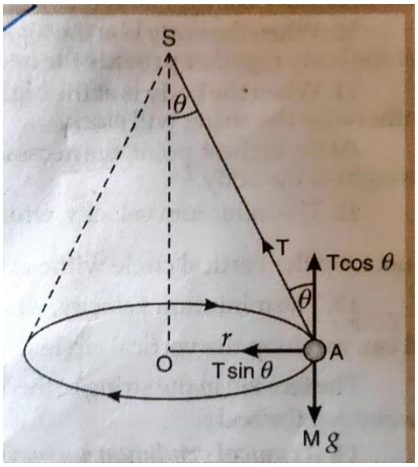
where θ= angle made by the cyclist with the vertical
v= linear velocity of the cyclist
r= radius of the circular path
g= acceleration due to gravity = 9.8 m/s^2 or 10 m/s^2
given: speed of the cyclist = $14\sqrt{3}$ m/s
radius of the circular path =$20\sqrt{3}$m
putting the values in equation 1

$$\tan\theta = \frac{\left(14\sqrt{3}\right)^2}{20\sqrt{3} \text{ x } 9.8} = \frac{14 \text{ x} 14 \text{ x } 3}{20\sqrt{3} \text{ x } 9.8} = \sqrt{3}$$

$\tan\theta = \tan 60^\circ$
so, $\theta = 60^\circ$
Hence, cyclist makes an angle of 60° with the vertical.

8. Correct option: (A)

A conical pendulum is a small heavy mass suspended by a string from a rigid support and whirled round in a horizontal circle with a constant speed.



Consider a heavy mass M tied to a string and suspended from a rigid support S. The mass is whirled round in a horizontal circle of radius r with a constant speed v, such that the string SA makes an angle θ with the vertical as shown in the above figure. When the mass is in position A, it is acted upon by the following forces:

(i) The weight Mg of the suspended mass acting vertically downward.

(ii) Tension T in the string acting along the length of the string towards the point of suspension. The tension T in the string can be resolved into two components Le. T sinθ along AO and T cos θ along vertically upwards. The component T cos θ balances the weight M g of the suspended body and the component T sinθ provides the necessary centripetal force to move it along horizontal circular path. Thus,

$T \cos \theta = Mg$--------(1)

$T \sin \theta = Mr\omega^2$-------(2)

Dividing equation 2 by 1

$$\frac{T \sin \theta}{T \cos \theta} = \frac{Mr\omega^2}{Mg}$$

$$\tan \theta = \frac{r\omega^2}{g}$$

$$\text{angular speed } (\omega) = \sqrt{\frac{g \tan\theta}{r}}$$

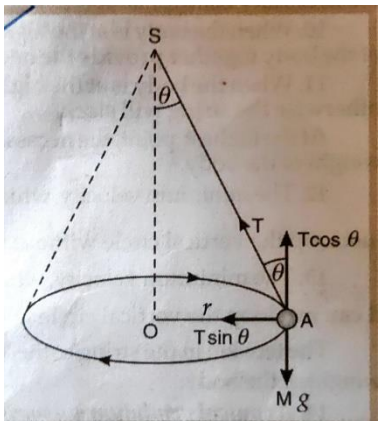
Therefore, time period of the suspended body (conical pendulum) is given by

$$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{g \tan\theta}{r}}}$$

$$\text{Time period}(T) = 2\pi\sqrt{\frac{r}{g\tan\theta}}$$

9. Correct option: ()

As the particle does not have any motion along the vertical



T cosθ =mg=constant

$T_1\cos\theta_1 = T_2\cos\theta_2$-------(1)

$\theta_1 = 45^\circ$

$\theta_2 = 60^\circ$

Putting the values in equation 1

$T_1\cos45° = T_2\cos60^\circ$

$$\frac{T_1}{T_2} = \frac{\cos60°}{\cos45^\circ} = \frac{\frac{1}{2}}{\frac{1}{\sqrt{2}}} = \frac{1}{\sqrt{2}}$$

10. Correct options: (A), (B), (C)

• In a uniform circular motion, the **linear speed remains constant, its linear velocity is not constant**. It is because, the linear velocity of the object at any point of its circular path is directed along the tangent to the path and therefore its direction keeps on changing continuously. The change of linear velocity results in acceleration, which is always directed along the radius of the circle and towards its centre. Thus, uniform circular motion is a case of accelerated motion.

• For a uniform circular motion

the angular velocity is constant.

So, angular acceleration $\alpha = \frac{d\omega}{dt} = 0$

As Tangential/linear acceleration$(a_T) = \alpha r = 0$

centripetal acceleration $(a_c) = \frac{v^2}{r}$ is constant.

$a_c \neq 0$, $a_T = 0$

Magnitude of Net acceleration is constant.

centripetal acceleration $(a_c) = \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

according to newton's second law of motion

F= m x a

Centripetal force(F)= $\frac{mv^2}{r}$ is also constant in magnitude.

11. Correct options(A), (B), (C), (D)

When an object is moving in a circular motion, its total acceleration consists of two components

• Radial component (centripetal acceleration) $a_c = \frac{v^2}{r}$ (where v= linear velocity, r= radius of the circle)

• Transverse component of the acceleration (tangential/linear acceleration)

a_T or a $= r\alpha$ (Where r= radius, α = angular acceleration)

Magnitude of the resultant acceleration = $\sqrt{(a_c)^2 + (a_T)^2}$

(A) For uniform translation, there is no change in the velocity, hence the acceleration is zero.

$a_c = 0$, $a_T = 0$, $a = 0$

(B) In case of accelerated translatory motion the tangential/linear acceleration is non zero

$a_t \neq 0$, $a_c = 0$, $a = a_t$

(C) For a uniform circular motion

the angular velocity is constant.

So, angular acceleration $\alpha = \frac{d\omega}{dt} = 0$

As Tangential/linear acceleration$(a_T) = \alpha r = 0$

centripetal acceleration $(a_c) = \frac{v^2}{r}$ is constant.

$a_c \neq 0$, $a_T = 0$

Magnitude of the resultant acceleration(a) = $\sqrt{(a_c)^2} = a_c$

(D) For a non-uniform circular motion radial component and the tangential component both are non zero.

$a_c \neq 0$, $a_T \neq 0$

Magnitude of the resultant acceleration(a) = $\sqrt{(a_c)^2 + (a_T)^2}$

12. Correct option: (A), (B), (D)

• A body in circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration. The force acting towards the centre/inwards is called the centripetal force.

centripetal acceleration $(a_c) = \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

(b) according to newton's second law of motion

F= m x a

Centripetal force(F)= $\frac{mv^2}{r}$

• Centrifugal force is not a real force. It is called fictitious or pseudo force because its origin cannot be identified. It arises in the non inertial frame of reference and acts in the outwards direction. For example. When a bus takes a turn, the passengers in the bus participate in rotation and experience centrifugal force outwards.

• For a uniform circular motion

the angular velocity is constant.

So, angular acceleration $\alpha = \frac{d\omega}{dt} = 0$

As Tangential/linear acceleration$(a_T) = \alpha r = 0$

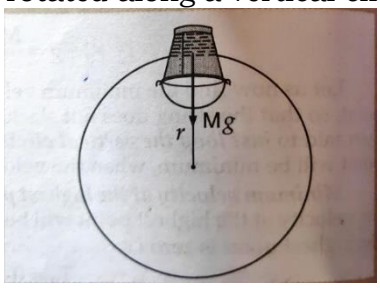
centripetal acceleration $(a_c) = \frac{v^2}{r}$ is constant.

Centripetal force = ma= $\frac{mv^2}{r}$ is constant.

Hence magnitude of centripetal force is constant only in uniform circular motion otherwise it will change if the motion is non-uniform.

13. Correct options: (A), (B), (C)

• If a bucket containing water is rotated along a vertical circle such that its

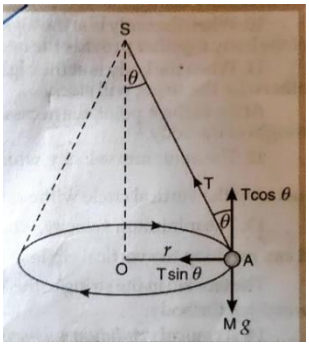


velocity at the lowest point is equal to or greater than $\sqrt{5gr}$, the water will not spill, even when bucket is at the highest point .

• When the water bucket is rotated along a vertical circle of radius r with velocity v, it requires a centripetal force $\frac{Mv^2}{r}$, where m is the mass of the water bucket. Let v be the velocity of the bucket at the top of the circular path.

• The weight of the water bucket is used up in providing the necessary centripetal force and hence the water does not spill.

14. Correct options(A), (C), (D)



A conical pendulum is a small heavy mass suspended by a string from a rigid support and whirled round in a horizontal circle with a constant speed.

Consider a heavy mass M tied to a string and suspended from a rigid support S. The mass is whirled round in a horizontal circle of radius r with a constant speed v, such that the string SA makes an angle θ with the vertical as shown in the above figure. When the mass is in position A, it is acted upon by the following forces:

(i) The weight Mg of the suspended mass acting vertically downward.

(ii) Tension T in the string acting along the length of the string towards the point of suspension. The tension T in the string can be resolved into two components Le. T sinθ along AO and T cos θ along vertically upwards. The component T cos θ balances the weight M g of the suspended body and the component T sinθ provides the necessary centripetal force to move it along horizontal circular path. Thus,

$T \cos \theta = Mg$--------(1)

$T \sin \theta = Mr\omega^2$-------(2)

$\frac{T \sin \theta}{T \cos \theta} = \frac{Mr\omega^2}{Mg}$

$\tan \theta = \frac{r\omega^2}{g}$

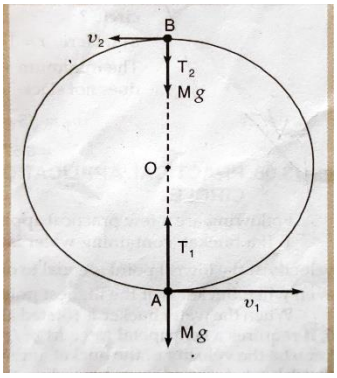
angular speed $(\omega) = \sqrt{\frac{g \tan\theta}{r}}$

Therefore, time period of the suspended body (conical pendulum) is given by

$T = \frac{2\pi}{\omega} = \frac{2\pi}{\sqrt{\frac{g \tan\theta}{r}}}$

Time period $(T) = 2\pi \sqrt{\frac{r}{g \tan\theta}}$

15. Correct option: ()



Given: radius of the circular path = 0.5 m

Acceleration due to gravity (g) = 9.8m/s^2

$v_2 = \sqrt{gr}$------(1)

This is the minimum velocity the body should possess at the top so that it can just loop the vertical circle without the slackening of the string.

Putting the given values in equation 1

$v_2 = \sqrt{9.8x0.5}$

$v_2 = \sqrt{4.9}$

$v_2 = 2.21$ m/s

Hence, minimum speed at the highest point = 2.21 m/s

16. Correct option: ()

$v_1 = \sqrt{5gr}$--------(2)

The equation gives the magnitude of the velocity at the lowest point with which body can safely go round the vertical circle of radius r or can loop the circle of radius r.

Putting the given values in equation 2

$v_2 = \sqrt{5x9.8x0.5} = \sqrt{24.5} = 4.9$ m/s

Hence, minimum speed at the lowest point = 4.9 m/s

17. Correct option: ()

At the lowest point A, a part of the tension T_1 balances the weight of the body and the remaining part provides the necessary centripetal force. Therefore,

$T_1 - mg = \frac{mv_1^2}{r}$----------(3) $v_1 = \sqrt{5gr}$

The equation gives the magnitude of the velocity at the lowest point with which body can safely go round the vertical circle of radius r or can loop the circle of radius r.

setting $v_1 = \sqrt{5gr}$

we have

$$T_1 - mg = \frac{m \times (\sqrt{5gr})^2}{r} = 5\ mg$$

$$T_1 = 6\ mg$$

Putting the given values

$T_1 = 6x\ 1\ x\ 9.8 = 58.8$ N

Hence, tension in the string at the lowest point = 58.8 N

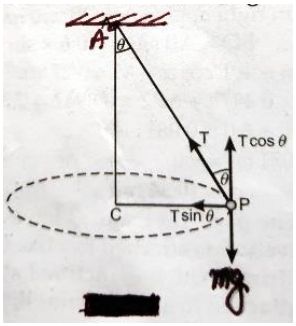
18. Correct option: (D)

Given: mass of the particle = 0.15 kg

length of the string= 2m

Acceleration (a)=10m/s^2

Let T be the tension in the strong. It can be resolved into components along the horizontal and vertical as below:



(i) T sinθ along the horizontal(radially inward).
(ii) T cosθ along the vertically upwards.
As the particle does not have any motion along the vertical
T cosθ =mg
Or T cosθ =mg
T cosθ = 0.15 x 10 ($g = 10m/s^2$)
T cosθ =1.5----------(1)
The component T sinθ provides the necessary centripetal force
T sinθ= ma
T sinθ= 0.15x 10
T sinθ= 1.5 ----------(2)
Dividing equation 2 by 1
$\tan\theta = \frac{1.5}{1.5} = 1 = \tan 45^\circ$
θ=45°

19. Correct option: (B)
From equation 1
T cos 45° =1.5
$T \times \frac{1}{\sqrt{2}} = 1.5$
$T = 1.5\sqrt{2}$ N

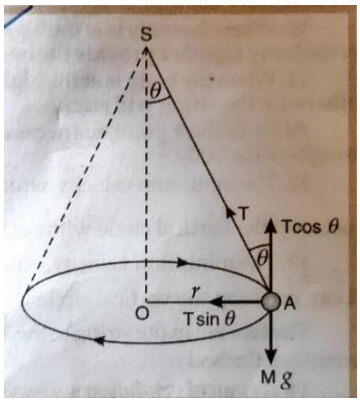
20. Correct option: ()
let the speed of the particle = v
radius (r)= base of the triangle APC=lsinθ
in triangle APC
r = lsin45°
$r = 2 \times \frac{1}{\sqrt{2}}$
$r = \sqrt{2}$ m
centripetal acceleration $(a_c) = \frac{v^2}{r}$
$a = \frac{v^2}{r}$
$10 = \frac{v^2}{\sqrt{2}}$
$10\sqrt{2} = v^2$
$10 \times 1.41 = v^2$
$14.1 = v^2$
$\sqrt{14.1}$m/s = v
Speed of the particle = $\sqrt{14.1}$ m/s

21. Correct Matching is: (a)---(R) (b)---(P) (c)---(Q) (d)---(T)
An external force required to make a body move along circular path with uniform speed is called centripetal force.

A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

centripetal Force (F_c) $= \frac{mv^2}{r}$

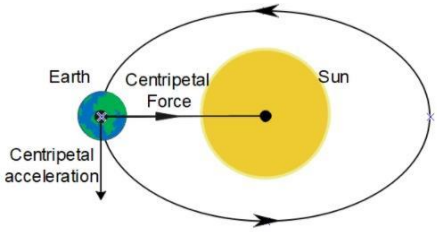
where m = mass of the body
v= linear velocity
r= radius of the circle



• When a stone is whirled in a horizontal circle
component T sinθ provides the necessary centripetal force to move it along horizontal circular path. Thus,

T sin θ = $\frac{mv^2}{r}$

• When the earth orbits a level curve towards the centre, the gravitational force exerted by the sun provides the necessary centripetal force.

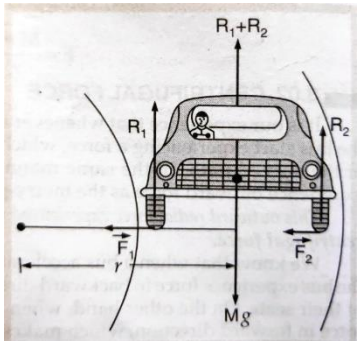


• when a car is taking a circular turn on a level road. While taking the round, the tyres of the vehicles try to leave the road and go away from the centre of the curve.
The total force of friction provides the necessary centripetal force

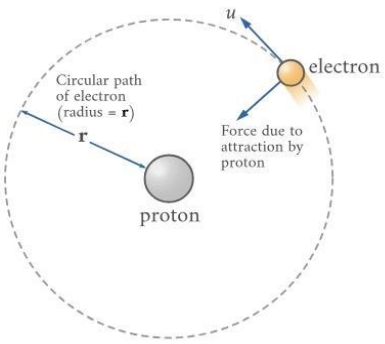
$F_1 + F_2 = \frac{mv^2}{r}$

where F_1 *and* F_2 *are the forces of friction*
$r =$ *radius of the circular turn*
v=linear velocity

- when an electron orbits in a hydrogen atom, the coulomb's attractive force exerted



by the proton on electron provides the necessary centripetal force.



22. Correct Matching is: (a)---(Q) (b)---(P) (c)---(R) (d)---(S)

(a) A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

centripetal acceleration $(a_c) = \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

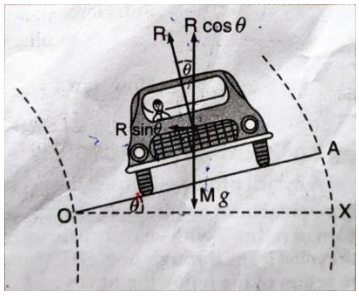
(b) according to newton's second law of motion

F= m x a

Centripetal force(F)= $\frac{mv^2}{r}$

(c)Consider a vehicle of weight Mg moving round a curved path of radius r with speed v on a road banked through angle θ. If OA is banked road and OX is a horizontal line, then angle AOX = θ is called angle of banking.

The vehicle is under the action of following forces:



(i) The weight Mg acting vertically downward i.e. perpendicular to the horizontal line OX.

(ii) The reaction R of the ground to the vehicle acting along normal to the banked road OA in upward direction.

The vertical component R cos θ of normal reaction R will balance the weight of the vehicle and the horizontal component R sin θ will provide the necessary centripetal force to the vehicle. Thus,

R cos θ = Mg--------(1)

R sin θ = $\frac{Mv^2}{r}$-------(2)

Dividing equation 2 by 1

$\frac{R\sin\theta}{R\cos\theta} = \frac{\frac{Mv^2}{r}}{Mg}$

$\tan\theta = \frac{v^2}{rg}$

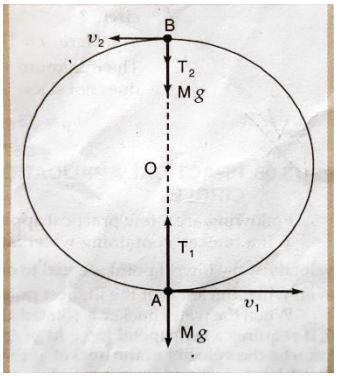
$\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$

Hence, the minimum angle of the road to be banked to take a safe turn $=\theta = \tan^{-1}\left(\frac{v^2}{rg}\right)$

(d) $v = \sqrt{\mu gr}$--------(1)

equation 1 gives the maximum velocity/minimum radius of curvature with which car can take the circular turn of radius r (without skidding)

23. Correct Answer is: 7



Given: mass (m) = 10kg

Length of the string = radius (r) of the circular path= 1m

Acceleration due to gravity (g) = 9.8 m/s^2 $v_1 = \sqrt{5gr}$

The equation gives the magnitude of the velocity at the lowest point

Putting the values

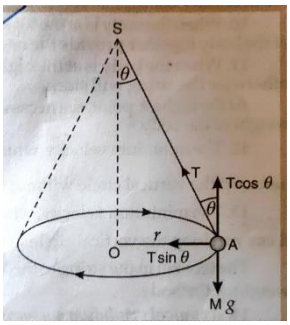
$v_1 = \sqrt{5x9.8 \times 1}$

$v_1 = \sqrt{49}$

$v_1 = 7m/s$

Hence, velocity of the particle at the bottom is 7 m/s

24. Correct Answer is: $\sqrt{\frac{9.8 \times \tan\theta}{r}}$



Given: mass of the ball =0.1 kg

Length of the string= 20 cm =0.20 m

Radius(r)=20cm=0.20m

g = $9.8m/s^2$

angular speed (ω) = $\sqrt{\frac{g\tan\theta}{r}}$-

where θ=angle made with the vertical

r = radius of the horizontal circle

$\sin\theta = \frac{OA}{SA} = \frac{0.20}{0.20} = 1 = \sin 90^\circ$

$\theta = 90^\circ$

angular speed (ω) = $\sqrt{\frac{9.8 \times \tan\theta}{r}}$

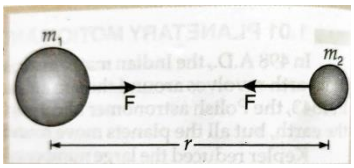
6. Gravitation

Solutions

LEVEL – I

1. Correct option: (B)

According to universal law of gravitation: Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of force is along the line joining the centres of the two bodies.



Suppose two bodies A and B of masses m_1 and m_2 are lying at a distance r from each other. Let the force of attraction between these two bodies be F. Now, according to the universal law of gravitation:

(i) the force between two bodies is directly proportional to the product of their masses. That is,

$F \propto m_1 \times m_2$

(ii) the force between two bodies is inversely proportional to the square of the distance between

them. That is,

$F \propto \frac{1}{r^2}$

Combining (1) and (2), we get:

$F \propto \frac{m_1 \times m_2}{r^2}$

Gravitational force, $F = G \times \frac{m_1 \times m_2}{r^2}$

where G is a constant known as "universal gravitational constant".

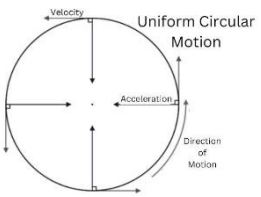
Since, the force of gravitation between two bodies is inversely proportional to the square of the distance between them. Therefore, if we double the distance between the two bodies, the force of gravitation decreases 4 times.

2. Correct option: (D)

Gravitational constant is constant throughout the universe and is independent of the nature and size of the bodies as well as the nature of medium between the bodies.

The value of gravitational constant (G) = $6.67 \times 10^{-11} Nm^2kg^{-2}$

3. Correct option: (B)



An external force required to make a body move along circular path with uniform speed is called centripetal force.

A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

centripetal Force (F_c) $= \frac{mv^2}{r}$

where m = mass of the body

v= linear velocity

r= radius of the circle

4. Correct option: (B)

- Mass of an object is a scalar quantity and it is constant.
- Weight is defined as the force exerted on a body by gravity.

Weight is often expressed as the product of the mass and the acceleration due to gravity.

W= mg

Where, w= weight of the body

m = mass of the body

g= acceleration due to gravity

acceleration due to gravity on earth is 6 times of that on moon.

$g_E = 6g_M$

Hence, the weight of an object on earth is 6 times of that on the moon

$W_E = 6W_M$

Where W_E = weight of object on earth

W_M = weight of object on moon.

Hence, the weight of the rock will become 6 times when it will be brought from the moon but its mass will remain the same.

5. Correct option: (A)

Weight is defined as the force exerted on a body by gravity.

Weight is often expressed as the product of the mass and the acceleration due to gravity.

W= mg

Where, w= weight of the body

m = mass of the body

g= acceleration due to gravity

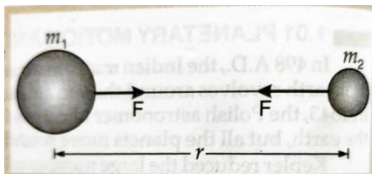
6. Correct option: (D)

The universal law of gravitation was given by Newton. So, it is also known as Newton's law of gravitation. According to universal law of gravitation: Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of force is along the line joining the centres of the two bodies.

Newton's law of gravitation is called universal law of gravitation because it is applicable to all the bodies having mass: whether the bodies are big or small; whether the bodies are terrestrial (which are on earth) or celestial (which are in outer space) such as the stars, planets and satellites.

7. Correct option: (D)

According to universal law of gravitation: Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of force is along the line joining the centres of the two bodies.



Suppose two bodies A and B of masses m_1 and m_2 are lying at a distance r from each other . Let the force of attraction between these two bodies be F. Now, according to the universal law of gravitation:

(i) the force between two bodies is directly proportional to the product of their masses. That is,

$F \propto m_1 \times m_2$

(ii) the force between two bodies is inversely proportional to the square of the distance between

them. That is,

$F \propto \frac{1}{r^2}$

Combining (1) and (2), we get:

$F \propto \frac{m_1 \times m_2}{r^2}$

Gravitational force, $F = G \times \frac{m_1 \times m_2}{r^2}$

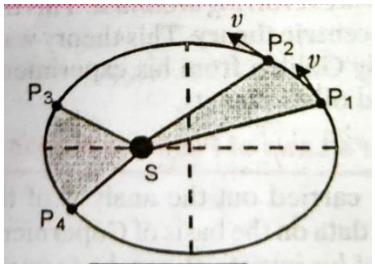
where G is a constant known as "universal gravitational constant".

8. Correct option: (C)

The radius of the earth = 6.4×10^6m

9. Correct option: (A)

Kepler's Second law (law of Areas): The line joining a planet to the sun sweeps out equal areas in equal intervals of time. Le. the areal velocity of the planet around the sun is constant.

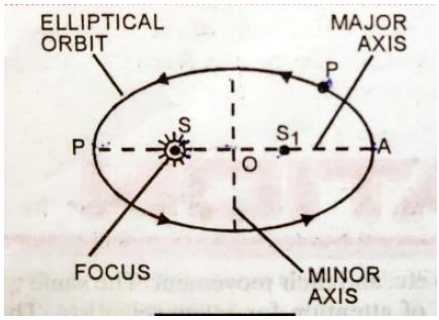


To keep the areal velocity constant, the linear speed of the planet goes on changing. The linear speed is more, when the planet is close to the sun and less, when it is away from the sun.

10. Correct option: (B)

Kepler's First law (law of orbits): Every planet revolves around the sun in an elliptical orbit. The sun is situated at one foci of the ellipse.

The ellipticity of the orbits is, however, very slight, and the orbits may be taken as nearly circular.



11. Correct option: (C)

Kepler's Second law (law of Areas): The line joining a planet to the sun sweeps out equal areas in equal intervals of time. Le. the areal velocity of the planet around the sun is constant.

To keep the areal velocity constant, the linear speed of the planet goes on changing. The linear speed is more, when the planet is close to the sun and less, when it is away from the sun.

as

$\text{kinetic energy K.E} = \frac{1}{2}mv^2$

kinetic energy is directly proportional to the speed.

Speed is minimum at the point C, hence, kinetic energy is minimum at the point C

12. Correct option: (B)

Kepler's third law (Law of periods): The square of the time period of revolution of á planet around the sun is directly proportional to the cube of semi major axis of its elliptical orbit

i.e. $T^2 \propto R^3$

where, T = time taken by the planet to go once around the sun,

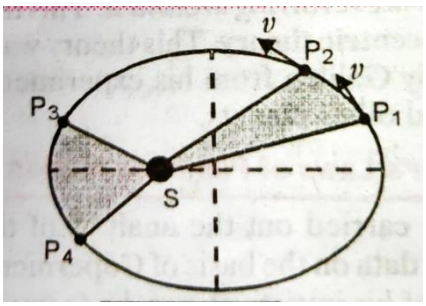
R = semi major axis of the elliptical orbit.

Or

$\frac{T^2}{R^3} = constant$

$$T^2R^{-3} = constant$$

13. Correct option: (C)



Kepler's Second law (law of Areas): The line joining a planet to the sun sweeps out equal areas in equal intervals of time. Le. the areal velocity of the planet around the sun is constant.

To keep the areal velocity constant, the linear speed of the planet goes on changing. The linear speed is more, when the planet is close to the sun and less, when it is away from the sun.

14. Correct option: (B)

The force that binds the neutrons and protons together in a nucleus is called the strong nuclear force. It acts between the two protons or two neutrons or a proton and a neutron, but only if the particles are very close together.

15. Correct option: (D)

Kepler's third law (Law of periods): The square of the time period of revolution of á planet around the sun is directly proportional to the cube of semi major axis of its elliptical orbit

i.e. $T^2 \propto R^3$

where, T = time taken by the planet to go once around the sun,

R= semi major axis of the elliptical orbit.

Or

$T = constant\ R^{\frac{3}{2}}$

$\frac{T_1}{T_2} = \left(\frac{R_1}{R_2}\right)^{\frac{3}{2}}$

Where T_1 = time period of revolution of earth

R_1 = distance of earth from the sun

T_2 = time period of revolution of planet

R_2 = distance of planet from the sun

Given : $R_2 = 5R_1$

$T_1 = 1$ year for earth

$\frac{1}{T_2} = \left(\frac{R_1}{5R_1}\right)^{\frac{3}{2}}$

$\frac{1}{T_2} = \left(\frac{1}{5}\right)^{\frac{3}{2}}$

$T_2 = 5^{\frac{3}{2}}$ years

Hence, time period of the planet = $5^{\frac{3}{2}}$ years

16. Correct option: (A)

acceleration due to gravity on earth is 6 times of that on moon.

$g_E = 6g_M$

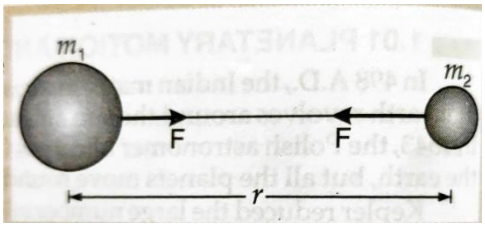
Hence, the gravitational force of attraction due to earth is 6 times that of moon.

While take off, the spacecraft has to do more work against the greater gravitational force of attraction of the earth. Hence, more fuel is spent while take off.

Whereas moon has lesser gravitational force of attraction, hence lesser fuel will be spent.

17. Correct option: (A)

According to universal law of gravitation: Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of force is along the line joining the centers of the two bodies.



Suppose two bodies A and B of masses m_1 and m_2 are lying at a distance r from each other . Let the force of attraction between these two bodies be F. Now, according to the universal law of gravitation:

(i) the force between two bodies is directly proportional to the product of their masses. That is,

$F \propto m_1 \times m_2$

(ii) the force between two bodies is inversely proportional to the square of the distance between

them. That is,

$F \propto \frac{1}{r^2}$

Combining (1) and (2), we get:

$F \propto \frac{m_1 \times m_2}{r^2}$

Gravitational force, $F = G \times \frac{m_1 \times m_2}{r^2}$

where G is a constant known as "universal gravitational constant".

We know that the masses of the bodies cannot be negative, hence the gravitational force cannot be repulsive.

Hence, the gravitational forces are always attractive.

18. Correct option: (B)

According to universal law of gravitation, the gravitational force F between two bodies of masses m_1 and m_2 placed at a distance r apart is given by:

$F = G \times \frac{m_1 \times m_2}{r^2}$

Gravitational constant (G) = $F \times \frac{r^2}{m_1 \times m_2}$

C.G.S unit of force F= dyne

C.G.S unit of distance r= cm

C.G.S unit of mass m = g

C.G.S unit of G = $\text{dyne} \times \frac{\text{cm}^2}{\text{g}^2} = \text{dyne cm}^2\text{g}^{-2}$

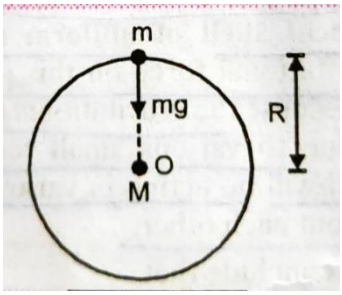
The value of gravitational constant (G) in C.G.S system = $6.67 \times 10^{-8} \text{dyne cm}^2 \text{ g}^{-2}$

19. Correct option: (B)

The atmosphere is held around the earth due to the gravitational force of attraction exerted by the earth on the atmosphere. The gases of the atmosphere have some mass and hence, they are attracted towards the earth.

According to universal law of gravitation: Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of force is along the line joining the centres of the two bodies.

20. Correct option: (C)



If we drop a body of mass m from a distance R from the centre of the earth of mass M, then the force exerted by the earth is given by the universal law of gravitation as:

$F = G \times \frac{M \times m}{R^2}$ --------(1)

where G is a constant known as "universal gravitational constant".

This force exerted by the earth produces an acceleration in the body due to which the body moves downwards

We also know that

Force = mass x acceleration

F= mx a

Acceleration (a) = $\frac{F}{m}$--------(2)

Putting the value of F from equation 1

$$a = \frac{GxM \times m}{R^2 \text{ x } m}$$

$$a = \frac{GM}{R^2}$$

this acceleration produced by the earth is known as the acceleration due to gravity and represented by the symbol (g)

so,

$$g = \frac{GM}{R^2}$$

where G = universal gravitational constant.

M= mass of the earth

R = radius of the earth

21. Correct option: (A)

Acceleration due to gravity at the latitude λ is given by

$g' = g - r\omega^2 \cos^2 \lambda$----------(1)

Where r = radius of the earth

At the equator $\lambda = 0^\circ$

Putting the value in equation 1

$g' = g - r\omega^2 \cos^2 0°$ ($\cos 0^\circ = 1$)

$g' = g - r\omega^2$--------(2)

If the earth stops rotating then the angular speed of the earth ω = 0

Putting in the equation 2

g' = g

hence, acceleration due to gravity will increase at the equator.

22. Correct option: (D)

Acceleration due to gravity:

$g = \frac{GM}{R^2}$----------(1)

where G = universal gravitational constant.

M= mass of the earth

R = radius of the earth

Let the new mass = M'

New radius = R'

$g' = \frac{GM'}{R'^2}$----------(2)

According to the question

M' =M

Radius shrinks by 1%

New radius

R' = R- 1% of R

R' = R- $\frac{1R}{100}$

R'= R- 0.01 R

R' = 0.99 R

Putting the values in equation 2

$g' = \frac{GM}{(0.99R)^2}$

$g' = \frac{GM}{0.9801(R)^2}$--------(3)

dividing equation 3 by 1

$$\frac{g'}{g} = \frac{\frac{G\,M}{0.9801(R)^2}}{\frac{GM}{R^2}} = \frac{1}{0.9801} = 1.02$$

$g' = 1.02g$

percentage change in the g $= \frac{g'-g}{g} x100\%$

$= \frac{1.02g-g}{g} x100\%$

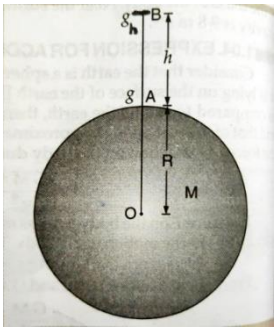
$=(1.02 - 1)x100\%$

$=0.02 \times 100\%$

$= 2\%$

Hence, acceleration due to gravity will increase by 2%

23. Correct option: (B)



Consider that earth is a sphere of radius R and mass M. Let g be the value of acceleration due to the gravity at the point A on the surface of earth. Then

$g = \frac{GM}{R^2}$--------(1)

If g_h is acceleration due to gravity at the point B at height h above the surface of earth, then

$g_h = \frac{GM}{(R+h)^2}$--------(2)

Dividing the equations 2 by 1 , we have

$$\frac{g_h}{g} = \frac{GM}{(R+h)^2} \times \frac{R^2}{GM}$$

$$\frac{g_h}{g} = \frac{R^2}{(R+h)^2}$$

$$\frac{g_h}{g} = \frac{R^2}{R^2\left(1+\frac{h}{R}\right)^2}$$

$\frac{g_h}{g} = \left(1+\frac{h}{R}\right)^{-2}$-------(3)

Expanding the equation 3 by using Binomial theorem, we have

$\frac{g_h}{g} = 1 - \frac{2h}{R}$ + terms containing higher powers of $\frac{h}{R}$

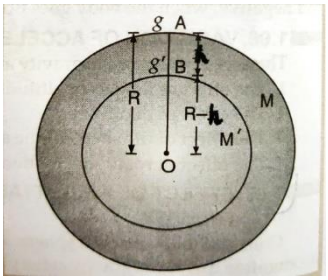
Usually, h is quite small as compared to R and therefore the higher powers of h/R can be neglected. Thus,

$\frac{g_h}{g} = 1 - \frac{2h}{R}$

$g_h = g(1 - \frac{2h}{R})$---------(4)

From the equation 4, it follows that as we go up, the value of acceleration due to gravity decreases.

24. Correct option: (A)



Consider that the earth is a sphere of radius R and mass M. Then, value of acceleration due to gravity at the point A on the surface of earth is given by

$g = \frac{GM}{R^2}$

If ρ is density of the material of earth, then

$M = \frac{4}{3}\pi R^3 \rho$

$g = G \times \frac{\frac{4}{3}\pi R^3 \rho}{R^2}$

$g = \frac{4}{3}\pi GR\rho$----------(1)

Let g_h be acceleration due to gravity at the point B at a depth h below the surface of earth. A body at the point B will experience force only due to the portion of the earth of radius OB (=R-h). The outer spherical shell, whose thickness is h, will not exert any force on body at point B. If M' is the mass of the portion of the earth, whose radius R-h, then

$g' = \frac{GM'}{(R-h)^2}$

Now,

$M' = \frac{4}{3}\pi(R-h)^3\rho$

$$g_h = \frac{G \times \frac{4}{3}\pi(R-h)^3\rho}{(R-h)^2}$$

$g_h = \frac{4}{3}\pi G(R-h)\rho$----------(2)

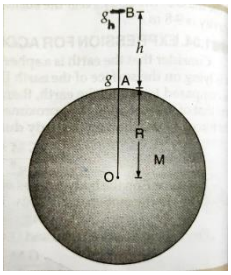
Dividing the equation 2 by 1, we have

$\frac{g_h}{g} = \frac{\frac{4}{3}\pi G(R-h)\rho}{\frac{4}{3}\pi GR\rho} = \frac{R-h}{R}$

$g_h = g\left(1-\frac{h}{R}\right)$

Therefore, the value of acceleration due to gravity decreases with depth also.

25. Correct option: (C)



Consider that earth is a sphere of radius R and mass M. Let g be the value of acceleration due to the gravity at the point A on the surface of earth. Then

$g = \frac{GM}{R^2}$--------(1)

If g_h is acceleration due to gravity at the point B at height h above the surface of earth, then

$g_h = \frac{GM}{(R+h)^2}$---------(2)

Dividing the equations 2 by 1 , we have

$$\frac{g_h}{g} = \frac{GM}{(R+h)^2} \times \frac{R^2}{GM}$$

$$\frac{g_h}{g} = \frac{R^2}{(R+h)^2}$$

$$\frac{g_h}{g} = \frac{R^2}{R^2\left(1+\frac{h}{R}\right)^2}$$

$\frac{g_h}{g} = \left(1+\frac{h}{R}\right)^{-2}$-------(3)

Expanding the equation 3 by using Binomial theorem, we have

$\frac{g_h}{g} = 1-\frac{2h}{R}$ + terms containing higher powers of $\frac{h}{R}$

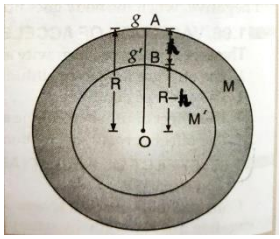
Usually, h is quite small as compared to R and therefore the higher powers of h/R can be neglected. Thus,

$\frac{g_h}{g} = 1-\frac{2h}{R}$

$g_h = g(1-\frac{2h}{R})$---------(4)

From the equation 4, it follows that as we go up, the value of acceleration due to gravity decreases.

26. Correct option: (A), (B)



Consider that the earth is a sphere of radius R and mass M. Then, value of acceleration due to gravity at the point A on the surface of earth is given by

$$g= \frac{GM}{R^2}$$

If ρ is density of the material of earth, then

$$M= \frac{4}{3}\pi R^3 \rho$$

$$g=G \times \frac{\frac{4}{3}\pi R^3 \rho}{R^2}$$

$$g= \frac{4}{3}\pi GR\rho \text{----------(1)}$$

Let g_h be acceleration due to gravity at the point B at a depth h below the surface of earth. A body at the point B will experience force only due to the portion of the earth of radius OB (=R-h). The outer spherical shell, whose thickness is h, will not exert any force on body at point B. If M' is the mass of the portion of the earth, whose radius R-h, then

$$g' = \frac{GM'}{(R-h)^2}$$

Now,

$$M' = \frac{4}{3}\pi (R-h)^3 \rho$$

$$g_h = \frac{G \times \frac{4}{3}\pi (R-h)^3 \rho}{(R-h)^2}$$

$$g_h = \frac{4}{3}\pi G(R-h)\rho \text{----------(2)}$$

Dividing the equation 2 by 1, we have

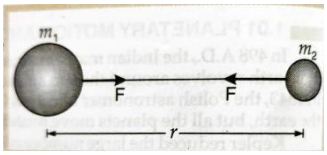
$$\frac{g_h}{g} = \frac{\frac{4}{3}\pi G(R-h)\rho}{\frac{4}{3}\pi GR\rho} = \frac{R-h}{R}$$

$$g_h = g\left(1-\frac{h}{R}\right)$$

Therefore, the value of acceleration due to gravity decreases with depth also.

27. Correct option: (C)

According to universal law of gravitation: Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of force is along the line joining the centres of the two bodies.



Suppose two bodies A and B of masses m_1 and m_2 are lying at a distance r from each other . Let the force of attraction between these two bodies be F. Now, according to the universal law of gravitation:

(i) the force between two bodies is directly proportional to the product of their masses. That is,

$F \propto m_1 \times m_2$

(ii) the force between two bodies is inversely proportional to the square of the distance between

them. That is,

$F \propto \frac{1}{r^2}$

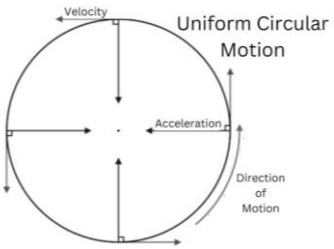
Combining (1) and (2), we get:

$F \propto \frac{m_1 \times m_2}{r^2}$

Gravitational force, $F = G \times \frac{m_1 \times m_2}{r^2}$

where G is a constant known as "universal gravitational constant".

28. Correct option: (B)



An external force required to make a body move along circular path with uniform speed is called centripetal force.

A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

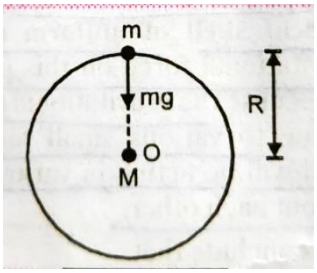
centripetal Force $(F_c) = \frac{mv^2}{R}$

where m = mass of the body

v= linear velocity

R= radius of the circle

29. Correct option: (C)



If we drop a body of mass m from a distance R from the centre of the earth of mass M, then the force exerted by the earth is given by the universal law of gravitation as:

$F = G \times \frac{M \times m}{R^2}$ --------(1)

where G is a constant known as "universal gravitational constant".

This force exerted by the earth produces an acceleration in the body due to which the body moves downwards

We also know that

Force = mass x acceleration

F= m x a

Acceleration (a) = $\frac{F}{m}$--------(2)

Putting the value of F from equation 1

$$a = \frac{GxM \times m}{R^2 \, x \, m}$$

$$a = \frac{GM}{R^2}$$

this acceleration produced by the earth is known as the acceleration due to gravity and represented by the symbol (g)

so,

$$g = \frac{GM}{R^2}$$

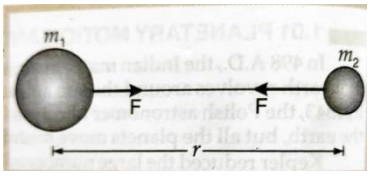
where G = universal gravitational constant.

M= mass of the earth

R = radius of the earth

30. Correct option: (B)

According to universal law of gravitation: Every body in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of force is along the line joining the centres of the two bodies.



Suppose two bodies A and B of masses m_1 and m_2 are lying at a distance r from each other . Let the force of attraction between these two bodies be F. Now, according to the universal law of gravitation:

$$F \propto \frac{m_1 \times m_2}{r^2}$$

Gravitational force, $F = G \times \frac{m_1 \times m_2}{r^2}$

where G is a constant known as "universal gravitational constant".

Since, the force of gravitation between two bodies is inversely proportional to the square of the distance between them. Therefore, if we double the distance between the two bodies, the force of gravitation decreases 4 times.

31. Correct option: (D)

Suppose two bodies A and B of masses M and m are lying at a distance R from each other . Let the force of attraction between these two bodies be F. Now, according to the universal law of gravitation:

$$F \propto \frac{M \times m}{R^2}$$

Gravitational force, F= $\frac{GMm}{R^2}$

where G is a constant known as "universal gravitational constant".

32. Correct option: (C)

According to universal law of gravitation: Everybody in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of force is along the line joining the centres of the two bodies.

according to the universal law of gravitation:

Gravitational force, F= G $\times \frac{m_1 \times m_2}{r^2}$

where G is a constant known as "universal gravitational constant".

the water up the hills has some mass and hence it is attracted towards the sea downwards due to the gravitational force of the earth.

33. Correct option: (D)

According to universal law of gravitation: Everybody in the universe attracts every other body with a force which is directly proportional to the product of their masses and inversely proportional to the square of the distance between them. The direction of force is along the line joining the centres of the two bodies.

according to the universal law of gravitation:

Gravitational force, F= G $\times \frac{m_1 \times m_2}{r^2}$

where G is a constant known as "universal gravitational constant".

We know that the masses of the bodies cannot be negative, hence the gravitational force cannot be repulsive.

Hence, the gravitational forces are always attractive.

LEVEL-II

1. Correct option: (A)

We know that,

The electrostatic force between two electrons is given by

$$F_e = \frac{Kq^2}{r^2}$$

Here, K =electrostatic force constant (K=$9 \times 10^9 N\, m^2/C^2$)

q = charge on the electrons ($q = 1.6 \times 10^{-19} C$)

r = separation between the electrons.

The gravitational force between two electrons is given by

$$F_g = \frac{G{m_e}^2}{r^2}$$

Here, G =universal gravitational constant (G=$6.67 \times 10^{-11} N \cdot m^2/kg^2$)

m_e = mass on an electron ($m_e = 9.1 \times 10^{-31} kg$)

r = distance of separation between the electrons.

The ratio of the electrostatic force to gravitational force between two electrons is,

$$\frac{F_e}{F_g} = \frac{\frac{Kq^2}{r^2}}{\frac{Gm_e^2}{r^2}} = \frac{Kq^2}{Gm_e^2}$$

Substituting all known values,

$$\frac{F_e}{F_g} = \frac{9 \times 10^9 N \cdot m^2/C^2 \times}{6.67 \times 10^{-11} N.m^2/kg^2 \times} \frac{(1.6 \times 10^{-19} C)^2}{(9.1 \times 10^{-31} kg)^2} = 4.17 \times 10^{42}$$

Hence, the ratio of the electrostatic to the gravitational force between two electrons is 4.17×10^{42}

2. Correct option: (C)

according to the universal law of gravitation:

Gravitational force, F= $G \times \frac{m_1 \times m_2}{r^2}$----------(1)

Given: mass (m_1) = 50kg

mass (m_2) = 60kg

distance between the masses(r)= 2m

value of gravitational constant G =$6.67 \text{ x } 10^{-11} Nm^2kg^{-2}$

putting the value sin equation 1

F= $6.67 \text{ x } 10^{-11} \times \frac{50 \times 60}{(2)^2}$

F= $6.67 \text{ x } 10^{-11} \times \frac{3000}{4}$

F= $6.67 \text{ x } 10^{-11} \times 750$

F= $5002 \text{ x } 10^{-11}$

F= $5 \text{ x } 10^{-8} N$

Hence, force of attraction between them = $5 \text{ x } 10^{-8} N$

3. Correct option: (A)

Acceleration due to gravity:

$g = \frac{GM}{R^2}$----------(1)

where G = universal gravitational constant.

M= mass of the earth

R = radius of the earth

Also

$g' = \frac{GM'}{R'^2}$----------(2)

where G = universal gravitational constant.

M'= mass of the planet

R' = radius of the planet

Given: M'= 4M

Diameter (D') = 2 Diameter of the earth

$D' = 2D$

$R' = \frac{D'}{2}$

$R' = \frac{2D}{2} = D$

$R = \frac{D}{2}$

Putting the values in 1 and 2

$g = \frac{GM}{\left(\frac{D}{2}\right)^2}$

$g = \frac{GM}{\frac{D^2}{4}}$

$g = \frac{4GM}{D^2}$----------(3)

$g' = \frac{G\ x4\ M}{D^2}$----------(4)

dividing equation 4 by 3

$\frac{g'}{g} = \frac{\frac{G\ x4\ M}{D^2}}{\frac{4GM}{D^2}} = 1$

$g' = g$

g' = 9.8 m/s^2

hence, acceleration due to gravity on the planet = 9.8 m/s^2

4. Correct option: (A)

Acceleration due to gravity:

$g = \frac{GM}{R^2}$----------(1)

where G = universal gravitational constant.

M= mass of the earth

R = radius of the earth

Let the new mass = M'

New radius = R'

$g' = \frac{GM'}{R'^2}$----------(2)

According to the question

M' =M

Radius shrinks by 1.5%

New radius

R' = R- 1.5% of R

R' = R- $\frac{1.5R}{100}$

R'= R- 0.015 R

R' = 0.985 R

Putting the values in equation 2

$g' = \frac{GM}{(0.985R)^2}$

$g' = \frac{GM}{0.970(R)^2}$--------(3)

dividing equation 3 by 1

$$\frac{g'}{g} = \frac{\frac{GM}{0.970(R)^2}}{\frac{GM}{R^2}} = \frac{1}{0.970} = 1.03$$

$g' = 1.03g$

percentage change in the g $= \frac{g'-g}{g} x100\%$

$= \frac{1.03g-g}{g} x100\%$

$=(1.03 - 1)x100\%$

$=0.03 \times 100\%$

$= 3\%$

Hence, percentage change in g = 3%

5. Correct option: (D)

Kepler's third law (Law of periods): The square of the time period of revolution of á planet around the sun is directly proportional to the cube of semi major axis of its elliptical orbit

i.e. $T^2 \propto R^3$

where, T = time taken by the planet to go once around the sun,

R= semi major axis of the elliptical orbit.

Or

$T = constant\ R^{\frac{3}{2}}$

$$\frac{T_1}{T_2} = \left(\frac{R_1}{R_2}\right)^{\frac{3}{2}}$$

Where T_1 = time period of revolution of earth

R_1 = distance of earth from the sun

T_2 = New duration of the year

R_2 = new distance of the earth from the sun

Given : $R_2 = 2R_1$

$T_1 = 1$ year for earth

$$\frac{1}{T_2} = \left(\frac{R_1}{2R_1}\right)^{\frac{3}{2}}$$

$$\frac{1}{T_2} = \left(\frac{1}{2}\right)^{\frac{3}{2}}$$

$T_2 = 2^{\frac{3}{2}}$ years

Hence, time period of revolution or new duration of the year = 2.82 years

6. Correct option: (D)

Acceleration due to gravity:

$g = \frac{GM}{R^2}$----------(1)

where G = universal gravitational constant.

M= mass of the earth

R = radius of the earth

Also

Mass(M)= volume(V) x density (ρ)

volume of the earth (sphere)= $\frac{4\pi}{3}R^3$

mass(M)= $\frac{4\pi}{3}R^3$ x ρ

putting the values in equation 1

$$g = \frac{G\frac{4\pi}{3}R^3 \text{ x } \rho}{R^2}$$

$g = \frac{G4\pi R\rho}{3}$----------(2)

for planet 1

$g_1 = \frac{G4\pi R_1\rho_1}{3}$----------(3)

For planet 2

$g_2 = \frac{G4\pi R_2\rho_2}{3}$----------(4)

dividing equation 4 by 3

$$\frac{g_1}{g_2} = \frac{\frac{G4\pi R_1\rho_1}{3}}{\frac{G4\pi R_2\rho_2}{3}} = \frac{R_1\rho_1}{R_2\rho_2}$$

$g_1 : g_2 = R_1\rho_1 : R_2\rho_2$

7. Correct option: (B)

Acceleration due to gravity at a depth (h) below the surface of the earth is given by

$g' = g\left(1 - \frac{h}{R}\right)$--------(1)

where g' = acceleration due to gravity at the depth h

g = acceleration due to gravity at earth surface

R = radius of the earth

Radius of the earth = $6.4 \text{ x } 10^6 m$

g' =1% of g

g' = $\frac{1}{100}$g

g' = 0.01 g

putting the value in equation 1

$$0.01g = g\left(1 - \frac{h}{R}\right)$$

$$0.01 = \left(1 - \frac{h}{R}\right)$$

$\frac{h}{R}$ =1-0.01

$\frac{h}{R}$ =0.99

h = 0.99R

h = $0.99 \text{ x } 6.4 \text{ x } 10^6 m$

h= $6336 \text{ x } 10^3 m$

h = 6336 km

hence, at a depth 6336 km below the surface of the earth, the acceleration due to gravity will become 1 % of its value at the earth's surface

8. Correct option: (C)

Acceleration due to gravity at a height (h) above the surface of the earth is given by

$\frac{g'}{g} = \frac{R^2}{(R+h)^2}$--------(1)

where g' = acceleration due to gravity at the depth h

g = acceleration due to gravity at earth surface

R = radius of the earth

Radius of the earth = 6400 km

We know that weight of the body = mg

Multiplying and dividing left side of equation 1 by m

$\frac{m\prime g'}{mg} = \frac{R^2}{(R+h)^2}$----------(2)

Given: weight on the surface of the earth(mg) = 60 N

weight at a height h(m'g') = 30 N

putting the values in equation 2

$\frac{30}{60} = \frac{(6400)^2}{(6400+h)^2}$

$\frac{1}{2} = \frac{(6400)^2}{(6400+h)^2}$

$(6400+h)^2 = 2x(6400)^2$

$6400 + h = 6400\sqrt{2}$

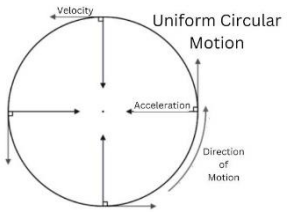
$h = 6400\sqrt{2} - 6400$

h = 6400 x 1.41 – 6400

h = 2624 km

hence, at a height 2624 km above the surface of the earth, his weight will become 30 kg.

9. Correct option: (B)



An external force required to make a body move along circular path with uniform speed is called centripetal force.

A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

centripetal Force $(F_c) = \frac{mv^2}{r}$

where m = mass of the body

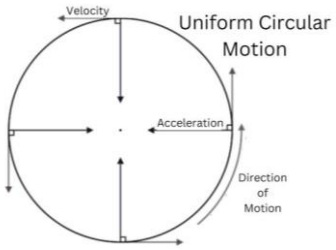
v= linear velocity

r= radius of the circle

The earth revolves around the sun due to the gravitational force of attraction of the sun which provides the necessary centripetal force for the circular motion.

Hence, revolution of earth around the sun is the evidence to show that there must be a force acting on earth and directed towards the sun.

10. Correct option: (D)



In a uniform circular motion, the linear speed remains constant, its linear velocity is not constant. It is because, the linear velocity of the object at any point of its circular path is directed along the tangent to the path and therefore its direction keeps on changing continuously. The change of linear velocity results in acceleration, which is always directed along the radius of the circle and towards its centre. Thus, uniform circular motion is a case of accelerated motion.

An external force required to make a body move along circular path with uniform speed is called centripetal force.

A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

centripetal acceleration $(a_c) = \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

• In case of uniform circular motion, the angular velocity is constant.

So, angular acceleration $\alpha = \frac{d\omega}{dt} = 0$

As Tangential/linear acceleration$(a_T) = \alpha r = 0$

• Magnitude of centripetal acceleration $(a_c) = \frac{v^2}{r}$ is constant.

Magnitude of net acceleration is constant but direction changes continuously.

11. Correct option: (C)

Time period of the satellite = it is the time taken by the satellite to complete one revolution around the earth and is denoted by T

$$T = \frac{\frac{2\pi}{R} r^{\frac{3}{2}}}{\sqrt{g}}$$

Where r = radius of the orbit of satellite

R = radius of the earth

Hence, the time period of the satellite varies as $r^{\frac{3}{2}}$.

12. Correct option: (B)

A body in uniform circular motion is in a continuous accelerated motion and the acceleration is directed along the radius of the circular path and towards the centre. It is called the centripetal acceleration

centripetal acceleration $(a_c) = \frac{v^2}{r}$

where v= linear velocity

r= radius of the circle

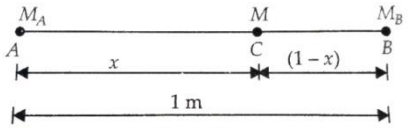
13. Correct option: (A)

according to the universal law of gravitation:

force of attraction between two masses placed at a distance R is given by

$F= \frac{GMm}{R^2}$ --------(1)

let the point mass C is placed at a distance x from the point mass A as shown in the figure.



Given: $\frac{M_A}{M_B} = \frac{4}{3}$

Force between A and C

$F_{AC} = \frac{GMM_A}{x^2}$ ----------(2)

Force between B and C

$F_{BC} = \frac{GMM_B}{(1-x)^2}$ ----------(2)

According to the question

$F_{AC} = \frac{1}{3}F_{BC}$

$\frac{GMM_A}{x^2} = \frac{1}{3}\left(\frac{GMM_B}{(1-x)^2}\right)$

$\frac{M_A}{x^2} = \left(\frac{M_B}{3(1-x)^2}\right)$

$\frac{M_A}{M_B} = \left(\frac{x^2}{3(1-x)^2}\right)$

$\frac{4}{3} = \left(\frac{x^2}{3(1-x)^2}\right)$

$4 = \left(\frac{x^2}{(1-x)^2}\right)$

$\sqrt{4} = \frac{x}{1-x}$

$2 = \frac{x}{1-x}$

$2(1 - x) = x$

$2 - 2x = x$

$2 = 3x$

$\frac{2}{3} = x$

Hence, $x= \frac{2}{3}m$

The distance of C from A = $\frac{2}{3}\ m$

14. Correct option: (A)

Acceleration due to gravity at a height (h) above the surface of the earth is given by

$\frac{g'}{g} = \frac{R^2}{(R+h)^2}$--------(1)

where g' = acceleration due to gravity at the depth h

g = acceleration due to gravity at earth surface

R = radius of the earth

g= 9.8 m/s^2

Given: Radius of the earth = 6400 km = 6400000 m

Height above the surface (h) = 1600 km

Putting the given values in equation 1

$$\frac{g'}{9.8} = \frac{(6400)^2}{(6400+1600)^2}$$

$$\frac{g'}{9.8} = \frac{40960000}{64000000}$$

$$g' = \frac{4096x9.8}{6400} = 6.27$$

Now according to the question

Acceleration due to gravity at a depth h will be half of the above value

$$\frac{g'}{2} = \frac{6.27}{2} = 3.13$$

Acceleration due to gravity at a depth (h) below the surface of the earth is given by

$$g_h = g\left(1 - \frac{h}{R}\right)\text{--------(2)}$$

where g_h = acceleration due to gravity at the depth h

g = acceleration due to gravity at earth surface

R = radius of the earth

Putting the values in equation 2

$$3.13 = 9.8\left(1 - \frac{h}{6400000}\right)$$

$$3.13 = 9.8\left(1 - \frac{h}{6400000}\right)$$

$$0.319 = \left(1 - \frac{h}{6400000}\right)$$

$$\frac{h}{6400000} = 1 - 0.319$$

$$\frac{h}{6400000} = 0.68$$

$$h = 0.68 \times 6400000$$

$$h = 4352000$$

$$h = 4.3 \times 10^6\ m$$

hence, at a depth $4.3 \times 10^6 m$ below the surface of the earth, Acceleration due to gravity will be half of the value at 1600 km above the surface of the earth.

15. Correct option: (A)

Acceleration due to gravity at a height (h) above the surface of the earth is given by

$$\frac{g'}{g} = \frac{R^2}{(R+h)^2}\text{--------(1)}$$

where g' = acceleration due to gravity at the depth h

g = acceleration due to gravity at earth surface

R = radius of the earth

We know that weight of the body = mg

Multiplying and dividing left side of equation 1 by m

$\frac{m\prime g'}{mg} = \frac{R^2}{(R+h)^2}$----------(2)

Given: weight on the surface of the earth(mg) = 72 N

To find: weight at a height R/2 =(m'g')

putting the values in equation 2

$\frac{m'g'}{72} = \frac{(R)^2}{\left(R+\frac{R}{2}\right)^2}$

$m'g' = \frac{72}{\left(1+\frac{1}{2}\right)^2} = \frac{72}{\left(\frac{3}{2}\right)^2} = \frac{\frac{72}{9}}{4} = \frac{72x4}{9} = 32$

m'g' = 32 N

hence, weight of the object at height R/2 = 32 N

16. Correct option: (A)

Acceleration due to gravity at the latitude λ is given by

$g' = g - R\omega^2 \cos^2 \lambda$----------(1)

Where R= radius of the earth

Given: λ= 60˚

g'=0

radius of the earth (R)= 6400km= 6400000 m

g= 10 m/s^2

Putting the value in equation 1

$0 = g - R\omega^2 \cos^2 60°$ ($\cos 60^\circ = \frac{1}{2}$)

$0 = g - R\omega^2 \text{ x } (\frac{1}{2})^2$

$0 = g - R\omega^2 \text{ x } \frac{1}{4}$

$R\omega^2 \text{ x } \frac{1}{4} = g$

$\omega^2 = \frac{4g}{R}$

$\omega = \sqrt{\frac{4g}{R}}$

putting the given values

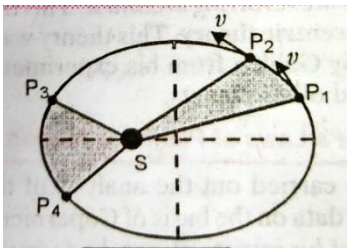
$\omega = \sqrt{\frac{4x\,10}{6400000}} = \frac{1}{\sqrt{160000}} = \frac{1}{400} = 2.5 \text{ x } 10^{-3} rad/s$

hence, angular velocity = $2.5 \text{ x } 10^{-3}\ rad/s$

17. Correct option: (B)

According to the Kepler's Second law (law of Areas): The line joining a planet to the sun sweeps out equal areas in equal intervals of time. Le. the areal velocity of the planet around the sun is constant.

To keep the areal velocity constant, the linear speed of the planet goes on changing. The linear speed is more, when the planet is close to the sun and less, when it is away from the sun.



angular momentum $\vec{L} = 2m \times \frac{d\vec{A}}{dt}$

from the kepler's second law the areal velocity remains constant

$\frac{d\vec{A}}{dt} = constant$

Since, no torque acts on the planet. Hence, the angular momentum remains constant in the planetary motion.

18. Correct option: (C)

Acceleration due to gravity at a height (h) above the surface of the earth is given by

$\frac{g'}{g} = \frac{R^2}{(R+h)^2}$---------(1)

where g' = acceleration due to gravity at the depth h

g = acceleration due to gravity at earth surface

R = radius of the earth

We know that weight of the body = mg

Multiplying and dividing left side of equation 1 by m

$\frac{m'g'}{mg} = \frac{R^2}{(R+h)^2}$----------(2)

Given: weight on the surface of the earth=mg

weight at a height h = m'g'

According to the question

$m'g' = \frac{1}{6}mg$

putting the values in equation 2

$\frac{\frac{1}{16}mg}{mg} = \frac{(R)^2}{(R+h)^2}$

$\frac{1}{16} = \frac{(R)^2}{(R+h)^2}$

$(R+h)^2 = 16x(R)^2$

$R + h = R\sqrt{16}$

$h = 4R - R$

h= 3R

hence, at a height 3R above the surface of the earth, his weight will become $\frac{1}{16}$ th of its weight on the surface of the earth.

19. Correct option: (C)

Acceleration due to gravity at the latitude λ is given by

$g' = g - R\omega^2 \cos^2 \lambda$----------(1)

Where R= radius of the earth

Given: $\lambda = 45^\circ$

g'=0

Putting the value in equation 1

$0 = g - R\omega^2 \cos^2 45°$ $\quad (\cos 45^\circ = \frac{1}{\sqrt{2}})$

$0 = g - R\omega^2 \text{ x } (\frac{1}{\sqrt{2}})^2$

$0 = g - R\omega^2 \text{ x } \frac{1}{2}$

$R\omega^2 \text{ x } \frac{1}{2} = g$

$\omega^2 = \frac{2g}{R}$

$\omega = \sqrt{\frac{2g}{R}}$

hence, angular velocity $= \sqrt{\frac{2g}{R}}$

20. Correct options(A), (B), (C)

Gravitational constant is constant throughout the universe and is independent of the nature and size of the bodies as well as the nature of medium between the bodies.

The value of gravitational constant (G) = $6.67 \text{ x } 10^{-11} Nm^2kg^{-2}$

21. Correct option: (B), (D)

according to the universal law of gravitation:

Gravitational force between the earth and the sun, $F = G \times \frac{M_s \times M_e}{r^2}$

where M_s = mass of the sun

$M_e = mass\ of\ the\ earth$

r = radius of the orbit

G is a constant known as "universal gravitational constant".

• Length of one day is due to rotation of the earth on its own axis, hence it will not be affected by G.

• Time period of the revolution of the earth around the sun decides the length of one year.

Time period (T) = $\frac{2\pi}{\omega}$

As the gravitational force provides the necessary centripetal force for the rotational motion of the earth around the sun

$F = G \times \frac{M_s \times M_e}{r^2} = M_e\omega^2 r$

As the G is directly proportional to angular speed

As G decreases value of ω will also decrease

Time period is inversely proportional to the ω

Hence the time period of revolution will increase if the value of G decreases.

So, the length of 1 year will also increase.

• As the gravitational force provides the necessary centripetal force for the rotational motion of the earth around the sun

$F = G \times \frac{M_s \times M_e}{r^2} = \frac{M_e v^2}{r}$

As the G is directly proportional to the velocity

As G decreases value of v will also decrease

Kinetic energy = $\frac{1}{2} M_e v^2$

Hence kinetic energy of the will also decrease.

22. Correct option: (D)

Acceleration due to gravity:

$g = \frac{GM}{R^2}$----------(1)

where G = universal gravitational constant.

M= mass of the earth

R = radius of the earth

Let the new mass = M'

New radius = R'

$g' = \frac{GM'}{R'^2}$----------(2)

According to the question

M' =M

Radius decreases to 80% of the original value

New radius

R' = 80% of R

$R' = \frac{80R}{100}$

R'= 0.8 R

Putting the values in equation 2

$g' = \frac{GM}{(0.8R)^2}$

$g' = \frac{GM}{0.64(R)^2}$--------(3)

dividing equation 3 by 1

$\frac{g'}{g} = \frac{\frac{GM}{0.64(R)^2}}{\frac{GM}{R^2}} = \frac{1}{0.64} = 1.56$

$g' = 1.56g$

percentage change in the g $= \frac{g'-g}{g} x100\%$

$= \frac{1.56g-g}{g} x100\%$

$= (1.56 - 1)x100\%$

$=0.56x\ 100\%$

$= 56\%$

Hence, percentage change in g = 56%

Hence, the new value of g increases by about 56%

23. Correct option: (A), (B)

- Gravitational force between the two bodies form an action- reaction pair that is the force of attraction exerted by body A on the body B is equal and opposite to the force exerted by the body B on body A. However the acceleration of the objects will not be equal.
- Gravitational constant G is a scalar having magnitude only and no direction whereas acceleration due to gravity (g) is a vector having both magnitude and direction.

• The value of gravitational constant (G) = $6.67 \times 10^{-11} Nm^2kg^{-2}$ and value of acceleration due to gravity is $9.8 m/s^2$.

• Gravitational constant is a constant throughout the universe and is independent of the nature and size of the bodies as well as the nature of medium between the bodies. whereas the value of g varies with the height, depth, shape and rotation of the earth.

24. Correct option: (A), (B)

Kepler's third law (Law of periods): The square of the time period of revolution of á planet around the sun is directly proportional to the cube of semi major axis of its elliptical orbit

i.e. $T^2 \propto R^3$

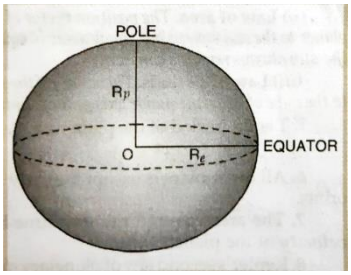
where, T = time taken by the planet to go once around the sun,

R = semi major axis of the elliptical orbit.

Or

$T = constant\ R^{\frac{3}{2}}$

$\frac{T_1}{T_2} = \left(\frac{R_1}{R_2}\right)^{\frac{3}{2}}$



25. Correct option: (C)

• **Effect of shape:**

Earth is not a perfect sphere. It is flattened at the poles and bulges out at the equator. Equatorial radius R_e of the earth is greater than the polar radius R_p

We know that

Acceleration due to gravity:

$g = \frac{GM}{R^2}$

where G = universal gravitational constant.

M= mass of the earth

R = radius of the earth

For the equator

$g_{equator} = \frac{GM}{R_e^2}$

For poles

$g_{poles} = \frac{GM}{R_p^2}$

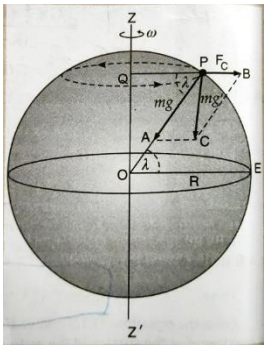
Since $R_e > R_p$

So, $g_{equator} < g_{pole}$

hence, g will be lesser at the equator as compared to that at the poles.

g is least at the equator and maximum at the poles.

• **Effect of rotation of the earth**



The earth possesses rotational motion about its axis through its poles. the value of acceleration due to gravity at a place(at a given latitude) is affected due to its rotational motion.

Acceleration due to gravity at the latitude λ is given by

$g' = g - R\omega^2 \cos^2 \lambda$----------(1)

As g' is less than g it follows that acceleration due to gravity decreases due to the rotation of the earth.

Where R = radius of the earth

At the equator λ= 0˚

Putting the value in equation 1

$g'_e = g - r\omega^2 \cos^2 0°$ (cos 0˚ = 1)

$g'_e = g - r\omega^2$--------(3)

At the poles λ= 90˚

Putting the value in equation 1

$g'_p = g - r\omega^2 \cos^2 90°$ (cos 90˚ = 0)

$g'_p = g$--------(4)

We can see from equation 1 and 2

Maximum effect of rotational motion of the earth on the value of g is at the equator.

The value of g increases with the increase in the latitude of a place.

26. Correct option: (A)

Acceleration due to gravity:

$g = \frac{GM}{r^2}$----------(1)

where G = universal gravitational constant.

M= mass of the Planet

r = radius of the planet

Also

Mass(M)= volume(V) x density (ρ)

volume of the earth (sphere)= $\frac{4\pi}{3}r^3$

mass(M)= $\frac{4\pi}{3}r^3$ x ρ

putting the values in equation 1

$$g = \frac{G\frac{4\pi}{3}r^3 \times \rho}{r^2}$$

$g = \frac{G4\pi r\rho}{3}$----------(2)

for planet 1

$g_1 = \frac{G4\pi r_1\rho_1}{3}$----------(3)

For planet 2

$g_2 = \frac{G4\pi r_2\rho_2}{3}$----------(4)

dividing equation 4 by 3

$$\frac{g_1}{g_2} = \frac{\frac{G4\pi r_1\rho_1}{3}}{\frac{G4\pi r_2\rho_2}{3}} = \frac{r_1\rho_1}{r_2\rho_2}$$

$$\frac{g_1}{g_2} = \frac{r_1\rho_1}{r_2\rho_2}$$

It is given that $\rho_1 = \rho_2$

$$\frac{g_1}{g_2} = \frac{r_1}{r_2}$$

27. Correct option: (B)

Acceleration due to gravity at the latitude λ is given by

$g' = g - r\omega^2 \cos^2 \lambda$----------(1)

Where r = radius of the earth

At the equator λ= 0˚

Putting the value in equation 1

$g' = g - r\omega^2 \cos^2 0°$ (cos 0˚ = 1)

$g' = g - r\omega^2$

28. Correct option: (C)

Acceleration due to gravity at the latitude λ is given by

$g' = g - r\omega^2 \cos^2 \lambda$----------(1)

Where r = radius of the earth

We know that weight = mg

$m'g' = mg - mr\omega^2 \cos^2 \lambda$----------(2)

At the equator λ= 0˚

Putting the value in equation 1

$m_e'g_e' = mg - mr\omega^2 \cos^2 0°$ (cos 0˚ = 1)

$m_e'g_e' = mg - mr\omega^2$--------(3)

At the poles λ= 90˚

Putting the value in equation 1

$m_p'g_p' = mg - mr\omega^2 \cos^2 90°$ (cos 90˚ = 0)

$m_p'g_p' = mg$--------(4)

Change in weight when body is moved from equator to the poles

$m_p'g_p' - m_e'g_e' = mg - mg + mr\omega^2$

$m_p'g_p' - m_e'g_e' = mr\omega^2$

29. Correct option: (B)

Acceleration due to gravity at the latitude λ is given by

$g' = g - r\omega^2 \cos^2 \lambda$----------(1)

Where r = radius of the earth

At the equator λ= 0˚

Putting the value in equation 1

$g' = g - r\omega^2 \cos^2 0°$ (cos 0˚ = 1)

$g' = g - r\omega^2$

when the earth becomes weightless g'= 0

$0 = g - r\omega^2$

$g = r\omega^2$

$\frac{g}{r} = \omega^2$

$\sqrt{\frac{g}{r}} = \omega$

Angular velocity of the rotation of the earth = $\sqrt{\frac{g}{r}}$

30. Correct Matching is: (a)----(P); (b)----(Q); (c)----(Q); (d)----(Q)

• Acceleration due to gravity at the latitude λ is given by

$g' = g - r\omega^2 \cos^2 \lambda$----------(1)

Where r = radius of the earth

At the poles λ= 90˚

Putting the value in equation 1

$g' = g - r\omega^2 \cos^2 90°$ (cos 90˚ = 0)

$g' = g$

Value of g at the poles is independent of the rotational speed.

• Acceleration due to gravity at the latitude λ is given by

$g' = g - r\omega^2 \cos^2 \lambda$----------(1)

Where r = radius of the earth

At the equator λ= 0˚

Putting the value in equation 1

$g' = g - r\omega^2 \cos^2 0°$ (cos 0˚ = 1)

$g' = g - r\omega^2$

as the value of rotational speed *ω* decreases, the value g' increases at the equator.

• Time period of the geostationary satellite is given by the kepler's law

According to the Kepler's third law (Law of periods): The square of the time period of revolution of á planet around the sun is directly proportional to the cube of semi major axis of its elliptical orbit

i.e. $T^2 \propto r^3$

where, T = time taken by the planet to go once around the sun,

r = distance of the satellite from the surface of the earth .

Or

$T = constant\ r^{\frac{3}{2}}$

$\frac{T_1}{T_2} = \left(\frac{r_1}{r_2}\right)^{\frac{3}{2}}$ ----------(1)

Also

Time period (T) = $\frac{2\pi}{\omega}$

Where **ω is the angular speed of rotation**

From equation 1

$\frac{\omega_2}{\omega_1} = \left(\frac{r_1}{r_2}\right)^{\frac{3}{2}}$

as the angular speed of rotation decreases $\frac{\omega_2}{\omega_1} < 1$

hence, $r_2 > r_1$

hence, distance of geostationary satellite will increase.

• Energy of the satellite = $-\frac{1}{2}\frac{GMm}{r}$

As the rotational speed of the earth decreases, distance increases

$r_2 > r_1$

As energy is inversely proportional to the distance.

As the distance increases, energy becomes less negative and hence increases.

31. Correct Answer is: $\frac{g'}{g} = 3$

Acceleration due to gravity:

$g = \frac{GM}{R^2}$----------(1)

where G = universal gravitational constant.

M= mass of the earth

R = radius of the earth

Also

Mass(M)= volume(V) x density (ρ)

volume of the earth (sphere)= $\frac{4\pi}{3}R^3$

mass(M)= $\frac{4\pi}{3}R^3$ x ρ

putting the values in equation 1

$g = \frac{G\frac{4\pi}{3}R^3 \text{ x } \rho}{R^2}$

$g = \frac{G4\pi R\rho}{3}$----------(2)

for new planet

density ρ is same as the earth

$g' = \frac{G4\pi R'\rho}{3}$----------(3)

R'= 3R

Putting the value of R' in equation 3

$g' = \frac{G4\pi 3R\rho}{3}$----------(4)

dividing equation 4 by 2

$\frac{g'}{g} = \frac{\frac{G4\pi 3R\rho}{3}}{\frac{G4\pi R\rho}{3}} = 3$

7. Oscillations

Solutions

LEVEL-I

Characteristics Of SHM:

1. Correct option: (B) Periodic but not simple harmonic.

According to SHM, a particle's acceleration should always be greater than its displacement. Otherwise put, T

$a \propto -x$

When a particle revolves, this is obviously not the case. Therefore, it won't be in SHM. When the speed is uniform, on the other hand, the particle always begins at one point in the circle and returns there after a predetermined amount of time T , also known as the time period. This motion is hence periodic.

We can infer that a particle's motion is periodic but not just harmonic when it moves in a circle at a constant speed.

2. Correct option: (C) Both periodic and simple harmonic.

If a magnet is suspended freely and is moved away from its N-S direction, the motion of the magnet is periodic because the magnet oscillates about its position over a set amount of time.

If the magnet behaves as a simple pendulum, it will oscillate in accordance with the laws of simple harmonic motion. A linear restoring force (proportional to displacement) plus a sinusoidal motion pattern define SHM, a type of periodic motion.

3. Correct option: (D) Motion of bob of simple pendulum.

When moved away from its equilibrium position, a simple pendulum displays motion that is both periodic and simple harmonic. It follows the rules of basic harmonic motion and oscillates back and forth with a regular period.

4. Correct option: (D) Proportionality between restoring force and displacement from equilibrium Position.

In SHM, the restoring force exerted on an object is geared towards the equilibrium position and is directly proportional to the displacement of that object from its equilibrium or rest position. The relationship between the restoring force and displacement is proportionate, and this relationship is a fundamental property of simple harmonic motion.

The restoring force is given by,

$F \propto -x$

5. Correct option: (B) $y = A\sin\omega t$ **.**

$y = A\sin\omega t$ does not represent the SHM.

$y = A\,sin\,\omega\,t \cdots\cdots (1)$

$\frac{dy}{dt} = 2A\omega\,cos\,2\,\omega t$

$\frac{d^2y}{dt^2} = -4A\omega^2\,sin\,2\,\omega t$

$\frac{d^2y}{dt^2} = -4\omega^2 y$

From the equation (1) this is not equation of S.H.M.

6. Correct option: (D) $\sqrt{A^2 + B^2}$ **.**

Given:

Equation of SHM,

$y = A\sin\omega t + B\cos\omega t$

Taking derivative both sides,

$y = A\sin\omega t + B\cos\omega t$

$y' = y - A_0 = A\sin\omega t + B\cos\omega t$

The resultant amplitude is given by as,

$R = \sqrt{A^2 + B^2 + 2Ab\cos 90^\circ}$

$R = \sqrt{A^2 + B^2}$

So, the amplitude of oscillation is $\sqrt{A^2 + B^2}$.

7. Correct option: (D) Akx **.**

Equation of simple harmonic motion is given by,

$\frac{d^2x}{dt^2} \propto -x$

So, the force acting on S.H.M along the x-axis will be $F = -Akx$.

Where, A and k positive constant.

8. Correct option: (B) 2:1.

Equations of SHM,

$y_1 = 10\sin\left(\omega t + \frac{\pi T}{4}\right)\cdots\cdots(1)$

$y_2 = 25\sin\left(\omega t + \frac{\sqrt{3}\pi T}{4}\right)\ldots\ldots(2)$

The amplitude of equation (1) is,

$a_1 = 10$

The amplitude of equation (2) is,

$a_2 = 25$

So, the ratio of their amplitude is,

$\frac{a_1}{a_2} = \frac{10}{5} = \frac{2}{1}$

So, the ratio of their amplitude is 2:1.

9. Correct option: (D) $0.5\cos 5\pi t$ **.**

Initial phase, $\phi = \frac{\pi}{2}$

Amplitude, $a = 0.5\text{cm}$

Now from the equation of SHM,

$$y = a\,sin(\omega t + \phi)$$

$$y = a\,sin\left(\frac{2\pi}{T}t + \phi\right)$$

$$y = -0.5\,sin\left(\frac{2\pi}{0.4}t + \frac{\pi}{2}\right)$$

$$y = 0.5\,sin\left(5\pi t + \frac{\pi}{2}\right)$$

$$y = 0.5\,cos\,5\,\pi t$$

So, the equation of SHM is $0.5\cos 5\pi t$.

10. Correct option: (C) $3\sqrt{3}\text{m}$**.**

The equation of simple harmonic oscillator is given by,

$$x = 6\sin\left[\frac{\pi}{8}t + \frac{\pi}{12}\right]$$

Time, $t = 2\text{sec}$

Put $t = 2\text{sec}$ in above equation we get,

$$x = 6\,sin\left[\frac{\pi}{8} \times 2 + \frac{\pi}{12}\right]$$
$$x = 6\,sin\left[\frac{\pi}{4} + \frac{\pi}{12}\right]$$
$$x = 6\,sin\left[\frac{\pi}{3}\right]$$
$$x = 6 \times \frac{\sqrt{3}}{2}$$
$$x = 3\sqrt{3}m$$

So, the displacement is $3\sqrt{3}\text{m}$.

11. Correct option: (A) 5m.

The equation of motion,

$$y = 10\sin(2\pi t + \frac{\pi}{6})$$

For initial velocity, $t = 0$

$$y = 10\,sin\left(2\pi \times 0 + \frac{\pi}{6}\right)$$
$$y = 10\,sin\left(\frac{\pi}{6}\right)$$
$$y = 10\left(\frac{1}{2}\right)$$
$$y = 5m$$

So, the initial displacement is 5m.

12. Correct option: (D) $\frac{T}{12}$**.**

Let's assume that the particle is in the mean position at time $t = 0$.

Consequently, the displacement equation for the particle running SHM is,

$$x = A\sin\omega t$$

Where A and ω, respectively, represent the oscillation's frequency and amplitude.

the time it takes a particle to transition from its mean location to its half-maximum displacement is given by,

$$\frac{A}{2} = A\sin\omega t$$
$$\sin\omega t = 0.5$$

Thus,

$$\omega t = \frac{\pi}{6}$$
$$\therefore \frac{2\pi}{T}t = \frac{\pi}{6}$$
$$t = \frac{T}{12}$$

So, the particle to cover half of maximum displacement is $\frac{T}{12}$.

13. Correct option: (D)3:4.

Given:

Equation of SHM,

$$y = 4\sin\left(80t + \frac{\pi}{2}\right)\text{m}$$

Here,

$$\omega_1 = 80$$
$$\omega_1 = \frac{2\pi}{T_1}$$
$$T_1 = \frac{2\pi}{\omega_1}$$
$$T_1 = \frac{2\pi}{80}$$
$$T_1 = \frac{\pi}{40}$$

For, $y = 2\cos\left(60t + \frac{\pi}{2}\right)$

$$\omega_2 = 60$$
$$\omega_2 = \frac{2\pi}{T_2}$$
$$T_2 = \frac{2\pi}{\omega_2}$$
$$T_2 = \frac{2\pi}{60}$$
$$T_2 = \frac{\pi}{30}$$

So, the ratio of time periods is given by,

$$\frac{T_1}{T_2} = \frac{\frac{\pi}{40}}{\frac{\pi}{30}}$$
$$\frac{T_1}{T_2} = \frac{30}{40}$$
$$\frac{T_1}{T_2} = \frac{3}{4}$$

So, the ratio of time periods is 3:4.

14. Correct option: (C) $\frac{\sqrt{3}}{2}v$.

At mean position,

$v = a\omega$

Velocity at half the amplitude is,

$$v' = \omega\sqrt{a^2 - y^2}$$
$$v' = \sqrt{a^2 - \frac{a^2}{4}} = \frac{\sqrt{3}}{2}a\omega = \frac{\sqrt{3}}{2}v$$

So, the particle at a distance equal to half of the amplitude will be $\frac{\sqrt{3}}{2}v$.

15. Correct option: (D) $0\frac{\text{m}}{\text{s}}$**.**

The particle's amplitude is 8 cm. Therefore, the particle is in its most extreme position if its displacement is 8 cm. Any simple harmonic motion has zero velocity at its most extreme position. Therefore, at that location, the particle's velocity is zero.

16. Correct option: (B) $\frac{1}{2}$**.**

We know that,

$a = -\omega^2 x$

$\omega^2 = 16\pi^2$

$\omega = 4\pi$

$T = \frac{2\pi}{\omega} = \frac{2\pi}{4\omega} = \frac{1}{2}$

So, the time period of SHM is $\frac{1}{2}$.

17. Correct option: (A) $50\frac{\text{m}}{\text{s}}$**.**

Given:

The equation of SHM is,

$y = 0.25\sin\left(200t + 0.5\right)$

Now , we know that,

$y = 0.25\sin\left(200t + 0.5\right)$

$v_{\max} = a\omega$

$\omega = 200, a = 0.25\text{m}$

$\text{v}_{\max} = 0.25 \times 200$

$v_{\max} = 50\frac{\text{m}}{\text{s}}$

So, the maximum velocity of particle is $50\frac{\text{m}}{\text{s}}$.

18. Correct option: (C)1.5.

$T = 628\text{ms}=0.628\text{s}$

$v_{\max} = 15\frac{\text{cm}}{\text{s}} = 0.15\frac{\text{m}}{\text{s}}$

The object's maximum speed is indicated by,

$v_{\max} = A\omega = A\frac{2\pi}{T}$

Amplitude is given by,

$A = \frac{v_{\max} T}{2\pi}$

$A = \frac{0.15 \times 0.628}{2 \times 3.14}$

$A = 1.5\text{cm}$

So, the amplitude of the motion in centimetres is 1.5 cm.

19. Correct option: (B) 2:1.

Velocity at mean position is given by as,

$v_{\max} = A\omega$

$\omega = \frac{2\pi}{T}$

Now,

$V_1 = \frac{A \times 2\pi}{T_1}$

$V_2 = \frac{A \times 2\pi}{T_2}$

$\frac{V_1}{V_2} = \frac{T_2}{T_1}$

$\frac{V_1}{V_2} = \frac{6}{3}$

$\frac{V_1}{V_2} = \frac{2}{1}$

So, the ratio of the velocities P and Q when they meet is 2:1.

20. Correct option: (B) π.

The maximum velocity is given by as,

$v_{\max} = A\omega$

$v_{\max} = A\frac{2\pi}{T}$

$v_{\max} = 4\frac{2\pi}{8}$

$v_{\max} = \frac{8\pi}{8} = \pi \frac{\text{cm}}{\text{s}}$

So, the maximum speed is $\pi \frac{\text{cm}}{\text{s}}$.

21. Correct option: (A) 10.

$m = 100gm = 0.1kg$

$T = 4.5sec$

$A = 7cm$

$\omega = \frac{2\pi}{T}$

$\omega = \frac{2\pi}{4.5}$

Now velocity is given by,

$v_{\max} = \omega A$

$v_{\max} = 2 \times 3.14 \times \frac{7}{4.5}$

$v_{\max} = 9.76$

$v_{\max} \approx 10 \frac{\text{cm}}{\text{s}}$

Velocity of the body at equilibrium position is $10 \frac{\text{cm}}{\text{s}}$.

22. Correct option: (C) 0.4ms^{-2}**.**

$T = \frac{11}{7}$

$\omega = \frac{2\pi}{T} = \frac{2\times 22}{7}\times\frac{7}{11} = 4\frac{\text{rad}}{\text{s}}$

$x = 0.025\text{m}$

We know that,

$a = -\omega^2 x = 16\times 0.025$

$a = -0.4\text{ms}^{-2}$

So, the maximum acceleration is 0.4ms^{-2}.

23. Correct option: (A) $\vec{F}.\vec{a}$**.**

$F.a = |F||a|\cos\theta$

Acceleration and force always move in the same direction. As a result, it will always be positive.

24. Correct option: (B) 2 N.

Amplitude, $A = 2\text{cm}$

Force, $F = 4\text{N}$

Now force is given by,

$F = mW^2A = 4$

$F_1 = mW^2x$

F being directly proportional to x, the force at halfway with an amplitude of 2 cm will be 2N.

Energy of SHM:

25. Correct option: (D) $\frac{3}{4}$**.**

We know that, Kinetic energy is given by,

$KE = \frac{1}{2}m\omega^2\left(a^2 - x^2\right)$

At, $x = \frac{a}{2}$

Now kinetic energy become,

$KE = \frac{1}{2}m\omega^2\left[a^2 - \left(\frac{a}{2}\right)^2\right]$

$KE = \frac{3}{4}\left[\frac{1}{2}m\omega^2a^2\right]$

Total kinetic energy $= \frac{1}{2}m\omega^2a^2$

Now fraction of total energy is,

$= \frac{KE}{Totalenergy} = \frac{\frac{3}{4}\left[\frac{1}{2}m\omega^2a^2\right]}{\frac{1}{2}m\omega a^2} = \frac{3}{4}$

Fraction of the total energy is kinetic is $\frac{3}{4}$.

26. Correct option: (C) Restoring force.

The restoring force is given by as,

$F_{\max} = k.y_{\max}$

$F_{\max} = kA$

Restoring force at,

$y = \frac{A}{2} = \frac{KA}{2}$

$F' = \frac{KA}{2}$

$F' = \frac{F_{\max}}{2}$

When a particle's displacement in simple harmonic motion (SHM) is equal to half of its amplitude ($y = \frac{A}{2}$), the restoring force (C) will actually be half of its highest magnitude. This is so because the restoring force in SHM, which reaches its maximum value at the amplitude, is precisely proportional to the displacement from the equilibrium position.

27. Correct option: (B) Amplitude and frequency.

Energy of the particle during the SHM is given by,

$E = \frac{1}{2} m\omega^2 A^2 + U_{mean}$

Where, U_{mean} is the (Potential energy) at mean position.

$E = \frac{1}{2} m(2\pi f)^2 A^2 + U_{mean}$

$\left(\because T = \frac{2\pi}{\omega} = \frac{1}{f} \right)$

$E = 2\pi^2 m f^2 A^2 + U_{mean}$

So, E is a frequency and amplitude dependent quantity.

28. Correct option: (B) $\frac{\pi}{\omega}$**.**

Given:

$x = A\sin(\omega + \phi)$

$T = \frac{2\pi}{\omega}$

Time period is given by,

$KE = \frac{T}{2} = \frac{\pi}{\omega}$

So, the kinetic energy oscillates with the period is $\frac{\pi}{\omega}$.

29. Correct option: (C) 4.

The conservation of total mechanical energy E in SHM is well known.

$$E = U + KE$$
$$For U = KE$$
$$U = \frac{E}{2}$$

$$\frac{1}{2}m\omega^2x^2 = \frac{1}{2}\left(\frac{1}{2}m\omega^2A^2\right)$$
$$x = \pm\frac{A}{\sqrt{2}}$$

These sites will be visited 4 times by the particle throughout one full cycle.

30. Correct option: (C) Independent of x.

The particle in simple harmonic motion is given by,

$x(t) = A\cos(\omega t + \phi)$

Therefore,

Potential energy is given by,

$$U = \frac{1}{2}kx^2$$

$$U = \frac{1}{2}kA^2\cos^2(\omega t + \phi)$$

Kinetic energy is given by,

$$KE = \frac{1}{2}mv^2$$

$$KE = \frac{1}{2}m\left(\frac{dx}{dt}\right)^2$$

$$KE = \frac{1}{2}kA^2\sin^2(\omega t + \phi)$$

Now, total energy is given by,

$E = U + KE$

$$E = \frac{1}{2}mA^2\cos^2(\omega t + \phi) + \frac{1}{2}mA^2\sin^2(\omega t + \phi)$$

$$E = \frac{1}{2}kA^2$$

Consequently, total energy is x independent.

Simple Pendulum:

31. Correct option: (D) $\frac{16}{9}$.

We know that,

$$T = 2\pi\sqrt{\frac{T}{g}} \cdots\cdots(1)$$

From the question: $\frac{T_1}{T_2} = \frac{3}{4}$

Now,

$$T \propto \frac{1}{\sqrt{g}}$$
$$T^2 \propto \frac{1}{g}$$
$$\frac{T_1^2}{T_2^2} = \frac{g_2}{g_1}$$
$$\frac{g_1}{g_2} = \frac{T_2^2}{T_1^2} = \frac{4^2}{3^2} = \frac{16}{9}$$

$$\frac{g_1}{g_2}=\frac{16}{9}$$

So, The acceleration due t on them are in the ratio is $\frac{16}{9}$.

32. Correct option: (A) 3.14sec.

Given:

Height, $h=8\text{m}$

Time period, $t=2\text{sec}$

Length of pendulum, $l=1\text{m}$

According to equation of motion,

$$s=ut+\frac{1}{2}gt^2$$

$$\therefore u=0$$

$$s=\frac{1}{2}gt^2$$

$$g=\frac{2h}{t^2}=\frac{2\times 8}{4}=4\text{ms}^{-1}$$

Now time period of pendulum is given by,

$$T=2\pi\sqrt{\frac{l}{g}}$$

$$T=2\pi\sqrt{\frac{1}{4}}$$

$$T=\frac{2\pi}{2}$$

$$T=\pi=3.14ec$$

So, the time period of pendulum is 3.14 sec.

33. Correct option: (C) 0.5 cm.

The following is the equation for the pendulum of time:

$$T=2\pi\sqrt{\frac{l}{g}}$$

Let the length of the pendulum be 'l' when g changes. Then,

$$T=2\pi\sqrt{\frac{l}{g}}\cdots\cdots(1)$$

$$T=2\pi\sqrt{\frac{l'}{g'}}\cdots\cdots(2)$$

From the equations (1) and (2),

$$\frac{l}{g}=\frac{l'}{g'}$$

$$l'=\frac{g'}{g}l$$

Pendulum length change is

$$\Delta l=l'-l$$

$$\Delta l=\frac{g'}{g}l-l$$

$$\Delta l = \left(\frac{g' - g}{g}\right) l$$

$$\Delta l = \left(\frac{g' - g}{g}\right)\left(\frac{2}{2\pi}\right)^2$$

$$\Delta l = \frac{983 - 978}{10}$$

$$\Delta l = \frac{5}{10}$$

$$\Delta l = 0.5\text{cm}$$

So, the increase length is 0.5 cm.

34. Correct option: (A) $\sqrt{6}$.

Time period of pendulum is given by,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Now, time period on earth is,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Time period on the moon is,

$$T' = 2\pi\sqrt{\frac{l}{g'}} = 2\pi\sqrt{\frac{l}{\frac{g}{6}}}$$

Therefore,

$$\frac{T'}{T} = \frac{2\pi\sqrt{\frac{l}{g}}}{2\pi\sqrt{\frac{l}{\frac{g}{6}}}}$$

$$\frac{T'}{T} = \frac{\sqrt{6}}{1}$$

$$T' = \sqrt{6}T$$

So, the surface of the moon w.r.t time period on the earth will be $\sqrt{6}$.

Springs:

35. Correct option: (B) $\frac{5g}{k}$m.

Time period of spring when mass m hangs from spring,

$$T = 2\pi\sqrt{\frac{m}{k}}$$

$$mg = kx$$

$$x = \frac{mg}{k}$$

$$x = \frac{5g}{k}$$

So, the required value is $\frac{5g}{k}$m.

36. Correct option: (A) $\sqrt{\frac{k}{m}}$**.**

Time period is given by as,

$$T = 2\pi\sqrt{\frac{m}{k}}$$

Therefore, natural angular frequency is given by,

$$\omega = \sqrt{\frac{k}{m}}$$

37. Correct option: (C) $\frac{16}{9}$**.**

The time period is given by,

$$T = 2\pi\sqrt{\frac{m}{k}} \cdots\cdots (1)$$

Also,

$$\frac{5T}{3} = 2\pi\sqrt{\frac{M+m}{k}} \cdots\cdots (2)$$

Using equation (1) and (2),

$$\frac{5}{3} = \sqrt{\frac{M+m}{M}} \cdots\cdots (2)$$

$$\frac{25}{9} = \frac{M+m}{M}$$

$$\frac{25}{9} = 1 + \frac{m}{M}$$

$$\frac{m}{M} = \frac{16}{9}$$

So, the ratio of $\frac{m}{M}$ is $\frac{16}{9}$.

38. Correct option: (B) $2\pi\sqrt{\frac{2m}{k}}$**.**

The spring constant in the equation is provided by:-

$$\frac{1}{k'} = \sum\frac{1}{k'} \Rightarrow \frac{1}{k'} + \frac{1}{k} + \frac{1}{2k} + \frac{1}{4k} + \frac{1}{8k} + \frac{1}{2048k}$$

$$\frac{1}{k'} = \frac{3841}{2048K}$$

Or, approximately,

$$k' \approx \frac{2048k}{3841} \approx \frac{k}{2}$$

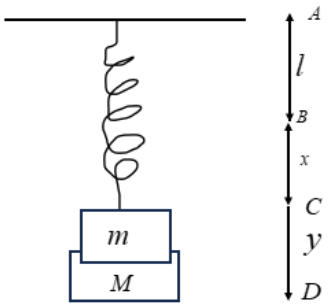
Now, time period is given by,

$$T' = 2\pi\sqrt{\frac{m}{k'}} = 2\pi\sqrt{\frac{2m}{k}}$$

So, the time period is approximately $2\pi\sqrt{\frac{2m}{k}}$.

39. Correct option: (B) $\frac{Mg}{k}$ **.**

Free body diagram:



Let l be the spring's original length. The string will stretch by an amount of x from B to C when one mass m is suspended, and by an amount of y from C to D when mass M is suspended.

Therefore, the system's equation of motion,

$(M+m)g = -k(x+y) \cdots\cdots (1)$

The motion equation will be when M is taken out.

$mg = -kx \cdots\cdots (2)$

Using equation (1) and (2), we get,

$Mg = -ky$

Or, $y = -\frac{Mg}{k}$

Whenever there is a negative sign, the displacement is pointed in the direction of the mean position.

Consequently, the oscillation's amplitude is $\frac{Mg}{k}$.

40. Correct option: (B) $T = 2\pi\sqrt{\frac{x(M+m)}{mg}}$

The time period given by as,

$T = 2\pi\sqrt{\frac{m}{k}}$

$\therefore kx = mg$

$\frac{1}{k} = \frac{x}{mg}$

when weight is added, total mass $= M+m$

Now,

$T = 2\pi\sqrt{\frac{M+m}{k}} = 2\pi\sqrt{\frac{x(M+m)}{mg}}$

So, the required time period is $2\pi\sqrt{\frac{x(M+m)}{mg}}$.

41. Correct option: (B) $\frac{2Mg}{k}$**.**

The potential energy is,

$$PE = \frac{1}{2}kx^2$$

We know that,

$$PE = Mgx$$

From the above two equation,

$$Mgx = \frac{1}{2}kx^2$$

$$x = \frac{2Mg}{k}$$

So, the maximum extension in the spring is $\frac{2Mg}{k}$.

42. Correct option: (B) $2.22\frac{\text{rad}}{\text{s}}$**.**

The mass of the body, $m = 0.98\text{kg}$

The spring's spring constant is,

$$k = 4.84\frac{\text{N}}{\text{m}}$$

The system's angular frequency is determined by:

$$\omega = \sqrt{\frac{k}{m}}$$

$$\omega = \sqrt{\frac{4.84}{0.98}}$$

$$\omega = 2.22\frac{\text{rad}}{s}$$

So, the angular frequency of the body is $2.22\frac{\text{rad}}{\text{s}}$.

LEVEL-II

1. Correct option: (C) $\frac{2\pi}{\omega}$**.**

$y = \sin\omega t + \sin 2\omega t = y_1 + y_2 + y_3$

$$T_1 = \frac{2\pi}{\omega}$$

$$T_2 = \frac{2\pi}{2\omega}$$

$$T_3 = \frac{2\pi}{3\omega}$$

In time $\frac{2\pi}{\omega}$ first function completes one oscillation, second function two oscillations and the third three so for the combination $T = \frac{2\pi}{\omega}$.

2. Correct option: (C) $-\frac{\pi}{6}$.

$y_1 = 0.sin\left(100\pi t + \frac{\pi}{3}\right)$

$y_2 = 0.1\,cos\,\pi\,t = 0.1\,sin\left(\frac{\pi}{2} + \pi t\right) = 0.1\,sin\,\pi\,t\left(\pi t + \frac{\pi}{2}\right)$

Now, distinguish between the two equations with respect to time to determine the particle's velocity.

$\frac{dy_1}{dt} = v_1 = 0.1 \times 100\pi\,cos\left(100\pi t + \frac{\pi}{3}\right) = 0.1\pi\,cos\left(\pi t + \frac{\pi}{3}\right)$

Similarly,

$\frac{dy_1}{dt} = v_2 = 0.1 \times \pi\,cos\left(\pi t + \frac{\pi}{2}\right) = 0.1\pi\,cos\left(\pi t + \frac{\pi}{2}\right)$

Now. If equation, $x = A\sin(\omega t + \phi)$ is given then, at $t = 0$ phase of motion is ϕ.

Similarly, at, $t = 0$ phase of 1st particle velocity is $\frac{\pi}{3}$.

And at, $t = 0$ phase velocity od 2nd particle is $\frac{\pi}{2}$.

Now we know that phase difference is given by,

Phase difference=Phase of 1st particle at $t = 0$ -Phase of 2nd particle at $t = 0$

$= \frac{\pi}{3} - \frac{\pi}{2} = -\frac{\pi}{6}$

So, the phase difference is $-\frac{\pi}{6}$.

3. Correct option: (C) $\frac{T}{8}$.

We know that,

$y = A\sin\omega t$

Let, $t = 0$, $x = 0$

Therefore,

$x = \frac{A}{\sqrt{2}}$

$\frac{A}{\sqrt{2}} = A\,sin\,\omega\,t$

$sin\,\omega\,t = \frac{1}{\sqrt{2}}$

For shortest time, $\omega t = \frac{n}{4}$

$\Rightarrow \left(\frac{2n}{T}\right)t = \frac{n}{4}$

$t = \frac{T}{8}$

So, the time required is $\frac{T}{8}$.

4. Correct option: (A) 300 units.

Simple harmonic equation:

$y = 3\sin\left(100 + \frac{\pi}{6}\right)$

The wave's amplitude is determined by the wave equation and is, $A = 3$

The wave's angular velocity can be calculated as,

$\omega = 100\dfrac{\text{rad}}{\text{sec}}$

We are aware that the velocity can be calculated as:

$v_{\max} = a\omega$

$v_{\max} = 3\times 100 = 300\text{units}$

So, the maximum velocity of SHM is 300 units.

5. Correct option: (D) $48\dfrac{\text{m}}{\text{s}^2}$**.**

Velocity, $V = \omega\sqrt{A^2 - x^2}$

Comparing, $\omega = 4\dfrac{\text{rad}}{\text{sec}}$

$A^2 = 9$

$A = 3\text{m}$

Now we know that maximum acceleration is given by,

$a = -\omega^2 A$

$a = -16\times 3 = -48\dfrac{\text{m}}{\text{s}^2}$

So, the maximum acceleration is $48\dfrac{\text{m}}{\text{s}^2}$.

6. Correct option: (D) $\dfrac{A^2 - X^2}{X^2}$**.**

Total energy is given by as,

$T = \dfrac{1}{2}m\omega^2(A^2 - X^2)$

$V = \dfrac{1}{2}m\omega^2 X^2$

Now,

$\dfrac{T}{V} = \dfrac{\frac{1}{2}m\omega^2(A^2 - X^2)}{\frac{1}{2}m\omega^2 X^2}$

$\dfrac{T}{V} = \dfrac{A^2 - X^2}{X^2}$

So, the ratio of T to U is $\dfrac{A^2 - X^2}{X^2}$.

7. Correct option: (B) 18J.

$y = 6\sin\left(100t + \dfrac{\pi}{4}\right)$

$A = 6\text{cm}$

$m = 1\text{kg}$

$\omega = 100\dfrac{\text{rad}}{\text{sec}}$

Now the maximum kinetic energy is given by,

$$KE_{\max} = \frac{1}{2} m\omega^2 A^2$$

$$KE_{\max} = \frac{1}{2} \times 1 \times (100)^2 \times (10^{-2})^2 \times 6^2$$

$$KE_{\max} = 18\text{J}$$

So, the maximum kinetic energy is 18 J.

8. Correct option: (B) $m\omega^2 A^2$**.**

Given: SHM equations,

$y_1 = A\sin\omega t$

$y_2 = A\cos\omega t$

Now,

$$y = y_1 + y_2 = A\sin\omega t + A\cos\omega t = \sqrt{2}A\sin\left(\omega t + \frac{\pi}{4}\right)$$

Amplitude is given by as,

$$\text{Amplitude} = \sqrt{2}A$$

So, the total energy of the system is,

$$\frac{1}{2} m\omega^2 \left(\sqrt{2}A\right)^2 = m\omega^2 A^2$$

So, the total mechanical energy of the particle is $m\omega^2 A^2$.

9. Correct option: (C) $\frac{\sqrt{3}}{2}T$**.**

Time period of pendulum is given by,

$$T_p = 2\pi\sqrt{\frac{\text{String length}}{\text{Downward acceleration}}} = 2\pi\sqrt{\frac{L}{a_y}}$$

At stationary lift, acceleration is , $a_y = g$

Time period is,

$$T = 2\pi\sqrt{\frac{L}{g}} \cdots\cdots (1)$$

Lift accelerates upward, $a = g + \frac{g}{3} = \frac{4g}{3}$

Time period is, $T' = 2\pi\sqrt{\frac{L}{\frac{4g}{3}}} = \frac{\sqrt{3}}{2}T$

So, the time period is $\frac{\sqrt{3}}{2}T$.

10. Correct option: (C) $T_1^2 + T_2^2 = 2T^2$**.**

The time period is given by as,

$$T = 2\pi\sqrt{\frac{l}{g}}$$

Now,

$T_1 = 2r\sqrt{\frac{1+10\text{cm}}{g}}$

$T_1 = 2r\sqrt{\frac{1-10\text{cm}}{g}}$

$T_1^2 + T_2^2 = 4\pi^2\left(\frac{2l+10\text{cm-10m}}{g}\right)$

$L = 2\times 30\text{cm}=60\text{cm}$

$T_1^2 + T_2^2 = 2T^2$

So, the required relation is $T_1^2 + T_2^2 = 2T^2$.

11. Correct option: (D) $\frac{T}{2}$.

Let, T_1 be time period in case(i),

We also know that,

$T = 2\pi\sqrt{\frac{m}{k}} \cdots\cdots (1)$

We know that if spring cut in x halves then,

k (for each spring)= $xk = 2k(x = 2)$

Case(ii):

$k_{net} = k_1 + k_2$

$k_{net} = 2k + 4k$

$k_{net} = 4k$

Now using equation (1) we get,

$T \propto \frac{1}{\sqrt{k}}$

$\frac{T_1}{T_2} = \sqrt{\frac{k_2}{k_1}}$

$\frac{T_1}{T_2} = \sqrt{\frac{4k}{k}}$

$T_2 = \frac{T}{2}$

So, time period will be $\frac{T}{2}$.

12. Correct option: (D) $2\pi\sqrt{\frac{m}{3k}}$.

The time period is given by as,

$T = 2\pi\sqrt{\frac{m}{k}}$

The equivalent spring constant= $2k + k = 3k$

Noe, the time period of these oscillations is $2\pi\sqrt{\frac{m}{3k}}$

13. Correct option: (A) $1:\sqrt{2}:\frac{1}{\sqrt{2}}$**.**

The time period is given by as,

$T=2\pi\sqrt{\frac{m}{k}}$

For case(i)

The time period is given by as,

$T_1=2\pi\sqrt{\frac{m}{k}}$

For case(ii):

$T_2=2\pi\sqrt{\frac{2m}{k}}$ (Series combination of springs)

For case(iii):

For case(ii):

$T_3=2\pi\sqrt{\frac{m}{2k}}$ (Parallel combination of springs).

Taking the ratio, $T_1:T_2:T_3=1:\sqrt{2}:\frac{1}{\sqrt{2}}$.

So, the ratio is $1:\sqrt{2}:\frac{1}{\sqrt{2}}$.

14. Correct option: (A) 1.1 sec.

The length of liquid column,

$L=2\times30\text{cm}=60\text{cm}$

The time period is given by as,

$T=2\pi\sqrt{\frac{L}{2g}}$

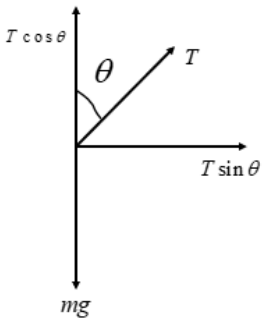
$T=2\pi\sqrt{\frac{60}{2\times9.8}}$

$T=1.1\text{sec}$

So, time period is 1.1 sec.

15. Correct options: (A) and (D).

Free body diagram:



From the free body diagram,

$T\cos\theta=mg\cdots\cdots(1)$

$T\sin\theta=ma\cdots\cdots(2)$

Using equation (1) and (2), we get,

$\tan\theta = \frac{a}{g}$

$\theta = \tan^{-1}\left(\frac{a}{g}\right)$

So, string makes an angle of $\tan^{-1}\left(\frac{a}{g}\right)$ with the vertical.

Now tension is given by as,

$T = \sqrt{(mg)^2 + (ma)^2}$

$T = m\sqrt{g^2 + a^2}$

So, the tension in the string is $m\sqrt{g^2 + a^2}$

16. Correct option: (B),(C) and (D).

17. Correct options: (B) and (C).

The highest velocity happens at mean position $(x = 0)$.

Thus velocity is maximum at $\frac{T}{4}$ and $\frac{3T}{4}$, not $\frac{T}{2}$.

Extreme positions—i.e., 0, $\frac{T}{2}$, T—have the highest rates of acceleration. B is right option.

At the mean position, which is at $\frac{T}{4}$, $\frac{3T}{4}$, there is no force or acceleration. C is right option.

KE=PE, when PE=0, which occurs at mean position.

$\frac{T}{2}$ is extreme not mean, thus D is incorrect.

18. Correct options: (B),(C) and (D).

We know that,

$x = A\sin(\omega t + b)$

$V = A\omega\cos(\omega t + b)$

$a = -A\omega^2\sin(\omega t + b)$

So, $x^2 + \frac{v^2}{\omega^2} = A^2$ This is the equation of ellipse.

$a = -\omega^2 x$ This is the equation of straight line .

$v^2 + \frac{a^2}{\omega^2} = A^2\omega^2$ This is the equation od ellipse.

19. Correct options: (B),(C) and (D).

$a = 0 \text{ at } x = 0.5\text{m}$

And particle is released from, $x = 2\text{m}$.

Hence, $A = 2 - 0.5 = 1.5\text{m}$

We know that,

$\omega^2 = 100$

$\omega = 10\frac{\text{rad}}{\text{sec}}$

Now, time period is given by as,

$T = \frac{2\pi}{\omega} = \frac{2\pi}{10}$

$T = 0.63sec$

v_{max}

$v \frac{m}{s}_{max}$

B) The particle will perform SHM of amplitude of 1.5 m.

The particle will perform SHM of time period 0.63.s.

So, the particle will have a maximum velocity of $15\frac{\text{m}}{\text{s}}$.

20. Correct options: (A),(B) and (C).

From the graph,

$x = A\cos\omega t$

(a) At, $t = \frac{3T}{4}$, The particle is positioned mean.

Acceleration =0, Force =0

(b) At, $t = T$ particle is again at extreme position. So acceleration is maximum

(c) At, $t = \frac{T}{4}$ The particle is positioned mean. Velocity is at its maximum.

(d) When KE=PE

$\frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2$

$\frac{1}{2}k(A^2 - x^2) = \frac{1}{2}kx^2$

$A^2 = 2x^2$

$x = \pm\frac{A}{\sqrt{2}}$

But at, $t = \frac{T}{2}, x = -A$

21. Correct options: (B) and (D).

Average kinetic energy per cycle,= $KE_{avg} = \frac{1}{4}m\omega^2A^2$

Maximum kinetic energy per cycle= $KE_{\max} = \frac{1}{2}m\omega^2A^2$

$KE_{avg} = \frac{1}{2} \times KE_{\max}$

For periodic curve,

$x_{rms} = \frac{x_{\max}}{\sqrt{2}}$

Since, velocity in SHM also varies periodically.

$v_{rms} = \frac{v_{\max}}{\sqrt{2}}$

22. Correct option:(B) $\frac{\pi}{2}\frac{\text{rad}}{\text{s}}$

The angular frequency is given by as,

$$\omega = \frac{2\pi}{T}$$

$$\omega = \frac{2\pi}{4} = \frac{\pi}{2}\frac{\text{rad}}{\text{s}}$$

So, the angular frequency of particle is $\frac{\pi}{2}\frac{\text{rad}}{\text{s}}$.

23. Correct option: (B) $\pi\frac{\text{cm}}{\text{s}}$**.**

Maximum speed is given by as,

$$v_{\max} = A\omega$$

$$v_{\max} = 2.\frac{\pi}{2} = \pi\frac{\text{cm}}{\text{s}}$$

So, maximum speed of particle is $\pi\frac{\text{cm}}{\text{s}}$.

24. Correct option: (B) 0 cm.

At , $t = 0$ particle starts from extreme position.

$$x = A\cos\omega t$$

$$x = 2\cos\left(\frac{\pi}{2}t\right)$$

$$At, t = 1\text{sec}$$

$$x = 0$$

So, the displacement of particle at t = 3 sec is 0.

25. Correct option: (A)

26. Correct option: (C)

27. Correct option: (A)

28. Correct Matching is: (a)-(Q),(b)-(R),(c)-(S),(d)-(T).

$$y = A\sin\omega t + A\sin\left(\omega t + \frac{2\pi}{3}\right)$$

This might also be expressed as,

$$y = A\left[\sin\omega t + \sin\left(\omega t + \frac{2\pi}{3}\right)\right]\cdots\cdots(i)$$

Using the trigonometry formula,

$$\sin C + \sin D = 2\sin\left(\frac{C+D}{2}\right)\cos\left(\frac{C-D}{2}\right)$$

Now, equation (1) become,

$$y = 2A\,sin\left(\omega t + \frac{\pi}{3}\right)cos\left(-\frac{\pi}{3}\right)$$

$$y = 2A\,sin\left(\omega t + \frac{\pi}{3}\right)cos\left(\frac{\pi}{3}\right)$$

$$y = 2A\,sin\left(\omega t + \frac{\pi}{3}\right)\cdots\cdots(ii)$$

Comparing the above equation with $y = A\sin\left(\omega t + \phi\right)$.

Now we can conclude that equation(ii) represents SHM with amplitude $= A$.

Initial phase, $\phi = \frac{\pi}{3}$

Now, differentiating equation(ii) with respect to on both sides we get,

$$v = \frac{dy}{dt} = \omega A \cos\left(\omega t + \frac{\pi}{3}\right)$$

Velocity (v) will be at its maximum when,

$$A\cos\left(\omega t + \frac{\pi}{3}\right) = 1$$

So, the maximum velocity is,

$v_{max} = \omega A$

29. Correct Matching is: (a)-(Q),(b)-(P),(c)-(S),(d)-(R).

(a) The velocity is maximum at mean position.

$v_{max} = A\omega$

(b) Kinetic energy is given by,

$KE = \frac{1}{2}mv^2 = \frac{1}{2}m\omega^2(A^2 - x^2)$

Total kinetic energy is given by,

$KE_{Total} = \frac{1}{2}m\omega^2 A^2$

$$KE = \frac{3}{4} \times KE_{Total}$$

$$A^2 - x^2 = \frac{3}{4}A^2$$

$$x^2 = \frac{A^2}{4}$$

$$x = \frac{A}{2}$$

(c) Potential energy, $PE = \frac{1}{2}m\omega^2 x^2 = \frac{3}{4} \times \frac{1}{2}m\omega^2 A^2$

$x = \frac{\sqrt{3}}{2}A$

(d) The maximum Acceleration at extreme position is given by, $a_{max} = -\omega^2 A$

30. Correct Matching is: (a)-(R),(b)-(S),(c)-(T),(d)-(P).

(a) The maximum velocity is given by, $v_{max} = A\omega$

(b) The maximum acceleration is given by, $a_{max} = \omega^2 A$

(c) The maximum force is given by,

$F_{max} = m \times a_{max}$

$F_{max} = -m\omega^2 A$

(d) The maximum total energy is given by, $E_{max} = \frac{1}{2}mv_{max}^2 = \frac{1}{2}m\omega^2 A^2$

31. Correct Answer is: 2

The energy is given by,

$$E = \frac{1}{2}m\omega^2 A^2$$

$$E = \frac{1}{2} \times 10 \times (2\pi \times 20)^2 \times (0.05)^2$$

$$E = 2J$$

So, the energy of oscillation is 2J.

8. Heat
Solutions

Temperature Scales and Thermal Expansion

LEVEL-I

Concept of temperature:

Measurement of temperature:

1. Correct option: (B) Temperature.

The direction of heat flow is determined by the difference in body temperatures. A body's heat transfers from one that is hotter to one that is colder.

2. Correct option: (D) Both A & C.

When a substance is heated, its volume grows while its mass stays constant; as a result, the density of the substance, which is the ratio of mass to volume, falls with an increase in temperature.

3. Correct option: (C) $160^{\circ}\mathrm{C}$ **.**

The relation between Fahrenheit scale and Celsius Scale.

$$F=\frac{9}{5}C+32$$

Given: $2C=F$

Now,

$$2C=\frac{9}{5}C+32$$

$$0.2C=32$$

$$C=16\times10$$

$$C=160^{\circ}\mathrm{C}$$

Therefore, at $160^{\circ}\mathrm{C}$, the reading on the Fahrenheit scale is twice that of the Celsius scale.

4. Correct option: (C) $90^{\circ}\mathrm{C}$ **.**

Utilize the formula below to translate the temperature from the specified thermometer scale to Celsius (°C).

$$\mathrm{C}=\frac{\text{Reading -Lower fixed Point}}{\text{Upper fixed point-Lower fixed point}}$$

$$\mathrm{C}=\frac{\text{X-LFP}}{\text{(UFP-LFP)}}\times100$$

$$C=\frac{87-6}{96-6}\times100$$

$$C=90^{\circ}\mathrm{C}$$

So, the temperature on the centigrade scale is $90^{\circ}\mathrm{C}$. $l_2=\alpha\left(l_1\right)(\ )$

Thermal expansion of solids:

5. Correct option: (C) a and b.

Solids expand linearly, superficially (in area), and cubically, but liquids and gases expand cubically.

Therefore, both the assertion "b) Liquids and gases undergo cubical expansion" and the statement "a) Solids undergo linear, superficial, and cubical expansions" are true. As a result, option C, which incorporates both assertions a and b, is the appropriate response.

6. Correct option: (A) 1:2:3.

We know that,

$\gamma = 3\alpha$

$\beta = 2\alpha$

So, the ratio of linear, superficial, and cubical expansion coefficients is $\alpha : \beta : \gamma = 1:2:3$.

Linear expansion coefficient (α):

7. Correct option: (A) $L_1\alpha_1 = L_2\alpha_2$ **.**

Difference of length at any temperature = $L_1(1+\alpha_1 T) - L_2(1+\alpha_2 T)$

Difference of length at any temperature= $(L_1 - L_2) + T(L_1\alpha_1 - L_2\alpha_2)$

The coefficient of temperature must now disappear because length is no longer dependent on temperature.

$(L_1\alpha_1 - L_2\alpha_2) = 0$

$L_1\alpha_1 = L_2\alpha_2$

So, the condition for the difference in lengths to be constant at all temperature is $L_1\alpha_1 = L_2\alpha_2$

8. Correct option: (B) $5^{\circ}\mathrm{C}$**.**

Range $= 5\times10^{5}\,\mathrm{mm}$

$\alpha = 10\times10^{-6}\,\mathrm{K^{-1}}$

The maximum length variation permitted at 1 mm intervals is,

$1\mathrm{mm}\times\alpha\Delta\theta = 5\times10^{-5}\,\mathrm{mm}$

So,

$\Delta\theta = \frac{5\times10^{-5}}{10\times10^{-6}} = 5^{\circ}C$

So, the maximum temperature variation allowable during the ruling is $5^{\circ}\mathrm{C}$.

9. Correct option: (C) $11\times10^{-5}\,\mathrm{cm}$ **shorter.**

Usually, as we lower the iron bar's temperature, it gets shorter than it was.

Change in length is given by,

$l_2 = \alpha(l_2)(t_2 - t_1)$

$l_2 = (11\times10^{-6})(10)(19-20)$

$l_2 = 11\times10^{-5}\mathrm{cm}$

So, length is shortened by shorter.

10. Correct option: (B) $6.45\times10^{-6}\,\mathrm{K^{-1}}$**.**

Initial temperature, $T_1 = 285\mathrm{K}$

Final temperature, $T_2 = 373\mathrm{K}$

Initial length, $L_0 = 3.521\mathrm{m}$

Change in length, $\Delta L = 3.523\mathrm{cm}$

The coefficient of linear expansion is given by,

$$\alpha_L = \frac{\Delta L}{L_0\Delta T}$$

$$\alpha_L = \frac{(3.523-3.521)}{3.521(373-285)}$$

$\alpha_L = 6.45\times10^{-6}\,\mathrm{K^{-1}}$

So, the coefficient of linear expansion for the metal is $6.45\times10^{-6}\,\mathrm{K^{-1}}$.

11. Correct option: (A)

12. Correct option: (B) 50.0928 m.

Given:

$L = 50.0\text{m}$

$\Delta T = 80\text{-}16 = 64^{\circ}\text{C}$

$\alpha = 29\times10^{-6}\,\text{K}$

We know that,

$dl = 29\times10^{-6}\times50\times(80-16)$

$dl = 0.0928\text{m}$

Now,

Length=L+dl

Length=50+0.0928

Lemght=50.0928m

So, increase in the length of the hot pipe 50.0928m.

Superficial expansion coefficient (β):

13. Correct option: (A) $-3.75^{\circ}\,\text{C}$**.**

Area of expansion is given by as:

$$\frac{\Delta A}{A_0} = \alpha_A \Delta T$$

$$\text{or } A_{t_2} = A_{t_1}\left[1+\alpha\left(t_2-t_1\right)\right]$$

Where, A_{t_2} =Area at temperature t_2

α = Area expansion coefficient

Given:

At $t_2 - t$, $A_{t_2}\left(\text{brass}\right) = A_{t_1}\left(\text{steel}\right)$

$$\alpha_{brass} = \alpha_B = 19\times10^{-6}\,\frac{1}{^{\circ}\text{C}}$$

$$\alpha_{steel} = 11\times10^{-6}\,\frac{1}{^{\circ}\text{C}}$$

For brass $t_1\left(\text{brass}\right) = t_{1B} = 10^{\circ}\text{C}$

For steel, $t_1\left(\text{steel}\right) = t_{1s} = 20^{\circ}\text{C}$

Also, $A_{t_1(\text{brass})}\left(\text{brass}\right) = A_{t_1(\text{steel})}(\text{steel}) = A$

Upon substitution we get,

$$A\left[1+\alpha_b(t-t_{1B}\right] = A\left[1+\alpha_s(t-t_{1s})\right]$$

$$\text{Or}, \alpha_B(t-t_B) = \alpha_s(t-t_s)$$

$$19\times10^{-6}\left(t-10\right) = 11\times10^{-6}\left(t-20\right)$$

$$8t = -30$$

$$t = -3.75^{\circ}\,\text{C}$$

So, the common temperature is $-3.75^{\circ}\,\text{C}$.

14. Correct option: (B) $2.584\,\text{mm}^2$**.**

$A_0 = 800\text{mm}^2$

$\Delta T = 100 - 15 = 85^{\circ}\text{C}$

$\alpha = 19 \times 10^{-6}\text{K}^{-1}$

$\beta = 2\alpha = 2 \times 19 \times 10^{-6} = 38 \times 10^{-6}\text{K}^{-1}$

The formula for area expansion can be used to calculate the increase in the silver plate's area as the temperature rises from 15 to 100 degrees Celsius:

$\Delta A = A_0 \times \beta \times \Delta T$

$\Delta A = 800 \times 38 \times 10^{-6} \times 85 = 2.584\,\text{mm}^2$

$2.584\,\text{mm}^2$

So, the increase in area is $2.584\,\text{mm}^2$

Cubical expansion coefficient (γ)**:**

15. Correct option: (B) Thrice the coefficient of linear expansion.

The coefficient of superficial expansion, or the increase in area with temperature, is roughly twice that of linear expansion. The cubic expansion coefficient is nearly three times greater than the linear expansion coefficient.

16. Correct option: (B)

17. Correct option: (A) 358.8 K.

We know that,

$V_2 = V_1[1 + \gamma(t_2 - t_1)]$

$480 \times 10^9 = 440 \times 10^9[1 + (12 \times 10^{-4})(t_2 - 283)]$

$480 = 440 + 440(12 \times 10^{-4})(t_2 - 283)$

$480 - 440 = 440(12 \times 10^{-4})(t_2 - 283)$

$(t_2 - 283) = \dfrac{440}{440 \times 12 \times 10^{-4}} = 75.76\text{K}$

And $t_2 = 75.76 + 283 = 358.8K$

So, the required temperature is 358.8 K.

18. Correct option: (B) $2.012l$**.**

The volumetric expansion formula can be used to calculate the volume of water at 40 degrees Celsius:

$V_f = V_i(1 + \beta\Delta T)$

Where, V_i =initial volume *at* 20°C =2 litres

V_f =Final volume *at* 40°C

β =The coefficient of volumetric expansion= $5 \times 10^{-5}\,\dfrac{1}{^{\circ}\text{C}}$

ΔT = The change in temperature= $= (40 - 20) = 20^{\circ}\text{C}$

Now, substitute the value we get,

$V_f = 2\left(1 + (5 \times 10^{-5} \times 20)\right)$

$V_f = 2.012l$

So, the volume *at* 40°C is $2.012l$.

Note: Two options are same.

Relation between expansion coefficients:

19. Correct option: ()

We know that,

$\Delta l = l\alpha\Delta T$

$\Rightarrow \alpha = \frac{\Delta l}{l\Delta T} = \frac{0.08\times10^{-1}}{10\times(100-0)} = 8\times10^{-6}$

Now,

$\gamma = 3\alpha = 24\times10^{-6}$

From the volumetric expansion:

$V = V_0(1+\gamma\Delta T) = 100\times\left(1+25\times10^{-6}\times(100-0)\right)$

$V = 100\times1.0024 = 100.24\text{cc}$

So, . The volume it measures at 100°C in C.C is 100.24cc.

Note: Option is not available.

20. Correct option: (D) 0.36.

We know that,

$\gamma = \frac{\Delta V}{V\Delta t}$

$\because \gamma = 3\alpha$

$V = l^3$

$3\alpha = \frac{\Delta V}{l^3\Delta t}$

$\Delta V = 3\alpha l^3\Delta t$

$\Delta V = 3\times12\times10^{-6}\times(10)^3\times10$

$\Delta V = 0.36\text{c.c}$

So, change in volume in c.c. is 0.36c.c.

21. Correct option: (C) $5\times10^{-5}\frac{1}{^{o}\text{C}}$**.**

Given:

$\Delta T = 20^{o}\text{C}$

$\frac{\Delta V}{V} = 0.3\% = \frac{0.3}{100} = 3\times10^{-3}$

In thermal expansion:

$\frac{\Delta V}{V} = 3\alpha\Delta T$

$3\times10^{-3} = 3\alpha\Delta T$

$3\times10^{-3} = 3\alpha\times20$

$\alpha = 5\times10^{-5}\frac{1}{^{o}\text{C}}$

So, coefficient of linear expansion is $5\times10^{-5}\frac{1}{^{o}\text{C}}$.

Applications of expansion of solids:

22. Correct option: (D) Will bend with steel on concave side.

We are aware that aluminium has a higher heat conductivity than steel. A bimetallic strip is originally bonded together using both steel and aluminium strips. Steel will expand less than aluminium when heated, on average.

23. Correct option: (A) A) Loses time.

The time period of pendulum is given by,

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Where, T is the time period.

$$g = 9.81\frac{\text{m}}{\text{s}^2}$$

Temperature-related changes in pendulum length (L) cause changes in both period and length. Higher temperatures cause the length to lengthen and the period to lengthen, both of which display the ticks' times. Thus, the clock will become inaccurate.

24. Correct option: (A) 500°C**.**

We know that,

$\Delta L = L\alpha\Delta T$

$6 = 994\times 3.6\times 10^{-5}\times\Delta T$

$$\Delta T = \frac{6}{994\times 3.6\times 10^{-5}} = 503^{\text{o}}\text{C}$$

$\Delta T \approx 500^{\circ}\text{C}$

So, the temperature increase is 500°C

25. Correct option: (A) 474.54°C**.**

Diameter of no-tyre=2m.

Its is fitted on to a wooden wheel of diameter 2.01 m.

$$\alpha = 11\times 10^{-6}\frac{1}{^{\text{o}}\text{C}}$$

Now strain is given by,

$$Strain = \frac{L_f - L_i}{L_i} = \frac{2.01 - 2}{2} = \frac{0.01}{2} = 5\times 10^{-3}$$

Now, we know that,

Strain $= \alpha\Delta T$

$5\times 10^{-3} = 11\times 10^{-6}(\Delta T)$

$$\Delta T = \frac{0.01}{2\times 11\times 10^{-6}}$$

$\Delta T = 454.54^{o}C$

$T_f - 20 = 454.54^{o}C$

$T_f = 474.54^{\circ}C$

So, the temperature to which the tyre must be heated is 474.54°C.

26. Correct option: (A) 74 k.

Given:

$L_1 = 15.02\text{mm} = 15.02\times 10^{-3}\text{m} = 0.01502\text{m}$

$L_2 = 15\text{mm} = 0.0015\text{m}$

$\alpha = 18\times 10^{-6}\text{K}^{-1}$

Now we know that,

$L_2 = L_1\left[1+\alpha\left(t_2-t_1\right)\right]$

$0.015 = 0.01502 + \left(0.01502\right)\left(18\times10^{-6}\right)\left(t_2-t_1\right)$

$0.015 = 0.01502 + (0.01502)(18\times10^{-6})(t_2-t_1)$

$0.015 - 0.01502 = 0.01502(18\times10^{-6})(t_2-t_1)$

$-0.00002 = 0.01502\times18\times10^{-6}\times(t_2-t_1)$

$t_2-t_1 = K\dfrac{-0.000012}{0.01502\times18\times10^{-6}}$

$t_2-t_1 = 74\text{K}$

So, For the insertion to be possible, the shaft needs to be cooled by 74 K.

27. Correct option: (D) 8.64sec**.**

Time period is given by as,

$T = 2\pi\sqrt{\dfrac{L}{g}}\cdots\cdots(1)$

Actual length is given by,

$L = L_0\left(1+\alpha\Delta\theta\right)$

$\dfrac{L-L_0}{L} = \dfrac{\Delta L}{L} = \alpha\Delta\theta\cdots\cdots(2)$

Using equation(1),

$\dfrac{\Delta T}{T} = \dfrac{1}{2}\times\dfrac{\Delta L}{L}$

Now, using equation(2),

$\dfrac{\Delta T}{T} = \dfrac{1}{2}\alpha\Delta\theta$

For a day, $T = 24\text{hours}$=86400sec

$\dfrac{\Delta t}{t} =$

So, the average temperature is 8.64sec.

28. Correct option: (A) 155.5 sec.

Time period is given by as,

$T = 2\pi\sqrt{\dfrac{L}{g}}\cdots\cdots(1)$

Using equation(1),

$\dfrac{\Delta T}{T} = \dfrac{1}{2}\times\dfrac{\Delta L}{L} = \dfrac{1}{2}\alpha\Delta\theta$

$\alpha_{brass} = 1.8\times10^{-5}\dfrac{1}{^{\circ}\text{C}}$

Therefore,

$\Delta T = \dfrac{86400}{2}\times1.8\times10^{-5}\times20$

$\Delta T = 155.5\text{sec}$

So, the required time is 155.5sec.

29. Correct option: (A) $\frac{1}{84600}/^{\circ}\mathrm{C}$.

Fractional time increase associated with temperature change,

$\frac{\Delta t}{t}=\frac{1}{2}\alpha\Delta T\cdots\cdots(1)$

Where,

$\frac{\Delta t}{t}=$ Fractional increment

$\alpha=$ coefficient of linear expansion

$\Delta T=$ change in temperature

$T_0=0^{\circ}\mathrm{C}$

clocks show correct time, and at $T=25^{\circ}\mathrm{C}$ it loses 12.5 sec.

$\frac{\Delta t}{t}=\frac{-12.5}{24\times60\times60}$

$\Delta T=T_0-T=0-25=-25^{\circ}\mathrm{C}$

Substitute the values in equation (1),

$\frac{-12.5}{24\times60\times60}=\frac{1}{2}\times\alpha\times(-25)$

$\frac{-12.5}{86400}=-12.5\times\alpha$

$\alpha=\frac{1}{86400}\frac{1}{^{\circ}\mathrm{C}}$

So, the coefficient of linear expansion of metal of pendulum is $\frac{1}{84600}/^{\circ}\mathrm{C}$.

30. Correct option: (D) 1×10^{-3}.

Fractional time increase associated with temperature change,

$\frac{\Delta t}{t}=\frac{1}{2}\alpha\Delta T\cdots\cdots(1)$

Where,

$\frac{\Delta t}{t}=$ Fractional increment

$\alpha=$ coefficient of linear expansion

$\Delta T=$ change in temperature

$\Delta T=10^{\circ}\mathrm{C}$

$\alpha=2\times10^{-6}\frac{1}{^{\circ}\mathrm{C}}$

Using equation(1),

$\frac{\Delta t}{t}=\frac{1}{2}\times2\times10^{-6}\times10$

$\frac{\Delta t}{t}=10^{-5}$

percentage change of period,

$$\%\frac{\Delta t}{t}=\frac{\Delta t}{t}\times 100$$

$$\frac{\Delta t}{t}=10^{-5}\times 100$$

$$\frac{\Delta t}{t}=1\times 10^{-3}$$

So, the period of the pendulum will have percentage increase of 1×10^{-3}.

LEVEL-II

1. Correct option: (C) 30.372 mm.

Given:

$R_1 = 30\text{mm}$

$T_1 = 20^{o}\text{C}=20+273=293\text{K}$

$T_2 = 420+273=693\text{K}$

$\alpha=31\times10^{-6}\text{K}^{-1}$

We know that,

$$R_2 = R_1\left[1+\alpha\left(T_2-T_1\right)\right]$$

$$R_2 = 30\left[1+31\times10^{-6}(692-293)\right]$$

$$R_2 = 30.372\text{mm}$$

So, the increase in the radius is 30.372 mm.

2. Correct option: (A) 280.5mm^2**.**

Given:

$R_1 = 30\text{mm}$

$T_1 = 20^{o}\text{C}=20+273=293\text{K}$

$T_2 = 420+273=693\text{K}$

$\alpha=31\times10^{-6}\text{K}^{-1}$

Initial area is ,

$$A_1 = 4\pi R^2$$

$$A_1 = 4\pi\left(30\right)^2 = 3600\pi\text{mm}^2$$

We know that,

$A_2 = A_1[1+\beta(T_2-T_1)]$

$A_2 = 3600\pi[1+2\times\alpha(693-293)]$

$A_2 = 3600\pi[1+2\times31\times10^{-6}\times(693-293)]$

$A_2 = 3600\pi+3600\pi(0.0248)$

$A_2 = 280.5\text{mm}^2$

So, the increase in the surface area is 280.5mm^2.

3. Correct option: (C) 4207mm^3**.**

Given:

$R_1 = 30\text{mm}$

$T_1 = 20^{o}\text{C}=20+273=293\text{K}$

$T_2 = 420+273=693\text{K}$

$\alpha=31\times10^{-6}\text{K}^{-1}$

Initial volume is,

$V_1 = \frac{4}{3}\pi R_1^3$

$V_1 = \frac{4}{3}\pi \times (30)^3 = 3769.91\text{mm}^3$

We know that,

$V_2 = V_1[1 + \gamma(T_2 - T_1)]$

$V_2 = 3769.91[1 + 3 \times \alpha(693 - 293)]$

$V_2 = 3769.91[1 + 3 \times 31 \times 10^{-6} \times (693 - 293)]$

$V_2 = 4207\text{mm}^3$

So, the increase in the volume 4207mm^3.

4. Correct option: (B) 4:3.

The expansion is given by,

Expansion= $l\alpha\Delta T$

Given that the expansions are the same for the same ΔT, we obtain

$l_1\alpha_1 = l_2\alpha_2$

So, $\frac{l_1}{l_2} = \frac{\alpha_2}{\alpha_1} = \frac{4}{3}$

So, In order for the expansions to be identical when heated through the same temperature range, the ratio of the rods' initially lengths is 4:3.

5. Correct option: (A) $2\times10^{-4}/\,^{o}\text{C}$**.**

The length of the wire lengthens when heated and stretches outward, increasing the size of the ring. Consequently, at the same proportional increase, the ring's gap widens.

So, if 1cm become 1.02cm, 60cm will become 61.2cm.

So, change in length = $61.2\text{cm} - 60\text{cm} = 1.2\text{cm}$.

original length of 60cm and temperature change $= 100^{o}\text{C}$

Putting these values in the formula of $\alpha = \frac{dl}{l \times dl}$, Where, $dl =$ Change is length and $dt =$ Change in temperature, we get, $\alpha = 2\times10^{-4}/\,^{o}\text{C}$.

So, the linear expansion coefficient of material of wire is $2\times10^{-4}/\,^{o}\text{C}$.

6. Correct option:(A) $30\times10^{-6}/\,^{o}\text{C}$**.**

Given:

$T_0 = 0$

$T_1 = 100^{o}\text{C}$

$r_0 = 100\text{cm}$

$r_t = 100.1\text{cm}$

Now according to thermal expansion:

$l_t = l_0(1+\alpha\Delta T)$

$2\pi r_t = 2\pi r_0(1+\alpha\Delta T)$

$r_t = r_0(1+\alpha\Delta T)$

$100.1 = 100(1+\alpha\Delta T)$

$\frac{100.1}{100} = 1+\alpha(100-0)$

$\alpha = 1\times10^{5}/\,^{o}\text{C}$

The sphere's cubical expansion coefficient is three times more than its linear expansion coefficient,

$\gamma = 3\alpha$

$\gamma = 3\times10^{-5} = 30\times10^{-6} / {}^{o}\text{C}$

So, Coefficient of cubical expansion of the sphere is $30\times10^{-6} / {}^{o}\text{C}$.

7. Correct option: (A) 2.59 sec.

$\Delta T = 20 - 25 = 5^{o}\text{C}$

$\alpha = 0.000012^{o}\text{C}^{-1} = 12\times10^{-6\,o}\text{C}^{-1}$

Time lost per day,

Time lost per day $= = \frac{1}{2}\alpha\Delta T \times 864000\,\text{sec}$

Time lost per day $= \frac{1}{2}\times12\times10^{-6}\times864000\,\text{sec}$

Time lost per day $= 2.590\,\text{sec}$

So, the error in seconds per day will be 2.59 sec.

8. Correct options: (A) and (D).

Since only the type of material, starting length, and temperature increase affect linear expansion.

9. Correct options: (A), (C) and (D).

Since the linear expansion coefficient depends solely on temperature. Different temperature ranges, material length, and material form have no effect on the coefficient of linear expansion.

10. Correct option: (ABD)

11. Correct options: (C) and (D).

Thermostats are modern devices that sense the surrounding temperature and maintain a preset temperature. The typical construction consists of two metallic strips with marginally varying thermal expansion coefficients. This causes the bimetallic strip to flex, which, at high enough temperatures, can be exploited to disrupt an electrical circuit.

12. Correct options: (A) and (D).

Increase in diameter, $\Delta T = D\alpha$

So, increase in circumference = $\pi\Delta D = \pi D\alpha$

We also see that the area expansion coefficient $= 2\alpha$

So, change in area =Initial area $\times 2\alpha = \frac{\pi D^2}{4}\times2\times\alpha = \frac{\pi D^2\alpha}{2}$

13. Correct option: (A)150.24 cm.

Initial length of composite, $L = 50+100 = 150\text{cm}$

Length of iron rod at 100°C $= 50\left[1+12\times10^{-6}\times\left(100-20\right)\right] = 50.048\text{cm}$

Length of aluminium rod at 100°C = $= 50\left[1+24\times10^{-6}\times\left(100-20\right)\right] = 100.192\text{cm}$

So, final length of the composite at $100^{\circ}\text{C} = 50.048+100.192 = 150.24\text{cm}$

14. Correct option: (A) 0.24cm**.**

Initial length of composite, $L = 50+100 = 150\text{cm}$

Length of iron rod at 100°C $= 50\left[1+12\times10^{-6}\times\left(100-20\right)\right] = 50.048\text{cm}$

Length of aluminium rod at 100°C = $= 50\left[1+24\times10^{-6}\times\left(100-20\right)\right] = 100.192\text{cm}$

So, final length of the composite at $100^{\circ}\text{C} = 50.048+100.192 = 150.24\text{cm}$

So, the change in length of the composite system, $\Delta L = L' - L = 150.24 - 150 = 0.24\text{cm}$

15. Correct option: (C) $20\times10^{-6}/\,^{o}\text{C}$

Initial length of composite, $L=50+100=150\text{cm}$

Length of iron rod at 100°C $=50\left[1+12\times10^{-6}\times\left(100-20\right)\right]=50.048\text{cm}$

Length of aluminium rod at $100^{\circ}\text{C}=$ $=50\left[1+24\times10^{-6}\times\left(100-20\right)\right]=100.192\text{cm}$

So, final length of the composite at $100^{\circ}\text{C}=50.048+100.192=150.24\text{cm}$

So, the change in length of the composite system, $\Delta L=L'-L=150.24-150=0.24\text{cm}$

Now, average coefficient of expansion is at 100°C given by,

$$\alpha=\frac{\Delta L}{L\times\Delta T}=\frac{0.24}{150\times\left(100-20\right)}=20\times10^{-6}/\,^{o}\text{C}$$

So, average coefficient of expansion is at 100°C is $20\times10^{-6}/\,^{o}\text{C}$.

16. Correct option: (B) $26\times10^{-6}\text{C}^{-1}$**.**

The coefficient of linear expansion of nickel is, $\alpha=13\times10^{-6}\text{C}^{-1}$

The coefficient of areal expansion of nickel,

$\beta=2\alpha$

$\beta=2\times13\times10^{-6}\text{C}^{-1}$

$\beta=26\times10^{-6}\text{C}^{-1}$

So, the coefficient of areal expansion is $26\times10^{-6}\text{C}^{-1}$.

17. Correct option: (C) $57\times10^{-6}/^{o}\text{C}$**.**

The coefficient of linear expansion of nickel is, $\alpha=19\times10^{-6}/^{o}\text{C}$

The coefficient of volume expansion (γ) for brass using the given values of α:

$\gamma=3\alpha$

$\gamma=3\times19\times10^{-6}/^{o}\text{C}$

$\gamma=57\times10^{-6}/^{o}\text{C}$

So, the coefficient of volume expansion is $57\times10^{-6}/^{o}\text{C}$.

18. Correct option: (B) 0.0051m**.**

The coefficient of linear expansion of nickel is, $\alpha=17\times10^{-6}/^{o}\text{C}$

Original length of the copper wire, $L=10\text{m}$

Change in temperature, $\Delta T=60-30=30^{o}\text{C}$

Using the formula for linear expansion, you can determine how much a copper wire lengthens when heated from 30 to 60 degrees Celsius:

$\Delta L=\alpha L\Delta T$

$\Delta L=17\times10^{-6}\times10\times30=0.0051\text{m}$

So, increase in length is 0.0051m.

19. Correct Matching is: a-S; b-R; c-PT; d-Q

20. Correct Matching is: (a)-(Q),(b(-(R),(c)-(P),(d)-(S).

(a) Thermal expansion is the change in dimensions (length, area, or volume) of a material due to temperature fluctuations. It varies with material properties, object size, and temperature change. Larger objects and materials with higher coefficients of thermal expansion expand more when heated, a fundamental consideration in engineering and construction to prevent structural issues.

(b) The three coefficients of thermal expansion ($\left(\alpha,\beta\text{ and }\gamma\right)$) are totally determined by the composition of the material. They are inherent qualities unique to each material

and are unaffected by changes in the size or temperature of the object. These coefficients, which are distinctive features of the material itself, indicate how a material reacts to temperature changes.

(c) A bimetallic strip has two separate metal layers that are joined together. Because each metal has a different coefficient of thermal expansion, they all expand differently when heated. The strip bends as a result, which makes it perfect for controlling a watch's balance wheel. The strip's curvature adapts to temperature fluctuations to maintain accurate timekeeping.

(d) Pendulum clocks are made of the low thermal expansion steel known as Invar. The pendulum's steady length and negligible expansion and contraction with temperature variations assure the clock's accuracy while enabling reliable timekeeping under a variety of environmental circumstances.

21. Correct Answer: 6

Linear expansion, $\alpha_1 = 6\times10^{-6}\ /\ ^{o}\text{C}$

Linear expansion, $\alpha_2 = 4\times10^{-6}\ /\ ^{o}\text{C}$

For, $\alpha_1 = 6\times10^{-6}\ /\ ^{o}\text{C}$

We know that,

$\Delta L = L\alpha\Delta T$

$\Delta L = 40\times6\times10^{-6}\times\Delta T\cdots\cdots(1)$

For, $\alpha_2 = 4\times10^{-6}\ /\ ^{o}\text{C}$

$\Delta L = L\alpha\Delta T$

$\Delta L = L\times4\times10^{-6}\times\Delta T\cdots\cdots(2)$

Now using equation (1) and (2),

$$\frac{\Delta L}{\Delta L} = \frac{40\times6\times10^{-6L}\times\Delta T}{L\times4\times10^{-6L}\times\Delta T}$$

$L\times4 = 40\times6$

$L = 60\text{cm}$

So, the length of the other rod is 60cm or $6\times10^{1}\text{cm}$.

The value of n is 6.

Expansion of Liquids and Gases

LEVEL-I

1. Correct option: (B) $1\times10^{-3}/^{o}\text{C}$**.**

We know that,

$V = V_0\left(1+\gamma\Delta T\right)$

$$\frac{1}{5}V = \frac{1}{6}V\left(1+\gamma\times200\right)$$

$$\frac{6}{5} = 1+\gamma\times200$$

$$200\gamma = \frac{6}{5}-1 = \frac{1}{5}$$

$$\gamma = \frac{1}{5\times200}$$

$\gamma = 1\times10^{-3}/^{o}\text{C}$

So, the coefficient of apparent expansion of the liquid is $1\times10^{-3}/^{o}\text{C}$.

2. Correct option: (D) $0.00005/^{\circ}\mathrm{C}$.

Given:

$$\text{Expelled mass}=\frac{1}{50}\text{mass remaining}$$

$$\Delta T = 40^{\circ}\mathrm{C}$$

We know that,

$$\gamma_{app}=\frac{\text{Mass expelled}}{\text{Mass remains}\times\Delta\text{T}}$$

Now,

$$\gamma_{app}=\frac{\text{Mass expelled}}{\text{Mass remains}\times\Delta\text{T}}$$

$$\gamma_{app}=\frac{\frac{x}{50}}{x\times 40}$$

$$\gamma_{app}=0.00005/^{\circ}\mathrm{C}$$

So, Coefficient of apparent expansion of the liquid is $0.00005/^{\circ}\mathrm{C}$.

3. Correct option: (A) $0.000155/^{\circ}\mathrm{C}$.

We know that,

$$\gamma_R=\gamma_a+3\alpha$$

$$0.000182=\gamma_a+3(0.000009)$$

$$\gamma_a=\frac{0.000182}{3(0.000009)}$$

$$\gamma_a=0.000155/^{\circ}\mathrm{C}$$

So, the coefficient of apparent expansion of liquid is $0.000155/^{\circ}\mathrm{C}$.

4. Correct option: (C) $10^3/^{\circ}\mathrm{C}$.

The amount of liquid expelled is $\frac{1}{40}$th.

$$\gamma_a=\frac{x}{(m-x)\Delta T}$$

Where, $x=$The amount of liquid expelled.

$\Delta T=$Temperature difference$=(100-75)=25^{\circ}\mathrm{C}$

$$\gamma_a=\frac{1}{40}\times\frac{1}{25}$$

$$\gamma_a=10^{-3}/^{\circ}C$$

So, the coefficient of apparent expansion of the liquid is$10^3/^{\circ}\mathrm{C}$

5. Correct option:

The ratio of liquid spilled over to liquid still inside the bottle for every degree that the temperature changes is known as the coefficient of apparent expansion.

$$\gamma_{app}=\frac{\text{Massexpelled}}{\text{Massremains}\times\Delta\text{T}}$$

Let us consider mass spilled is x.

$$\gamma_{app}=\frac{x}{(400-x)(90)}$$

$$x=5.46g$$

So, the mass of the mercury 5.46 g.

Note: Option is not available.

6. Correct option: (A) 15.2 ml.

Amount spilled out=Difference in volumes increased.

Volume of mercury increased $= V\gamma\Delta T$

$\Rightarrow 1000\times 1.82\times 10^{-4}\times 100 = 18.2\text{ml}$

So increased volume is 18.2 ml.

Cubical expansion coefficient is given by,

Cubical expansion coefficient= $3\times$ linear coefficient

Increase in capacity of glass $= 1000\times 3\times 10^{-6}\times 10 = 3\text{ml}$

Mercury spilled out= $= 18.2-3 = 15.2\text{ml}$

So, the amount of mercury expelled is 15.2ml.

7. Correct option: (D) $0.00033\%/^{\circ}\text{C}$**.**

Mass = density$\times$volume

$m = \rho\times V$

$V = \dfrac{m}{\rho}$

$\ln V = \ln m - \ln \rho$ Now,

$\Rightarrow 0 = \dfrac{\Delta m}{m} - \dfrac{\Delta\rho}{\rho}$

$\Delta\rho = \rho\alpha_{\rho}\Delta T$

$\Rightarrow \dfrac{\Delta m}{m} = \dfrac{\Delta\rho}{\rho}$

$\Rightarrow \dfrac{\Delta m}{m} = \alpha_{\rho}\Delta T$

$\Rightarrow \dfrac{2}{82} = \alpha_{\rho}\Delta T$

$\Rightarrow \alpha_{\rho} = \dfrac{2}{82\Delta T}\cdots\cdots(1)$

$\Delta T = 85-10 = 75$

Putting in equation(1),

$\alpha_{\rho} = \dfrac{2}{82\times 75} = 0.00033\%/^{\circ}\text{C}$

So, Coefficient of apparent expansion of liquid is $0.00033\%/^{\circ}\text{C}$.

8. Correct option: (B) $\dfrac{1}{21}\gamma_r$**.**

We know that,

$\gamma_r = \gamma_a + \gamma_g$

$\gamma_r = \gamma_a + 3\alpha_g\cdots\cdots(1)$

$\gamma_a = \dfrac{6}{7}\gamma_r$

Substitute γ_a in equation (1),

$\dfrac{1}{7}\gamma_r = 3\alpha_g$

$\alpha_g = \dfrac{1}{3\times 7}\gamma_r = \dfrac{1}{21}\gamma_r$

So, the coefficient of linear expansions of the vessel is $\frac{1}{21}\gamma_r$.

9. Correct option: (D) $0.0001/^{o}\mathrm{C}$**.**

Coefficient of real expansion, $\gamma_{\text{real}} = \gamma_{\text{apparent}} + \gamma_g$

The ratio of liquid spilled over to liquid still inside the bottle for every degree that the temperature changes is known as the coefficient of apparent expansion.

$$\gamma_{app} = \frac{\text{Mass expelled}}{\text{Mass remains} \times \Delta T}$$

$$\gamma_{app} = \frac{1000-990}{990\times 100}$$

$$\gamma_{app} = 0.01\times 10^{-2} / {}^{o}\mathrm{C}$$

$$\gamma_{glass} = \gamma_{real} - \gamma_{apparent}$$

$$\gamma_{glass} = 0.0002 - 0.0001$$

$$\gamma_{glass} = 0.0001/^{o}\mathrm{C}$$

So, the coefficient of cubical expansion of glass is $0.0001/^{o}\mathrm{C}$.

10. Correct option: (D) $18\times 10^{-5}/^{o}\mathrm{C}$**.**

AU-tube is filled with mercury.

Let P_0 be atmospheric pressure.

Condition of pressure equilibrium:

$$P_0 + \rho_0 g h_1 = P_0 + \rho_{100} g h_2$$

$$\rho_0 h_1 = \rho_{100} h_2 \cdots\cdots (1)$$

If there is change in density.

$$\rho_0 h_1 = \frac{\rho_0}{1+\gamma\Delta\theta} h_2 \cdots\cdots (1)$$

$$h_1 = \frac{h_2}{1+\gamma(100-0)}$$

$$1+\gamma\times 100 = \frac{h_2}{h_1} = \frac{101.8}{100}$$

$$\gamma = 18\times 10^{-5}/^{o}\mathrm{C}$$

So, the coefficient of real expansion of mercury is $18\times 10^{-5}/^{o}\mathrm{C}$.

11. Correct option: (B) 48 gm.

Original volume , $V_0 = 50\text{gm}$

We know that coefficient volume expansion is given by,

$\gamma = 3\alpha$

Therefore,

$$\left(V_{glass}\right)_{fixed} = 3\alpha V_0 \Delta T + V_0$$

$$\left(V_{final}\right)_{liquid} = 3\times 8\times 10^{-6}\times 80 V_0 + V_0$$

$$\left(V_{final}\right)_{liquid} = 192\times 10^{-5} V_0 + V_0$$

$$\left(V_{final}\right)_{liquid} = 40\times 10^{-3} V_0 + V_0$$

Now,

$\Delta V = \left\{\left(4000\times10^{-5}\right)-\left(192\times10^{-5}\right)\right\}V_0$

$\Delta V = 3508\times10^{-5} = 0.038V_0$

$\Delta V = 1.9\text{gm}$

Now mass left is ,

$\text{Mass left} = 50-1.9 = 48.1\text{gm} \approx 48\text{gm}$

So, the mass of the liquid it holds at 80°C is 48 gm.

12. Correct option: (D) $1\times10^{-4}/^{o}\text{C}$**.**

We know that,

$\gamma_r = \gamma_a + \gamma_g = \gamma_a + 3\alpha_g$

For both vessel A and B,

$\gamma_r = \gamma_{a1} + 3\alpha_{g1} = \gamma_{a2} + 3\alpha_2$

$\Rightarrow 18\times10^{-4} + 3\alpha_1 = 21\times10^{-4} + 3\alpha_2$

$\Rightarrow 3(\alpha_1 - \alpha_2) = 3\times10^{-4}$

$\Rightarrow \alpha_1 - \alpha_2 = 1\times10^{-4}/\,^{o}C$

So, the difference in the coefficients of linear expansions of A and B is $1\times10^{-4}/^{o}\text{C}$.

Anomalous Expansion of Water:

13. Correct option: (B) 4°C**.**

Water is forced into a standard cage-like structure with small gaps below zero degrees Celsius, which causes the volume of the ice to increase.

At 4°C, the ice's cage-like structure begins to break down, allowing molecules to move past one another and fill in any vacant spaces. As a result, water volume temporarily falls. Density is inversely related to volume, therefore as volume decreases, density rises.

When the temperature exceeds 4°C, the extra space required by the water molecules' increased motion begins to exceed the space gained as a result of structural changes, and the

At 4 degrees Celsius, water has its highest density.

14. Correct option: (D)First decreases and then increases.

Water (ρ) has a maximum density of $1\frac{\text{g}}{\text{cm}^3}$ at 4°C. While the mass of water does not change with temperature, the density will decrease as the temperature goes from greater than or below 4°C.

15. Correct option: (B) More dense.

When water in a lake cools to the point when it is about to freeze, the more dense water sinks to the bottom.

At about 4 degrees Celsius (39.2 degrees Fahrenheit), water becomes the densest. Below this temperature, water begins to constrict and get denser. It expands and loses density as it approaches the freezing point and starts to solidify into ice, which is why ice floats on the water's surface. Water cools and becomes denser before to freezing, which causes it to sink to the lake's bottom. Thermal stratification is the term for this process.

16. Correct option: (D) Anomalous expansion of water.

When liquid is frozen, it has the ability to expand.

Longer storage in the freezer will cause the water in the bottle to expand. The bottle's walls may experience significant strain from this expansion, which leads to the bottle breaking. This is because water expands abnormally when it freezes, contrary to the behaviour of most other substances, which shrinks as they are cooled.

17. Correct option: (B) Water expands from $4^{o}C$ to $0^{o}C$.

Water experiences abnormal expansion when it is cooled from $4^{o}C$ to $0^{o}C$.
In turn, the glass jar will break when it is sealed.

18. Correct option: (A) Falls on heating.

When a vessel is heated, it expands, causing the liquid level to decrease.

Variation of Density with Temperature:

19. Correct option:

The density is given by as,

$$\rho = \frac{\rho_0}{1 + r_0 t}$$

$$\rho = \frac{13.6}{1 + 0.18 \times 10^{-6} \times 473}$$

$$\rho = 12.53 \frac{\text{gm}}{\text{c.c}}$$

So, the density is $12.53 \frac{\text{gm}}{\text{c.c}}$.

Note: option is not available.

20. Correct option: (C)1.0006.

For this case:

$$\frac{d_{20}}{d_{40}} = 1 + \gamma dt$$

Where,

$d_{20} = \text{Density at } 20^{\circ}C$

$d_{40} = \text{Density at } 40^{\circ}C$

$dt = 20$

$\gamma = 3\alpha$

Substitute the all value we get,

$$\frac{d_{20}}{d_{40}} = 1.0006$$

So, the ratio is 1.0006.

21. Correct option: (B) $8.99 \frac{\text{gm}}{\text{cc}}$.

Given:

$$\rho_1 = 9 \frac{\text{gm}}{\text{cc}}$$

$$\alpha = 17 \times 10^{-6} / {}^{o}\text{C}$$

$$\Delta T = 20 - 0 = 20^{\circ}C$$

The formula for volume expansion due to temperature change and density can be used to determine the density of copper at 20°C given its density at 0°C and the coefficient of linear expansion:

$$\rho_2 = \frac{\rho_1}{(1+\alpha\Delta T)}$$

$$\rho_2 = \frac{9}{(1+17\times10^{-6}\times20)}$$

$$\rho_2 = 8.99\frac{\text{gm}}{\text{cc}}$$

So, the density is $8.99\frac{\text{gm}}{\text{cc}}$.

22. Correct option: (A)

Coefficient of Volume & Pressure Expansion of Gases:

23. Correct option: (C) -123°C**.**

Given:

$p_1 = 30\text{atm}$

$p_2 = 1\text{atm}$

$v_1 = v$

$v_2 = 15v$

$T_1 = 27^{o}\text{C}=27_273=300^{o}\text{K}$

We know that,

$$\frac{p_1v_1}{T_1} = \frac{p_2v_2}{T_2}$$

$$T_2 = \frac{p_2v_2v_1}{p_1v_1}$$

Substituting the value we get,

$$T_2 = \frac{1\times15v\times300}{30\times v} = 150^{\circ}\text{K}$$

$$T_2 = 150-273 = -123^{\circ}\text{C}$$

So, the final temperature of the gas is -123°C.

24. Correct option: (D) 927°C**.**

We know that,

$$\frac{p_1v_1}{T_1} = \frac{p_2v_2}{T_2}$$

$$\frac{pv}{300} = \frac{2p\times2v}{T_2}$$

$$T_2 = 1200K$$

$$T_2 = 1200-273 = 927^{\circ}C$$

So, the final temperature attained by the gas is 927°C.

25. Correct option: (B) $\frac{1}{320}/^{o}\text{C}$**.**

Gas has a $100\,\text{cc}$ volume at 20°C and a $125\,\text{cc}$ volume at 100°C. The process is conducted under constant pressure.

Let v_i and v_f represent the beginning and final volumes, respectively, and let T represent the temperature difference.

$\Delta T = 100 - 20 = 80^{\circ}\text{C}$

We know that,

$v_f = v_i\left(1+\gamma\Delta T\right)$

Put,

$v_f = 100\,\text{cc}$

$v_i = 125\,\text{cc}$

$\Delta T = 100 - 20 = 80^{\circ}\text{C}$

Now we get,

$125 = 100\left(1+80\gamma\right)$

$\gamma = \frac{1}{320}/^{\circ}\text{C}$

So, the volume coefficient of the gas is $\frac{1}{320}/^{\circ}\text{C}$

26. Correct option: (B) 327°C**.**

Given:

$p_1 = p_2 = 75\text{cm}$

$p_2 = 2p_0 = 2p_1$

We know that,

$\frac{T_1}{p_1} = \frac{T_2}{p_2}$

As, $p_2 = 2p_1$

$\frac{T_1}{p_1} = \frac{T_2}{2p_1}$

$T_2 = 2T_1$

$T_2 = 2\times 300 = 600\text{K}$

$T_2 = 600\text{-}273 = 327^{\circ}\text{C}$

So, the required temperature is 327°C.

27. Correct option: (C) 25cm^3**.**

Given:

$p_1 = 1\text{atm}$

$v_1 = 100\text{cc}$

$p_2 = 4\text{atm}$

We know that,

$\frac{1\times 100}{T} = \frac{4\times v_2}{T}\left(T_1 = T_2 = T\right)$

$v_2 = 25\,\text{cm}^3$

So, the volume is 25cm^3.

28. Correct option: (C) $\frac{1}{273}$**.**

At constant pressure is,

$V \propto T$

$\frac{\Delta V}{V} = \frac{\Delta T}{T}$

Consequently, the volume rise per degree ratio. kelvin temperature increase to its initial volume,

$\frac{\frac{\Delta V}{\Delta T}}{V} = \frac{1}{T} = \frac{1}{273}$

So, the ratio of increase in volume of an ideal gas per degree rise in Kelvin temperature to its original volume is

$\frac{1}{273}$.

29. Correct option: (B) $\frac{1}{300}/^{\circ}\mathrm{C}$**.**

Gas has a $100\,\mathrm{cc}$ volume at $20^{\circ}\mathrm{C}$ and a $125\mathrm{cc}$ volume at $100^{\circ}\mathrm{C}$. The process is conducted under constant pressure.

Let v_i and v_f represent the beginning and final volumes, respectively, and let T represent the temperature difference.

$\Delta T = 100 - 20 = 80^{o}C$

We know that,

$v_f = v_i\left(1+\gamma\Delta T\right)$

Put,

$v_f = 100\mathrm{cc}$

$v_i = 125\mathrm{cc}$

$\Delta T = 100 - 20 = 80^{o}C$

Now we get,

$125 = 100\left(1+80\gamma\right)$

$\gamma = \frac{1}{320}/^{\circ}\mathrm{C}$

So, the volume coefficient of the gas is $\frac{1}{320}/^{\circ}\mathrm{C}$

The option's closest match is (C) $\frac{1}{300}/^{\circ}\mathrm{C}$.

30. Correct option: (B) 1000 cc.

Using the ideal gas equation:

$Pv = nRt$

$PV = \text{constant}$

$760 \times 500 = 380 \times v$

$v = \frac{760 \times 500}{380}$

$v = 1000\mathrm{cc}$

So, the volume is 1000 cc.

LEVEL-II

1. Correct option: (B) $\gamma = 2\alpha$ **.**

$A_0 =$ Area of the X-direction at $0^{\circ}\mathrm{C}$

$A_\theta =$ Area of the X-direction $\theta^{\circ}\mathrm{C}$

$V_0 =$ Volume of liquid at $0^{\circ}\mathrm{C}$

$V_\theta =$ volume of liquid at $\theta^\circ C$

$l =$ length of liquid column(constant)

$V_0 = lA_0$

$V_\theta = lA_\theta$

$A_\theta = A_0(1+2\alpha\theta)$

$V_\theta = V_0(1+\lambda\theta)$

$lA_\theta = lA_0(1+\gamma\theta)$

$A_\theta = A_0(1+2\alpha\theta) =_0 (1+\gamma\theta)$

$\gamma = 2\alpha$

So, the relation between the coefficient of volume expansion and be the coefficient of linear expansion is $\gamma = 2\alpha$.

2. Correct option: (B) 1.016m.

$A_0 =$ Area cross section of the tube X-direction at $0^\circ C$

$A_\theta =$ Area-cross section of the tube X-direction $\theta^\circ C$

$V_0 =$ Volume of mercury thread at $0^\circ C$

$V_\theta =$ volume of mercury thread at $\theta^\circ C$

We know that, length of thread at $\theta^\circ C$ is given by as:

$$l_\theta = \frac{V_\theta}{A_\theta} = \frac{V_0(1+r_m\theta)}{A_0(1+\beta\theta)}$$

$$= l_0(1+r_m\theta)(1+\beta\theta)^{-1}$$

$$= l_0[1+(r_m-\beta)\theta]$$

$$= l_0\left[1+(r_m-\frac{2r_g}{3})\theta\right]$$

$$= 100\left[1+(0.000182-\frac{2}{3}(0.00025)100\right]$$

$= 101.6cm = 1.016m$

So, the required length is 1.016m.

3. Correct option: (A) $627^\circ C$**.**

We know that,

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

Here,

$V_1 = V$

$V_2 = 3V$

Substitute all the value we get,

$$T_2 = \frac{V_2T_1}{V_1}$$

$$T_2 = \frac{3V\times 300}{V}\left(T_1 = 27+273 = 300K\right)$$

$T_2 = 900K$

$T_2 = 900-273 = 627^\circ C$

So, the final temperature of the gas will be $627^\circ C$.

4. Correct option: (D) 100K.

We know that,

$$\frac{p_1 v_1}{T_1} = \frac{p_2 v_2}{T_2}$$

$$T_2 = \frac{p_2 v_2 v_1}{p_1 v_1}$$

Given:

$p_1 = 30atm$

$p_2 = 1atm$

$T_1 = 27 + 273 = 300^o K$

$v_1 = v$

$v_2 = 10v$

Substitute the value we get,

$$\frac{30v}{300} = \frac{1 \times 10V}{T_2}$$

$$T_2 = 100\text{K}$$

So, the final temperature is 100 K.

5. Correct options: (A) and (C).

Here volume is constant,

$p \propto T$

$$\frac{p_1}{p_2} = \frac{T_1}{T_2}$$

$$\frac{p}{p + 0.04p} = \frac{T + 273}{T + 283}$$

$$\frac{1}{1.04} = \frac{T + 273}{T + 284}$$

$T = 250^o C$

$T = 273 + 250 = 523K$

So, , the initial temperature of the gas is 250°C or 523K.

6. Correct options: (B) and (D).

Here volume is constant,

$p \propto T$

$$\frac{p_1}{p_2} = \frac{T_1}{T_2}$$

$$\frac{p}{p + 0.005p} = \frac{T + 273}{T + 275}$$

$$\frac{1}{1.005} = \frac{T + 273}{T + 275}$$

$T = 400^{\circ}\text{C}$

$T = 273 + 400 = 673\text{K}$

So, , the initial temperature of the gas is 400°C or 673K.

7. Correct options: (B) and (D).

Water steadily becomes denser as it cools until it reaches a temperature of 4°C. Water has the maximum density and minimum volume at 4°C. Water expands as it cools further, causing its density to drop from 4°C to 0°C.

4°C in kelvin scale= $= 4 + 273 = 277\text{K}$

8. Correct options: (A),(B) and (D).

A) The reason the water in the can freezes is because the ice around it causes the temperature of the water to go below the freezing point.

B) Heat transfer from the water to the surrounding ice causes a drop in water temperature.

D) As water molecules lose energy to the cooler environment, their kinetic energy drops.

9. Correct options: (C) Volume increases and density decreases with temperature.

The density is given by as,

$$\text{Density}=\frac{\text{mass}}{\text{volume}}$$

Density will decrease if mass remains constant while volume rises.

A liquid's molecules are always moving at random, or in Brownian motion. Given that the motion is random, the liquid's overall effective motion is zero.

When a liquid is heated, the molecules that make up the liquid absorb the heat, gain kinetic energy, and move more quickly. The molecules will thus clash more frequently and with greater intensity. In other words, when the temperature rises, the liquid will occupy more volume while the amount of molecules making up the liquid stays the same. This is because the intermolecular separation will also increase.

The volume of the liquid grows with temperature while the liquid's mass stays constant. As a result, raising the temperature will cause the density to drop.

10. Correct option: (A) $\frac{(1+60\gamma Fe)}{(1+60\gamma Hg)}$**.**

Let us consider volume at $T=0^{o}\text{C}$ is V_0 and volume at $T=60^{o}\text{C}$ is V.

$V=V_0(1+60\gamma Fe)\cdots\cdots(1)$

We can write if $V_0^{'}$ and V' are the volumes of the iron block submerged in the mercury at $T=0^{o}\text{C}$ and $T=60^{o}\text{C}$, respectively.

$V'=V_0^{'}(1+60\gamma Hg)\cdots\cdots(2)$

Dividing (1) by (2),

$$\frac{V}{V'}=\frac{V_0(1+60\gamma Fe)}{V_0^{'}(1+60\gamma Hg)}$$

$$\frac{1}{k_2}=\frac{1}{k_1}\frac{(1+60\gamma Fe)}{(1+60\gamma Hg)}\left(\because \frac{V_0^{'}}{V_0}=k_1,\frac{V'}{V}=k_2\right)$$

$$\frac{k_1}{k_2}=\frac{(1+60\gamma Fe)}{(1+60\gamma Hg)}$$

11. Correct option: (B)

12. Correct option: (B) $8.33\times10^{-4}/^{o}\text{C}$**.**

The buoyant force at that temperature must be equal to the body's weight for it to float with 75% of its weight submerged in liquid.

$\text{Mg}=0.75\text{V}\rho_{20^{o}}g$

Similarly, the buoyant force at that temperature must be equal to the body's weight for it to float with 80% of its weight submerged in liquid.

$$\frac{\rho_{100^\circ}}{\rho_{20^\circ}} = \frac{0.75}{0.80} = \frac{1}{1+\gamma\Delta T}$$

$$\gamma = 8.33\times10^{-4}/^{\circ}\mathrm{C}$$

So, the coefficient of real expansion of the liquid is $8.33\times10^{-4}/^{\circ}\mathrm{C}$.

13. Correct option: (A) 3.44 cc.

Given:

$V_m = 250\text{c.c}$

$\gamma_m = 1.8\times10^{-4}/^{\circ}\mathrm{C}$

$\Delta T = 110-20 = 90^{\circ}\mathrm{C}$

$\gamma_g = 9\times10^{-6}/^{\circ}\mathrm{C}$

Expansion of mercury= $V_m\times\gamma_m\times\Delta T = 250\times1.8\times10^{-4}\times90\cdots\cdots(1)$

where V_m, γ_m, and ΔT stand for the mercury's initial volume, absolute expansion coefficient, and temperature increase, respectively.

Expansion of glass container= $V_g\times\gamma_3\times\Delta T = 250\times9\times10^{-6}\times90\cdots\cdots(2)$

where V_g, γ_g, and ΔT stand for the container's initial volume, the glass's coefficient of volume expansion, and the temperature increase.

The mercury spill over volume, ΔV, is calculated by deducting (2) from (1).

$$\Delta V = 250\times90\times\{1.8-0.27\}\times10^{-4} = 3.44\,\text{cc}$$

So, the coefficient of absolute expansion of mercury is 3.44 cc.

14. Correct option: (A) 4 gm.

We know that,

$$\gamma_a = \frac{m_2-m_1}{m_1\Delta T}$$

For given problem:

$$\frac{m_2-m_1}{m_1(T_2-T_1)} = \frac{m_2-m_1}{m_1(T_2-T_1)}$$

$$\frac{2}{52} = \frac{m_2-m_1}{104}$$

$$m_2-m_1 = 4\text{gm}$$

So, the mass of the liquid expelled is 4 gm.

15. Correct option: (B)

16. Correct option: (B) 400 cc.

Given:

$V = 300\text{cc}$

$\alpha = \frac{1}{300}\,^{\circ}\mathrm{C}^{-1}$

$\Delta T = 100-0 = 100^{\circ}\mathrm{C}$

Volume is given by as,

$V^1 = V(1+\alpha\Delta T)$

$$V^1 = 300\left(1+\frac{1}{300}\times100\right)$$

$V^1 = 400\text{cc}$

So, the volume is 400cc.

17. Correct option: (C)

18. Correct option: (a-R; b-S, c-P, d-Q)

19. Correct option: (a-S, b-Q, c-P, d-R)

20. Correct Answer is: 7

Given:

$\gamma_L = 7\gamma_s$

Apparent coefficient of volume expansion:

$\gamma = \gamma_L - \gamma_S = 7\gamma_S - \gamma_S = 6\gamma_S$

Ratio between the liquid's perceived and absolute expansion:

$$\frac{\gamma_L}{\gamma_{app}} = \frac{7\gamma_s}{6\gamma_6} = \frac{7}{6}$$

the ratio of absolute and apparent expansion of the liquid is $\frac{7}{6}$, then $x = 7$.

21. Correct Answer is: 1

Both glass and mercury will expand when heated, but mercury does so more so than glass, causing some mercury to flow out.

Coefficient of volume expansion of glass will be, $\gamma = 3\alpha = 0.3\times10^{-4}/^{\circ}\text{C}$

Now we know that,

$\Delta V = \gamma \times V \times \Delta T$

Mercury expansion will is, $1.30\times10^{-4}\times100\times1000=13\text{cc}$

Vessel expansion is $0.3\times10^{-4}\times100\times1000=3\text{cc}$

Hence, volume of mercury spilled is, $13\text{cc}-3\text{cc}=10\text{cc}$.

the volume of mercury spill out is $10\times1\text{cc}$ .Then $x = 1$.

Heat Capacity and Calorimetry

LEVEL-I

1. Correct option:(B) Specific heat capacity.

The term "specific heat capacity" refers to the quantity of heat energy needed to raise a substance's temperature per unit mass by one degree Celsius.

If a substance of mass m temperature rises by t when the heat of Q is applied, then,

$Q = cm\Delta t$

Where, $c =$ Specific heat capacity

If, $m = 1, \Delta t = 1^{\circ}C$ then, $Q = c$

Therefore, specific heat is the amount of energy needed to increase a unit mass of a substance's temperature by one degree Celsius.

2. Correct option: (C) $\frac{J}{\text{kgK}}$.

B) $\frac{J}{\text{kgK}}$, which stands for joules per kilogramme per kelvin, is the SI unit for specific heat.

3. Correct option: (D) All the above.

If a substance of mass m temperature rises by t when the heat of Q is applied, then,

$Q = cm\Delta t$

So, the specific heat capacity is given by,

$$c = \frac{1}{m}\frac{Q}{t}$$

We can infer from the relationship above that the specific heat is inversely proportional to temperature change, or more specifically, its temperature and mass, and proportionate to the quantity of heat absorbed or rejected. However, we are also aware that different materials have varying specific heats. Thus, it relies on its mass, temperature, and nature, we can conclude.

4. Correct option: (C) Both A and B.

Nature of the Substance: Due to variances in their specific heat capacities, different substances have varied heat capacities.

Body mass - The heat capacity is inversely correlated with body mass. In order to raise a mass' temperature by a specific amount, more heat is needed.

5. Correct option: (B) High specific heat.

Since water has a high specific heat capacity and can absorb a lot of heat, it is utilised as coolant in automobile radiators. Due to its high specific heat capacity, water requires more heat to get hot than other liquids do.

6. Correct option: (D) Mercury.

For mercury, the minimum specific heat is in liquids.

7. Correct option: (B) 1:16.

Thermal capacity=volume $\times$Specific heat

Mass = volume $\times$density

Volume for a sphere is inversely proportional to radius cube.

$TC \propto r^3$

$$\frac{TC_1}{TC_2} = \frac{r_1^3 \rho_1 C_1}{r_2^3 \rho_2 C_2}$$

$$\frac{TC_1}{TC_2} = \left(\frac{1}{2}\right)^3 \left(\frac{2}{3}\right)\left(\frac{3}{4}\right)$$

$$\frac{TC_1}{TC_2} = \frac{1}{16}$$

So, the ratio of their thermal capacities 1:16.

8. Correct option: (A) $50\frac{\text{cal}}{^{o}\text{C}}$**.**

$m = 500g$

$s = 0.1\frac{\text{cal}}{\text{g}^{\circ}\text{C}}$

$T_1 = 30^{\circ}C$

$T_2 = 290^{\circ}C$

Thermal heat capacity is given by ,

$H = ms$

$H = 500 \times 0.1$

$H = 50\frac{\text{cal}}{^{o}\text{C}}$

So, the thermal capacity of the block is $50\frac{\text{cal}}{^{o}\text{C}}$.

9. Correct option: (B) 3:2.

Given:

$\rho_A = 0.5\frac{\text{g}}{\text{cc}^3} = 5000\frac{\text{kg}}{\text{m}^3}$

$\rho_A = 0.6\frac{\text{g}}{\text{cc}^3} = 6000\frac{\text{kg}}{\text{m}^3}$

Mass of A= $=8\times10^{-3}\times5000=40\text{kg}$

Mass of B $=10\times10^{-3}\times6000=60\text{kg}$

So,

$C_A \times 40 = C_B \times 60$

$\frac{C_A}{C_B} = \frac{6}{4} = \frac{3}{2}$

So, The specific heat ratio is 3:2.

10. Correct option: (C)0.05.

Given:

Specific Heat= $0.09\frac{\text{cal}}{\text{gm}^\circ\text{C}}$

Now we know that,

$T(c) = \frac{5}{9}\left(T(f) - 32\right)$

$\frac{9T(c)}{5} = T(f) - 32$

If, $T(c) = 1^\circ\text{C}$

$T(f) - 32 = \frac{9}{5} = 1.8$

If, $0.09\frac{\text{cal}}{\text{gm}^\circ\text{C}}$,therefore,

$\frac{0.09}{1.8} = 0.05\frac{\text{cal}}{\text{gm}^\circ\text{F}}$

If the temperature is measured in Fahrenheit scale the value of its specific heat in $\frac{\text{cal}}{\text{gm}^\circ\text{F}}$ is $0.05\frac{\text{cal}}{\text{gm}^\circ\text{F}}$.

Change Of State:

11. Correct option: (B) Latent heat.

The statement is accurate; latent heat is the amount of heat a substance absorbs during a change in condition without increasing its temperature.

Latent heat is thermal energy that is heated or absorbed by a body or thermodynamic system while the temperature remains constant—typically during a first order phase transition.

12. Correct option: (C) $L = \frac{Q}{m}$**.**

latent heat is the amount of heat a substance absorbs during a change in condition without increasing its temperature.

The latent heat is given by as,

$L = \frac{Q}{m}$

13. Correct option: (C) Remains same.

The Archimedes principle states that a floating object displaces some liquid, thus as the ice melts, the water level won't change since the melted ice will occupy the same volume as it did before.

14. Correct option: (B) Regelation.

The act of pressing two pieces of ice together causes the ice to melt owing to the increased pressure, fusing the two surfaces together. This process, known as regelation, results in the ice melting due to the increased pressure and solidifying into a single piece.

15. Correct option: (D) B & C.

Molar heat capacity, which can be positive or negative, determines the amount of heat supplied throughout a process. If positive, then heat is gained otherwise it is lost, e.g for polytropic process $PV^x = \text{Constant}$ with $x > \gamma$ temperature increase is accompanied by heat release.

Now according to ideal gas equation, $PV = nRT$. An rise in T indicates an increase in P or V, whichever is greater.

PV work is negative when P increases and V decreases.

16. Correct option: (B) Specific heat.

If Q is the heat delivered to a mass m, specific heat s, and temperature rise, respectively, Then,

$$Q = ms\Delta t$$

$$\Delta t = \frac{Q}{ms}$$

So a substance's temperature rise is caused by the quantity "ms". Specific capacity is the name given to this amount.

17. Correct option: (B) Due to evaporation of sweat.

The comforting effect of a fan is due to perspiration evaporation. As perspiration evaporates, thermal energy is released through evaporation. results in those physical regions feeling cool.

18. Correct option: (C) Increasing the average distance between the molecules.

The heat applied to a body during the melting process is used to lengthen the average distance between molecules, altering the physical state or phase of the body. Latent heat of fusion of the body is the name given to this heat that does not cause an increase in body temperature.

19. Correct option: (C) Lowering of melting point of ice at the bottom due to high pressure.

The tremendous pressure at an iceberg's base actually decreases rather than raises the melting point of ice. This behaviour is in line with the laws of thermodynamics and the way pressure affects the water phase diagram.

With increasing pressure, water's melting point drops.

$$P = \rho gh$$

Because of the intense pressure near the glacier's base, the melting point drops there. This causes the ice to melt at the base while remaining frozen at the top. The melting point of ice can be decreased by increasing pressure.

20. Correct option: (D) 30136J**.**

The amount of heat needed to turn ice into water at $0^\circ C$,

$$Q_1 = mL = 10^{-2} \times 0.355 \times 10^{-6} = 3350\text{J}$$

The amount of heat needed to raise water's temperature from $0^\circ\text{C}\,\text{to}\,100^\circ\text{C}$,

$Q_2 = mS(12-1) = 10^{-2} \times 4186 \times 100 = 4186\text{J}$

The amount of heat needed to turn water into steam at 100°C,

$Q_3 = mL_2 = 10^{-2} \times 2.26 \times 100 = 22600\text{J}$

Total amount oh heat is given by,

$Q = Q_1 + Q_2 + Q_3$

$Q = 3350 + 4185 + 22600$

$Q = 30136J$

So, the specific heat of water is 30136J.

21. Correct option: (B) 0.813g.

Given:

Latent heat of steam, $L_s = 540\frac{\text{cal}}{g}$

Specific heat of water, $L_w = 1\frac{\text{cal}}{\text{g}^{\circ}\text{C}}$

Mass of water, $m_w = 100g$

According to the combination approach or using the calorimetry principle, we can write,

Heat lost by steam =heat gained by water

$m_s L_s + m_s S_w(100 - t) = m_s S_w(t - 20)$

$m_s \times 540 + m \times 1(100 - 25)$

$100 \times 1 \times (25 - 20)$

$615m_s = 500$

$m_s = \frac{500}{615} = 0.813\text{g}$

So, the amount of steam is 0.813g.

22. Correct option: (B) 4 gm.

Let us consider $x\,g$ steam get condensed into water at 100°C.

Q=mL

$\text{Q=x} \times 540\,\text{cal} \cdots\cdots (1)$

Ice at 0°C is currently turning water at 100°C into ice by absorbing heat. necessary heat is,

$\Rightarrow 6 \times 80 + 6(100) = x \times 540$

$\Rightarrow 480 + 600 = x \times 540$

$\Rightarrow x = \frac{1089}{540}$

$\Rightarrow x = 2g$

Mass of steam left un condensed =6-2- =4gm.

23. Correct option:(A) $\frac{M_1S_1+M_2S_2}{M_1+M_2}$.

Let us consider, M_1, M_2 are masses and S_1, S_2 are their respective specific heats.

Total mass of mixture $= M_1 + M_2$

Now, we can write this,

$(M_1 + M_2)S = M_1S_1 + M_2S_2$

Where, S is the specific heat of the mixture.

$S = \frac{M_1S_1 + M_2S_2}{M_1 + M_2}$

So, the specific heat of the mixture is $\frac{M_1S_1+M_2S_2}{M_1+M_2}$

24. Correct option: (A) 3:2.

Let us consider liquid-1 has specific heat s_1 at $40^\circ C$.

Similarly, liquid-2 has specific heat s_2 at $30^\circ C$.

Common temperature is $36^\circ C$ and both have equal mass.

Now,

Heat lost by liquid-1 is,

$Q_1 = ms_1\Delta t = ms_1(40-60) = 4ms_1$

Heat gained by liquid -2 is $Q_2 = ms_2\Delta t$

$Q_2 = ms_2 \times (36-30) = 6m_2$

According to the principle of energy conservation, both should be equal.

$Q_1 = Q_2$

$4ms_1 = 6ms_2$

$\frac{s_1}{s_2} = \frac{6}{4} = \frac{3}{2}$

$s_1 : s_2 = 3:2$

So, the ratio of their specific heats is 3:2.

25. Correct option: (C) $40^\circ C$**.**

Latent heat is ice, $L_{ice} = 366\frac{KJ}{kg}$

Latent heat of steam is , $L_{steam} = 2260\frac{KJ}{kg}$

Mass of steam, $M_{steam} = 0.01kg$

Specific heat of water, $S_{water} = 4.18\frac{KJ}{kg^oC}$

Mass of ice, $M_{ice} = 0.05kg$

We know that,

Heat Loos from steam = heat gain by ice

$M_{steam}L_{steam} + M_{steam}S_{water}\left(100 - T_f\right) = M_{ice}L_{ice} + M_{ice}S_{water}$

$T_f = \frac{M_{steam}L_{steam} - M_{ice}L_{ice} + M_{steam}S_{water} \times 100}{\left(M_{ice} + M_{steam}\right)S_{water}}$

$T_f = \frac{0.0 \times 2260 - 0.05 \times 336 + 0.01 \times 4.2 \times 100}{(0.01 + 0.05) \times 4.2}$

$T_f = 39.6^oC \approx 40^oC$

So, final temperature is $40^\circ C$.

26. Correct option: (C) $10^\circ C$**.**

Let us consider ice mass is m.

$L_f = 80\frac{cal}{g}$

Initial temperature $=0^oC$

Resulting temperature $=t$

Mass of wate is m.

Initial temperature = 100^oC

Final temperature $= t^{\circ}\text{C}$

heat lost by water = heat gained by ice to become water at 0°C + heat gained by 0°C water to raise its temperature $t^{\circ}\text{C}$.

$m \times 1 \times (100 - t) = 80m + m \times 1 \times (t - 0)$

$m(100 - t) = m(80 + t)$

$100 - t = 80 + t$

$200 = 2t$

$t = 10^{\circ}\text{C}$

So, the resultant temperature is 10°C.

27. Correct option: (A) 45°C**.**

Given:

$t_1 = 70^{\circ}C$

$t_2 = 20^{\circ}C$

Here, $m_1 = m_2 = 10\text{kg}$

Since mass of 1 litre of water is 1kg.

$s_1 = s_2 = 4200\dfrac{\text{J}}{\text{kgK}}$

We know that,

$t = \dfrac{m_1 s_1 t_1 + m_2 s_2 t_2}{m_1 s_1 + m_2 s_2}$

$t = \dfrac{10 \times 4200 \times 70 + 10 \times 4200 \times 20}{10 \times 4200 + 10 \times 4200}$

$t = 45^{\circ}\text{C}$

So, the resultant temperature of the water is 45°C

LEVEL-II

1. Correct option: (A) 3360.

Given:

$m = 10\text{g} = 10 \times 10^{-3}\text{kg} = 0.01\text{kg}$

$\Delta Q = 8\dfrac{\text{cal}}{\text{c}}$

Specific heat is given by,

Specific heat,

$s = \dfrac{T(c)}{m}$

$s = \dfrac{8}{0.01}$

$s = 800\dfrac{\text{cal}}{\text{kg-K}}$

We know that,

1cal=4.186J

Now,

$s=800\times4.186$

$s=3360\frac{\text{J}}{\text{kg-k}}$

So, the specific heat is $3360\frac{\text{J}}{\text{kg-k}}$.

2. Correct option: (B) $z=8y$**.**

We know that,
Thermal capacity is given by,

$H=ms$

$H=\rho\frac{4}{3}\pi r^2 s$

$\frac{H_1}{H_2}=\frac{\rho_1\times s_1\times r_1^3}{\rho\times s_2\times r_2^3}$

$\frac{H_1}{H_2}=\frac{z}{x}\times\frac{x}{y}\times\frac{1}{8}$

$z=8y$

So, the ratio of their thermal capacities is $z=8y$.

3. Correct option: (C) $\frac{yt_2}{xt_1}$**.**

We know that,
For substance:

$\Delta Q=ms_1\Delta T=x\times s_1\times t_1\cdots\cdots(1)$

For water:

$\Delta Q=ms_2\Delta T=x\times s_2\times t_2\cdots\cdots(2)$

Using equation (1) and (2) we get,

$x\times s_1\times t_1=x\times s_2\times t_2$

$\frac{s_1}{s_2}=\frac{yt_2}{xt_1}$

Where, $s_1=$water specific heat of substance

$s_2=$Specific heat of water

So, The ratio of specific heats of the substances is $\frac{yt_2}{xt_1}$.

4. Correct option: (D) 35°C**.**

We know that,

$\theta=\frac{m_1s_1\theta_1+m_2s_2\theta_2}{m_1s_1+m_2s_2}$

$\theta=\frac{(3m\times4s\times60)+(4m\times20\times5s)}{(3m\times4s)+(4m\times5s)}$

$\theta=\frac{(720+400)}{(12+20)}=35^\circ\text{C}$

So, the two liquids are mixed, the resultant temperature is 35°C.

5. Correct option: (A) 0°C **.**

Heat is released when 50 g of water heated to 60°C is transformed into water cooled to 30°C .

$Q = ms\Delta\theta = 50\times1\times60 = 3000\text{cal}$

A mass of ice that can melt as a result of this heat.

$m = \frac{Q}{L} = \frac{3000}{80} = 37.5\text{g}$

Specifically, the entire block of ice is not melted, and the combination will be 0°C .

6. Correct option: (B) 0.1.

Latent heat of fusion , $m = 80\frac{\text{cal}}{\text{g}} = 336\frac{\text{J}}{\text{g}}$

$m_{ice} = 25g = 25\times10^{-3}$

Formula for balancing the heat is,

$mc\Delta T = L\times m_{ice}$

$0.2\times100\times C = 336\times10^{3}\times25\times10^{-3}$

$C = 420\frac{\text{J}}{\text{kg}^{\circ}\text{C}}$

$C = 0.1\frac{\text{cal}}{\text{g}^{\circ}\text{C}}$

So, then the specific heat of iron is $0.1\frac{\text{cal}}{\text{g}^{\circ}\text{C}}$.

7. Correct option: (A) 30 g.

Since, the ice block is very large, so all the poured water cools down to 0°C and the heat rejected in cooling process is taken by ice and it will melt from 0°C ice to 0°C water.

Thus, according to the principle of energy conservation, if m mass of ice melts,

$m\times L = m_{w}(T_{i} - T_{f})$

$m\times80 = 80\times1\times30$

$m = 30g$

So, the amount of ice that melts is 30g.

8. Correct option: (A) 10°C **.**

θ is temperature.

$\theta_{\max}$ is given by: $\theta_{\max} = \frac{\theta_{w} - \frac{L_{i}}{c_{w}}}{2}$

Putting values:

$\theta_{\max} = \frac{\theta_{w} - \frac{L_{i}}{c_{w}}}{2} = \frac{100 - \frac{80}{1}}{2} = 10^{o}\text{C}$

So, the final temperature of the mixture is 10°C .

9. Correct option: (A).

During the conversion process:

i) increasing the temperature of ice from -10 °C to 0 °C; gradual T.

ii) The transition from ice to water at 0 degrees Celsius with no change in temperature

iii) iii) Raise water temperature from zero degrees Celsius to one hundred degrees Celsius; raise T

iv) Changing from liquid to steam at 100 degrees Celsius with no change in temperature

10. Correct option: (B) and (D).

Let final temperature be T.

$ms(2\theta - T) = m \times 2s(T - \theta)$

$T = \frac{4\theta}{3}$

Heat capacity $= ms$

Ratio of heat capacities for same mass = 1:2.

11. Correct option: (A) and (B).

The latent heat of ice fusion is $80\frac{\text{cal}}{\text{g}}$ and the latent heat of water vaporisation is $540\frac{\text{cal}}{\text{g}}$.

12. Correct option: (A)

The assertion is accurate since the calorimetric law states that heat released by hot bodies equals heat received by cold bodies.

The idea of energy conservation can also be used to understand it.

A) Heat lost = heat gained by the body

B) It represents the law of conservation of heat energy

Statement B is true because calorimetry is founded on the idea of energy conservation.

Though it appears to be an imperfect representation of the entire equation, Statement C is also true. The full equation for calorimetry is frequently expressed as:

$Q = mc\Delta T$

Where:

The quantity Q stands for the volume of heat absorbed or emitted.

The substance's mass is m.

The substance's specific heat capacity is given as c.

ΔT stands for temperature change.

13. Correct option: (B) $2520\frac{\text{J}}{\text{kgK}}$**.**

Mass of lead piece, $m = 25\text{g} = 0.025\text{kg}$

Amount of heat given out=1200 calories

Heat energy given out, $dQ = 1200 \times 4.2\text{J}$

Specific heat,

$s = \frac{1}{m}\frac{dQ}{dT} = \frac{1}{0.025} \times \frac{1200 \times 4.2}{80} = \frac{5040}{2} = 2520\frac{\text{J}}{\text{kgK}}$

So, the specific heat is $2520\frac{\text{J}}{\text{kgK}}$.

14. Correct option: (C) $63\frac{J}{K}$.

Mass of lead piece, $m = 25\text{g}=0.025\text{kg}$

Amount of heat given out=1200 calories

Heat energy given out, $dQ = 1200\times 4.2\text{J}$

Specific heat,

$$s = \frac{1}{m}\frac{dQ}{dT} = \frac{1}{0.025}\times\frac{1200\times 4.2}{80} = \frac{5040}{2} = 2520\frac{J}{\text{kgK}}$$

Thermal capacity $= ms = 0.025\times 2520 = 63\dfrac{\text{J}}{\text{K}}$

15. Correct option: (D) 0.015kg **.**

Mass of lead piece, $m = 25\text{g}=0.025\text{kg}$

Amount of heat given out=1200 calories

Heat energy given out, $dQ = 1200\times 4.2\text{J}$

Specific heat,

$$s = \frac{1}{m}\frac{dQ}{dT} = \frac{1}{0.025}\times\frac{1200\times 4.2}{80} = \frac{5040}{2} = 2520\frac{J}{\text{kgK}}$$

Thermal capacity $= ms = 0.025\times 2520 = 63\frac{J}{K}$

Water equivalent= $\dfrac{63}{4200}\text{kg}=0.015\text{kg}$

16. Correct Matching is: (a)-(R),(b)-(T),(c)-(S),(d)-(Q).

(a) AB represents the rise in temperature of solid from 0°C to $t_1^{o}\text{C}$.

(b) CD represents the rise in temperature of liquid from $t_1^{\circ}\text{C}$ to $t_3^{o}\text{C}$.

(c) BC represents the melting at temperature $t_1^{o}\text{C}$.

(d) DE represents boiling at temperature $t_3^{o}\text{C}$.

17. Correct Answer is:

According to thermodynamics,

Heat loss by water = heat gain by ice

It is given that,

Mass of water, $m = 80\text{gm}$

Specific heat of water, $s = 1\dfrac{\text{cal}}{\text{g}^{\circ}\text{C}}$

So, heat lost by water during cooling from 30°C to 0°C.

$$\Delta H = ms\Delta T$$
$$\Delta H = 80\times 1\times(30-0)$$
$$\Delta H = 2400\text{cal}\cdots\cdots(1)$$

Heat gain by ice, $m_1 l = m_1\times 80\dfrac{\text{cal}}{\text{gm}}\cdots\cdots(2)$

Where, L is the latent heat of fusion.

m_l is the mass of ice melted.

Using equation (1) and (2),

$2400 = 80m_l$

$m_l = 30\text{g}$

Hence,30g of ice is melted.